

AlgoVision

Generated by Doxygen 1.9.8

1 Namespace Index	1
1.1 Namespace List	1
2 Hierarchical Index	3
2.1 Class Hierarchy	3
3 Class Index	5
3.1 Class List	5
4 File Index	7
4.1 File List	7
5 Namespace Documentation	9
5.1 AppConstants Namespace Reference	9
5.1.1 Variable Documentation	9
5.1.1.1 BaseEdgeWidth	9
5.1.1.2 btnSize	10
5.1.1.3 btnSpacing	10
5.1.1.4 createGraphDialogMinHeight	10
5.1.1.5 createGraphDialogMinWidth	10
5.1.1.6 defaultBorderWidth	10
5.1.1.7 defaultRadius	10
5.1.1.8 EdgePadding	10
5.1.1.9 graphPageDefaultTitle	10
5.1.1.10 graphTypeDialogErrorMinHeight	10
5.1.1.11 graphTypeDialogErrorMinWidth	10
5.1.1.12 legendMinHeight	11
5.1.1.13 MaxNodeScale	11
5.1.1.14 MinNodeScale	11
5.1.1.15 NodeScale	11
5.1.1.16 optionBtnSize	11
5.1.1.17 optionIconSize	11
5.1.1.18 startPageTitle	11
5.1.1.19 windowHeight	11
5.1.1.20 windowMinHeight	11
5.1.1.21 windowMinWidth	11
5.1.1.22 windowWidth	11
5.2 Ui Namespace Reference	11
6 Class Documentation	13
6.1 AddEdgeCommand Class Reference	13
6.1.1 Constructor & Destructor Documentation	15
6.1.1.1 AddEdgeCommand()	15
6.1.2 Member Function Documentation	15

6.1.2.1 redo()	15
6.1.2.2 undo()	15
6.1.3 Member Data Documentation	15
6.1.3.1 m_c	15
6.1.3.2 m_edgId	15
6.1.3.3 m_from	15
6.1.3.4 m_hasId	16
6.1.3.5 m_to	16
6.1.3.6 m_weight	16
6.2 AddNodeAndEdgeCommand Class Reference	16
6.2.1 Constructor & Destructor Documentation	18
6.2.1.1 AddNodeAndEdgeCommand()	18
6.2.2 Member Function Documentation	18
6.2.2.1 redo()	18
6.2.2.2 undo()	18
6.2.3 Member Data Documentation	18
6.2.3.1 m_c	18
6.2.3.2 m_fromId	18
6.2.3.3 m_hasId	18
6.2.3.4 m_newId	18
6.2.3.5 m_pos	19
6.3 AddNodeCommand Class Reference	19
6.3.1 Constructor & Destructor Documentation	21
6.3.1.1 AddNodeCommand()	21
6.3.2 Member Function Documentation	21
6.3.2.1 redo()	21
6.3.2.2 undo()	21
6.3.3 Member Data Documentation	21
6.3.3.1 m_c	21
6.3.3.2 m_hasId	21
6.3.3.3 m_id	21
6.3.3.4 m_pos	21
6.4 Algorithm Class Reference	22
6.4.1 Constructor & Destructor Documentation	23
6.4.1.1 Algorithm()	23
6.4.1.2 ~Algorithm()	23
6.4.2 Member Function Documentation	23
6.4.2.1 addStep()	23
6.4.2.2 clearSteps()	23
6.4.2.3 execute()	23
6.4.2.4 getSteps()	23
6.4.2.5 resultString()	23

6.4.3 Member Data Documentation	24
6.4.3.1 m_graph	24
6.4.3.2 m_resultString	24
6.4.3.3 m_steps	24
6.5 AlgorithmTab::AlgorithmConfig Struct Reference	24
6.5.1 Member Function Documentation	25
6.5.1.1 operator!=(())	25
6.5.1.2 operator==(())	25
6.5.2 Member Data Documentation	25
6.5.2.1 m_algorithmName	25
6.5.2.2 m_endNode	25
6.5.2.3 m_startNode	25
6.6 AlgorithmController Class Reference	25
6.6.1 Detailed Description	28
6.6.2 Constructor & Destructor Documentation	28
6.6.2.1 AlgorithmController()	28
6.6.2.2 ~AlgorithmController()	28
6.6.3 Member Function Documentation	29
6.6.3.1 clear()	29
6.6.3.2 isFinished	29
6.6.3.3 loadSteps	29
6.6.3.4 nextStep	29
6.6.3.5 onAlgorithmError	29
6.6.3.6 prevStep	29
6.6.3.7 redo()	29
6.6.3.8 requestErrorDialog	29
6.6.3.9 reset	29
6.6.3.10 resultString()	30
6.6.3.11 setResultString	30
6.6.3.12 undo()	30
6.6.4 Member Data Documentation	30
6.6.4.1 m_applier	30
6.6.4.2 m_currentIndex	30
6.6.4.3 m_redoStack	30
6.6.4.4 m_resultString	30
6.6.4.5 m_steps	30
6.6.4.6 m_undoStack	30
6.7 AlgorithmError Struct Reference	31
6.7.1 Detailed Description	31
6.7.2 Member Data Documentation	31
6.7.2.1 m_message	31
6.7.2.2 m_type	31

6.8 AlgorithmStep Struct Reference	32
6.8.1 Detailed Description	32
6.8.2 Member Data Documentation	32
6.8.2.1 m_from	32
6.8.2.2 m_node	33
6.8.2.3 m_prevEdgeState	33
6.8.2.4 m_prevNodeState	33
6.8.2.5 m_to	33
6.8.2.6 m_type	33
6.9 AlgorithmStepApplier Class Reference	33
6.9.1 Detailed Description	34
6.9.2 Constructor & Destructor Documentation	34
6.9.2.1 AlgorithmStepApplier()	34
6.9.3 Member Function Documentation	34
6.9.3.1 apply()	34
6.9.3.2 check()	34
6.9.3.3 resetGraphState()	34
6.9.3.4 setGraph()	35
6.9.3.5 stepToEdgeState()	35
6.9.3.6 stepToNodeState()	35
6.9.3.7 undo()	35
6.9.4 Member Data Documentation	35
6.9.4.1 m_graph	35
6.10 AlgorithmTab Class Reference	35
6.10.1 Member Enumeration Documentation	38
6.10.1.1 RunState	38
6.10.2 Constructor & Destructor Documentation	39
6.10.2.1 AlgorithmTab()	39
6.10.2.2 ~AlgorithmTab()	39
6.10.3 Member Function Documentation	39
6.10.3.1 graphEditAllowedChanged	39
6.10.3.2 initIcons()	39
6.10.3.3 initLayout()	39
6.10.3.4 makeLegendItem()	39
6.10.3.5 selectedConfig()	39
6.10.3.6 showAlgorithmErrorDialog	40
6.10.3.7 startTimerForPlay()	40
6.10.3.8 updateControls()	40
6.10.3.9 updateLegendForAlgorithm()	40
6.10.3.10 updateUiForAlgorithm()	40
6.10.4 Member Data Documentation	40
6.10.4.1 m_algorithmCombo	40

6.10.4.2 m_algorithmController	40
6.10.4.3 m_applier	40
6.10.4.4 m_currentConfig	40
6.10.4.5 m_endLabel	41
6.10.4.6 m_endNodeEdit	41
6.10.4.7 m_endRow	41
6.10.4.8 m_graphController	41
6.10.4.9 m_graphEditAllowed	41
6.10.4.10 m_helpBtn	41
6.10.4.11 m_legendContainer	41
6.10.4.12 m_legendLayout	41
6.10.4.13 m_legendScroll	41
6.10.4.14 m_nextBtn	41
6.10.4.15 m_noInputLabel	42
6.10.4.16 m_pauseBtn	42
6.10.4.17 m_playBtn	42
6.10.4.18 m_prevBtn	42
6.10.4.19 m_restartBtn	42
6.10.4.20 m_startLabel	42
6.10.4.21 m_startNodeEdit	42
6.10.4.22 m_startRow	42
6.10.4.23 m_state	42
6.10.4.24 m_timer	42
6.10.4.25 m_worker	43
6.11 AlgorithmWorker Class Reference	43
6.11.1 Detailed Description	45
6.11.2 Constructor & Destructor Documentation	45
6.11.2.1 AlgorithmWorker()	45
6.11.3 Member Function Documentation	45
6.11.3.1 algorithmErrorOccurred	45
6.11.3.2 resultReady	45
6.11.3.3 run()	45
6.11.3.4 stepsReady	45
6.11.4 Member Data Documentation	46
6.11.4.1 m_algorithm	46
6.11.4.2 m_end	46
6.11.4.3 m_graph	46
6.11.4.4 m_start	46
6.12 AStar Class Reference	46
6.12.1 Constructor & Destructor Documentation	49
6.12.1.1 AStar()	49
6.12.2 Member Function Documentation	49

6.12.2.1 aStar()	49
6.12.2.2 checkConditions()	49
6.12.2.3 execute()	50
6.12.2.4 getPath()	50
6.12.2.5 getTotalCost()	50
6.12.2.6 heuristic()	50
6.12.2.7 resultString()	50
6.12.3 Member Data Documentation	50
6.12.3.1 m_path	50
6.12.3.2 m_totalCost	50
6.13 BellmanFord Class Reference	51
6.13.1 Constructor & Destructor Documentation	53
6.13.1.1 BellmanFord()	53
6.13.2 Member Function Documentation	53
6.13.2.1 bellmanFord()	53
6.13.2.2 checkConditions()	53
6.13.2.3 execute()	54
6.13.2.4 hasNegativeCycle()	54
6.13.2.5 resultString()	54
6.13.3 Member Data Documentation	54
6.13.3.1 m_hasNegativeCycle	54
6.13.3.2 m_minDistance	54
6.13.3.3 m_resultString	54
6.14 BFS Class Reference	55
6.14.1 Constructor & Destructor Documentation	57
6.14.1.1 BFS()	57
6.14.2 Member Function Documentation	57
6.14.2.1 bfs()	57
6.14.2.2 checkConditions()	57
6.14.2.3 execute()	57
6.14.2.4 getVisited()	58
6.14.2.5 resultString()	58
6.14.3 Member Data Documentation	58
6.14.3.1 m_order	58
6.14.3.2 m_visited	58
6.15 ClearGraphCommand Class Reference	58
6.15.1 Constructor & Destructor Documentation	60
6.15.1.1 ClearGraphCommand()	60
6.15.2 Member Function Documentation	60
6.15.2.1 capture()	60
6.15.2.2 redo()	60
6.15.2.3 undo()	60

6.15.3 Member Data Documentation	60
6.15.3.1 m_c	60
6.15.3.2 m_edges	60
6.15.3.3 m_nodes	60
6.16 DFS Class Reference	61
6.16.1 Constructor & Destructor Documentation	63
6.16.1.1 DFS()	63
6.16.2 Member Function Documentation	63
6.16.2.1 checkConditions()	63
6.16.2.2 dfs()	63
6.16.2.3 execute()	63
6.16.2.4 getVisited()	64
6.16.2.5 resultString()	64
6.16.3 Member Data Documentation	64
6.16.3.1 m_order	64
6.16.3.2 m_visited	64
6.17 Dijkstra Class Reference	64
6.17.1 Constructor & Destructor Documentation	67
6.17.1.1 Dijkstra()	67
6.17.2 Member Function Documentation	67
6.17.2.1 checkConditions()	67
6.17.2.2 dijkstra()	67
6.17.2.3 execute()	67
6.17.2.4 resultString()	68
6.17.3 Member Data Documentation	68
6.17.3.1 m_resultString	68
6.18 DirectedEdgeItem Class Reference	68
6.18.1 Constructor & Destructor Documentation	72
6.18.1.1 DirectedEdgeItem()	72
6.18.2 Member Function Documentation	72
6.18.2.1 arrowPath()	72
6.18.2.2 boundingRect()	72
6.18.2.3 edgePath()	72
6.18.2.4 paint()	72
6.18.2.5 shape()	73
6.18.3 Member Data Documentation	73
6.18.3.1 m_arrowAngle	73
6.18.3.2 m_arrowSize	73
6.18.3.3 m_skewness	73
6.19 Edge Class Reference	73
6.19.1 Constructor & Destructor Documentation	76
6.19.1.1 Edge()	76

6.19.1.2 <code>~Edge()</code>	76
6.19.2 Member Function Documentation	76
6.19.2.1 <code>endNode()</code>	76
6.19.2.2 <code>getId()</code>	77
6.19.2.3 <code>getState()</code>	77
6.19.2.4 <code>getWeight()</code>	77
6.19.2.5 <code>setState()</code>	77
6.19.2.6 <code>setWeight()</code>	77
6.19.2.7 <code>startNode()</code>	77
6.19.3 Member Data Documentation	77
6.19.3.1 <code>m_id</code>	77
6.19.3.2 <code>m_idEndNode</code>	77
6.19.3.3 <code>m_idStartNode</code>	77
6.19.3.4 <code>m_state</code>	78
6.19.3.5 <code>m_weight</code>	78
6.20 EdgItem Class Reference	78
6.20.1 Constructor & Destructor Documentation	82
6.20.1.1 <code>EdgItem()</code>	82
6.20.1.2 <code>~EdgItem()</code>	82
6.20.2 Member Function Documentation	82
6.20.2.1 <code>adjust()</code>	82
6.20.2.2 <code>adjustPointsGeometry()</code>	82
6.20.2.3 <code>adjustWeightGeometry()</code>	82
6.20.2.4 <code>boundingRect()</code>	82
6.20.2.5 <code>calculateColor()</code>	82
6.20.2.6 <code>calculateNormal()</code>	82
6.20.2.7 <code>connectNodes()</code>	83
6.20.2.8 <code>disconnectNodes()</code>	83
6.20.2.9 <code>edgePath()</code>	83
6.20.2.10 <code>edgeSelected</code>	83
6.20.2.11 <code>editEdgeWeightRequest</code>	83
6.20.2.12 <code>getEdgeCenter()</code>	83
6.20.2.13 <code>getWeightPosition()</code>	83
6.20.2.14 <code>hasWeight()</code>	83
6.20.2.15 <code>initEdgeWeight()</code>	83
6.20.2.16 <code>modelEdge()</code>	84
6.20.2.17 <code>mouseDoubleClickEvent()</code>	84
6.20.2.18 <code>mousePressEvent()</code>	84
6.20.2.19 <code>onEdgeUpdated()</code>	84
6.20.2.20 <code>onEdgeWeightChanged</code>	84
6.20.2.21 <code>paint()</code>	84
6.20.2.22 <code>setHasWeight()</code>	84

6.20.2.23 setModelEdge()	84
6.20.2.24 setWeight()	85
6.20.2.25 shape()	85
6.20.2.26 updateSize()	85
6.20.2.27 weight()	85
6.20.3 Member Data Documentation	85
6.20.3.1 m_destNode	85
6.20.3.2 m_destPoint	85
6.20.3.3 m_hasWeight	85
6.20.3.4 m_modelEdge	85
6.20.3.5 m_observerId	85
6.20.3.6 m_penWidth	86
6.20.3.7 m_shapeStroke	86
6.20.3.8 m_sourceNode	86
6.20.3.9 m_sourcePoint	86
6.20.3.10 m_weight	86
6.21 EdgeSnapshot Struct Reference	86
6.21.1 Member Data Documentation	87
6.21.1.1 m_from	87
6.21.1.2 m_id	87
6.21.1.3 m_to	87
6.21.1.4 m_weight	87
6.22 EditableTextItem Class Reference	87
6.22.1 Constructor & Destructor Documentation	90
6.22.1.1 EditableTextItem()	90
6.22.2 Member Function Documentation	90
6.22.2.1 center()	90
6.22.2.2 centerText()	90
6.22.2.3 finishEditing()	90
6.22.2.4 keyPressEvent()	90
6.22.2.5 oldText()	90
6.22.2.6 setCenter() [1/2]	90
6.22.2.7 setCenter() [2/2]	91
6.22.2.8 setEditGraphSceneState	91
6.22.2.9 setOldText()	91
6.22.2.10 startEditing()	91
6.22.2.11 textCommitted	91
6.22.3 Member Data Documentation	91
6.22.3.1 m_center	91
6.22.3.2 m_oldText	91
6.23 EditEdgeWeightCommand Class Reference	92
6.23.1 Constructor & Destructor Documentation	94

6.23.1.1 EditEdgeWeightCommand()	94
6.23.2 Member Function Documentation	94
6.23.2.1 redo()	94
6.23.2.2 undo()	94
6.23.3 Member Data Documentation	94
6.23.3.1 m_afterText	94
6.23.3.2 m_afterW	94
6.23.3.3 m_beforeText	94
6.23.3.4 m_beforeW	95
6.23.3.5 m_c	95
6.23.3.6 m_edgeld	95
6.24 EditNodeNameCommand Class Reference	95
6.24.1 Constructor & Destructor Documentation	97
6.24.1.1 EditNodeNameCommand()	97
6.24.2 Member Function Documentation	97
6.24.2.1 redo()	97
6.24.2.2 undo()	97
6.24.3 Member Data Documentation	97
6.24.3.1 m_afterName	97
6.24.3.2 m_beforeName	97
6.24.3.3 m_c	97
6.24.3.4 m_nodeld	98
6.25 FloydWarshall Class Reference	98
6.25.1 Constructor & Destructor Documentation	100
6.25.1.1 FloydWarshall()	100
6.25.2 Member Function Documentation	100
6.25.2.1 checkConditions()	100
6.25.2.2 execute()	100
6.25.2.3 floydWarshall()	100
6.25.2.4 resultString()	101
6.25.3 Member Data Documentation	101
6.25.3.1 m_distances	101
6.26 Graph Class Reference	101
6.26.1 Constructor & Destructor Documentation	103
6.26.1.1 Graph()	103
6.26.1.2 ~Graph()	103
6.26.2 Member Function Documentation	103
6.26.2.1 addEdge()	103
6.26.2.2 addEdgeSerialized()	104
6.26.2.3 addNode() [1/2]	104
6.26.2.4 addNode() [2/2]	104
6.26.2.5 clear()	104

6.26.2.6 fromVariant()	104
6.26.2.7 getAdjacencyList()	104
6.26.2.8 getEdge() [1/4]	104
6.26.2.9 getEdge() [2/4]	104
6.26.2.10 getEdge() [3/4]	105
6.26.2.11 getEdge() [4/4]	105
6.26.2.12 getEdges()	105
6.26.2.13 getEdgesMutable()	105
6.26.2.14 getNode() [1/2]	105
6.26.2.15 getNode() [2/2]	105
6.26.2.16 getNodes()	105
6.26.2.17 getNodesMutable()	105
6.26.2.18 isDirected()	105
6.26.2.19 isWeighted()	106
6.26.2.20 removeEdge()	106
6.26.2.21 removeNode()	106
6.26.2.22 toVariant()	106
6.26.3 Member Data Documentation	106
6.26.3.1 m_adjacencyList	106
6.26.3.2 m_edgId	106
6.26.3.3 m_edges	106
6.26.3.4 m_nodeId	106
6.26.3.5 m_nodes	107
6.26.3.6 m_numOfEdges	107
6.26.3.7 m_numOfNodes	107
6.27 GraphController Class Reference	107
6.27.1 Constructor & Destructor Documentation	111
6.27.1.1 GraphController()	111
6.27.2 Member Function Documentation	111
6.27.2.1 addEdge	111
6.27.2.2 addEdgeNoHistory	111
6.27.2.3 addEdgeNoHistoryById	111
6.27.2.4 addNode	111
6.27.2.5 addNodeAndEdge	111
6.27.2.6 addNodeNoHistory	111
6.27.2.7 addNodeWithIdNoHistory	112
6.27.2.8 buildScene	112
6.27.2.9 clear	112
6.27.2.10 clearNoHistory	112
6.27.2.11 clearScene	112
6.27.2.12 connectScene()	112
6.27.2.13 createGraph	112

6.27.2.14	disableScene	112
6.27.2.15	editEdgeWeight	112
6.27.2.16	editNodeName	113
6.27.2.17	enableScene	113
6.27.2.18	graph()	113
6.27.2.19	moveNode	113
6.27.2.20	moveNodeNoHistory	113
6.27.2.21	removeEdge	113
6.27.2.22	removeEdgeNoHistory	113
6.27.2.23	removeEdgeNoHistoryByld	113
6.27.2.24	removeNode	114
6.27.2.25	removeNodeNoHistory	114
6.27.2.26	resetScene	114
6.27.2.27	restoreEdgeNoHistory	114
6.27.2.28	scene()	114
6.27.2.29	sceneModified	114
6.27.2.30	setAddSceneState()	114
6.27.2.31	setEdgeWeightNoHistory	114
6.27.2.32	setEdgeWeightNoHistoryByld	115
6.27.2.33	setGraph()	115
6.27.2.34	setNodeNameNoHistory	115
6.27.2.35	setRemoveSceneState()	115
6.27.2.36	undoStack()	115
6.27.2.37	updateNodePosition()	115
6.27.3	Member Data Documentation	115
6.27.3.1	m_graph	115
6.27.3.2	m_scene	115
6.27.3.3	m_undoStack	116
6.28	GraphEditor Class Reference	116
6.28.1	Constructor & Destructor Documentation	118
6.28.1.1	GraphEditor()	118
6.28.1.2	~GraphEditor()	118
6.28.2	Member Function Documentation	118
6.28.2.1	graphController()	118
6.28.2.2	onAddRequestTrigger	118
6.28.2.3	onClearRequestTrigger	119
6.28.2.4	onPanRequestTrigger	119
6.28.2.5	onRedoRequestTrigger	119
6.28.2.6	onRemoveRequestTrigger	119
6.28.2.7	onUndoRequestTrigger	119
6.28.2.8	onZoomInRequestTrigger	119
6.28.2.9	onZoomOutRequestTrigger	119

6.28.2.10 redoRequested	119
6.28.2.11 undoRequested	119
6.28.3 Member Data Documentation	119
6.28.3.1 m_editTab	119
6.28.3.2 m_graphController	120
6.28.3.3 m_leftPlaceholder	120
6.28.3.4 m_undoStack	120
6.28.3.5 m_view	120
6.29 GraphEditTab Class Reference	120
6.29.1 Constructor & Destructor Documentation	123
6.29.1.1 GraphEditTab()	123
6.29.1.2 ~GraphEditTab()	123
6.29.2 Member Function Documentation	123
6.29.2.1 addRequested	123
6.29.2.2 applyEditButtonsState()	124
6.29.2.3 clearRequested	124
6.29.2.4 getNodeSizeSlider()	124
6.29.2.5 initLayout()	124
6.29.2.6 panRequested	124
6.29.2.7 redoRequested	124
6.29.2.8 removeRequested	124
6.29.2.9 setGraphEditsAllowed	124
6.29.2.10 setRedoEnabled	124
6.29.2.11 setUndoEnabled	124
6.29.2.12 undoRequested	125
6.29.2.13 zoomInRequested	125
6.29.2.14 zoomOutRequested	125
6.29.3 Member Data Documentation	125
6.29.3.1 m_addBtn	125
6.29.3.2 m_canRedo	125
6.29.3.3 m_canUndo	125
6.29.3.4 m_clearBtn	125
6.29.3.5 m_graphEditsAllowed	125
6.29.3.6 m_nodeSizeSlider	125
6.29.3.7 m_panBtn	125
6.29.3.8 m_redoBtn	126
6.29.3.9 m_removeBtn	126
6.29.3.10 m_undoBtn	126
6.29.3.11 m_zoomInBtn	126
6.29.3.12 m_zoomOutBtn	126
6.30 GraphScene Class Reference	126
6.30.1 Member Enumeration Documentation	129

6.30.1.1 State	129
6.30.2 Constructor & Destructor Documentation	130
6.30.2.1 GraphScene()	130
6.30.3 Member Function Documentation	130
6.30.3.1 addEdge()	130
6.30.3.2 addEdgeRequest	130
6.30.3.3 addNode()	130
6.30.3.4 addNodeAndEdge()	130
6.30.3.5 addNodeAndEdgeRequest	130
6.30.3.6 addNodeRequest	131
6.30.3.7 applyTheme()	131
6.30.3.8 clear()	131
6.30.3.9 disableScene()	131
6.30.3.10 editEdgeWeightRequest	131
6.30.3.11 editNodeNameRequest	131
6.30.3.12 enableScene()	131
6.30.3.13 findEdgeItemById()	131
6.30.3.14 findEdgeItemByNodes()	131
6.30.3.15 findNodeItemById()	132
6.30.3.16 mouseDoubleClickEvent()	132
6.30.3.17 mousePressEvent()	132
6.30.3.18 moveNodeRequest	132
6.30.3.19 onEdgeSelectTrigger	132
6.30.3.20 onNodeSelectTrigger	132
6.30.3.21 removeEdge()	132
6.30.3.22 removeEdgeRequest	132
6.30.3.23 removeNode()	133
6.30.3.24 removeNodeRequest	133
6.30.3.25 resetScene()	133
6.30.3.26 selectNode()	133
6.30.3.27 setEditGraphSceneTrigger	133
6.30.3.28 setState()	133
6.30.3.29 updateNodeScalling()	133
6.30.4 Member Data Documentation	133
6.30.4.1 m_editLabel	133
6.30.4.2 m_firstNodeSelect	133
6.30.4.3 m_nodeItems	134
6.30.4.4 m_previousState	134
6.30.4.5 m_state	134
6.31 GraphView Class Reference	134
6.31.1 Member Enumeration Documentation	137
6.31.1.1 State	137

6.31.2 Constructor & Destructor Documentation	137
6.31.2.1 GraphView() [1/2]	137
6.31.2.2 GraphView() [2/2]	137
6.31.3 Member Function Documentation	138
6.31.3.1 init()	138
6.31.3.2 mouseDoubleClickEvent()	138
6.31.3.3 mouseMoveEvent()	138
6.31.3.4 mousePressEvent()	138
6.31.3.5 mouseReleaseEvent()	138
6.31.3.6 resetState()	138
6.31.3.7 setState()	138
6.31.3.8 state()	138
6.31.3.9 wheelEvent()	138
6.31.3.10 zoomIn()	139
6.31.3.11 zoomOut()	139
6.31.4 Member Data Documentation	139
6.31.4.1 m_lastMousePos	139
6.31.4.2 m_oldState	139
6.31.4.3 m_state	139
6.31.4.4 m_zoomFactor	139
6.32 Kahn Class Reference	140
6.32.1 Constructor & Destructor Documentation	142
6.32.1.1 Kahn()	142
6.32.2 Member Function Documentation	142
6.32.2.1 checkConditions()	142
6.32.2.2 execute()	142
6.32.2.3 getSorted()	142
6.32.2.4 kahn()	142
6.32.2.5 resultString()	143
6.32.3 Member Data Documentation	143
6.32.3.1 m_sorted	143
6.33 LoadFileWorker Class Reference	143
6.33.1 Constructor & Destructor Documentation	144
6.33.1.1 LoadFileWorker()	144
6.33.2 Member Function Documentation	145
6.33.2.1 failed	145
6.33.2.2 loaded	145
6.33.2.3 run()	145
6.33.3 Member Data Documentation	145
6.33.3.1 m_filePath	145
6.33.3.2 m_serializer	145
6.34 MainWindow Class Reference	146

6.34.1 Constructor & Destructor Documentation	148
6.34.1.1 MainWindow()	148
6.34.1.2 ~MainWindow()	148
6.34.2 Member Function Documentation	148
6.34.2.1 closeEvent()	148
6.34.2.2 connectGraphModifiedSignal	149
6.34.2.3 initMenuToolBar()	149
6.34.2.4 onChangeThemeTriggered	149
6.34.2.5 onCreateGraphTriggered	149
6.34.2.6 onGraphLoadedNewTab	149
6.34.2.7 onGraphLoadFailed	149
6.34.2.8 onHelpTriggered	149
6.34.2.9 onOpenGraphTriggered	149
6.34.2.10 onSaveGraphTriggered	149
6.34.2.11 onSaveImageTriggered	150
6.34.2.12 showStartPage()	150
6.34.3 Member Data Documentation	150
6.34.3.1 m_menuToolBar	150
6.34.3.2 m_serializer	150
6.34.3.3 m_tabs	150
6.34.3.4 m_tabWidget	150
6.34.3.5 m_themeManager	150
6.34.3.6 m_ui	150
6.35 MenuToolBar Class Reference	151
6.35.1 Constructor & Destructor Documentation	153
6.35.1.1 MenuToolBar()	153
6.35.1.2 ~MenuToolBar()	153
6.35.2 Member Function Documentation	153
6.35.2.1 changeThemeAction()	153
6.35.2.2 createGraphAction()	153
6.35.2.3 helpAction()	153
6.35.2.4 openGraphAction()	153
6.35.2.5 saveGraphAction()	153
6.35.2.6 saveImageAction()	154
6.35.2.7 setOptionShortcuts()	154
6.35.2.8 setStyle()	154
6.35.3 Member Data Documentation	154
6.35.3.1 m_changeThemeAction	154
6.35.3.2 m_createGraphAction	154
6.35.3.3 m_helpAction	154
6.35.3.4 m_openGraphAction	154
6.35.3.5 m_saveGraphAction	154

6.35.3.6 m_saveImageAction	154
6.36 MoveNodeCommand Class Reference	155
6.36.1 Constructor & Destructor Documentation	157
6.36.1.1 MoveNodeCommand()	157
6.36.2 Member Function Documentation	157
6.36.2.1 redo()	157
6.36.2.2 undo()	157
6.36.3 Member Data Documentation	157
6.36.3.1 m_c	157
6.36.3.2 m_newPos	157
6.36.3.3 m_nodeId	157
6.36.3.4 m_oldPos	158
6.37 Node Class Reference	158
6.37.1 Constructor & Destructor Documentation	160
6.37.1.1 Node()	160
6.37.1.2 ~Node()	160
6.37.2 Member Function Documentation	160
6.37.2.1 getComponentColor()	160
6.37.2.2 getId()	161
6.37.2.3 getName()	161
6.37.2.4 getPosition()	161
6.37.2.5 getState()	161
6.37.2.6 setComponentColor()	161
6.37.2.7 setName()	161
6.37.2.8 setPosition()	161
6.37.2.9 setState()	161
6.37.3 Member Data Documentation	161
6.37.3.1 m_componentColor	161
6.37.3.2 m_id	162
6.37.3.3 m_name	162
6.37.3.4 m_position	162
6.37.3.5 m_state	162
6.38 NodeItem Class Reference	162
6.38.1 Constructor & Destructor Documentation	166
6.38.1.1 NodeItem()	166
6.38.1.2 ~NodeItem()	166
6.38.2 Member Function Documentation	166
6.38.2.1 addEdge()	166
6.38.2.2 boundingRect()	166
6.38.2.3 calculateColor()	166
6.38.2.4 edges()	166
6.38.2.5 editNodeNameRequest	166

6.38.2.6	itemChange()	166
6.38.2.7	label()	167
6.38.2.8	modelNode()	167
6.38.2.9	mouseMoveEvent()	167
6.38.2.10	mousePressEvent()	167
6.38.2.11	mouseReleaseEvent()	167
6.38.2.12	moveNodeRequest	167
6.38.2.13	nodeSelected	167
6.38.2.14	onNameChanged	167
6.38.2.15	onNodeUpdated()	167
6.38.2.16	paint()	168
6.38.2.17	radius()	168
6.38.2.18	removeEdge()	168
6.38.2.19	setLabel()	168
6.38.2.20	setModelNode()	168
6.38.2.21	setNodeSelected()	168
6.38.2.22	shape()	168
6.38.2.23	updateNodePosition()	168
6.38.2.24	updateSize()	168
6.38.3	Member Data Documentation	169
6.38.3.1	m_borderWidth	169
6.38.3.2	m_edges	169
6.38.3.3	m_hasChangePosition	169
6.38.3.4	m_ignoreNextMouseRelease	169
6.38.3.5	m_label	169
6.38.3.6	m_modelNode	169
6.38.3.7	m_nodeSelected	169
6.38.3.8	m_observerId	169
6.38.3.9	m_oldCenter	169
6.38.3.10	m_radius	170
6.39	NodeSnapshot Struct Reference	170
6.39.1	Member Data Documentation	170
6.39.1.1	m_id	170
6.39.1.2	m_x	170
6.39.1.3	m_y	171
6.40	Observable< T > Class Template Reference	171
6.40.1	Detailed Description	172
6.40.2	Member Typedef Documentation	172
6.40.2.1	Observer	172
6.40.3	Member Function Documentation	172
6.40.3.1	addObserver()	172
6.40.3.2	notifyObservers()	173

6.40.3.3 removeObserver()	173
6.40.4 Member Data Documentation	173
6.40.4.1 m_id	173
6.40.4.2 m_observers	173
6.41 Prim Class Reference	174
6.41.1 Constructor & Destructor Documentation	176
6.41.1.1 Prim()	176
6.41.2 Member Function Documentation	176
6.41.2.1 checkConditions()	176
6.41.2.2 execute()	176
6.41.2.3 prim()	176
6.41.2.4 resultString()	177
6.41.3 Member Data Documentation	177
6.41.3.1 m_totalWeight	177
6.42 RemoveEdgeCommand Class Reference	177
6.42.1 Constructor & Destructor Documentation	179
6.42.1.1 RemoveEdgeCommand()	179
6.42.2 Member Function Documentation	179
6.42.2.1 redo()	179
6.42.2.2 undo()	179
6.42.3 Member Data Documentation	179
6.42.3.1 m_c	179
6.42.3.2 m_edgId	179
6.42.3.3 m_from	179
6.42.3.4 m_to	179
6.42.3.5 m_weight	180
6.43 RemoveNodeCommand Class Reference	180
6.43.1 Constructor & Destructor Documentation	182
6.43.1.1 RemoveNodeCommand()	182
6.43.2 Member Function Documentation	182
6.43.2.1 capture()	182
6.43.2.2 redo()	182
6.43.2.3 undo()	182
6.43.3 Member Data Documentation	182
6.43.3.1 m_c	182
6.43.3.2 m_edges	182
6.43.3.3 m_nodeId	182
6.43.3.4 m_x	183
6.43.3.5 m_y	183
6.44 SaveFileWorker Class Reference	183
6.44.1 Constructor & Destructor Documentation	185
6.44.1.1 SaveFileWorker()	185

6.44.2 Member Function Documentation	185
6.44.2.1 failed	185
6.44.2.2 finished	185
6.44.2.3 run()	185
6.44.3 Member Data Documentation	185
6.44.3.1 m_filePath	185
6.44.3.2 m_isDirected	185
6.44.3.3 m_isWeighted	185
6.44.3.4 m_serializable	185
6.44.3.5 m_serializer	186
6.45 Serializable Class Reference	186
6.45.1 Constructor & Destructor Documentation	187
6.45.1.1 ~Serializable()	187
6.45.2 Member Function Documentation	187
6.45.2.1 fromVariant()	187
6.45.2.2 toVariant()	187
6.46 Serializer Class Reference	187
6.46.1 Constructor & Destructor Documentation	188
6.46.1.1 ~Serializer()	188
6.46.2 Member Function Documentation	188
6.46.2.1 load()	188
6.46.2.2 save()	188
6.47 MainWindow::TabInfo Struct Reference	188
6.47.1 Member Data Documentation	190
6.47.1.1 m_editor	190
6.47.1.2 m_filePath	190
6.47.1.3 m_imagePath	190
6.47.1.4 m_isImageModified	190
6.47.1.5 m_isModified	190
6.48 Tarjan Class Reference	190
6.48.1 Constructor & Destructor Documentation	193
6.48.1.1 Tarjan()	193
6.48.2 Member Function Documentation	193
6.48.2.1 checkConditions()	193
6.48.2.2 execute()	193
6.48.2.3 init()	194
6.48.2.4 resultString()	194
6.48.2.5 tarjan()	194
6.48.3 Member Data Documentation	194
6.48.3.1 m_arrivalTime	194
6.48.3.2 m_components	194
6.48.3.3 m_incomingNumbering	194

6.48.3.4 m_lowLink	194
6.48.3.5 m_numComponents	194
6.48.3.6 m_onStack	194
6.48.3.7 m_tourOrder	195
6.49 ThemeManager Class Reference	195
6.49.1 Member Enumeration Documentation	196
6.49.1.1 Theme	196
6.49.2 Constructor & Destructor Documentation	196
6.49.2.1 ThemeManager()	196
6.49.3 Member Function Documentation	196
6.49.3.1 currentTheme()	196
6.49.3.2 loadTheme()	196
6.49.3.3 saveTheme()	196
6.49.3.4 setTheme()	196
6.49.3.5 styleSheet()	197
6.49.4 Member Data Documentation	197
6.49.4.1 m_currentTheme	197
6.49.4.2 m_settings	197
6.50 UndirectedEdgeItem Class Reference	197
6.50.1 Constructor & Destructor Documentation	201
6.50.1.1 UndirectedEdgeItem()	201
6.50.2 Member Function Documentation	201
6.50.2.1 boundingRect()	201
6.50.2.2 edgePath()	201
6.50.2.3 paint()	201
6.50.2.4 shape()	201
6.51 UnweightedDirectedGraph Class Reference	202
6.51.1 Constructor & Destructor Documentation	204
6.51.1.1 UnweightedDirectedGraph()	204
6.51.1.2 ~UnweightedDirectedGraph()	204
6.51.2 Member Function Documentation	205
6.51.2.1 addEdge()	205
6.51.2.2 isDirected()	205
6.51.2.3 isWeighted()	205
6.51.2.4 removeEdge()	205
6.52 UnweightedUndirectedGraph Class Reference	206
6.52.1 Constructor & Destructor Documentation	208
6.52.1.1 UnweightedUndirectedGraph()	208
6.52.1.2 ~UnweightedUndirectedGraph()	208
6.52.2 Member Function Documentation	209
6.52.2.1 addEdge()	209
6.52.2.2 isDirected()	209

6.52.2.3 isWeighted()	209
6.52.2.4 removeEdge()	209
6.53 WeightedDirectedGraph Class Reference	210
6.53.1 Constructor & Destructor Documentation	212
6.53.1.1 WeightedDirectedGraph()	212
6.53.1.2 ~WeightedDirectedGraph()	212
6.53.2 Member Function Documentation	213
6.53.2.1 addEdge()	213
6.53.2.2 isDirected()	213
6.53.2.3 isWeighted()	213
6.53.2.4 removeEdge()	213
6.54 WeightedUndirectedGraph Class Reference	214
6.54.1 Constructor & Destructor Documentation	216
6.54.1.1 WeightedUndirectedGraph()	216
6.54.1.2 ~WeightedUndirectedGraph()	216
6.54.2 Member Function Documentation	217
6.54.2.1 addEdge()	217
6.54.2.2 isDirected()	217
6.54.2.3 isWeighted()	217
6.54.2.4 removeEdge()	217
7 File Documentation	219
7.1 Headers/Algorithm/Algorithm.h File Reference	219
7.2 Algorithm.h	220
7.3 Headers/Algorithm/AlgorithmError.h File Reference	220
7.3.1 Enumeration Type Documentation	221
7.3.1.1 AlgorithmErrorType	221
7.4 AlgorithmError.h	222
7.5 Headers/Algorithm/AlgorithmStep.h File Reference	222
7.5.1 Detailed Description	223
7.5.2 Enumeration Type Documentation	223
7.5.2.1 StepType	223
7.6 AlgorithmStep.h	224
7.7 Headers/Algorithm/AlgorithmStepApplier.h File Reference	224
7.8 AlgorithmStepApplier.h	225
7.9 Headers/Algorithm/AStar.h File Reference	226
7.10 AStar.h	226
7.11 Headers/Algorithm/BellmanFord.h File Reference	227
7.12 BellmanFord.h	228
7.13 Headers/Algorithm/BFS.h File Reference	228
7.14 BFS.h	229
7.15 Headers/Algorithm/DFS.h File Reference	230

7.16 DFS.h	231
7.17 Headers/Algorithm/Dijkstra.h File Reference	231
7.18 Dijkstra.h	232
7.19 Headers/Algorithm/FloydWarshall.h File Reference	232
7.20 FloydWarshall.h	233
7.21 Headers/Algorithm/Kahn.h File Reference	234
7.22 Kahn.h	235
7.23 Headers/Algorithm/Prim.h File Reference	235
7.24 Prim.h	236
7.25 Headers/Algorithm/Tarjan.h File Reference	236
7.26 Tarjan.h	237
7.27 Headers/Controllers/AlgorithmController.h File Reference	238
7.28 AlgorithmController.h	239
7.29 Headers/Controllers/GraphController.h File Reference	240
7.30 GraphController.h	241
7.31 Headers/Graph/Edge.h File Reference	242
7.31.1 Enumeration Type Documentation	243
7.31.1.1 EdgeState	243
7.32 Edge.h	243
7.33 Headers/Graph/Graph.h File Reference	243
7.34 Graph.h	244
7.35 Headers/Graph/Node.h File Reference	245
7.35.1 Enumeration Type Documentation	246
7.35.1.1 NodeState	246
7.36 Node.h	246
7.37 Headers/Graph/Observable.h File Reference	247
7.38 Observable.h	248
7.39 Headers/Graph/UnweightedDirectedGraph.h File Reference	248
7.40 UnweightedDirectedGraph.h	249
7.41 Headers/Graph/UnweightedUndirectedGraph.h File Reference	250
7.42 UnweightedUndirectedGraph.h	250
7.43 Headers/Graph/WeightedDirectedGraph.h File Reference	251
7.44 WeightedDirectedGraph.h	252
7.45 Headers/Graph/WeightedUndirectedGraph.h File Reference	252
7.46 WeightedUndirectedGraph.h	253
7.47 Headers/MainWindow.h File Reference	253
7.48 MainWindow.h	254
7.49 Headers/Serialization/Serializable.h File Reference	256
7.50 Serializable.h	256
7.51 Headers/Serialization/Serializer.h File Reference	257
7.52 Serializer.h	257
7.53 Headers/Threads/AlgorithmWorker.h File Reference	258

7.54 AlgorithmWorker.h	259
7.55 Headers/Threads/LoadFileWorker.h File Reference	260
7.56 LoadFileWorker.h	261
7.57 Headers/Threads/SaveFileWorker.h File Reference	261
7.58 SaveFileWorker.h	262
7.59 Headers/UI/AlgorithmTab.h File Reference	262
7.60 AlgorithmTab.h	263
7.61 Headers/UI/AppConstants.h File Reference	265
7.62 AppConstants.h	266
7.63 Headers/UI/GraphEditor.h File Reference	266
7.64 GraphEditor.h	267
7.65 Headers/UI/GraphEditTab.h File Reference	268
7.66 GraphEditTab.h	269
7.67 Headers/UI/MenuToolBar.h File Reference	270
7.68 MenuToolBar.h	271
7.69 Headers/UI/ThemeManager.h File Reference	271
7.70 ThemeManager.h	272
7.71 Headers/View/DirectedEdgeItem.h File Reference	273
7.72 DirectedEdgeItem.h	274
7.73 Headers/View/EdgeItem.h File Reference	274
7.74 EdgeItem.h	275
7.75 Headers/View/EditableTextItem.h File Reference	276
7.76 EditableTextItem.h	277
7.77 Headers/View/GraphScene.h File Reference	278
7.78 GraphScene.h	279
7.79 Headers/View/GraphView.h File Reference	280
7.80 GraphView.h	281
7.81 Headers/View/NodeItem.h File Reference	281
7.82 NodeItem.h	282
7.83 Headers/View/UndirectedEdgeItem.h File Reference	283
7.84 UndirectedEdgeItem.h	285
7.85 Source/Algorithm/Algorithm.cpp File Reference	285
7.86 Source/Algorithm/AlgorithmStepApplier.cpp File Reference	285
7.87 Source/Algorithm/AStar.cpp File Reference	286
7.88 Source/Algorithm/BellmanFord.cpp File Reference	286
7.89 Source/Algorithm/BFS.cpp File Reference	286
7.90 Source/Algorithm/DFS.cpp File Reference	287
7.91 Source/Algorithm/Dijkstra.cpp File Reference	287
7.92 Source/Algorithm/FloydWarshall.cpp File Reference	287
7.93 Source/Algorithm/Kahn.cpp File Reference	288
7.94 Source/Algorithm/Prim.cpp File Reference	288
7.95 Source/Algorithm/Tarjan.cpp File Reference	289

7.96 Source/Controllers/AlgorithmController.cpp File Reference	289
7.97 Source/Controllers/GraphController.cpp File Reference	289
7.98 Source/Graph/Edge.cpp File Reference	290
7.99 Source/Graph/Graph.cpp File Reference	290
7.100 Source/Graph/Node.cpp File Reference	291
7.101 Source/Graph/UnweightedDirectedGraph.cpp File Reference	291
7.102 Source/Graph/UnweightedUndirectedGraph.cpp File Reference	292
7.103 Source/Graph/WeightedDirectedGraph.cpp File Reference	292
7.104 Source/Graph/WeightedUndirectedGraph.cpp File Reference	293
7.105 Source/main.cpp File Reference	293
7.105.1 Function Documentation	294
7.105.1.1 main()	294
7.106 Source/MainWindow.cpp File Reference	294
7.107 Source/Serialization/Serializer.cpp File Reference	294
7.108 Source/Threads/AlgorithmWorker.cpp File Reference	295
7.109 Source/Threads/LoadFileWorker.cpp File Reference	295
7.110 Source/Threads/SaveFileWorker.cpp File Reference	295
7.111 Source/UI/AlgorithmTab.cpp File Reference	296
7.112 Source/UI/GraphEditor.cpp File Reference	296
7.113 Source/UI/GraphEditTab.cpp File Reference	296
7.114 Source/UI/MenuToolBar.cpp File Reference	297
7.115 Source/UI/ThemeManager.cpp File Reference	297
7.116 Source/View/DirectedEdgItem.cpp File Reference	298
7.117 Source/View/EdgItem.cpp File Reference	298
7.118 Source/View/EditableTextItem.cpp File Reference	299
7.119 Source/View/GraphScene.cpp File Reference	299
7.120 Source/View/GraphView.cpp File Reference	300
7.121 Source/View/NodeItem.cpp File Reference	300
7.122 Source/View/UndirectedEdgItem.cpp File Reference	301

Chapter 1

Namespace Index

1.1 Namespace List

Here is a list of all namespaces with brief descriptions:

AppConstants		9
Ui		11

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Algorithm	22
AStar	46
BFS	55
BellmanFord	51
DFS	61
Dijkstra	64
FloydWarshall	98
Kahn	140
Prim	174
Tarjan	190
AlgorithmTab::AlgorithmConfig	24
AlgorithmError	31
AlgorithmStep	32
AlgorithmStepApplier	33
EdgeSnapshot	86
NodeSnapshot	170
Observable< T >	171
Observable< Edge >	171
Edge	73
Observable< Node >	171
Node	158
QGraphicsObject	
Edgeltem	78
DirectedEdgeltem	68
UndirectedEdgeltem	197
Nodeltem	162
QGraphicsScene	
GraphScene	126
QGraphicsTextItem	
EditableTextItem	87
QGraphicsView	
GraphView	134
QMainWindow	
MainWindow	146

QObject	
AlgorithmController	25
GraphController	107
QThread	
AlgorithmWorker	43
LoadFileWorker	143
SaveFileWorker	183
QToolBar	
MenuToolBar	151
QUndoCommand	
AddEdgeCommand	13
AddNodeAndEdgeCommand	16
AddNodeCommand	19
ClearGraphCommand	58
EditEdgeWeightCommand	92
EditNodeNameCommand	95
MoveNodeCommand	155
RemoveEdgeCommand	177
RemoveNodeCommand	180
QWidget	
AlgorithmTab	35
GraphEditTab	120
GraphEditor	116
Serializable	186
Graph	101
UnweightedDirectedGraph	202
UnweightedUndirectedGraph	206
WeightedDirectedGraph	210
WeightedUndirectedGraph	214
Serializer	187
MainWindow::TabInfo	188
ThemeManager	195

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

AddEdgeCommand	13
AddNodeAndEdgeCommand	16
AddNodeCommand	19
Algorithm	22
AlgorithmTab::AlgorithmConfig	24
AlgorithmController	
Controller that connects algorithm execution with the UI	25
AlgorithmError	
Represents an error that occurred during algorithm execution	31
AlgorithmStep	
Represents a single step of an algorithm for visualization	32
AlgorithmStepApplier	
Applies algorithm steps to the graph	33
AlgorithmTab	35
AlgorithmWorker	
Executes graph algorithms in a separate thread and generates visualization steps	43
AStar	46
BellmanFord	51
BFS	55
ClearGraphCommand	58
DFS	61
Dijkstra	64
DirectedEdgeItem	68
Edge	73
EdgeItem	78
EdgeSnapshot	86
EditableTextItem	87
EditEdgeWeightCommand	92
EditNodeNameCommand	95
FloydWarshall	98
Graph	101
GraphController	107
GraphEditor	116
GraphEditTab	120
GraphScene	126

GraphView	134
Kahn	140
LoadFileWorker	143
MainWindow	146
MenuToolBar	151
MoveNodeCommand	155
Node	158
NodeItem	162
NodeSnapshot	170
Observable< T >	
Base class implementing the Observer pattern for any subject type	171
Prim	174
RemoveEdgeCommand	177
RemoveNodeCommand	180
SaveFileWorker	183
Serializable	186
Serializer	187
MainWindow::TabInfo	188
Tarjan	190
ThemeManager	195
UndirectedEdgeItem	197
UnweightedDirectedGraph	202
UnweightedUndirectedGraph	206
WeightedDirectedGraph	210
WeightedUndirectedGraph	214

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

Headers/ MainWindow.h	253
Headers/Algorithm/ Algorithm.h	219
Headers/Algorithm/ AlgorithmError.h	220
Headers/Algorithm/ AlgorithmStep.h Defines structures and enums for representing and visualizing steps of graph algorithms	222
Headers/Algorithm/ AlgorithmStepApplier.h	224
Headers/Algorithm/ AStar.h	226
Headers/Algorithm/ BellmanFord.h	227
Headers/Algorithm/ BFS.h	228
Headers/Algorithm/ DFS.h	230
Headers/Algorithm/ Dijkstra.h	231
Headers/Algorithm/ FloydWarshall.h	232
Headers/Algorithm/ Kahn.h	234
Headers/Algorithm/ Prim.h	235
Headers/Algorithm/ Tarjan.h	236
Headers/Controllers/ AlgorithmController.h	238
Headers/Controllers/ GraphController.h	240
Headers/Graph/ Edge.h	242
Headers/Graph/ Graph.h	243
Headers/Graph/ Node.h	245
Headers/Graph/ Observable.h	247
Headers/Graph/ UnweightedDirectedGraph.h	248
Headers/Graph/ UnweightedUndirectedGraph.h	250
Headers/Graph/ WeightedDirectedGraph.h	251
Headers/Graph/ WeightedUndirectedGraph.h	252
Headers/Serialization/ Serializable.h	256
Headers/Serialization/ Serializer.h	257
Headers/Threads/ AlgorithmWorker.h	258
Headers/Threads/ LoadFileWorker.h	260
Headers/Threads/ SaveFileWorker.h	261
Headers/UI/ AlgorithmTab.h	262
Headers/UI/ AppConstants.h	265
Headers/UI/ GraphEditor.h	266
Headers/UI/ GraphEditTab.h	268
Headers/UI/ MenuToolBar.h	270

Headers/UI/ThemeManager.h	271
Headers/View/DirectedEdgeItem.h	273
Headers/View/EdgeItem.h	274
Headers/View/EditableTextItem.h	276
Headers/View/GraphScene.h	278
Headers/View/GraphView.h	280
Headers/View/NodeItem.h	281
Headers/View/UndirectedEdgeItem.h	283
Source/main.cpp	293
Source/MainWindow.cpp	294
Source/Algorithm/Algorithm.cpp	285
Source/Algorithm/AlgorithmStepApplier.cpp	285
Source/Algorithm/AStar.cpp	286
Source/Algorithm/BellmanFord.cpp	286
Source/Algorithm/BFS.cpp	286
Source/Algorithm/DFS.cpp	287
Source/Algorithm/Dijkstra.cpp	287
Source/Algorithm/FloydWarshall.cpp	287
Source/Algorithm/Kahn.cpp	288
Source/Algorithm/Prim.cpp	288
Source/Algorithm/Tarjan.cpp	289
Source/Controllers/AlgorithmController.cpp	289
Source/Controllers/GraphController.cpp	289
Source/Graph/Edge.cpp	290
Source/Graph/Graph.cpp	290
Source/Graph/Node.cpp	291
Source/Graph/UnweightedDirectedGraph.cpp	291
Source/Graph/UnweightedUndirectedGraph.cpp	292
Source/Graph/WeightedDirectedGraph.cpp	292
Source/Graph/WeightedUndirectedGraph.cpp	293
Source/Serialization/Serializer.cpp	294
Source/Threads/AlgorithmWorker.cpp	295
Source/Threads/LoadFileWorker.cpp	295
Source/Threads/SaveFileWorker.cpp	295
Source/UI/AlgorithmTab.cpp	296
Source/UI/GraphEditor.cpp	296
Source/UI/GraphEditTab.cpp	296
Source/UI/MenuToolBar.cpp	297
Source/UI/ThemeManager.cpp	297
Source/View/DirectedEdgeItem.cpp	298
Source/View/EdgeItem.cpp	298
Source/View/EditableTextItem.cpp	299
Source/View/GraphScene.cpp	299
Source/View/GraphView.cpp	300
Source/View/NodeItem.cpp	300
Source/View/UndirectedEdgeItem.cpp	301

Chapter 5

Namespace Documentation

5.1 AppConstants Namespace Reference

Variables

- constexpr int [windowMinWidth](#) = 800
- constexpr int [windowMinHeight](#) = 600
- constexpr int [windowWidth](#) = 1200
- constexpr int [windowHeight](#) = 900
- constexpr int [btnSize](#) = 120
- constexpr int [btnSpacing](#) = 24
- constexpr auto [startPageTitle](#) = "AlgoVision"
- constexpr int [optionBtnSize](#) = 90
- constexpr int [optionIconSize](#) = 64
- constexpr auto [graphPageDefaultTitle](#) = "Untitled"
- constexpr double [defaultRadius](#) = 30.0
- constexpr double [defaultBorderWidth](#) = 3.0
- double [NodeScale](#) = 1.0
- constexpr double [MinNodeScale](#) = 0.4
- constexpr double [MaxNodeScale](#) = 3.0
- constexpr double [EdgePadding](#) = 6.0
- constexpr double [BaseEdgeWidth](#) = 2.0
- constexpr int [createGraphDialogMinWidth](#) = 400
- constexpr int [createGraphDialogMinHeight](#) = 250
- constexpr int [graphTypeDialogErrorMinWidth](#) = 500
- constexpr int [graphTypeDialogErrorMinHeight](#) = 250
- constexpr int [legendMinHeight](#) = 400

5.1.1 Variable Documentation

5.1.1.1 BaseEdgeWidth

```
constexpr double AppConstants::BaseEdgeWidth = 2.0 [constexpr]
```

5.1.1.2 btnSize

```
constexpr int AppConstants::btnSize = 120 [constexpr]
```

5.1.1.3 btnSpacing

```
constexpr int AppConstants::btnSpacing = 24 [constexpr]
```

5.1.1.4 createGraphDialogMinHeight

```
constexpr int AppConstants::createGraphDialogMinHeight = 250 [constexpr]
```

5.1.1.5 createGraphDialogMinWidth

```
constexpr int AppConstants::createGraphDialogMinWidth = 400 [constexpr]
```

5.1.1.6 defaultBorderWidth

```
constexpr double AppConstants::defaultBorderWidth = 3.0 [constexpr]
```

5.1.1.7 defaultRadius

```
constexpr double AppConstants::defaultRadius = 30.0 [constexpr]
```

5.1.1.8 EdgePadding

```
constexpr double AppConstants::EdgePadding = 6.0 [constexpr]
```

5.1.1.9 graphPageDefaultTitle

```
constexpr auto AppConstants::graphPageDefaultTitle = "Untilted" [constexpr]
```

5.1.1.10 graphTypeDialogErrorMinHeight

```
constexpr int AppConstants::graphTypeDialogErrorMinHeight = 250 [constexpr]
```

5.1.1.11 graphTypeDialogErrorMinWidth

```
constexpr int AppConstants::graphTypeDialogErrorMinWidth = 500 [constexpr]
```

5.1.1.12 legendMinHeight

```
constexpr int AppConstants::legendMinHeight = 400 [constexpr]
```

5.1.1.13 MaxNodeScale

```
constexpr double AppConstants::MaxNodeScale = 3.0 [constexpr]
```

5.1.1.14 MinNodeScale

```
constexpr double AppConstants::MinNodeScale = 0.4 [constexpr]
```

5.1.1.15 NodeScale

```
double AppConstants::NodeScale = 1.0 [inline]
```

5.1.1.16 optionBtnSize

```
constexpr int AppConstants::optionBtnSize = 90 [constexpr]
```

5.1.1.17 optionIconSize

```
constexpr int AppConstants::optionIconSize = 64 [constexpr]
```

5.1.1.18 startPageTitle

```
constexpr auto AppConstants::startPageTitle = "AlgoVision" [constexpr]
```

5.1.1.19 windowHeight

```
constexpr int AppConstants::windowHeight = 900 [constexpr]
```

5.1.1.20 windowMinHeight

```
constexpr int AppConstants::windowMinHeight = 600 [constexpr]
```

5.1.1.21 windowMinWidth

```
constexpr int AppConstants::windowMinWidth = 800 [constexpr]
```

5.1.1.22 windowWidth

```
constexpr int AppConstants::windowWidth = 1200 [constexpr]
```

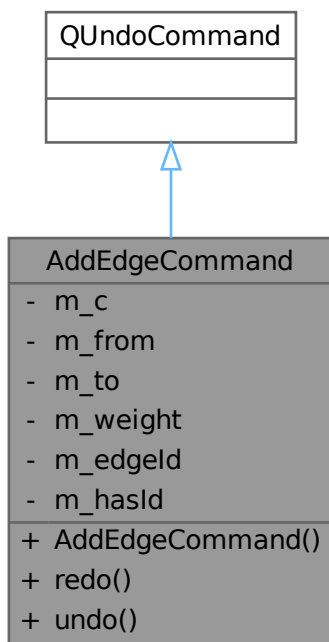
5.2 Ui Namespace Reference

Chapter 6

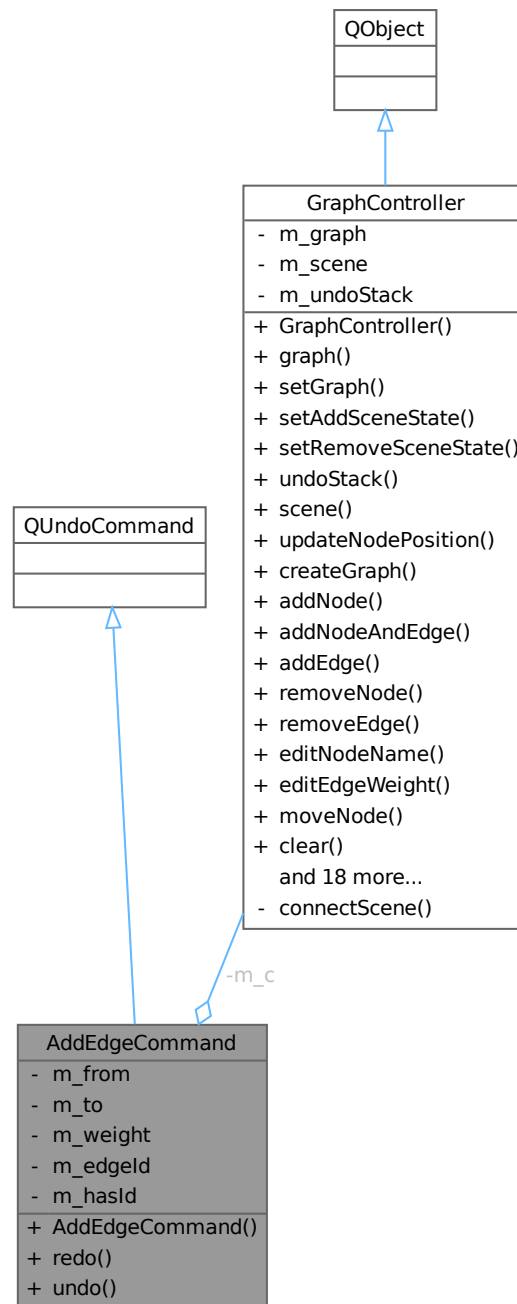
Class Documentation

6.1 AddEdgeCommand Class Reference

Inheritance diagram for AddEdgeCommand:



Collaboration diagram for AddEdgeCommand:



Public Member Functions

- [AddEdgeCommand](#) ([GraphController](#) *c, unsigned from, unsigned to, int weight)
- void [redo](#) () override
- void [undo](#) () override

Private Attributes

- [GraphController](#) * [m_c](#)
- unsigned [m_from](#)
- unsigned [m_to](#)
- int [m_weight](#)
- unsigned [m_edgId](#) = 0
- bool [m_hasId](#) = false

6.1.1 Constructor & Destructor Documentation

6.1.1.1 AddEdgeCommand()

```
AddEdgeCommand::AddEdgeCommand (
    GraphController * c,
    unsigned from,
    unsigned to,
    int weight ) [inline]
```

6.1.2 Member Function Documentation

6.1.2.1 redo()

```
void AddEdgeCommand::redo ( ) [inline], [override]
```

6.1.2.2 undo()

```
void AddEdgeCommand::undo ( ) [inline], [override]
```

6.1.3 Member Data Documentation

6.1.3.1 m_c

```
GraphController* AddEdgeCommand::m_c [private]
```

6.1.3.2 m_edgId

```
unsigned AddEdgeCommand::m_edgeId = 0 [private]
```

6.1.3.3 m_from

```
unsigned AddEdgeCommand::m_from [private]
```

6.1.3.4 m_hasId

```
bool AddEdgeCommand::m_hasId = false [private]
```

6.1.3.5 m_to

```
unsigned AddEdgeCommand::m_to [private]
```

6.1.3.6 m_weight

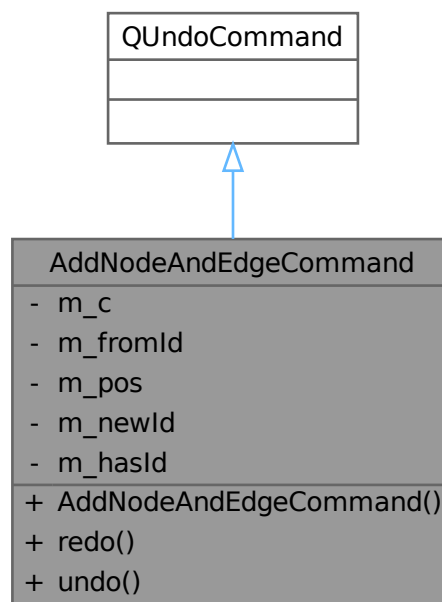
```
int AddEdgeCommand::m_weight [private]
```

The documentation for this class was generated from the following file:

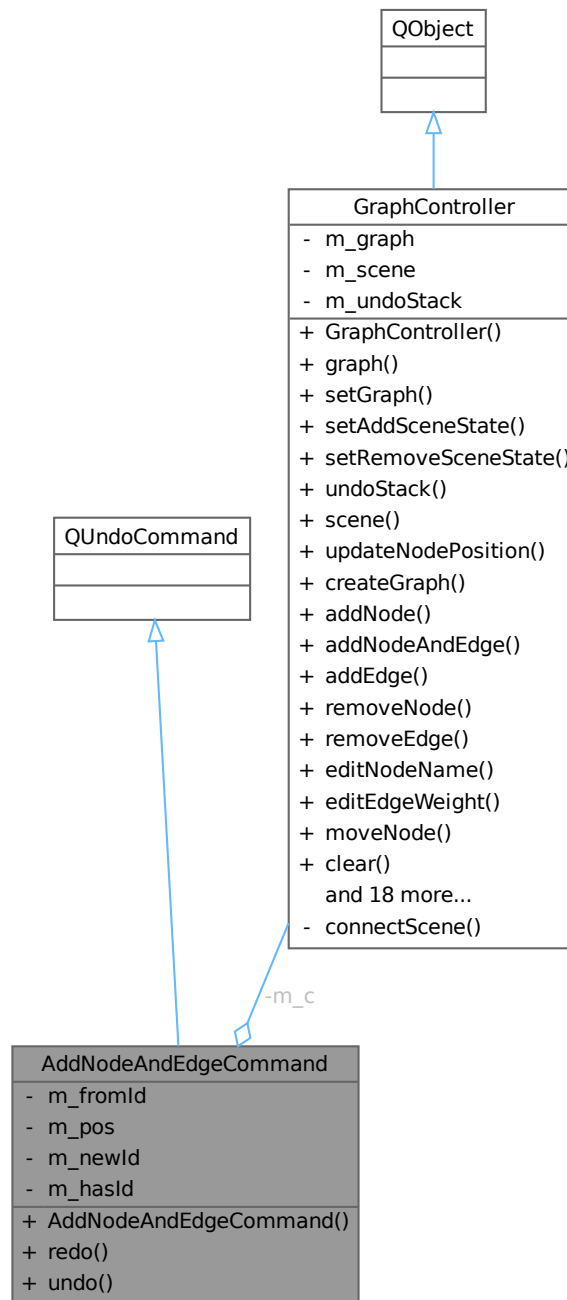
- Source/Controllers/[GraphController.cpp](#)

6.2 AddNodeAndEdgeCommand Class Reference

Inheritance diagram for AddNodeAndEdgeCommand:



Collaboration diagram for AddNodeAndEdgeCommand:



Public Member Functions

- [AddNodeAndEdgeCommand](#) ([GraphController](#) *c, unsigned fromId, const QPointF &pos)
- void [redo](#) () override
- void [undo](#) () override

Private Attributes

- [GraphController](#) * [m_c](#)
- unsigned [m_fromId](#)
- [QPointF](#) [m_pos](#)
- unsigned [m_newId](#) = 0
- bool [m_hasId](#) = false

6.2.1 Constructor & Destructor Documentation

6.2.1.1 AddNodeAndEdgeCommand()

```
AddNodeAndEdgeCommand::AddNodeAndEdgeCommand (
    GraphController * c,
    unsigned fromId,
    const QPointF & pos ) [inline]
```

6.2.2 Member Function Documentation

6.2.2.1 redo()

```
void AddNodeAndEdgeCommand::redo ( ) [inline], [override]
```

6.2.2.2 undo()

```
void AddNodeAndEdgeCommand::undo ( ) [inline], [override]
```

6.2.3 Member Data Documentation

6.2.3.1 m_c

```
GraphController* AddNodeAndEdgeCommand::m_c [private]
```

6.2.3.2 m_fromId

```
unsigned AddNodeAndEdgeCommand::m_fromId [private]
```

6.2.3.3 m_hasId

```
bool AddNodeAndEdgeCommand::m_hasId = false [private]
```

6.2.3.4 m_newId

```
unsigned AddNodeAndEdgeCommand::m_newId = 0 [private]
```

6.2.3.5 m_pos

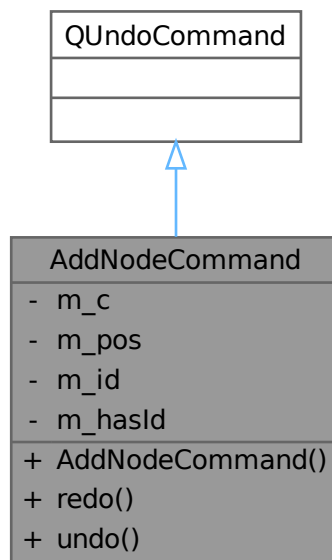
```
QPointF AddNodeAndEdgeCommand::m_pos [private]
```

The documentation for this class was generated from the following file:

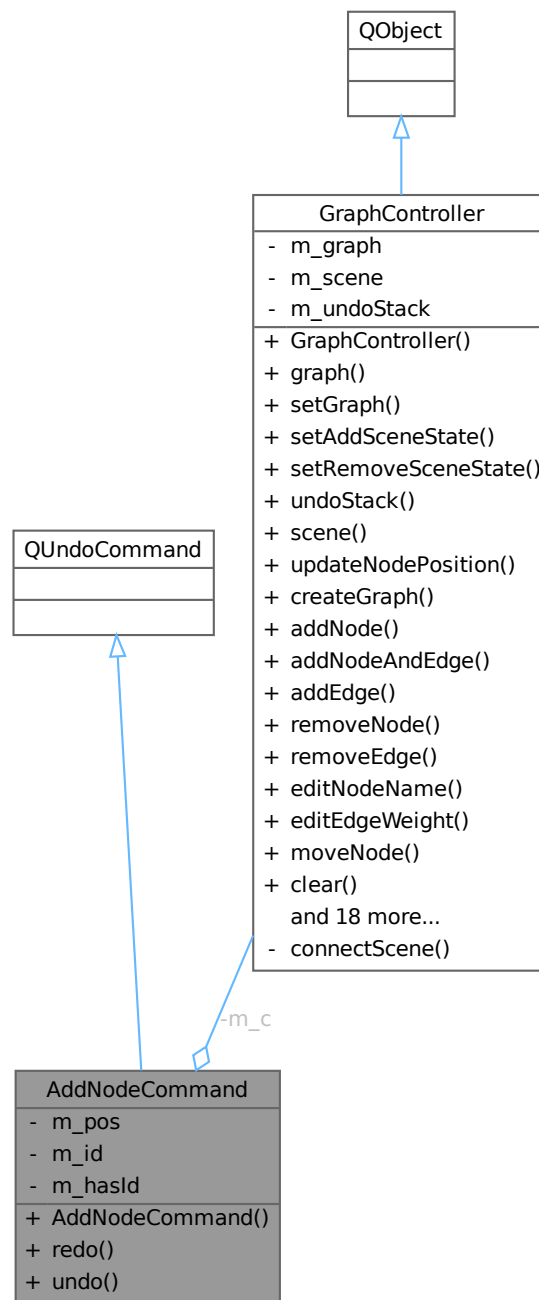
- Source/Controllers/[GraphController.cpp](#)

6.3 AddNodeCommand Class Reference

Inheritance diagram for AddNodeCommand:



Collaboration diagram for AddNodeCommand:



Public Member Functions

- [AddNodeCommand](#) ([GraphController](#) *c, const QPointF &pos)
- void [redo](#) () override
- void [undo](#) () override

Private Attributes

- [GraphController](#) * `m_c`
- `QPointF` `m_pos`
- unsigned `m_id` = 0
- bool `m_hasId` = false

6.3.1 Constructor & Destructor Documentation

6.3.1.1 AddNodeCommand()

```
AddNodeCommand::AddNodeCommand (
    GraphController * c,
    const QPointF & pos ) [inline]
```

6.3.2 Member Function Documentation

6.3.2.1 redo()

```
void AddNodeCommand::redo ( ) [inline], [override]
```

6.3.2.2 undo()

```
void AddNodeCommand::undo ( ) [inline], [override]
```

6.3.3 Member Data Documentation

6.3.3.1 m_c

```
GraphController* AddNodeCommand::m_c [private]
```

6.3.3.2 m_hasId

```
bool AddNodeCommand::m_hasId = false [private]
```

6.3.3.3 m_id

```
unsigned AddNodeCommand::m_id = 0 [private]
```

6.3.3.4 m_pos

```
QPointF AddNodeCommand::m_pos [private]
```

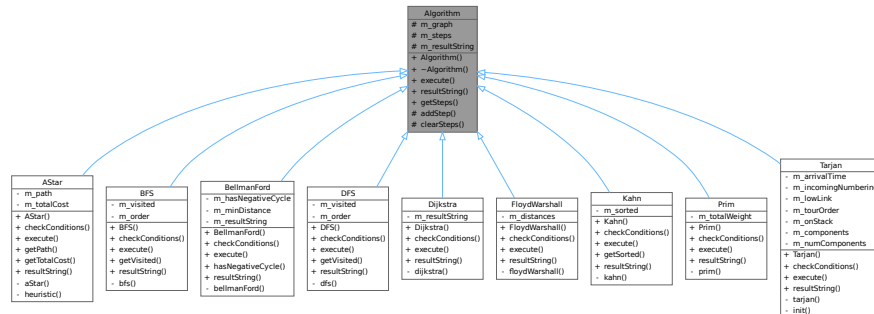
The documentation for this class was generated from the following file:

- Source/Controllers/[GraphController.cpp](#)

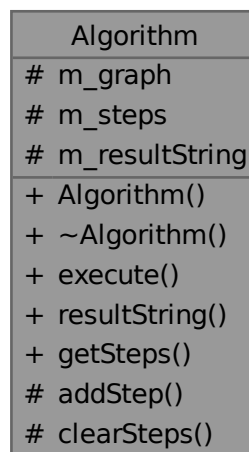
6.4 Algorithm Class Reference

```
#include <Algorithm.h>
```

Inheritance diagram for Algorithm:



Collaboration diagram for Algorithm:



Public Member Functions

- [Algorithm](#) (const std::shared_ptr< [Graph](#) >)
- virtual [~Algorithm](#) ()=default
- virtual std::optional< [AlgorithmError](#) > [execute](#) (unsigned=0, unsigned=0)=0
- virtual QString [resultString](#) () const =0
- const std::vector< [AlgorithmStep](#) > & [getSteps](#) () const

Protected Member Functions

- void [addStep](#) (const [AlgorithmStep](#) &)
- void [clearSteps](#) ()

Protected Attributes

- `std::shared_ptr< Graph > m_graph`
- `std::vector< AlgorithmStep > m_steps`
- `QString m_resultString`

6.4.1 Constructor & Destructor Documentation**6.4.1.1 Algorithm()**

```
Algorithm::Algorithm (
    const std::shared_ptr< Graph > g ) [explicit]
```

6.4.1.2 ~Algorithm()

```
virtual Algorithm::~Algorithm ( ) [virtual], [default]
```

6.4.2 Member Function Documentation**6.4.2.1 addStep()**

```
void Algorithm::addStep (
    const AlgorithmStep & step ) [protected]
```

6.4.2.2 clearSteps()

```
void Algorithm::clearSteps ( ) [protected]
```

6.4.2.3 execute()

```
virtual std::optional< AlgorithmError > Algorithm::execute (
    unsigned = 0,
    unsigned = 0 ) [pure virtual]
```

Implemented in [AStar](#), [BellmanFord](#), [BFS](#), [DFS](#), [Dijkstra](#), [FloydWarshall](#), [Kahn](#), [Prim](#), and [Tarjan](#).

6.4.2.4 getSteps()

```
const std::vector< AlgorithmStep > & Algorithm::getSteps ( ) const
```

6.4.2.5 resultString()

```
virtual QString Algorithm::resultString ( ) const [pure virtual]
```

Implemented in [AStar](#), [BellmanFord](#), [BFS](#), [DFS](#), [Dijkstra](#), [FloydWarshall](#), [Kahn](#), [Prim](#), and [Tarjan](#).

6.4.3 Member Data Documentation

6.4.3.1 m_graph

`std::shared_ptr<Graph> Algorithm::m_graph` [protected]

6.4.3.2 m_resultString

`QString Algorithm::m_resultString` [protected]

6.4.3.3 m_steps

`std::vector<AlgorithmStep> Algorithm::m_steps` [protected]

The documentation for this class was generated from the following files:

- Headers/Algorithm/[Algorithm.h](#)
- Source/Algorithm/[Algorithm.cpp](#)

6.5 AlgorithmTab::AlgorithmConfig Struct Reference

Collaboration diagram for AlgorithmTab::AlgorithmConfig:

AlgorithmTab::AlgorithmConfig	
+	m_algorithmName
+	m_startNode
+	m_endNode
+	operator==()
+	operator!=()

Public Member Functions

- bool [operator==](#) (const [AlgorithmConfig](#) &) const
- bool [operator!=](#) (const [AlgorithmConfig](#) &) const

Public Attributes

- QString [m_algorithmName](#)
- int [m_startNode](#)
- int [m_endNode](#)

6.5.1 Member Function Documentation

6.5.1.1 operator"!=()

```
bool AlgorithmTab::AlgorithmConfig::operator!= (
    const AlgorithmConfig & other ) const
```

6.5.1.2 operator==(

```
bool AlgorithmTab::AlgorithmConfig::operator== (
    const AlgorithmConfig & other ) const
```

6.5.2 Member Data Documentation

6.5.2.1 m_algorithmName

```
QString AlgorithmTab::AlgorithmConfig::m_algorithmName
```

6.5.2.2 m_endNode

```
int AlgorithmTab::AlgorithmConfig::m_endNode
```

6.5.2.3 m_startNode

```
int AlgorithmTab::AlgorithmConfig::m_startNode
```

The documentation for this struct was generated from the following files:

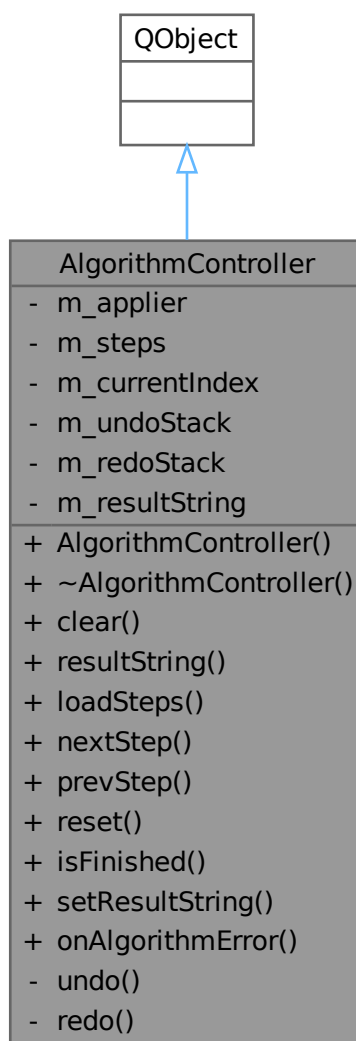
- [Headers/UI/AlgorithmTab.h](#)
- [Source/UI/AlgorithmTab.cpp](#)

6.6 AlgorithmController Class Reference

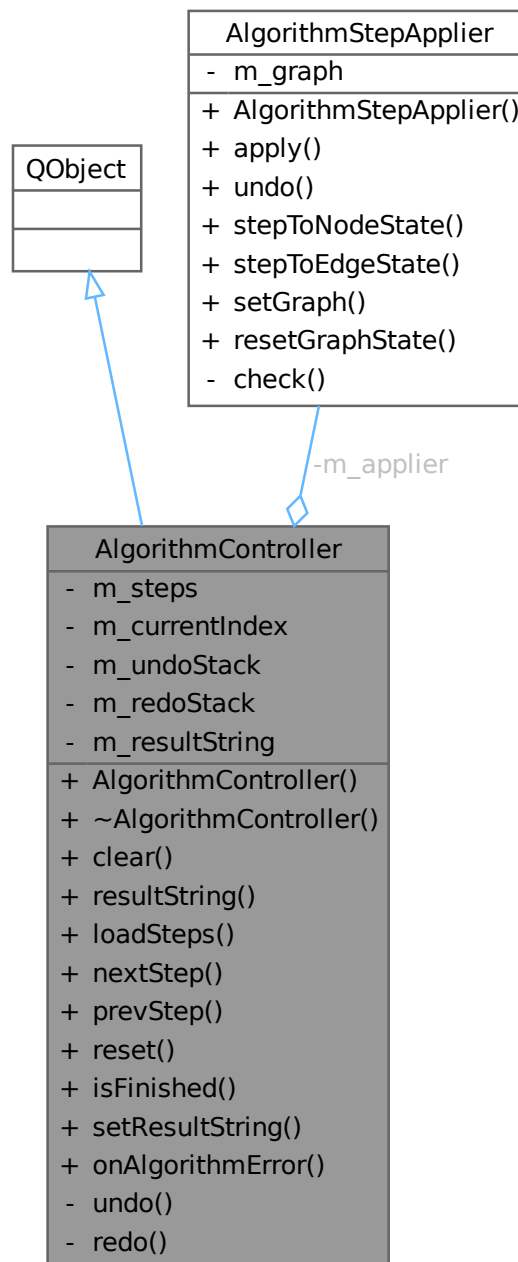
Controller that connects algorithm execution with the UI.

```
#include <AlgorithmController.h>
```

Inheritance diagram for AlgorithmController:



Collaboration diagram for AlgorithmController:



Public Slots

- void [loadSteps](#) (const std::vector< [AlgorithmStep](#) > &)
- void [nextStep](#) ()
- void [prevStep](#) ()
- void [reset](#) ()
- bool [isFinished](#) () const
- void [setResultString](#) (const QString &)
- void [onAlgorithmError](#) (const [AlgorithmError](#) &)

Signals

- void [requestErrorDialog](#) (const [AlgorithmError](#) &)

Public Member Functions

- [AlgorithmController](#) ([AlgorithmStepApplier](#) &, QObject **parent* = nullptr)
- [~AlgorithmController](#) () *override*=default
- void [clear](#) ()
- QString [resultString](#) () const

Private Member Functions

- void [undo](#) ()
- void [redo](#) ()

Private Attributes

- [AlgorithmStepApplier](#) & [m_applier](#)
- std::vector< [AlgorithmStep](#) > [m_steps](#)
- int [m_currentIndex](#) = -1
- std::vector< [AlgorithmStep](#) > [m_undoStack](#)
- std::vector< [AlgorithmStep](#) > [m_redoStack](#)
- QString [m_resultString](#)

6.6.1 Detailed Description

Controller that connects algorithm execution with the UI.

This class acts as a "bridge" between the algorithm and UI controls such as play, pause, next, previous, and reset. It receives algorithm steps and manages their execution.

6.6.2 Constructor & Destructor Documentation

6.6.2.1 AlgorithmController()

```
AlgorithmController::AlgorithmController (  
    AlgorithmStepApplier & applier,  
    QObject * parent = nullptr ) [explicit]
```

6.6.2.2 ~AlgorithmController()

```
AlgorithmController::~AlgorithmController ( ) [override], [default]
```


6.6.3 Member Function Documentation

6.6.3.1 clear()

```
void AlgorithmController::clear ( )
```

6.6.3.2 isFinished

```
bool AlgorithmController::isFinished ( ) const [slot]
```

6.6.3.3 loadSteps

```
void AlgorithmController::loadSteps (
    const std::vector< AlgorithmStep > & steps ) [slot]
```

6.6.3.4 nextStep

```
void AlgorithmController::nextStep ( ) [slot]
```

6.6.3.5 onAlgorithmError

```
void AlgorithmController::onAlgorithmError (
    const AlgorithmError & error ) [slot]
```

6.6.3.6 prevStep

```
void AlgorithmController::prevStep ( ) [slot]
```

6.6.3.7 redo()

```
void AlgorithmController::redo ( ) [private]
```

6.6.3.8 requestErrorDialog

```
void AlgorithmController::requestErrorDialog (
    const AlgorithmError & ) [signal]
```

6.6.3.9 reset

```
void AlgorithmController::reset ( ) [slot]
```

6.6.3.10 resultString()

```
QString AlgorithmController::resultString ( ) const
```

6.6.3.11 setResultString

```
void AlgorithmController::setResultString (
    const QString & s ) [slot]
```

6.6.3.12 undo()

```
void AlgorithmController::undo ( ) [private]
```

6.6.4 Member Data Documentation

6.6.4.1 m_applier

```
AlgorithmStepApplier& AlgorithmController::m_applier [private]
```

6.6.4.2 m_currentIndex

```
int AlgorithmController::m_currentIndex = -1 [private]
```

6.6.4.3 m_redoStack

```
std::vector<AlgorithmStep> AlgorithmController::m_redoStack [private]
```

6.6.4.4 m_resultString

```
QString AlgorithmController::m_resultString [private]
```

6.6.4.5 m_steps

```
std::vector<AlgorithmStep> AlgorithmController::m_steps [private]
```

6.6.4.6 m_undoStack

```
std::vector<AlgorithmStep> AlgorithmController::m_undoStack [private]
```

The documentation for this class was generated from the following files:

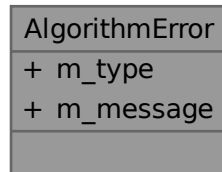
- Headers/Controllers/[AlgorithmController.h](#)
- Source/Controllers/[AlgorithmController.cpp](#)

6.7 AlgorithmError Struct Reference

Represents an error that occurred during algorithm execution.

```
#include <AlgorithmError.h>
```

Collaboration diagram for AlgorithmError:



Public Attributes

- [AlgorithmErrorType m_type](#)
- [std::string m_message](#)

6.7.1 Detailed Description

Represents an error that occurred during algorithm execution.

Contains the error type and an optional descriptive message.

6.7.2 Member Data Documentation

6.7.2.1 m_message

```
std::string AlgorithmError::m_message
```

6.7.2.2 m_type

```
AlgorithmErrorType AlgorithmError::m_type
```

The documentation for this struct was generated from the following file:

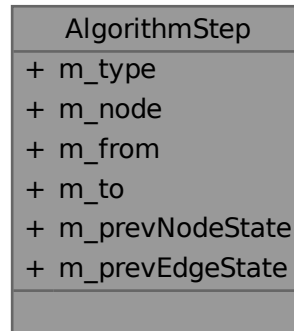
- Headers/Algorithm/[AlgorithmError.h](#)

6.8 AlgorithmStep Struct Reference

Represents a single step of an algorithm for visualization.

```
#include <AlgorithmStep.h>
```

Collaboration diagram for AlgorithmStep:



Public Attributes

- [StepType](#) `m_type` {[StepType::DefaultType](#)}
- `std::optional< unsigned >` [m_node](#)
- `std::optional< unsigned >` [m_from](#)
- `std::optional< unsigned >` [m_to](#)
- `std::optional< NodeState >` [m_prevNodeState](#)
- `std::optional< EdgeState >` [m_prevEdgeState](#)

6.8.1 Detailed Description

Represents a single step of an algorithm for visualization.

Contains the type of the step, optionally a node or edge affected, and optionally the previous states of nodes/edges before the step.

6.8.2 Member Data Documentation

6.8.2.1 `m_from`

```
std::optional<unsigned> AlgorithmStep::m_from
```

6.8.2.2 m_node

```
std::optional<unsigned> AlgorithmStep::m_node
```

6.8.2.3 m_prevEdgeState

```
std::optional<EdgeState> AlgorithmStep::m_prevEdgeState
```

6.8.2.4 m_prevNodeState

```
std::optional<NodeState> AlgorithmStep::m_prevNodeState
```

6.8.2.5 m_to

```
std::optional<unsigned> AlgorithmStep::m_to
```

6.8.2.6 m_type

```
StepType AlgorithmStep::m_type {StepType::DefaultType}
```

The documentation for this struct was generated from the following file:

- Headers/Algorithm/[AlgorithmStep.h](#)

6.9 AlgorithmStepApplier Class Reference

Applies algorithm steps to the graph.

```
#include <AlgorithmStepApplier.h>
```

Collaboration diagram for AlgorithmStepApplier:

AlgorithmStepApplier
- m_graph
+ AlgorithmStepApplier()
+ apply()
+ undo()
+ stepToNodeState()
+ stepToEdgeState()
+ setGraph()
+ resetGraphState()
- check()

Public Member Functions

- [AlgorithmStepApplier](#) (const std::shared_ptr< [Graph](#) >)
- void [apply](#) ([AlgorithmStep](#) &)
- void [undo](#) (const [AlgorithmStep](#) &)
- [NodeState](#) [stepToNodeState](#) (const [StepType](#)) const
- [EdgeState](#) [stepToEdgeState](#) (const [StepType](#)) const
- void [setGraph](#) (std::shared_ptr< [Graph](#) >)
- void [resetGraphState](#) ()

Private Member Functions

- void [check](#) (const [AlgorithmStep](#) &) const

Private Attributes

- std::shared_ptr< [Graph](#) > [m_graph](#)

6.9.1 Detailed Description

Applies algorithm steps to the graph.

This class takes individual algorithm steps and maps them to changes in the state of nodes and edges in the graph.

6.9.2 Constructor & Destructor Documentation

6.9.2.1 AlgorithmStepApplier()

```
AlgorithmStepApplier::AlgorithmStepApplier (  
    const std::shared_ptr< Graph > g ) [explicit]
```

6.9.3 Member Function Documentation

6.9.3.1 apply()

```
void AlgorithmStepApplier::apply (  
    AlgorithmStep & step )
```

6.9.3.2 check()

```
void AlgorithmStepApplier::check (  
    const AlgorithmStep & step ) const [private]
```

6.9.3.3 resetGraphState()

```
void AlgorithmStepApplier::resetGraphState ( )
```

6.9.3.4 setGraph()

```
void AlgorithmStepApplier::setGraph (
    std::shared_ptr< Graph > graph )
```

6.9.3.5 stepToEdgeState()

```
EdgeState AlgorithmStepApplier::stepToEdgeState (
    const StepType t ) const
```

6.9.3.6 stepToNodeState()

```
NodeState AlgorithmStepApplier::stepToNodeState (
    const StepType t ) const
```

6.9.3.7 undo()

```
void AlgorithmStepApplier::undo (
    const AlgorithmStep & step )
```

6.9.4 Member Data Documentation

6.9.4.1 m_graph

```
std::shared_ptr<Graph> AlgorithmStepApplier::m_graph [private]
```

The documentation for this class was generated from the following files:

- Headers/Algorithm/[AlgorithmStepApplier.h](#)
- Source/Algorithm/[AlgorithmStepApplier.cpp](#)

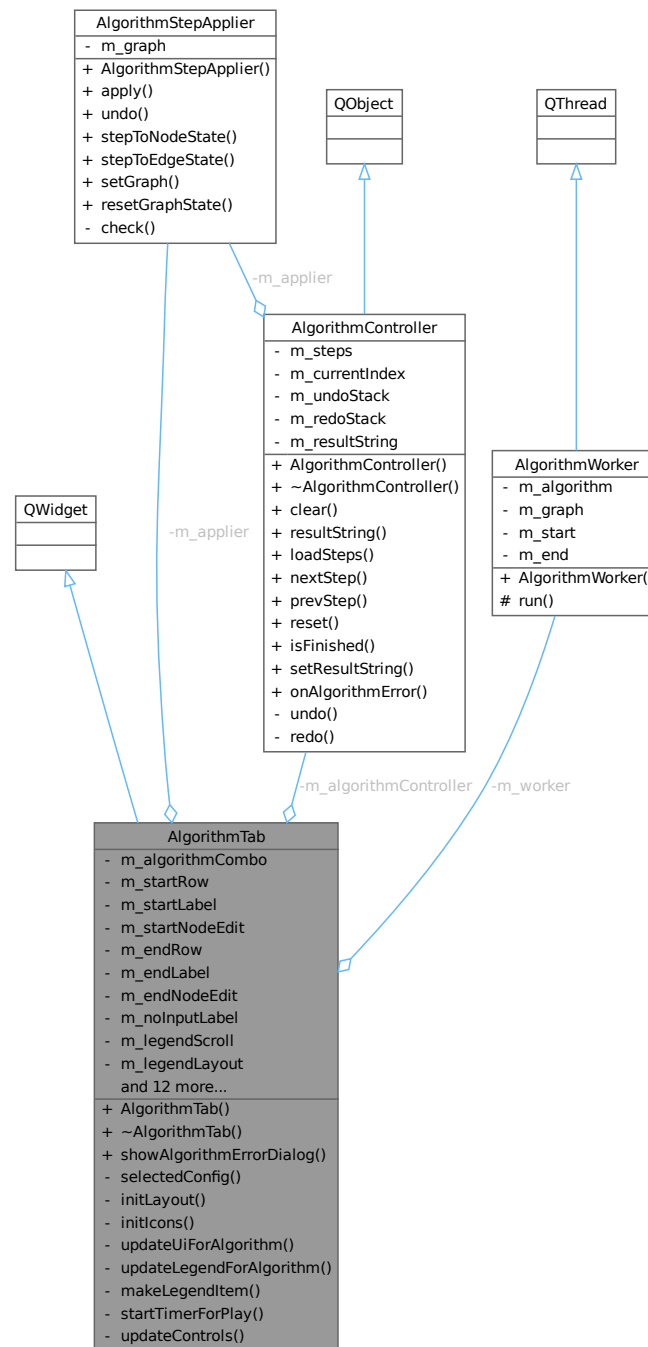
6.10 AlgorithmTab Class Reference

```
#include <AlgorithmTab.h>
```

Inheritance diagram for AlgorithmTab:



Collaboration diagram for AlgorithmTab:



Classes

- struct [AlgorithmConfig](#)

Public Slots

- void [showAlgorithmErrorDialog](#) (const [AlgorithmError](#) &)

Signals

- void [graphEditAllowedChanged](#) (bool)

Public Member Functions

- [AlgorithmTab](#) (std::shared_ptr< [GraphController](#) >, QWidget * = nullptr)
- [~AlgorithmTab](#) () override=default

Private Types

- enum class [RunState](#) { [Idle](#) , [Playing](#) , [Paused](#) , [Finished](#) }

Private Member Functions

- [AlgorithmConfig](#) [selectedConfig](#) () const
- void [initLayout](#) ()
- void [initIcons](#) ()
- void [updateUiForAlgorithm](#) (const QString &)
- void [updateLegendForAlgorithm](#) (const QString &)
- QWidget * [makeLegendItem](#) (const QColor &, const QString &)
- void [startTimerForPlay](#) ()
- void [updateControls](#) ()

Private Attributes

- QComboBox * [m_algorithmCombo](#)
- QWidget * [m_startRow](#)
- QLabel * [m_startLabel](#)
- QLineEdit * [m_startNodeEdit](#)
- QWidget * [m_endRow](#)
- QLabel * [m_endLabel](#)
- QLineEdit * [m_endNodeEdit](#)
- QLabel * [m_noInputLabel](#)
- QScrollArea * [m_legendScroll](#)
- QVBoxLayout * [m_legendLayout](#)
- QWidget * [m_legendContainer](#)
- QPushButton * [m_helpBtn](#)
- QToolButton * [m_prevBtn](#)
- QToolButton * [m_playBtn](#)
- QToolButton * [m_pauseBtn](#)
- QToolButton * [m_nextBtn](#)
- QToolButton * [m_restartBtn](#)
- std::shared_ptr< [GraphController](#) > [m_graphController](#)
- [AlgorithmStepApplier](#) [m_applier](#)
- [AlgorithmController](#) [m_algorithmController](#)
- [AlgorithmWorker](#) * [m_worker](#) = nullptr
- QTimer * [m_timer](#) = nullptr
- [RunState](#) [m_state](#) = [RunState::Idle](#)
- std::optional< [AlgorithmConfig](#) > [m_currentConfig](#)
- bool [m_graphEditAllowed](#) = true

6.10.1 Member Enumeration Documentation

6.10.1.1 RunState

```
enum class AlgorithmTab::RunState [strong], [private]
```

Enumerator

Idle	
Playing	
Paused	
Finished	

6.10.2 Constructor & Destructor Documentation

6.10.2.1 AlgorithmTab()

```
AlgorithmTab::AlgorithmTab (
    std::shared_ptr< GraphController > graphController,
    QWidget * parent = nullptr ) [explicit]
```

6.10.2.2 ~AlgorithmTab()

```
AlgorithmTab::~AlgorithmTab ( ) [override], [default]
```

6.10.3 Member Function Documentation

6.10.3.1 graphEditAllowedChanged

```
void AlgorithmTab::graphEditAllowedChanged (
    bool ) [signal]
```

6.10.3.2 initIcons()

```
void AlgorithmTab::initIcons ( ) [private]
```

6.10.3.3 initLayout()

```
void AlgorithmTab::initLayout ( ) [private]
```

6.10.3.4 makeLegendItem()

```
QWidget * AlgorithmTab::makeLegendItem (
    const QColor & color,
    const QString & text ) [private]
```

6.10.3.5 selectedConfig()

```
AlgorithmTab::AlgorithmConfig AlgorithmTab::selectedConfig ( ) const [private]
```

6.10.3.6 showAlgorithmErrorDialog

```
void AlgorithmTab::showAlgorithmErrorDialog (
    const AlgorithmError & error ) [slot]
```

6.10.3.7 startTimerForPlay()

```
void AlgorithmTab::startTimerForPlay ( ) [private]
```

6.10.3.8 updateControls()

```
void AlgorithmTab::updateControls ( ) [private]
```

6.10.3.9 updateLegendForAlgorithm()

```
void AlgorithmTab::updateLegendForAlgorithm (
    const QString & name ) [private]
```

6.10.3.10 updateUiForAlgorithm()

```
void AlgorithmTab::updateUiForAlgorithm (
    const QString & algorithmName ) [private]
```

6.10.4 Member Data Documentation

6.10.4.1 m_algorithmCombo

```
QComboBox\* AlgorithmTab::m_algorithmCombo [private]
```

6.10.4.2 m_algorithmController

```
AlgorithmController AlgorithmTab::m_algorithmController [private]
```

6.10.4.3 m_applier

```
AlgorithmStepApplier AlgorithmTab::m_applier [private]
```

6.10.4.4 m_currentConfig

```
std::optional<AlgorithmConfig> AlgorithmTab::m_currentConfig [private]
```

6.10.4.5 m_endLabel

```
QLabel* AlgorithmTab::m_endLabel [private]
```

6.10.4.6 m_endNodeEdit

```
QLineEdit* AlgorithmTab::m_endNodeEdit [private]
```

6.10.4.7 m_endRow

```
QWidget* AlgorithmTab::m_endRow [private]
```

6.10.4.8 m_graphController

```
std::shared_ptr<GraphController> AlgorithmTab::m_graphController [private]
```

6.10.4.9 m_graphEditAllowed

```
bool AlgorithmTab::m_graphEditAllowed = true [private]
```

6.10.4.10 m_helpBtn

```
QPushButton* AlgorithmTab::m_helpBtn [private]
```

6.10.4.11 m_legendContainer

```
QWidget* AlgorithmTab::m_legendContainer [private]
```

6.10.4.12 m_legendLayout

```
QVBoxLayout* AlgorithmTab::m_legendLayout [private]
```

6.10.4.13 m_legendScroll

```
QScrollArea* AlgorithmTab::m_legendScroll [private]
```

6.10.4.14 m_nextBtn

```
QToolButton* AlgorithmTab::m_nextBtn [private]
```

6.10.4.15 m_noInputLabel

QLabel* AlgorithmTab::m_noInputLabel [private]

6.10.4.16 m_pauseBtn

QToolButton* AlgorithmTab::m_pauseBtn [private]

6.10.4.17 m_playBtn

QToolButton* AlgorithmTab::m_playBtn [private]

6.10.4.18 m_prevBtn

QToolButton* AlgorithmTab::m_prevBtn [private]

6.10.4.19 m_restartBtn

QToolButton* AlgorithmTab::m_restartBtn [private]

6.10.4.20 m_startLabel

QLabel* AlgorithmTab::m_startLabel [private]

6.10.4.21 m_startNodeEdit

QLineEdit* AlgorithmTab::m_startNodeEdit [private]

6.10.4.22 m_startRow

QWidget* AlgorithmTab::m_startRow [private]

6.10.4.23 m_state

RunState AlgorithmTab::m_state = RunState::Idle [private]

6.10.4.24 m_timer

QTimer* AlgorithmTab::m_timer = nullptr [private]

6.10.4.25 m_worker

```
AlgorithmWorker* AlgorithmTab::m_worker = nullptr [private]
```

The documentation for this class was generated from the following files:

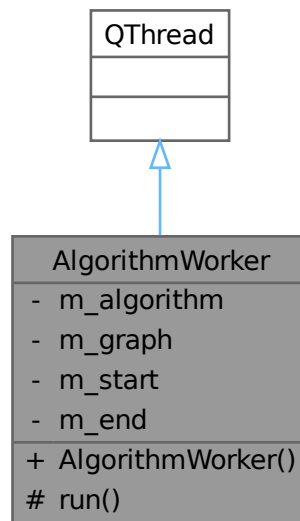
- [Headers/UI/AlgorithmTab.h](#)
- [Source/UI/AlgorithmTab.cpp](#)

6.11 AlgorithmWorker Class Reference

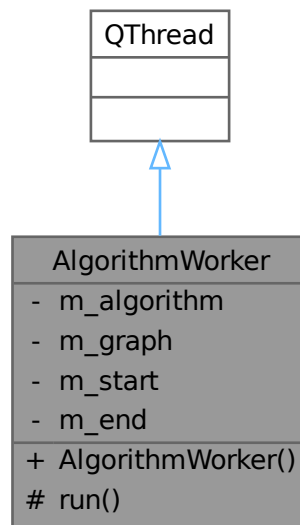
Executes graph algorithms in a separate thread and generates visualization steps.

```
#include <AlgorithmWorker.h>
```

Inheritance diagram for AlgorithmWorker:



Collaboration diagram for AlgorithmWorker:



Signals

- void `stepsReady` (const std::vector< `AlgorithmStep` > &)
- void `resultReady` (const QString &)
- void `algorithmErrorOccurred` (const `AlgorithmError` &)

Public Member Functions

- `AlgorithmWorker` (const QString &, const std::shared_ptr< `Graph` >, unsigned=0, unsigned=0)

Protected Member Functions

- void `run` () override

Private Attributes

- QString `m_algorithm`
- std::shared_ptr< `Graph` > `m_graph`
- unsigned `m_start`
- unsigned `m_end`

6.11.1 Detailed Description

Executes graph algorithms in a separate thread and generates visualization steps.

This class inherits from QThread to run the algorithm asynchronously. During execution, it emits signals to communicate:

- when the steps are ready for visualization,
- when the algorithm result is ready,
- if an error occurs during execution.

6.11.2 Constructor & Destructor Documentation

6.11.2.1 AlgorithmWorker()

```
AlgorithmWorker::AlgorithmWorker (
    const QString & algorithm,
    const std::shared_ptr< Graph > graph,
    unsigned start = 0,
    unsigned end = 0 )
```

6.11.3 Member Function Documentation

6.11.3.1 algorithmErrorOccurred

```
void AlgorithmWorker::algorithmErrorOccurred (
    const AlgorithmError & ) [signal]
```

6.11.3.2 resultReady

```
void AlgorithmWorker::resultReady (
    const QString & ) [signal]
```

6.11.3.3 run()

```
void AlgorithmWorker::run ( ) [override], [protected]
```

6.11.3.4 stepsReady

```
void AlgorithmWorker::stepsReady (
    const std::vector< AlgorithmStep > & ) [signal]
```

6.11.4 Member Data Documentation

6.11.4.1 m_algorithm

```
QString AlgorithmWorker::m_algorithm [private]
```

6.11.4.2 m_end

```
unsigned AlgorithmWorker::m_end [private]
```

6.11.4.3 m_graph

```
std::shared_ptr<Graph> AlgorithmWorker::m_graph [private]
```

6.11.4.4 m_start

```
unsigned AlgorithmWorker::m_start [private]
```

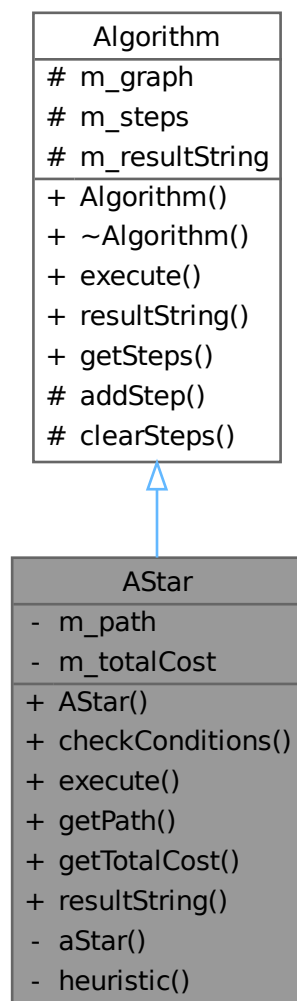
The documentation for this class was generated from the following files:

- Headers/Threads/[AlgorithmWorker.h](#)
- Source/Threads/[AlgorithmWorker.cpp](#)

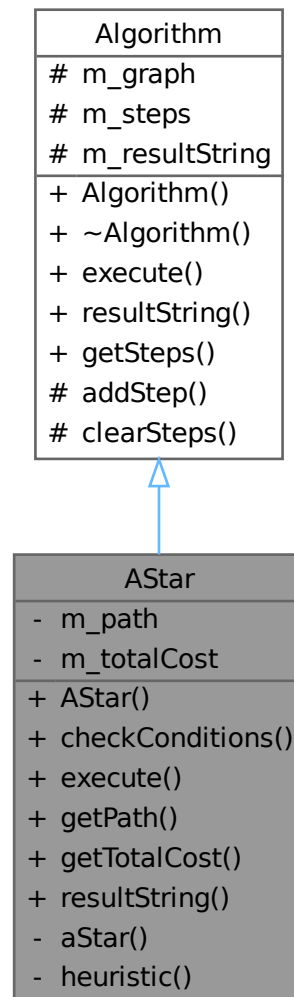
6.12 AStar Class Reference

```
#include <AStar.h>
```

Inheritance diagram for AStar:



Collaboration diagram for AStar:



Public Member Functions

- [AStar](#) (const std::shared_ptr< [Graph](#) >)
- std::optional< [AlgorithmError](#) > [checkConditions](#) (unsigned, unsigned) const
- std::optional< [AlgorithmError](#) > [execute](#) (unsigned, unsigned) override
- const std::vector< unsigned > & [getPath](#) () const
- int [getTotalCost](#) () const
- QString [resultString](#) () const override

Public Member Functions inherited from [Algorithm](#)

- [Algorithm](#) (const std::shared_ptr< [Graph](#) >)
- virtual [~Algorithm](#) ()=default
- const std::vector< [AlgorithmStep](#) > & [getSteps](#) () const

Private Member Functions

- `std::optional< AlgorithmError > aStar` (unsigned, unsigned)
- `int heuristic` (unsigned, unsigned) const

Private Attributes

- `std::vector< unsigned > m\_path`
- `int m\_totalCost = 0`

Additional Inherited Members

Protected Member Functions inherited from [Algorithm](#)

- `void addStep` (const [AlgorithmStep](#) &)
- `void clearSteps` ()

Protected Attributes inherited from [Algorithm](#)

- `std::shared_ptr< Graph > m\_graph`
- `std::vector< AlgorithmStep > m\_steps`
- `QString m\_resultString`

6.12.1 Constructor & Destructor Documentation

6.12.1.1 AStar()

```
AStar::AStar (  
    const std::shared_ptr< Graph > g ) [explicit]
```

6.12.2 Member Function Documentation

6.12.2.1 aStar()

```
std::optional< AlgorithmError > AStar::aStar (  
    unsigned start,  
    unsigned goal ) [private]
```

6.12.2.2 checkConditions()

```
std::optional< AlgorithmError > AStar::checkConditions (  
    unsigned start,  
    unsigned goal ) const
```

6.12.2.3 execute()

```
std::optional< AlgorithmError > AStar::execute (
    unsigned start,
    unsigned goal ) [override], [virtual]
```

Implements [Algorithm](#).

6.12.2.4 getPath()

```
const std::vector< unsigned > & AStar::getPath ( ) const
```

6.12.2.5 getTotalCost()

```
int AStar::getTotalCost ( ) const
```

6.12.2.6 heuristic()

```
int AStar::heuristic (
    unsigned node,
    unsigned goal ) const [private]
```

6.12.2.7 resultString()

```
QString AStar::resultString ( ) const [override], [virtual]
```

Implements [Algorithm](#).

6.12.3 Member Data Documentation

6.12.3.1 m_path

```
std::vector<unsigned> AStar::m_path [private]
```

6.12.3.2 m_totalCost

```
int AStar::m_totalCost = 0 [private]
```

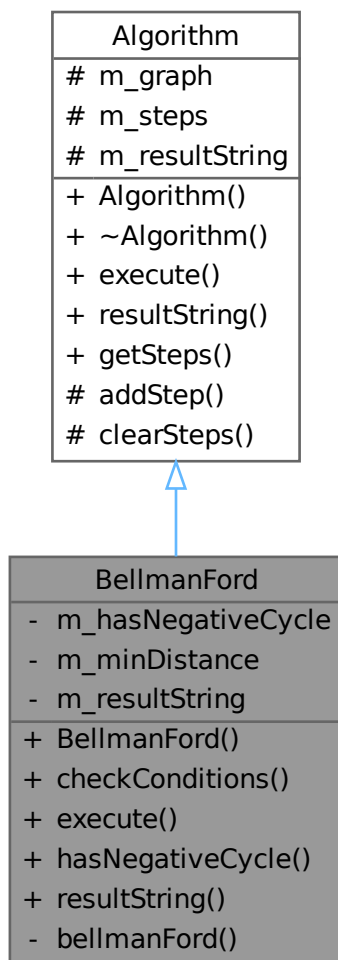
The documentation for this class was generated from the following files:

- Headers/Algorithm/[AStar.h](#)
- Source/Algorithm/[AStar.cpp](#)

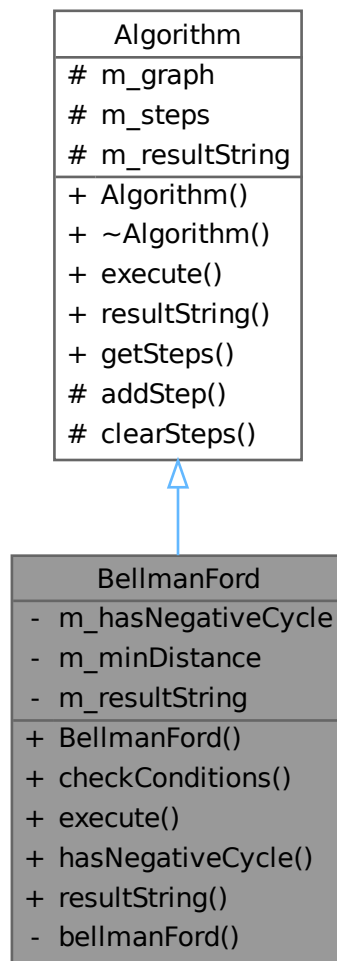
6.13 BellmanFord Class Reference

```
#include <BellmanFord.h>
```

Inheritance diagram for BellmanFord:



Collaboration diagram for BellmanFord:



Public Member Functions

- [BellmanFord](#) (const std::shared_ptr< [Graph](#) >)
- std::optional< [AlgorithmError](#) > [checkConditions](#) (unsigned) const
- std::optional< [AlgorithmError](#) > [execute](#) (unsigned, unsigned=0) override
- bool [hasNegativeCycle](#) () const
- QString [resultString](#) () const override

Public Member Functions inherited from [Algorithm](#)

- [Algorithm](#) (const std::shared_ptr< [Graph](#) >)
- virtual [~Algorithm](#) ()=default
- const std::vector< [AlgorithmStep](#) > & [getSteps](#) () const

Private Member Functions

- void [bellmanFord](#) (unsigned)

Private Attributes

- bool [m_hasNegativeCycle](#) = false
- std::map< unsigned, int > [m_minDistance](#)
- QString [m_resultString](#)

Additional Inherited Members

Protected Member Functions inherited from [Algorithm](#)

- void [addStep](#) (const [AlgorithmStep](#) &)
- void [clearSteps](#) ()

Protected Attributes inherited from [Algorithm](#)

- std::shared_ptr< [Graph](#) > [m_graph](#)
- std::vector< [AlgorithmStep](#) > [m_steps](#)
- QString [m_resultString](#)

6.13.1 Constructor & Destructor Documentation

6.13.1.1 [BellmanFord](#)()

```
BellmanFord::BellmanFord (
    const std::shared_ptr< Graph > g ) [explicit]
```

6.13.2 Member Function Documentation

6.13.2.1 [bellmanFord](#)()

```
void BellmanFord::bellmanFord (
    unsigned start ) [private]
```

6.13.2.2 [checkConditions](#)()

```
std::optional< AlgorithmError > BellmanFord::checkConditions (
    unsigned start ) const
```

6.13.2.3 execute()

```
std::optional< AlgorithmError > BellmanFord::execute (
    unsigned idStartNode,
    unsigned  = 0 ) [override], [virtual]
```

Implements [Algorithm](#).

6.13.2.4 hasNegativeCycle()

```
bool BellmanFord::hasNegativeCycle ( ) const
```

6.13.2.5 resultString()

```
QString BellmanFord::resultString ( ) const [override], [virtual]
```

Implements [Algorithm](#).

6.13.3 Member Data Documentation

6.13.3.1 m_hasNegativeCycle

```
bool BellmanFord::m_hasNegativeCycle = false [private]
```

6.13.3.2 m_minDistance

```
std::map<unsigned, int> BellmanFord::m_minDistance [private]
```

6.13.3.3 m_resultString

```
QString BellmanFord::m_resultString [private]
```

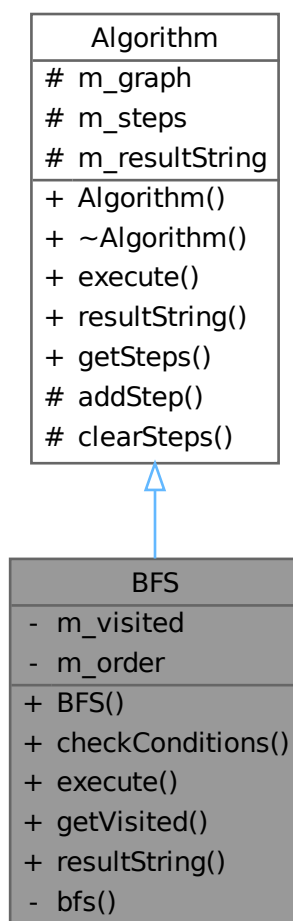
The documentation for this class was generated from the following files:

- Headers/Algorithm/[BellmanFord.h](#)
- Source/Algorithm/[BellmanFord.cpp](#)

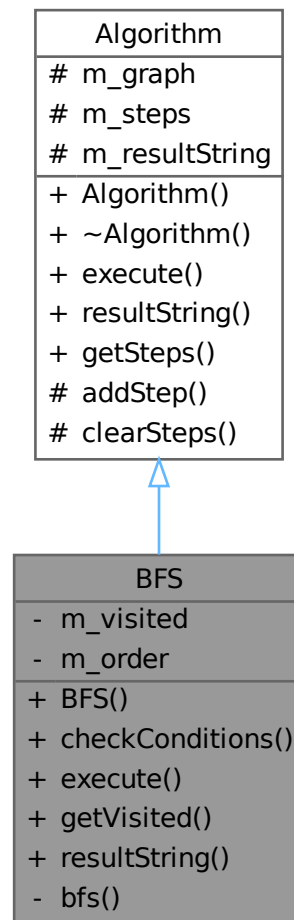
6.14 BFS Class Reference

```
#include <BFS.h>
```

Inheritance diagram for BFS:



Collaboration diagram for BFS:



Public Member Functions

- [BFS](#) (const std::shared_ptr< [Graph](#) >)
- std::optional< [AlgorithmError](#) > [checkConditions](#) (unsigned) const
- std::optional< [AlgorithmError](#) > [execute](#) (unsigned, unsigned=0) override
- const std::map< unsigned, bool > & [getVisited](#) () const
- QString [resultString](#) () const override

Public Member Functions inherited from [Algorithm](#)

- [Algorithm](#) (const std::shared_ptr< [Graph](#) >)
- virtual [~Algorithm](#) ()=default
- const std::vector< [AlgorithmStep](#) > & [getSteps](#) () const

Private Member Functions

- void [bfs](#) (unsigned)

Private Attributes

- `std::map< unsigned, bool >` `m_visited`
- `std::vector< unsigned >` `m_order`

Additional Inherited Members**Protected Member Functions inherited from [Algorithm](#)**

- `void` `addStep` (`const AlgorithmStep &`)
- `void` `clearSteps` ()

Protected Attributes inherited from [Algorithm](#)

- `std::shared_ptr< Graph >` `m_graph`
- `std::vector< AlgorithmStep >` `m_steps`
- `QString` `m_resultString`

6.14.1 Constructor & Destructor Documentation**6.14.1.1 `BFS()`**

```
BFS::BFS (
    const std::shared_ptr< Graph > g ) [explicit]
```

6.14.2 Member Function Documentation**6.14.2.1 `bfs()`**

```
void BFS::bfs (
    unsigned start ) [private]
```

6.14.2.2 `checkConditions()`

```
std::optional< AlgorithmError > BFS::checkConditions (
    unsigned start ) const
```

6.14.2.3 `execute()`

```
std::optional< AlgorithmError > BFS::execute (
    unsigned idStartNode,
    unsigned  = 0 ) [override], [virtual]
```

Implements [Algorithm](#).

6.14.2.4 getVisited()

```
const std::map< unsigned, bool > & BFS::getVisited ( ) const
```

6.14.2.5 resultString()

```
QString BFS::resultString ( ) const [override], [virtual]
```

Implements [Algorithm](#).

6.14.3 Member Data Documentation

6.14.3.1 m_order

```
std::vector<unsigned> BFS::m_order [private]
```

6.14.3.2 m_visited

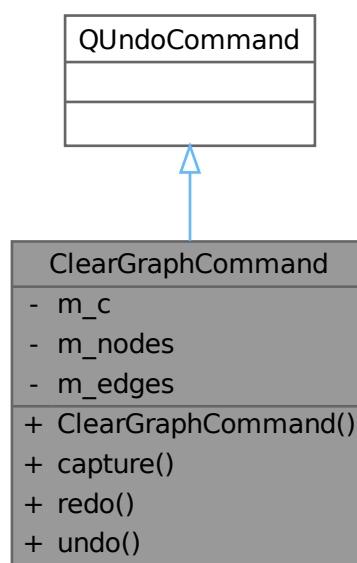
```
std::map<unsigned, bool> BFS::m_visited [private]
```

The documentation for this class was generated from the following files:

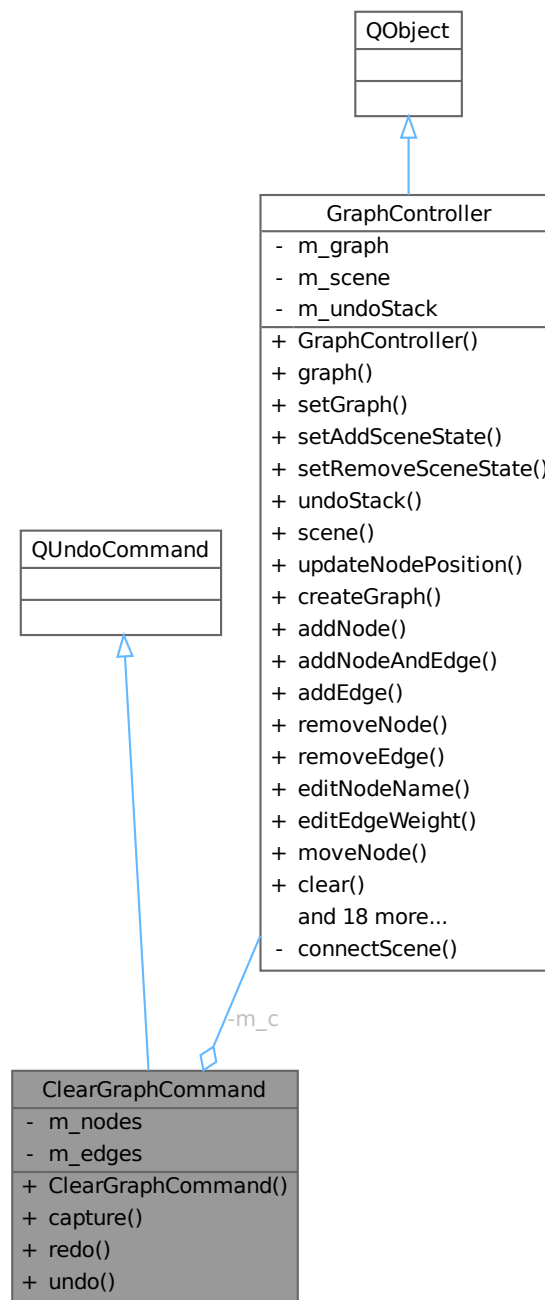
- Headers/Algorithm/[BFS.h](#)
- Source/Algorithm/[BFS.cpp](#)

6.15 ClearGraphCommand Class Reference

Inheritance diagram for ClearGraphCommand:



Collaboration diagram for ClearGraphCommand:



Public Member Functions

- [ClearGraphCommand](#) ([GraphController](#) *c)
- void [capture](#) ()
- void [redo](#) () override
- void [undo](#) () override

Private Attributes

- [GraphController](#) * `m_c`
- `std::vector< NodeSnapshot >` `m_nodes`
- `std::vector< EdgeSnapshot >` `m_edges`

6.15.1 Constructor & Destructor Documentation

6.15.1.1 ClearGraphCommand()

```
ClearGraphCommand::ClearGraphCommand (
    GraphController * c ) [inline], [explicit]
```

6.15.2 Member Function Documentation

6.15.2.1 capture()

```
void ClearGraphCommand::capture ( ) [inline]
```

6.15.2.2 redo()

```
void ClearGraphCommand::redo ( ) [inline], [override]
```

6.15.2.3 undo()

```
void ClearGraphCommand::undo ( ) [inline], [override]
```

6.15.3 Member Data Documentation

6.15.3.1 m_c

```
GraphController* ClearGraphCommand::m_c [private]
```

6.15.3.2 m_edges

```
std::vector<EdgeSnapshot> ClearGraphCommand::m_edges [private]
```

6.15.3.3 m_nodes

```
std::vector<NodeSnapshot> ClearGraphCommand::m_nodes [private]
```

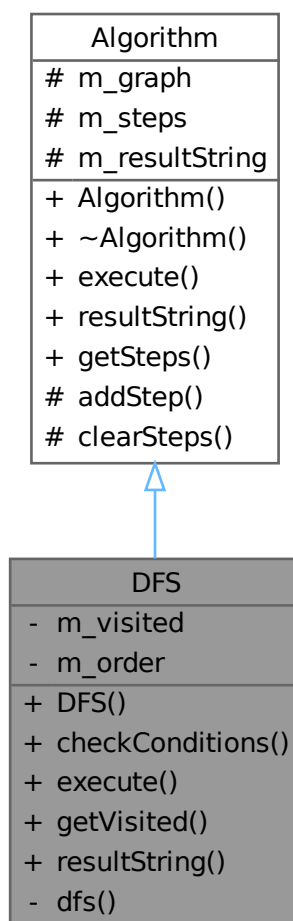
The documentation for this class was generated from the following file:

- Source/Controllers/[GraphController.cpp](#)

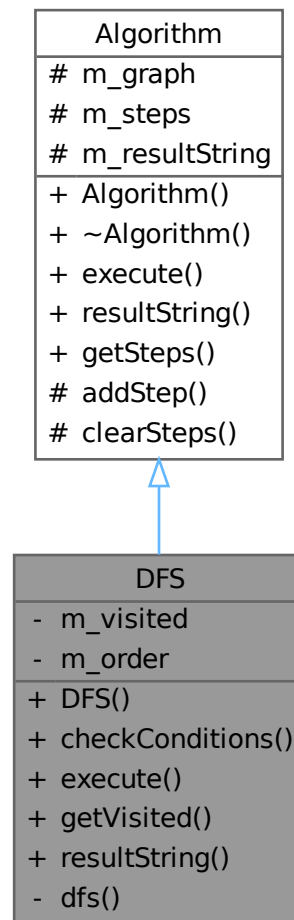
6.16 DFS Class Reference

```
#include <DFS.h>
```

Inheritance diagram for DFS:



Collaboration diagram for DFS:



Public Member Functions

- [DFS](#) (const std::shared_ptr< [Graph](#) >)
- std::optional< [AlgorithmError](#) > [checkConditions](#) (unsigned) const
- std::optional< [AlgorithmError](#) > [execute](#) (unsigned, unsigned=0) override
- const std::map< unsigned, bool > & [getVisited](#) () const
- QString [resultString](#) () const override

Public Member Functions inherited from [Algorithm](#)

- [Algorithm](#) (const std::shared_ptr< [Graph](#) >)
- virtual [~Algorithm](#) ()=default
- const std::vector< [AlgorithmStep](#) > & [getSteps](#) () const

Private Member Functions

- void [dfs](#) (unsigned)

Private Attributes

- `std::map< unsigned, bool >` `m_visited`
- `std::vector< unsigned >` `m_order`

Additional Inherited Members**Protected Member Functions inherited from [Algorithm](#)**

- `void` `addStep` (`const AlgorithmStep &`)
- `void` `clearSteps` ()

Protected Attributes inherited from [Algorithm](#)

- `std::shared_ptr< Graph >` `m_graph`
- `std::vector< AlgorithmStep >` `m_steps`
- `QString` `m_resultString`

6.16.1 Constructor & Destructor Documentation**6.16.1.1 DFS()**

```
DFS::DFS (
    const std::shared_ptr< Graph > g ) [explicit]
```

6.16.2 Member Function Documentation**6.16.2.1 checkConditions()**

```
std::optional< AlgorithmError > DFS::checkConditions (
    unsigned start ) const
```

6.16.2.2 dfs()

```
void DFS::dfs (
    unsigned nodeId ) [private]
```

6.16.2.3 execute()

```
std::optional< AlgorithmError > DFS::execute (
    unsigned idStartNode,
    unsigned  = 0 ) [override], [virtual]
```

Implements [Algorithm](#).

6.16.2.4 getVisited()

```
const std::map< unsigned, bool > & DFS::getVisited ( ) const
```

6.16.2.5 resultString()

```
QString DFS::resultString ( ) const [override], [virtual]
```

Implements [Algorithm](#).

6.16.3 Member Data Documentation

6.16.3.1 m_order

```
std::vector<unsigned> DFS::m_order [private]
```

6.16.3.2 m_visited

```
std::map<unsigned, bool> DFS::m_visited [private]
```

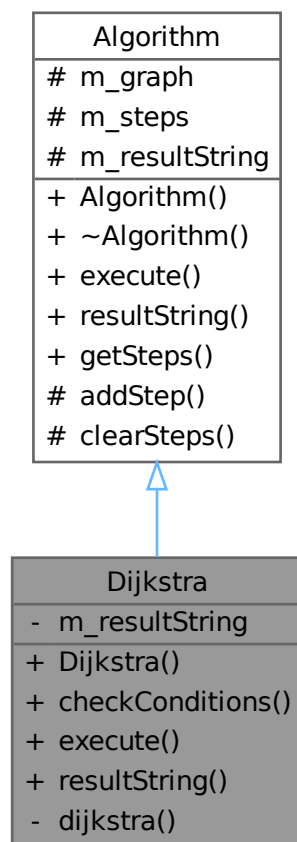
The documentation for this class was generated from the following files:

- Headers/Algorithm/[DFS.h](#)
- Source/Algorithm/[DFS.cpp](#)

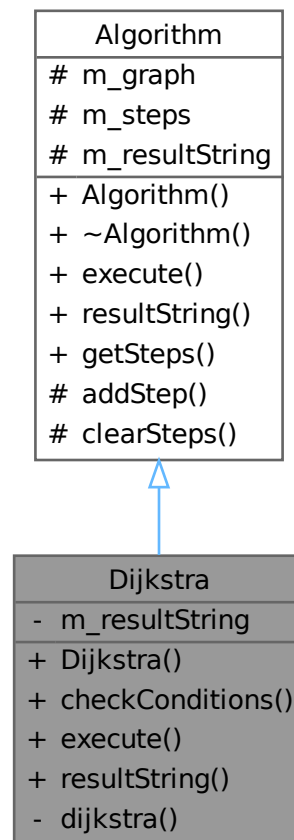
6.17 Dijkstra Class Reference

```
#include <Dijkstra.h>
```

Inheritance diagram for Dijkstra:



Collaboration diagram for Dijkstra:



Public Member Functions

- [Dijkstra](#) (const std::shared_ptr< [Graph](#) >)
- std::optional< [AlgorithmError](#) > [checkConditions](#) (unsigned) const
- std::optional< [AlgorithmError](#) > [execute](#) (unsigned, unsigned=0) override
- QString [resultString](#) () const override

Public Member Functions inherited from [Algorithm](#)

- [Algorithm](#) (const std::shared_ptr< [Graph](#) >)
- virtual [~Algorithm](#) ()=default
- const std::vector< [AlgorithmStep](#) > & [getSteps](#) () const

Private Member Functions

- void [dijkstra](#) (unsigned)

Private Attributes

- QString [m_resultString](#)

Additional Inherited Members

Protected Member Functions inherited from [Algorithm](#)

- void [addStep](#) (const [AlgorithmStep](#) &)
- void [clearSteps](#) ()

Protected Attributes inherited from [Algorithm](#)

- std::shared_ptr< [Graph](#) > [m_graph](#)
- std::vector< [AlgorithmStep](#) > [m_steps](#)
- QString [m_resultString](#)

6.17.1 Constructor & Destructor Documentation

6.17.1.1 Dijkstra()

```
Dijkstra::Dijkstra (
    const std::shared_ptr< Graph > g ) [explicit]
```

6.17.2 Member Function Documentation

6.17.2.1 checkConditions()

```
std::optional< AlgorithmError > Dijkstra::checkConditions (
    unsigned start ) const
```

6.17.2.2 dijkstra()

```
void Dijkstra::dijkstra (
    unsigned start ) [private]
```

6.17.2.3 execute()

```
std::optional< AlgorithmError > Dijkstra::execute (
    unsigned idStartNode,
    unsigned = 0 ) [override], [virtual]
```

Implements [Algorithm](#).

6.17.2.4 resultString()

```
QString Dijkstra::resultString ( ) const [override], [virtual]
```

Implements [Algorithm](#).

6.17.3 Member Data Documentation

6.17.3.1 m_resultString

```
QString Dijkstra::m_resultString [private]
```

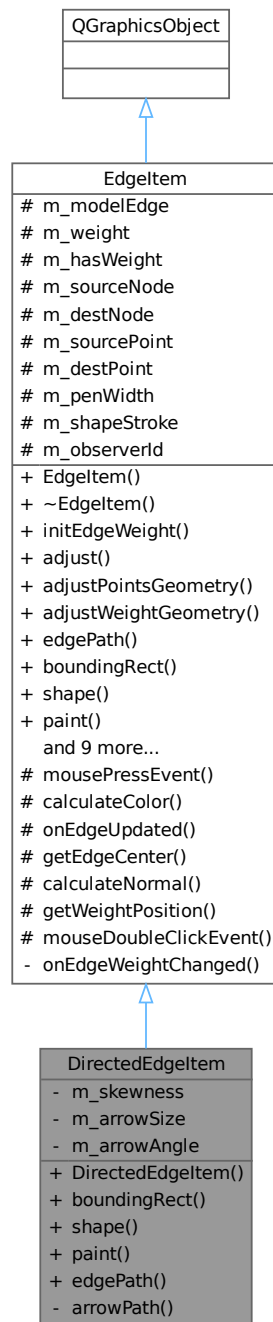
The documentation for this class was generated from the following files:

- Headers/Algorithm/[Dijkstra.h](#)
- Source/Algorithm/[Dijkstra.cpp](#)

6.18 DirectedEdgeItem Class Reference

```
#include <DirectedEdgeItem.h>
```


Inheritance diagram for DirectedEdgelItem:



Collaboration diagram for DirectedEdgeItem:



Public Member Functions

- [DirectedEdgeItem](#) ([Edge](#) *, [NodeItem](#) *, [NodeItem](#) *, bool=false)
- [QRectF](#) [boundingRect](#) () const override
- [QPainterPath](#) [shape](#) () const override
- void [paint](#) ([QPainter](#) *, const [QStyleOptionGraphicsItem](#) *, [QWidget](#) *) override
- [QPainterPath](#) [edgePath](#) () const override

Public Member Functions inherited from [EdgeItem](#)

- [EdgeItem](#) ([Edge](#) *, [NodeItem](#) *, [NodeItem](#) *, bool=false)
- [~EdgeItem](#) () override
- void [initEdgeWeight](#) ()
- void [adjust](#) ()
- void [adjustPointsGeometry](#) ()
- void [adjustWeightGeometry](#) () const
- void [connectNodes](#) ()
- void [disconnectNodes](#) ()
- [Edge](#) * [modelEdge](#) () const
- void [setModelEdge](#) ([Edge](#) *)
- bool [hasWeight](#) () const
- void [setHasWeight](#) (bool)
- [EditableTextItem](#) * [weight](#) () const
- void [setWeight](#) ([EditableTextItem](#) *)
- void [updateSize](#) ()

Private Member Functions

- QPainterPath [arrowPath](#) (const QPainterPath &) const

Private Attributes

- qreal [m_skewness](#) {70.0}
- qreal [m_arrowSize](#) {20.0}
- qreal [m_arrowAngle](#) {M_PI / 6.0}

Additional Inherited Members

Signals inherited from [EdgeItem](#)

- void [edgeSelected](#) ([EdgeItem](#) *)
- void [editEdgeWeightRequest](#) (const [EdgeItem](#) *, const QString &) const

Protected Member Functions inherited from [EdgeItem](#)

- void [mousePressEvent](#) (QGraphicsSceneMouseEvent *) override
- const QColor [calculateColor](#) () const
- void [onEdgeUpdated](#) ()
- QPointF [getEdgeCenter](#) () const
- QPointF [calculateNormal](#) () const
- QPointF [getWeightPosition](#) () const
- void [mouseDoubleClickEvent](#) (QGraphicsSceneMouseEvent *) override

Protected Attributes inherited from [EdgeItem](#)

- [Edge](#) * [m_modelEdge](#)
- [EditableTextItem](#) * [m_weight](#) {nullptr}
- bool [m_hasWeight](#)
- [NodeItem](#) * [m_sourceNode](#)
- [NodeItem](#) * [m_destNode](#)
- QPointF [m_sourcePoint](#)
- QPointF [m_destPoint](#)
- qreal [m_penWidth](#) {2}
- qreal [m_shapeStroke](#) {20.0}
- unsigned [m_observerId](#) {0}

6.18.1 Constructor & Destructor Documentation

6.18.1.1 DirectedEdgeItem()

```
DirectedEdgeItem::DirectedEdgeItem (
    Edge * modelEdge,
    NodeItem * sourceNode,
    NodeItem * destNode,
    bool hasWeight = false )
```

6.18.2 Member Function Documentation

6.18.2.1 arrowPath()

```
QPainterPath DirectedEdgeItem::arrowPath (
    const QPainterPath & edgePath ) const [private]
```

6.18.2.2 boundingRect()

```
QRectF DirectedEdgeItem::boundingRect ( ) const [override], [virtual]
```

Implements [EdgeItem](#).

6.18.2.3 edgePath()

```
QPainterPath DirectedEdgeItem::edgePath ( ) const [override], [virtual]
```

Implements [EdgeItem](#).

6.18.2.4 paint()

```
void DirectedEdgeItem::paint (
    QPainter * painter,
    const QStyleOptionGraphicsItem * ,
    QWidget * ) [override], [virtual]
```

Implements [EdgeItem](#).

6.18.2.5 shape()

```
QPainterPath DirectedEdgeItem::shape ( ) const [override], [virtual]
```

Implements [EdgeItem](#).

6.18.3 Member Data Documentation

6.18.3.1 m_arrowAngle

```
qreal DirectedEdgeItem::m_arrowAngle {M_PI / 6.0} [private]
```

6.18.3.2 m_arrowSize

```
qreal DirectedEdgeItem::m_arrowSize {20.0} [private]
```

6.18.3.3 m_skewness

```
qreal DirectedEdgeItem::m_skewness {70.0} [private]
```

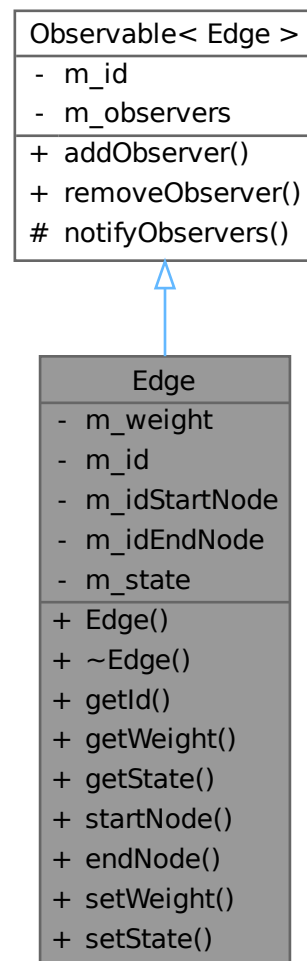
The documentation for this class was generated from the following files:

- Headers/View/[DirectedEdgeItem.h](#)
- Source/View/[DirectedEdgeItem.cpp](#)

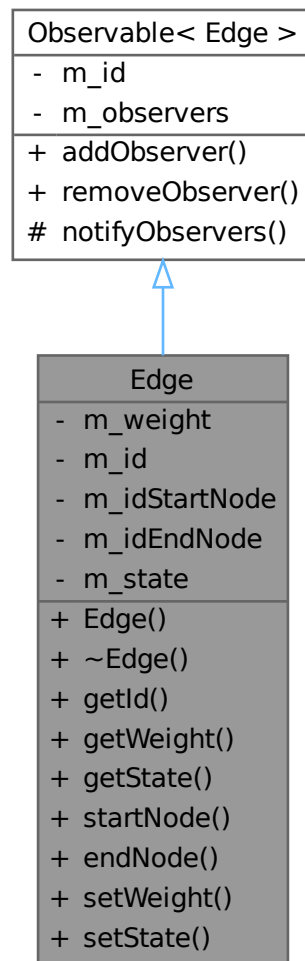
6.19 Edge Class Reference

```
#include <Edge.h>
```

Inheritance diagram for Edge:



Collaboration diagram for Edge:



Public Member Functions

- `Edge` (unsigned, unsigned, unsigned, int=1)
- `~Edge` ()=default
- unsigned `getId` () const
- int `getWeight` () const
- `EdgeState` `getState` () const
- unsigned `startNode` () const
- unsigned `endNode` () const
- void `setWeight` (int)
- void `setState` (`EdgeState`)

Public Member Functions inherited from `Observable< Edge >`

- unsigned `addObserver` (`Observer` observer)

Adds a new observer.

- void [removeObserver](#) (unsigned id)

Removes an observer by its unique ID.

Private Attributes

- int [m_weight](#)
- unsigned [m_id](#)
- unsigned [m_idStartNode](#)
- unsigned [m_idEndNode](#)
- [EdgeState](#) [m_state](#) = [EdgeState::Default](#)

Additional Inherited Members

Public Types inherited from [Observable< Edge >](#)

- using [Observer](#) = std::function< void([Edge](#) &)>
Type of the observer function: takes a reference to the subject.

Protected Member Functions inherited from [Observable< Edge >](#)

- void [notifyObservers](#) ([Edge](#) &subject)
Notifies all registered observers of a state change.

6.19.1 Constructor & Destructor Documentation

6.19.1.1 [Edge\(\)](#)

```
Edge::Edge (
    unsigned id,
    unsigned idStartNode,
    unsigned idEndNode,
    int weight = 1 )
```

6.19.1.2 [~Edge\(\)](#)

```
Edge::~Edge ( ) [default]
```

6.19.2 Member Function Documentation

6.19.2.1 [endNode\(\)](#)

```
unsigned Edge::endNode ( ) const
```


6.19.2.2 getId()

```
unsigned Edge::getId ( ) const
```

6.19.2.3 getState()

```
EdgeState Edge::getState ( ) const
```

6.19.2.4 getWeight()

```
int Edge::getWeight ( ) const
```

6.19.2.5 setState()

```
void Edge::setState (
    EdgeState state )
```

6.19.2.6 setWeight()

```
void Edge::setWeight (
    int w )
```

6.19.2.7 startNode()

```
unsigned Edge::startNode ( ) const
```

6.19.3 Member Data Documentation

6.19.3.1 m_id

```
unsigned Edge::m_id [private]
```

6.19.3.2 m_idEndNode

```
unsigned Edge::m_idEndNode [private]
```

6.19.3.3 m_idStartNode

```
unsigned Edge::m_idStartNode [private]
```

6.19.3.4 m_state

```
EdgeState Edge::m_state = EdgeState::Default [private]
```

6.19.3.5 m_weight

```
int Edge::m_weight [private]
```

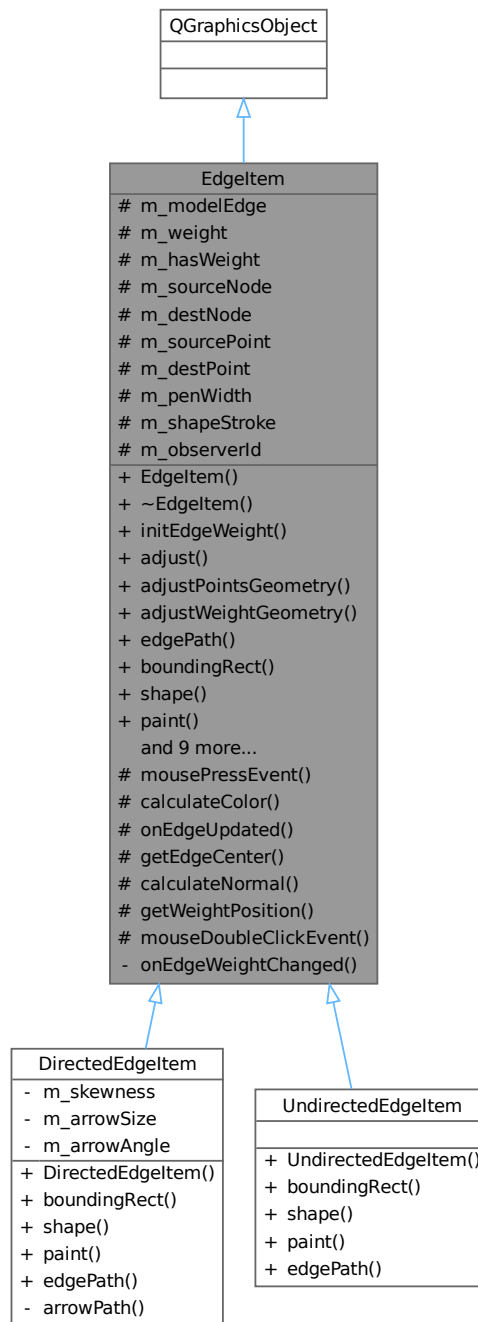
The documentation for this class was generated from the following files:

- Headers/Graph/[Edge.h](#)
- Source/Graph/[Edge.cpp](#)

6.20 Edgeltem Class Reference

```
#include <EdgeItem.h>
```

Inheritance diagram for Edgelitem:



Collaboration diagram for EdgelItem:



Signals

- void [edgeSelected](#) ([EdgelItem](#) *)
- void [editEdgeWeightRequest](#) (const [EdgelItem](#) *, const QString &) const

Public Member Functions

- [EdgelItem](#) ([Edge](#) *, [NodeItem](#) *, [NodeItem](#) *, bool=false)

- [~EdgItem](#) () override
- void [initEdgeWeight](#) ()
- void [adjust](#) ()
- void [adjustPointsGeometry](#) ()
- void [adjustWeightGeometry](#) () const
- virtual QPainterPath [edgePath](#) () const =0
- QRectF [boundingRect](#) () const override=0
- QPainterPath [shape](#) () const override=0
- void [paint](#) (QPainter *, const QStyleOptionGraphicsItem *, QWidget *) override=0
- void [connectNodes](#) ()
- void [disconnectNodes](#) ()
- [Edge](#) * [modelEdge](#) () const
- void [setModelEdge](#) ([Edge](#) *)
- bool [hasWeight](#) () const
- void [setHasWeight](#) (bool)
- [EditableTextItem](#) * [weight](#) () const
- void [setWeight](#) ([EditableTextItem](#) *)
- void [updateSize](#) ()

Protected Member Functions

- void [mousePressEvent](#) (QGraphicsSceneMouseEvent *) override
- const QColor [calculateColor](#) () const
- void [onEdgeUpdated](#) ()
- QPointF [getEdgeCenter](#) () const
- QPointF [calculateNormal](#) () const
- QPointF [getWeightPosition](#) () const
- void [mouseDoubleClickEvent](#) (QGraphicsSceneMouseEvent *) override

Protected Attributes

- [Edge](#) * [m_modelEdge](#)
- [EditableTextItem](#) * [m_weight](#) {nullptr}
- bool [m_hasWeight](#)
- [NodeItem](#) * [m_sourceNode](#)
- [NodeItem](#) * [m_destNode](#)
- QPointF [m_sourcePoint](#)
- QPointF [m_destPoint](#)
- qreal [m_penWidth](#) {2}
- qreal [m_shapeStroke](#) {20.0}
- unsigned [m_observerId](#) {0}

Private Slots

- void [onEdgeWeightChanged](#) (const QString &) const

6.20.1 Constructor & Destructor Documentation

6.20.1.1 EdgeItem()

```
EdgeItem::EdgeItem (
    Edge * modelEdge,
    NodeItem * sourceNode,
    NodeItem * destNode,
    bool hasWeight = false )
```

6.20.1.2 ~EdgeItem()

```
EdgeItem::~EdgeItem ( ) [override]
```

6.20.2 Member Function Documentation

6.20.2.1 adjust()

```
void EdgeItem::adjust ( )
```

6.20.2.2 adjustPointsGeometry()

```
void EdgeItem::adjustPointsGeometry ( )
```

6.20.2.3 adjustWeightGeometry()

```
void EdgeItem::adjustWeightGeometry ( ) const
```

6.20.2.4 boundingRect()

```
QRectF EdgeItem::boundingRect ( ) const [override], [pure virtual]
```

Implemented in [DirectedEdgeItem](#), and [UndirectedEdgeItem](#).

6.20.2.5 calculateColor()

```
const QColor EdgeItem::calculateColor ( ) const [protected]
```

6.20.2.6 calculateNormal()

```
QPointF EdgeItem::calculateNormal ( ) const [protected]
```

6.20.2.7 connectNodes()

```
void EdgItem::connectNodes ( )
```

6.20.2.8 disconnectNodes()

```
void EdgItem::disconnectNodes ( )
```

6.20.2.9 edgePath()

```
virtual QPainterPath EdgItem::edgePath ( ) const [pure virtual]
```

Implemented in [DirectedEdgItem](#), and [UndirectedEdgItem](#).

6.20.2.10 edgeSelected

```
void EdgItem::edgeSelected (
    EdgItem * ) [signal]
```

6.20.2.11 editEdgeWeightRequest

```
void EdgItem::editEdgeWeightRequest (
    const EdgItem * ,
    const QString & ) const [signal]
```

6.20.2.12 getEdgeCenter()

```
QPointF EdgItem::getEdgeCenter ( ) const [protected]
```

6.20.2.13 getWeightPosition()

```
QPointF EdgItem::getWeightPosition ( ) const [protected]
```

6.20.2.14 hasWeight()

```
bool EdgItem::hasWeight ( ) const
```

6.20.2.15 initEdgeWeight()

```
void EdgItem::initEdgeWeight ( )
```

6.20.2.16 modelEdge()

```
Edge * EdgeItem::modelEdge ( ) const
```

6.20.2.17 mouseDoubleClickEvent()

```
void EdgeItem::mouseDoubleClickEvent (
    QGraphicsSceneMouseEvent * event ) [override], [protected]
```

6.20.2.18 mousePressEvent()

```
void EdgeItem::mousePressEvent (
    QGraphicsSceneMouseEvent * event ) [override], [protected]
```

6.20.2.19 onEdgeUpdated()

```
void EdgeItem::onEdgeUpdated ( ) [protected]
```

6.20.2.20 onEdgeWeightChanged

```
void EdgeItem::onEdgeWeightChanged (
    const QString & name ) const [private], [slot]
```

6.20.2.21 paint()

```
void EdgeItem::paint (
    QPainter * ,
    const QStyleOptionGraphicsItem * ,
    QWidget * ) [override], [pure virtual]
```

Implemented in [DirectedEdgeItem](#), and [UndirectedEdgeItem](#).

6.20.2.22 setHasWeight()

```
void EdgeItem::setHasWeight (
    bool newHasWeight )
```

6.20.2.23 setModelEdge()

```
void EdgeItem::setModelEdge (
    Edge * newModelEdge )
```


6.20.2.24 setWeight()

```
void EdgItem::setWeight (
    EditableTextItem * newWeight )
```

6.20.2.25 shape()

```
QPainterPath EdgItem::shape ( ) const [override], [pure virtual]
```

Implemented in [DirectedEdgItem](#), and [UndirectedEdgItem](#).

6.20.2.26 updateSize()

```
void EdgItem::updateSize ( )
```

6.20.2.27 weight()

```
EditableTextItem * EdgItem::weight ( ) const
```

6.20.3 Member Data Documentation

6.20.3.1 m_destNode

```
NodeItem* EdgItem::m_destNode [protected]
```

6.20.3.2 m_destPoint

```
QPointF EdgItem::m_destPoint [protected]
```

6.20.3.3 m_hasWeight

```
bool EdgItem::m_hasWeight [protected]
```

6.20.3.4 m_modelEdge

```
Edge* EdgItem::m_modelEdge [protected]
```

6.20.3.5 m_observerId

```
unsigned EdgItem::m_observerId {0} [protected]
```

6.20.3.6 m_penWidth

```
qreal EdgeItem::m_penWidth {2} [protected]
```

6.20.3.7 m_shapeStroke

```
qreal EdgeItem::m_shapeStroke {20.0} [protected]
```

6.20.3.8 m_sourceNode

```
NodeItem* EdgeItem::m_sourceNode [protected]
```

6.20.3.9 m_sourcePoint

```
QPointF EdgeItem::m_sourcePoint [protected]
```

6.20.3.10 m_weight

```
EditableTextItem* EdgeItem::m_weight {nullptr} [protected]
```

The documentation for this class was generated from the following files:

- Headers/View/[EdgeItem.h](#)
- Source/View/[EdgeItem.cpp](#)

6.21 EdgeSnapshot Struct Reference

Collaboration diagram for EdgeSnapshot:



Public Attributes

- unsigned [m_id](#)
- unsigned [m_from](#)
- unsigned [m_to](#)
- int [m_weight](#)

6.21.1 Member Data Documentation

6.21.1.1 m_from

```
unsigned EdgeSnapshot::m_from
```

6.21.1.2 m_id

```
unsigned EdgeSnapshot::m_id
```

6.21.1.3 m_to

```
unsigned EdgeSnapshot::m_to
```

6.21.1.4 m_weight

```
int EdgeSnapshot::m_weight
```

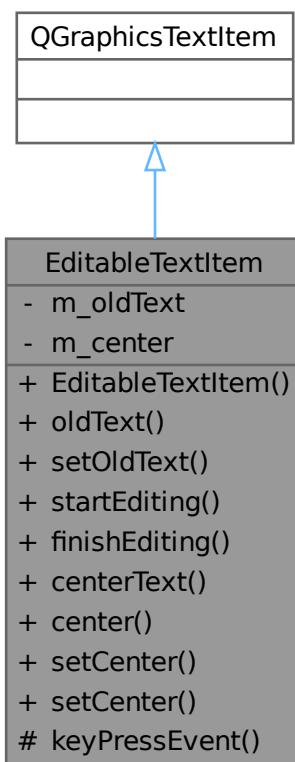
The documentation for this struct was generated from the following file:

- Source/Controllers/[GraphController.cpp](#)

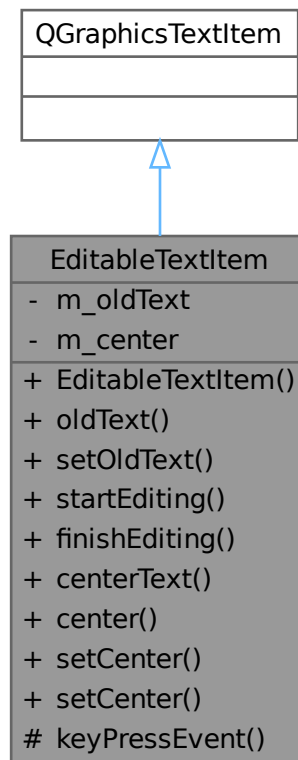
6.22 EditableTextItem Class Reference

```
#include <EditableTextItem.h>
```

Inheritance diagram for EditableTextItem:



Collaboration diagram for EditableTextItem:



Signals

- void [textCommitted](#) (const QString &text)
- void [setEditGraphSceneState](#) (bool, [EditableTextItem](#) *!=nullptr)

Public Member Functions

- [EditableTextItem](#) (QGraphicsItem *!=nullptr)
- QString [oldText](#) () const
- void [setOldText](#) (const QString &)
- void [startEditing](#) ()
- void [finishEditing](#) (bool)
- void [centerText](#) ()
- QPointF [center](#) () const
- void [setCenter](#) (const QPointF &)
- void [setCenter](#) (qreal, qreal)

Protected Member Functions

- void [keyPressEvent](#) (QKeyEvent *) override

Private Attributes

- QString [m_oldText](#)
- QPointF [m_center](#)

6.22.1 Constructor & Destructor Documentation

6.22.1.1 EditableTextItem()

```
EditableTextItem::EditableTextItem (
    QGraphicsItem * parent = nullptr ) [explicit]
```

6.22.2 Member Function Documentation

6.22.2.1 center()

```
QPointF EditableTextItem::center ( ) const
```

6.22.2.2 centerText()

```
void EditableTextItem::centerText ( )
```

6.22.2.3 finishEditing()

```
void EditableTextItem::finishEditing (
    bool commit )
```

6.22.2.4 keyPressEvent()

```
void EditableTextItem::keyPressEvent (
    QKeyEvent * event ) [override], [protected]
```

6.22.2.5 oldText()

```
QString EditableTextItem::oldText ( ) const
```

6.22.2.6 setCenter() [1/2]

```
void EditableTextItem::setCenter (
    const QPointF & newCenter )
```

6.22.2.7 setCenter() [2/2]

```
void EditableTextItem::setCenter (
    qreal x,
    qreal y )
```

6.22.2.8 setEditGraphSceneState

```
void EditableTextItem::setEditGraphSceneState (
    bool ,
    EditableTextItem * = nullptr ) [signal]
```

6.22.2.9 setOldText()

```
void EditableTextItem::setOldText (
    const QString & newOldText )
```

6.22.2.10 startEditing()

```
void EditableTextItem::startEditing ( )
```

6.22.2.11 textCommitted

```
void EditableTextItem::textCommitted (
    const QString & text ) [signal]
```

6.22.3 Member Data Documentation

6.22.3.1 m_center

```
QPointF EditableTextItem::m_center [private]
```

6.22.3.2 m_oldText

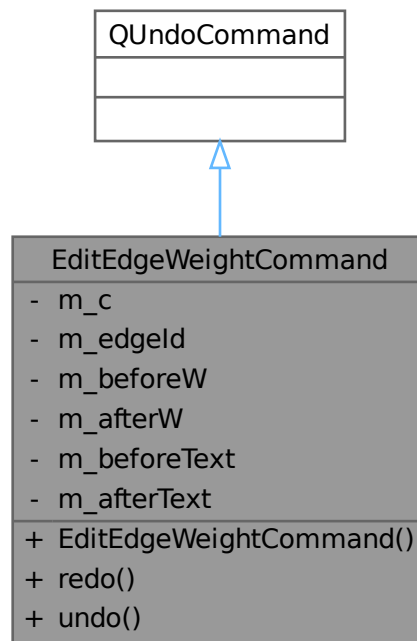
```
QString EditableTextItem::m_oldText [private]
```

The documentation for this class was generated from the following files:

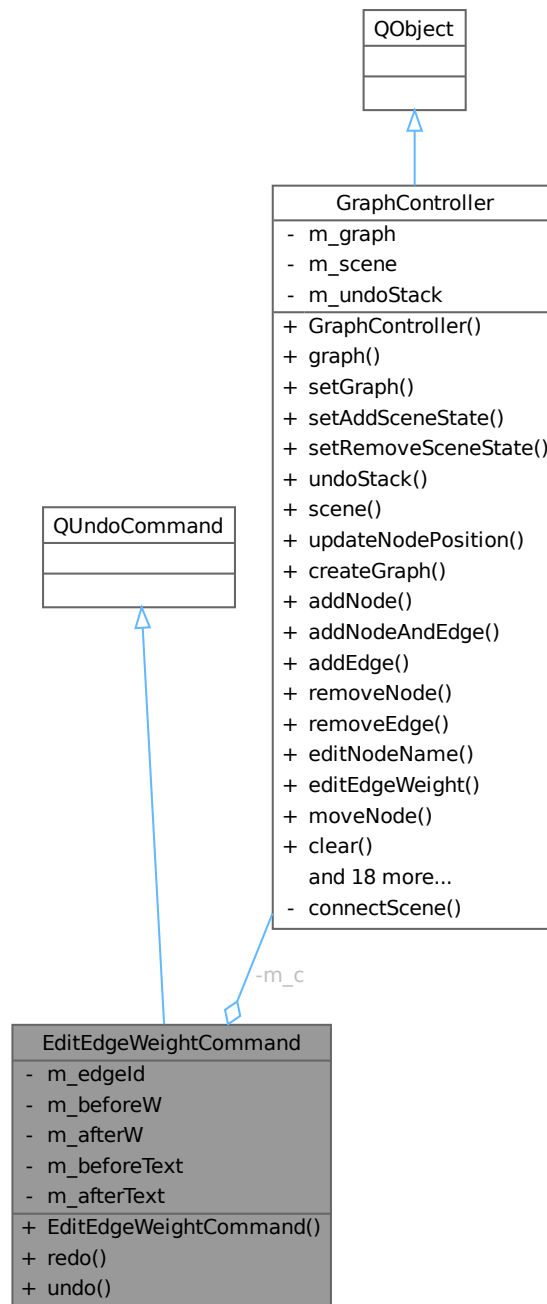
- Headers/View/[EditableTextItem.h](#)
- Source/View/[EditableTextItem.cpp](#)

6.23 EditEdgeWeightCommand Class Reference

Inheritance diagram for EditEdgeWeightCommand:



Collaboration diagram for EditEdgeWeightCommand:



Public Member Functions

- [EditEdgeWeightCommand](#) ([GraphController](#) *c, unsigned edgeId, int beforeW, QString beforeText, int afterW, QString afterText)
- void [redo](#) () override
- void [undo](#) () override

Private Attributes

- [GraphController](#) * [m_c](#) = nullptr
- unsigned [m_edgId](#) = 0
- int [m_beforeW](#) = 0
- int [m_afterW](#) = 0
- QString [m_beforeText](#)
- QString [m_afterText](#)

6.23.1 Constructor & Destructor Documentation

6.23.1.1 EditEdgeWeightCommand()

```
EditEdgeWeightCommand::EditEdgeWeightCommand (
    GraphController * c,
    unsigned edgeId,
    int beforeW,
    QString beforeText,
    int afterW,
    QString afterText ) [inline]
```

6.23.2 Member Function Documentation

6.23.2.1 redo()

```
void EditEdgeWeightCommand::redo ( ) [inline], [override]
```

6.23.2.2 undo()

```
void EditEdgeWeightCommand::undo ( ) [inline], [override]
```

6.23.3 Member Data Documentation

6.23.3.1 m_afterText

```
QString EditEdgeWeightCommand::m_afterText [private]
```

6.23.3.2 m_afterW

```
int EditEdgeWeightCommand::m_afterW = 0 [private]
```

6.23.3.3 m_beforeText

```
QString EditEdgeWeightCommand::m_beforeText [private]
```

6.23.3.4 m_beforeW

```
int EditEdgeWeightCommand::m_beforeW = 0 [private]
```

6.23.3.5 m_c

```
GraphController* EditEdgeWeightCommand::m_c = nullptr [private]
```

6.23.3.6 m_edgId

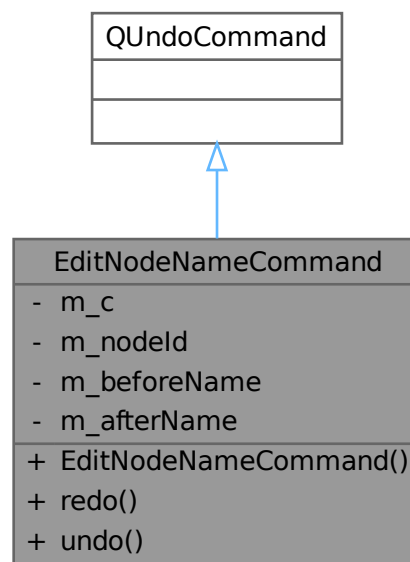
```
unsigned EditEdgeWeightCommand::m_edgeId = 0 [private]
```

The documentation for this class was generated from the following file:

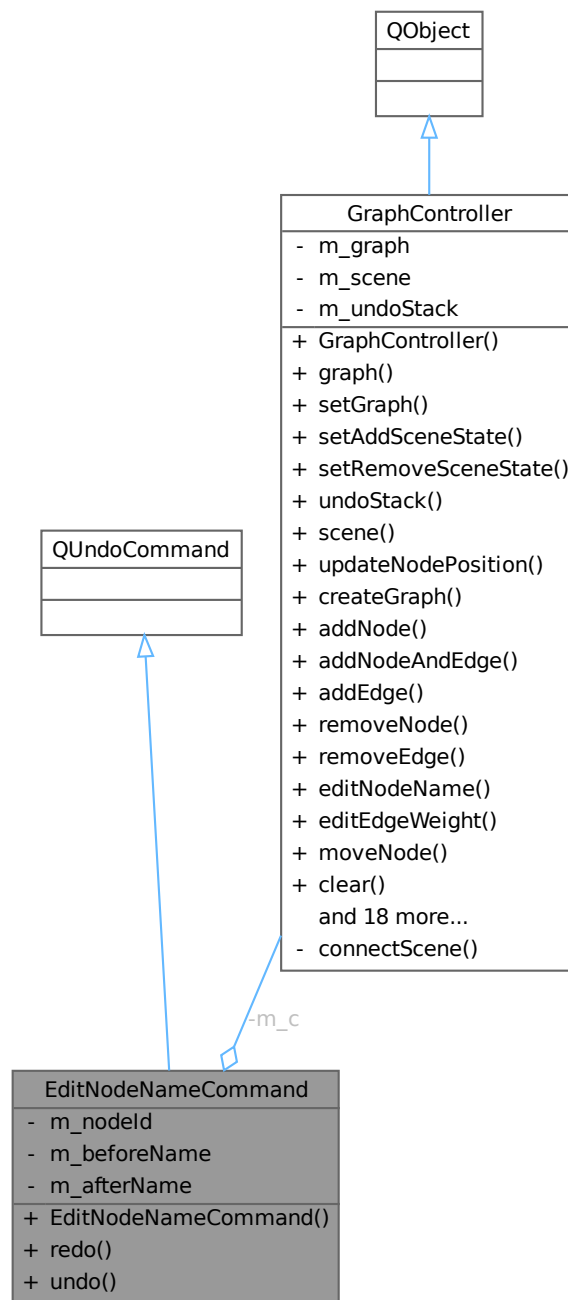
- Source/Controllers/[GraphController.cpp](#)

6.24 EditNodeNameCommand Class Reference

Inheritance diagram for EditNodeNameCommand:



Collaboration diagram for EditNodeNameCommand:



Public Member Functions

- [EditNodeNameCommand](#) ([GraphController](#) *c, unsigned nodeId, QString beforeName, QString afterName)
- void [redo](#) () override
- void [undo](#) () override

Private Attributes

- [GraphController](#) * `m_c`
- unsigned `m_nodeId`
- [QString](#) `m_beforeName`
- [QString](#) `m_afterName`

6.24.1 Constructor & Destructor Documentation

6.24.1.1 EditNodeNameCommand()

```
EditNodeNameCommand::EditNodeNameCommand (
    GraphController * c,
    unsigned nodeId,
    QString beforeName,
    QString afterName ) [inline]
```

6.24.2 Member Function Documentation

6.24.2.1 redo()

```
void EditNodeNameCommand::redo ( ) [inline], [override]
```

6.24.2.2 undo()

```
void EditNodeNameCommand::undo ( ) [inline], [override]
```

6.24.3 Member Data Documentation

6.24.3.1 m_afterName

```
QString EditNodeNameCommand::m_afterName [private]
```

6.24.3.2 m_beforeName

```
QString EditNodeNameCommand::m_beforeName [private]
```

6.24.3.3 m_c

```
GraphController* EditNodeNameCommand::m_c [private]
```

6.24.3.4 m_nodeId

```
unsigned EditNodeNameCommand::m_nodeId [private]
```

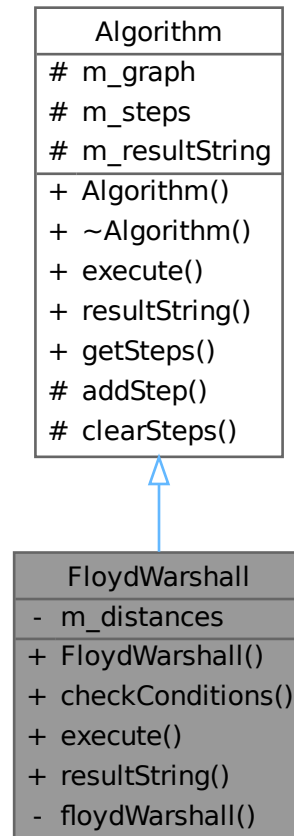
The documentation for this class was generated from the following file:

- Source/Controllers/[GraphController.cpp](#)

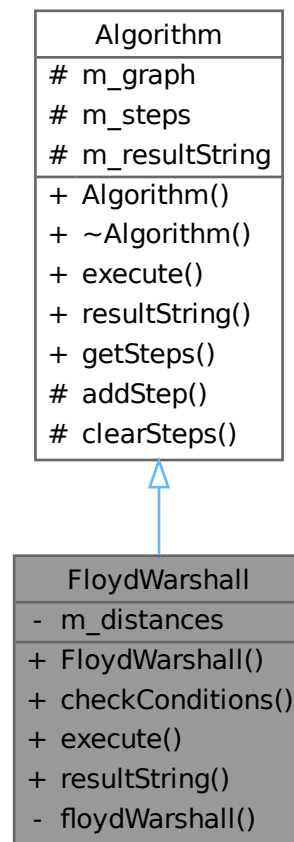
6.25 FloydWarshall Class Reference

```
#include <FloydWarshall.h>
```

Inheritance diagram for FloydWarshall:



Collaboration diagram for FloydWarshall:



Public Member Functions

- [FloydWarshall](#) (const std::shared_ptr< [Graph](#) >)
- std::optional< [AlgorithmError](#) > [checkConditions](#) () const
- std::optional< [AlgorithmError](#) > [execute](#) (unsigned=0, unsigned=0) override
- QString [resultString](#) () const override

Public Member Functions inherited from [Algorithm](#)

- [Algorithm](#) (const std::shared_ptr< [Graph](#) >)
- virtual [~Algorithm](#) ()=default
- const std::vector< [AlgorithmStep](#) > & [getSteps](#) () const

Private Member Functions

- std::optional< [AlgorithmError](#) > [floydWarshall](#) ()

Private Attributes

- `std::map< unsigned, std::map< unsigned, int > >` [m_distances](#)

Additional Inherited Members

Protected Member Functions inherited from [Algorithm](#)

- void [addStep](#) (const [AlgorithmStep](#) &)
- void [clearSteps](#) ()

Protected Attributes inherited from [Algorithm](#)

- `std::shared_ptr< Graph >` [m_graph](#)
- `std::vector< AlgorithmStep >` [m_steps](#)
- `QString` [m_resultString](#)

6.25.1 Constructor & Destructor Documentation

6.25.1.1 FloydWarshall()

```
FloydWarshall::FloydWarshall (
    const std::shared_ptr< Graph > g ) [explicit]
```

6.25.2 Member Function Documentation

6.25.2.1 checkConditions()

```
std::optional< AlgorithmError > FloydWarshall::checkConditions ( ) const
```

6.25.2.2 execute()

```
std::optional< AlgorithmError > FloydWarshall::execute (
    unsigned = 0,
    unsigned = 0 ) [override], [virtual]
```

Implements [Algorithm](#).

6.25.2.3 floydWarshall()

```
std::optional< AlgorithmError > FloydWarshall::floydWarshall ( ) [private]
```


6.25.2.4 resultString()

```
QString FloydWarshall::resultString ( ) const [override], [virtual]
```

Implements [Algorithm](#).

6.25.3 Member Data Documentation

6.25.3.1 m_distances

```
std::map<unsigned, std::map<unsigned, int> > FloydWarshall::m_distances [private]
```

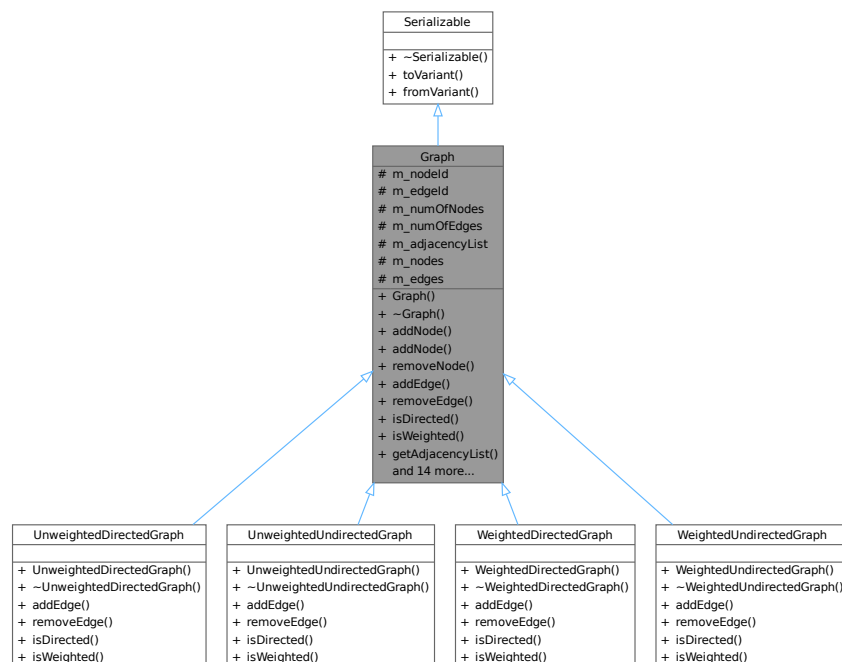
The documentation for this class was generated from the following files:

- Headers/Algorithm/[FloydWarshall.h](#)
- Source/Algorithm/[FloydWarshall.cpp](#)

6.26 Graph Class Reference

```
#include <Graph.h>
```

Inheritance diagram for Graph:



Collaboration diagram for Graph:



Public Member Functions

- [Graph](#) ()=default
- [~Graph](#) () override=default
- [Node * addNode](#) (double, double)
- void [addNode](#) (unsigned, double=0.0, double=0.0)
- void [removeNode](#) (unsigned)
- virtual void [addEdge](#) (unsigned, unsigned, int=1)=0
- virtual void [removeEdge](#) (unsigned)=0
- virtual bool [isDirected](#) () const =0
- virtual bool [isWeighted](#) () const =0
- std::map< unsigned, std::map< unsigned, unsigned > > [getAdjacencyList](#) () const

- `std::map< unsigned, Node > getNodes () const`
- `std::map< unsigned, Edge > getEdges () const`
- `std::vector< Node * > getNodesMutable ()`
- `std::vector< Edge * > getEdgesMutable ()`
- `Node * getNode (unsigned)`
- `const Node * getNode (unsigned) const`
- `Edge * getEdge (unsigned)`
- `const Edge * getEdge (unsigned) const`
- `Edge * getEdge (unsigned, unsigned)`
- `const Edge * getEdge (unsigned, unsigned) const`
- `void addEdgeSerialized (unsigned, unsigned, unsigned, int)`
- `QVariant toVariant () const override`
- `void fromVariant (const QVariant &) override`
- `void clear ()`

Public Member Functions inherited from [Serializable](#)

- `virtual ~Serializable ()=default`

Protected Attributes

- unsigned `m\_nodeId = 0`
- unsigned `m\_edgeId = 0`
- unsigned `m\_numOfNodes = 0`
- unsigned `m\_numOfEdges = 0`
- `std::map< unsigned, std::map< unsigned, unsigned > > m\_adjacencyList`
- `std::map< unsigned, Node > m\_nodes`
- `std::map< unsigned, Edge > m\_edges`

6.26.1 Constructor & Destructor Documentation

6.26.1.1 [Graph](#)()

```
Graph::Graph ( ) [default]
```

6.26.1.2 [~Graph](#)()

```
Graph::~~Graph ( ) [override], [default]
```

6.26.2 Member Function Documentation

6.26.2.1 [addEdge](#)()

```
virtual void Graph::addEdge (
    unsigned ,
    unsigned ,
    int = 1 ) [pure virtual]
```

Implemented in [WeightedDirectedGraph](#), [WeightedUndirectedGraph](#), [UnweightedDirectedGraph](#), and [UnweightedUndirectedGraph](#).

6.26.2.2 addEdgeSerialized()

```
void Graph::addEdgeSerialized (
    unsigned edgeId,
    unsigned from,
    unsigned to,
    int w )
```

6.26.2.3 addNode() [1/2]

```
Node * Graph::addNode (
    double x,
    double y )
```

6.26.2.4 addNode() [2/2]

```
void Graph::addNode (
    unsigned id,
    double x = 0.0,
    double y = 0.0 )
```

6.26.2.5 clear()

```
void Graph::clear ( )
```

6.26.2.6 fromVariant()

```
void Graph::fromVariant (
    const QVariant & variant ) [override], [virtual]
```

Implements [Serializable](#).

6.26.2.7 getAdjacencyList()

```
std::map< unsigned, std::map< unsigned, unsigned > > Graph::getAdjacencyList ( ) const
```

6.26.2.8 getEdge() [1/4]

```
Edge * Graph::getEdge (
    unsigned id )
```

6.26.2.9 getEdge() [2/4]

```
const Edge * Graph::getEdge (
    unsigned id ) const
```

6.26.2.10 getEdge() [3/4]

```
Edge * Graph::getEdge (
    unsigned from,
    unsigned to )
```

6.26.2.11 getEdge() [4/4]

```
const Edge * Graph::getEdge (
    unsigned from,
    unsigned to ) const
```

6.26.2.12 getEdges()

```
std::map< unsigned, Edge > Graph::getEdges ( ) const
```

6.26.2.13 getEdgesMutable()

```
std::vector< Edge * > Graph::getEdgesMutable ( )
```

6.26.2.14 getNode() [1/2]

```
Node * Graph::getNode (
    unsigned id )
```

6.26.2.15 getNode() [2/2]

```
const Node * Graph::getNode (
    unsigned id ) const
```

6.26.2.16 getNodes()

```
std::map< unsigned, Node > Graph::getNodes ( ) const
```

6.26.2.17 getNodesMutable()

```
std::vector< Node * > Graph::getNodesMutable ( )
```

6.26.2.18 isDirected()

```
virtual bool Graph::isDirected ( ) const [pure virtual]
```

Implemented in [UnweightedDirectedGraph](#), [UnweightedUndirectedGraph](#), [WeightedDirectedGraph](#), and [WeightedUndirectedGraph](#).

6.26.2.19 isWeighted()

```
virtual bool Graph::isWeighted ( ) const [pure virtual]
```

Implemented in [UnweightedDirectedGraph](#), [UnweightedUndirectedGraph](#), [WeightedDirectedGraph](#), and [WeightedUndirectedGraph](#).

6.26.2.20 removeEdge()

```
virtual void Graph::removeEdge (
    unsigned ) [pure virtual]
```

Implemented in [UnweightedDirectedGraph](#), [UnweightedUndirectedGraph](#), [WeightedDirectedGraph](#), and [WeightedUndirectedGraph](#).

6.26.2.21 removeNode()

```
void Graph::removeNode (
    unsigned id )
```

6.26.2.22 toVariant()

```
QVariant Graph::toVariant ( ) const [override], [virtual]
```

Implements [Serializable](#).

6.26.3 Member Data Documentation

6.26.3.1 m_adjacencyList

```
std::map<unsigned, std::map<unsigned, unsigned> > Graph::m_adjacencyList [protected]
```

6.26.3.2 m_edgeld

```
unsigned Graph::m_edgeId = 0 [protected]
```

6.26.3.3 m_edges

```
std::map<unsigned, Edge> Graph::m_edges [protected]
```

6.26.3.4 m_nodeId

```
unsigned Graph::m_nodeId = 0 [protected]
```

6.26.3.5 m_nodes

```
std::map<unsigned, Node> Graph::m_nodes [protected]
```

6.26.3.6 m_numOfEdges

```
unsigned Graph::m_numOfEdges = 0 [protected]
```

6.26.3.7 m_numOfNodes

```
unsigned Graph::m_numOfNodes = 0 [protected]
```

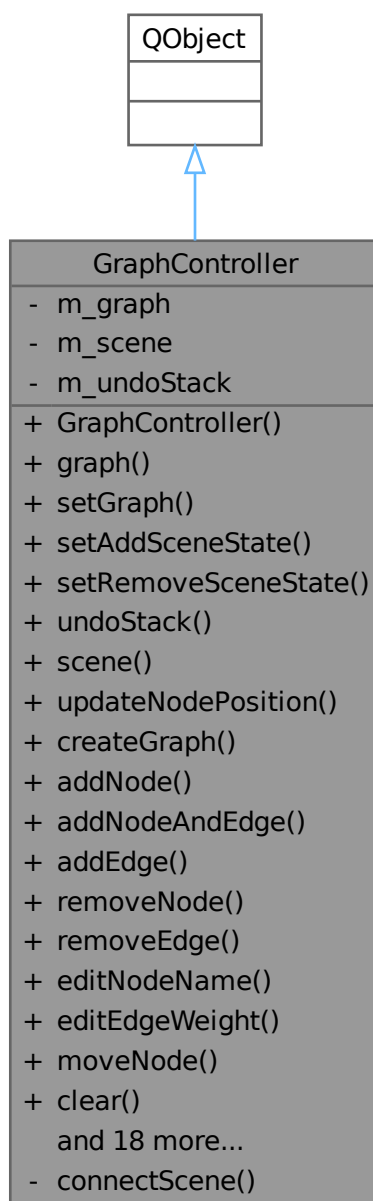
The documentation for this class was generated from the following files:

- Headers/Graph/[Graph.h](#)
- Source/Graph/[Graph.cpp](#)

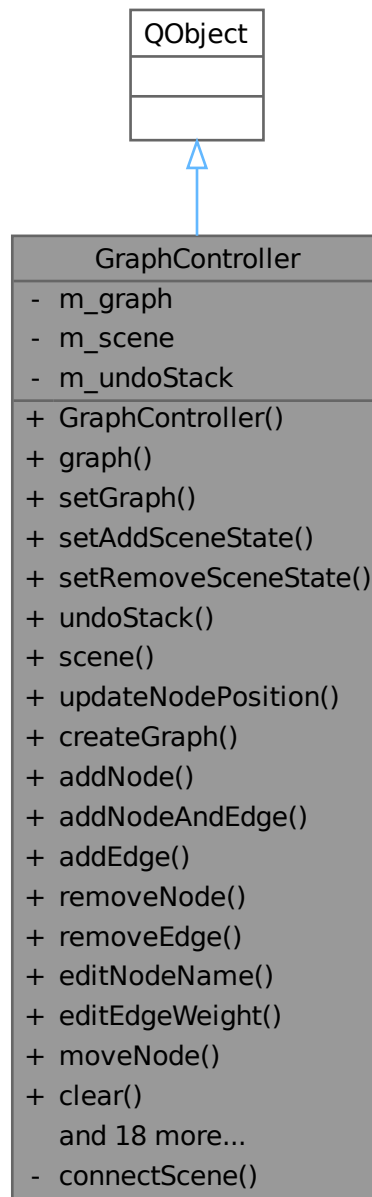
6.27 GraphController Class Reference

```
#include <GraphController.h>
```

Inheritance diagram for GraphController:



Collaboration diagram for GraphController:



Public Slots

- void [createGraph](#) (bool, bool)
- void [addNode](#) (const QPointF &)
- void [addNodeAndEdge](#) (const QPointF &, [NodeItem](#) *)
- void [addEdge](#) ([NodeItem](#) *, [NodeItem](#) *)
- void [removeNode](#) ([NodeItem](#) *)
- void [removeEdge](#) ([EdgeItem](#) *)
- void [editNodeName](#) (const [NodeItem](#) *, const QString &)

- void [editEdgeWeight](#) (const [EdgeItem](#) *, const QString &)
- void [moveNode](#) (const [NodeItem](#) *, const QPointF &, const QPointF &)
- void [clear](#) ()
- void [clearScene](#) () const
- void [buildScene](#) () const
- void [disableScene](#) () const
- void [enableScene](#) () const
- void [resetScene](#) () const
- void [addNodeNoHistory](#) (const QPointF &, unsigned &)
- void [addNodeWithIdNoHistory](#) (unsigned, const QPointF &)
- void [removeNodeNoHistory](#) (unsigned)
- void [addEdgeNoHistory](#) (unsigned, unsigned, int)
- void [removeEdgeNoHistory](#) (unsigned, unsigned)
- void [setEdgeWeightNoHistory](#) (unsigned, unsigned, int, const QString &)
- void [setNodeNameNoHistory](#) (unsigned, const QString &)
- void [removeEdgeNoHistoryById](#) (unsigned)
- void [addEdgeNoHistoryById](#) (unsigned)
- void [restoreEdgeNoHistory](#) (unsigned, unsigned, unsigned, int)
- void [setEdgeWeightNoHistoryById](#) (unsigned, int, const QString &)
- void [moveNodeNoHistory](#) (unsigned, const QPointF &)
- void [clearNoHistory](#) ()

Signals

- void [sceneModified](#) ()

Public Member Functions

- [GraphController](#) (QObject * = nullptr)
- std::shared_ptr< [Graph](#) > [graph](#) () const
- void [setGraph](#) (const std::shared_ptr< [Graph](#) > &)
- void [setAddSceneState](#) () const
- void [setRemoveSceneState](#) () const
- QUndoStack * [undoStack](#) () const
- [GraphScene](#) * [scene](#) () const
- void [updateNodePosition](#) ([NodeItem](#) *, const QPointF &)

Private Member Functions

- void [connectScene](#) () const

Private Attributes

- std::shared_ptr< [Graph](#) > [m_graph](#)
- std::unique_ptr< [GraphScene](#) > [m_scene](#) = std::make_unique<[GraphScene](#)>()
- QUndoStack * [m_undoStack](#) = nullptr

6.27.1 Constructor & Destructor Documentation

6.27.1.1 GraphController()

```
GraphController::GraphController (
    QObject * parent = nullptr ) [explicit]
```

6.27.2 Member Function Documentation

6.27.2.1 addEdge

```
void GraphController::addEdge (
    NodeItem * source,
    NodeItem * dest ) [slot]
```

6.27.2.2 addEdgeNoHistory

```
void GraphController::addEdgeNoHistory (
    unsigned from,
    unsigned to,
    int weight ) [slot]
```

6.27.2.3 addEdgeNoHistoryById

```
void GraphController::addEdgeNoHistoryById (
    unsigned edgeId ) [slot]
```

6.27.2.4 addNode

```
void GraphController::addNode (
    const QPointF & position ) [slot]
```

6.27.2.5 addNodeAndEdge

```
void GraphController::addNodeAndEdge (
    const QPointF & point,
    NodeItem * selectedNodeItem ) [slot]
```

6.27.2.6 addNodeNoHistory

```
void GraphController::addNodeNoHistory (
    const QPointF & pos,
    unsigned & outId ) [slot]
```

6.27.2.7 addNodeWithIdNoHistory

```
void GraphController::addNodeWithIdNoHistory (
    unsigned id,
    const QPointF & pos ) [slot]
```

6.27.2.8 buildScene

```
void GraphController::buildScene ( ) const [slot]
```

6.27.2.9 clear

```
void GraphController::clear ( ) [slot]
```

6.27.2.10 clearNoHistory

```
void GraphController::clearNoHistory ( ) [slot]
```

6.27.2.11 clearScene

```
void GraphController::clearScene ( ) const [slot]
```

6.27.2.12 connectScene()

```
void GraphController::connectScene ( ) const [private]
```

6.27.2.13 createGraph

```
void GraphController::createGraph (
    bool isDirected,
    bool isWeighted ) [slot]
```

6.27.2.14 disableScene

```
void GraphController::disableScene ( ) const [slot]
```

6.27.2.15 editEdgeWeight

```
void GraphController::editEdgeWeight (
    const EdgeItem * edgeItem,
    const QString & weight ) [slot]
```

6.27.2.16 editNodeName

```
void GraphController::editNodeName (
    const NodeItem * nodeItem,
    const QString & name ) [slot]
```

6.27.2.17 enableScene

```
void GraphController::enableScene ( ) const [slot]
```

6.27.2.18 graph()

```
std::shared_ptr< Graph > GraphController::graph ( ) const
```

6.27.2.19 moveNode

```
void GraphController::moveNode (
    const NodeItem * nodeItem,
    const QPointF & oldPos,
    const QPointF & newPos ) [slot]
```

6.27.2.20 moveNodeNoHistory

```
void GraphController::moveNodeNoHistory (
    unsigned nodeId,
    const QPointF & pos ) [slot]
```

6.27.2.21 removeEdge

```
void GraphController::removeEdge (
    EdgeItem * edgeItem ) [slot]
```

6.27.2.22 removeEdgeNoHistory

```
void GraphController::removeEdgeNoHistory (
    unsigned from,
    unsigned to ) [slot]
```

6.27.2.23 removeEdgeNoHistoryById

```
void GraphController::removeEdgeNoHistoryById (
    unsigned edgeId ) [slot]
```

6.27.2.24 removeNode

```
void GraphController::removeNode (
    NodeItem * nodeItem ) [slot]
```

6.27.2.25 removeNodeNoHistory

```
void GraphController::removeNodeNoHistory (
    unsigned id ) [slot]
```

6.27.2.26 resetScene

```
void GraphController::resetScene ( ) const [slot]
```

6.27.2.27 restoreEdgeNoHistory

```
void GraphController::restoreEdgeNoHistory (
    unsigned edgeId,
    unsigned from,
    unsigned to,
    int weight ) [slot]
```

6.27.2.28 scene()

```
GraphScene * GraphController::scene ( ) const
```

6.27.2.29 sceneModified

```
void GraphController::sceneModified ( ) [signal]
```

6.27.2.30 setAddSceneState()

```
void GraphController::setAddSceneState ( ) const
```

6.27.2.31 setEdgeWeightNoHistory

```
void GraphController::setEdgeWeightNoHistory (
    unsigned from,
    unsigned to,
    int weight,
    const QString & weightText ) [slot]
```

6.27.2.32 setEdgeWeightNoHistoryById

```
void GraphController::setEdgeWeightNoHistoryById (
    unsigned edgeId,
    int weight,
    const QString & text ) [slot]
```

6.27.2.33 setGraph()

```
void GraphController::setGraph (
    const std::shared_ptr< Graph > & newGraph )
```

6.27.2.34 setNodeNameNoHistory

```
void GraphController::setNodeNameNoHistory (
    unsigned nodeId,
    const QString & name ) [slot]
```

6.27.2.35 setRemoveSceneState()

```
void GraphController::setRemoveSceneState ( ) const
```

6.27.2.36 undoStack()

```
QUndoStack * GraphController::undoStack ( ) const
```

6.27.2.37 updateNodePosition()

```
void GraphController::updateNodePosition (
    NodeItem * nodeItem,
    const QPointF & point )
```

6.27.3 Member Data Documentation

6.27.3.1 m_graph

```
std::shared_ptr<Graph> GraphController::m_graph [private]
```

6.27.3.2 m_scene

```
std::unique_ptr<GraphScene> GraphController::m_scene = std::make_unique<GraphScene>() [private]
```

6.27.3.3 m_undoStack

```
QUndoStack* GraphController::m_undoStack = nullptr [private]
```

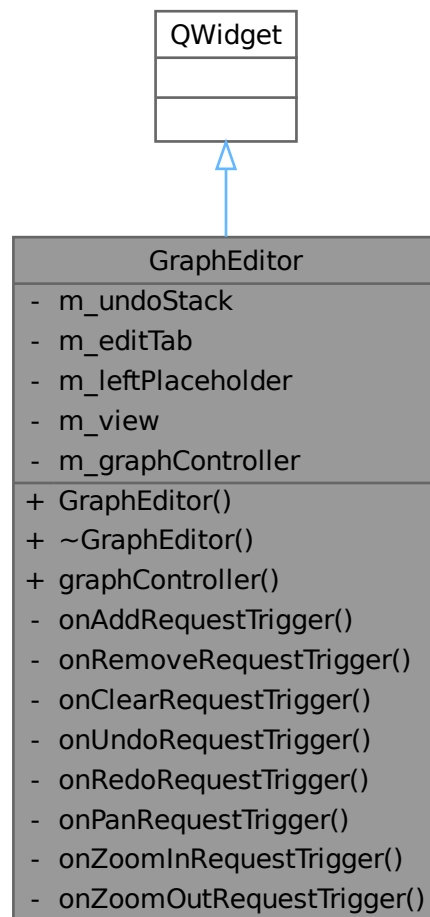
The documentation for this class was generated from the following files:

- Headers/Controllers/[GraphController.h](#)
- Source/Controllers/[GraphController.cpp](#)

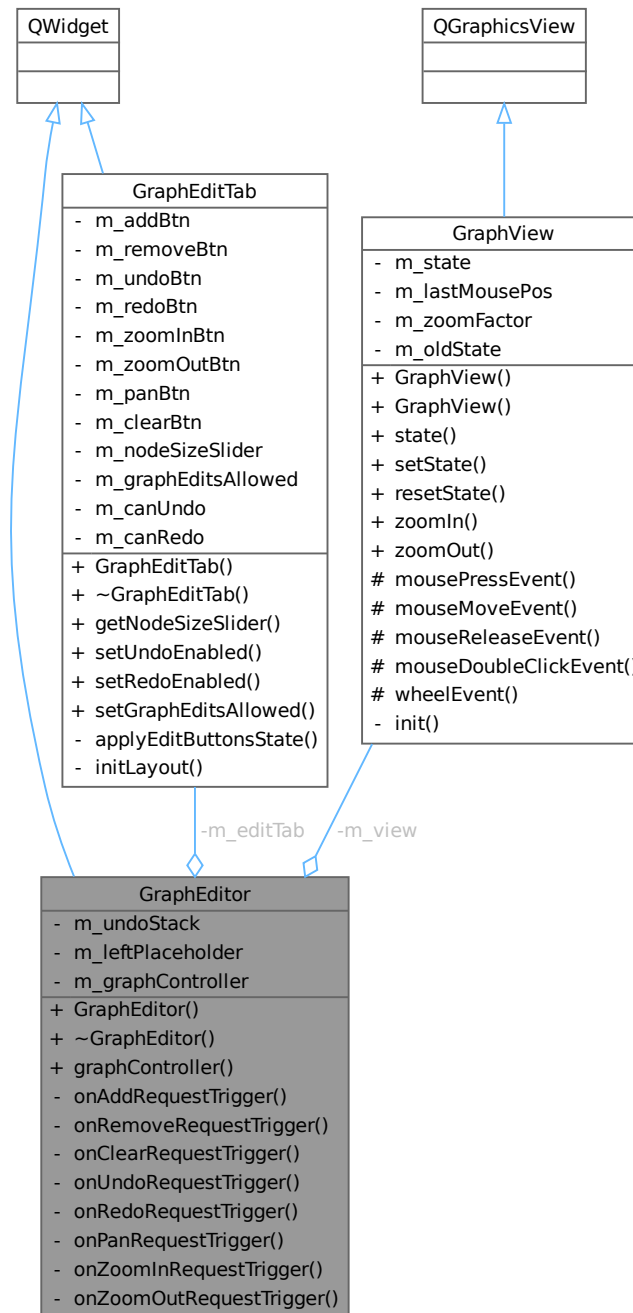
6.28 GraphEditor Class Reference

```
#include <GraphEditor.h>
```

Inheritance diagram for GraphEditor:



Collaboration diagram for GraphEditor:



Signals

- void `undoRequested` ()
- void `redoRequested` ()

Public Member Functions

- `GraphEditor` (const std::shared_ptr< `GraphController` > &, QWidget *!=nullptr)

- [~GraphEditor](#) () override=default
- std::shared_ptr< [GraphController](#) > [graphController](#) () const

Private Slots

- void [onAddRequestTrigger](#) ()
- void [onRemoveRequestTrigger](#) ()
- void [onClearRequestTrigger](#) ()
- void [onUndoRequestTrigger](#) ()
- void [onRedoRequestTrigger](#) ()
- void [onPanRequestTrigger](#) ()
- void [onZoomInRequestTrigger](#) ()
- void [onZoomOutRequestTrigger](#) ()

Private Attributes

- QUndoStack * [m_undoStack](#) = nullptr
- [GraphEditTab](#) * [m_editTab](#) = nullptr
- QLabel * [m_leftPlaceholder](#) = nullptr
- [GraphView](#) * [m_view](#) = nullptr
- std::shared_ptr< [GraphController](#) > [m_graphController](#)

6.28.1 Constructor & Destructor Documentation

6.28.1.1 GraphEditor()

```
GraphEditor::GraphEditor (
    const std::shared_ptr< GraphController > & graphController,
    QWidget * parent = nullptr ) [explicit]
```

6.28.1.2 ~GraphEditor()

```
GraphEditor::~~GraphEditor ( ) [override], [default]
```

6.28.2 Member Function Documentation

6.28.2.1 graphController()

```
std::shared_ptr< GraphController > GraphEditor::graphController ( ) const
```

6.28.2.2 onAddRequestTrigger

```
void GraphEditor::onAddRequestTrigger ( ) [private], [slot]
```

6.28.2.3 onClearRequestTrigger

```
void GraphEditor::onClearRequestTrigger ( ) [private], [slot]
```

6.28.2.4 onPanRequestTrigger

```
void GraphEditor::onPanRequestTrigger ( ) [private], [slot]
```

6.28.2.5 onRedoRequestTrigger

```
void GraphEditor::onRedoRequestTrigger ( ) [private], [slot]
```

6.28.2.6 onRemoveRequestTrigger

```
void GraphEditor::onRemoveRequestTrigger ( ) [private], [slot]
```

6.28.2.7 onUndoRequestTrigger

```
void GraphEditor::onUndoRequestTrigger ( ) [private], [slot]
```

6.28.2.8 onZoomInRequestTrigger

```
void GraphEditor::onZoomInRequestTrigger ( ) [private], [slot]
```

6.28.2.9 onZoomOutRequestTrigger

```
void GraphEditor::onZoomOutRequestTrigger ( ) [private], [slot]
```

6.28.2.10 redoRequested

```
void GraphEditor::redoRequested ( ) [signal]
```

6.28.2.11 undoRequested

```
void GraphEditor::undoRequested ( ) [signal]
```

6.28.3 Member Data Documentation

6.28.3.1 m_editTab

```
GraphEditTab* GraphEditor::m_editTab = nullptr [private]
```

6.28.3.2 m_graphController

```
std::shared_ptr<GraphController> GraphEditor::m_graphController [private]
```

6.28.3.3 m_leftPlaceholder

```
QLabel* GraphEditor::m_leftPlaceholder = nullptr [private]
```

6.28.3.4 m_undoStack

```
QUndoStack* GraphEditor::m_undoStack = nullptr [private]
```

6.28.3.5 m_view

```
GraphView* GraphEditor::m_view = nullptr [private]
```

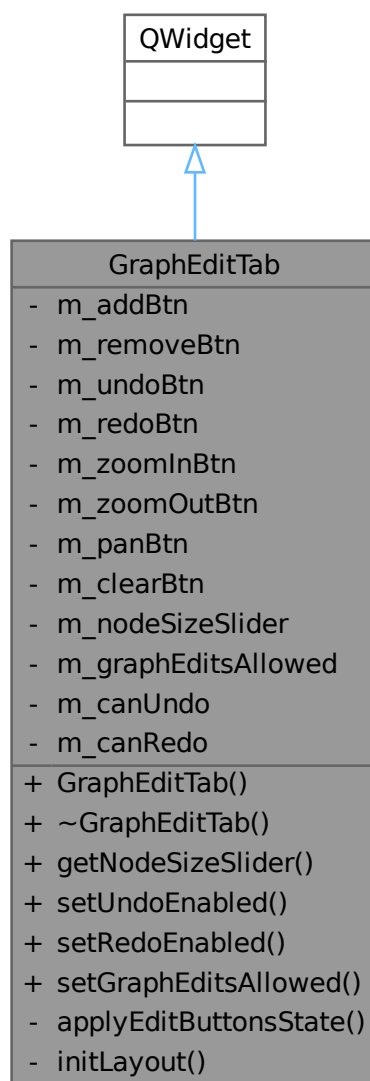
The documentation for this class was generated from the following files:

- [Headers/UI/GraphEditor.h](#)
- [Source/UI/GraphEditor.cpp](#)

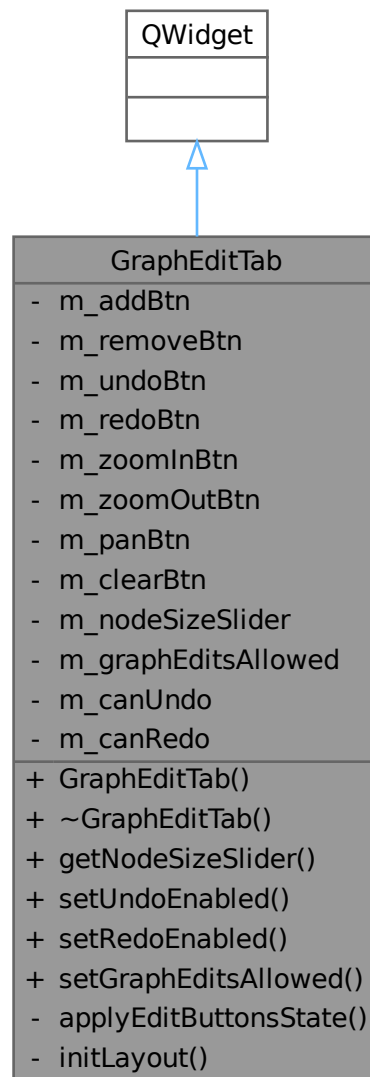
6.29 GraphEditTab Class Reference

```
#include <GraphEditTab.h>
```

Inheritance diagram for GraphEditTab:



Collaboration diagram for GraphEditTab:



Public Slots

- void [setUndoEnabled](#) (bool)
- void [setRedoEnabled](#) (bool)
- void [setGraphEditsAllowed](#) (bool)

Signals

- void [undoRequested](#) ()
- void [redoRequested](#) ()
- void [addRequested](#) ()

- void [removeRequested](#) ()
- void [clearRequested](#) ()
- void [panRequested](#) ()
- void [zoomInRequested](#) ()
- void [zoomOutRequested](#) ()

Public Member Functions

- [GraphEditTab](#) (QWidget **parent*=nullptr)
- [~GraphEditTab](#) () override=default
- QSlider * [getNodeSizeSlider](#) () const

Private Member Functions

- void [applyEditButtonsState](#) ()
- void [initLayout](#) ()

Private Attributes

- QPushButton * [m_addBtn](#)
- QPushButton * [m_removeBtn](#)
- QPushButton * [m_undoBtn](#)
- QPushButton * [m_redoBtn](#)
- QPushButton * [m_zoomInBtn](#)
- QPushButton * [m_zoomOutBtn](#)
- QPushButton * [m_panBtn](#)
- QPushButton * [m_clearBtn](#)
- QSlider * [m_nodeSizeSlider](#)
- bool [m_graphEditsAllowed](#) = true
- bool [m_canUndo](#) = false
- bool [m_canRedo](#) = false

6.29.1 Constructor & Destructor Documentation

6.29.1.1 GraphEditTab()

```
GraphEditTab::GraphEditTab (
    QWidget * parent = nullptr ) [explicit]
```

6.29.1.2 ~GraphEditTab()

```
GraphEditTab::~GraphEditTab ( ) [override], [default]
```

6.29.2 Member Function Documentation

6.29.2.1 addRequested

```
void GraphEditTab::addRequested ( ) [signal]
```

6.29.2.2 applyEditButtonsState()

```
void GraphEditTab::applyEditButtonsState ( ) [private]
```

6.29.2.3 clearRequested

```
void GraphEditTab::clearRequested ( ) [signal]
```

6.29.2.4 getNodeSizeSlider()

```
QSlider * GraphEditTab::getNodeSizeSlider ( ) const
```

6.29.2.5 initLayout()

```
void GraphEditTab::initLayout ( ) [private]
```

6.29.2.6 panRequested

```
void GraphEditTab::panRequested ( ) [signal]
```

6.29.2.7 redoRequested

```
void GraphEditTab::redoRequested ( ) [signal]
```

6.29.2.8 removeRequested

```
void GraphEditTab::removeRequested ( ) [signal]
```

6.29.2.9 setGraphEditsAllowed

```
void GraphEditTab::setGraphEditsAllowed (
    bool allowed ) [slot]
```

6.29.2.10 setRedoEnabled

```
void GraphEditTab::setRedoEnabled (
    bool enabled ) [slot]
```

6.29.2.11 setUndoEnabled

```
void GraphEditTab::setUndoEnabled (
    bool enabled ) [slot]
```


6.29.2.12 undoRequested

```
void GraphEditTab::undoRequested ( ) [signal]
```

6.29.2.13 zoomInRequested

```
void GraphEditTab::zoomInRequested ( ) [signal]
```

6.29.2.14 zoomOutRequested

```
void GraphEditTab::zoomOutRequested ( ) [signal]
```

6.29.3 Member Data Documentation

6.29.3.1 m_addBtn

```
QPushButton* GraphEditTab::m_addBtn [private]
```

6.29.3.2 m_canRedo

```
bool GraphEditTab::m_canRedo = false [private]
```

6.29.3.3 m_canUndo

```
bool GraphEditTab::m_canUndo = false [private]
```

6.29.3.4 m_clearBtn

```
QPushButton* GraphEditTab::m_clearBtn [private]
```

6.29.3.5 m_graphEditsAllowed

```
bool GraphEditTab::m_graphEditsAllowed = true [private]
```

6.29.3.6 m_nodeSizeSlider

```
QSlider* GraphEditTab::m_nodeSizeSlider [private]
```

6.29.3.7 m_panBtn

```
QPushButton* GraphEditTab::m_panBtn [private]
```

6.29.3.8 m_redoBtn

```
QPushButton* GraphEditTab::m_redoBtn [private]
```

6.29.3.9 m_removeBtn

```
QPushButton* GraphEditTab::m_removeBtn [private]
```

6.29.3.10 m_undoBtn

```
QPushButton* GraphEditTab::m_undoBtn [private]
```

6.29.3.11 m_zoomInBtn

```
QPushButton* GraphEditTab::m_zoomInBtn [private]
```

6.29.3.12 m_zoomOutBtn

```
QPushButton* GraphEditTab::m_zoomOutBtn [private]
```

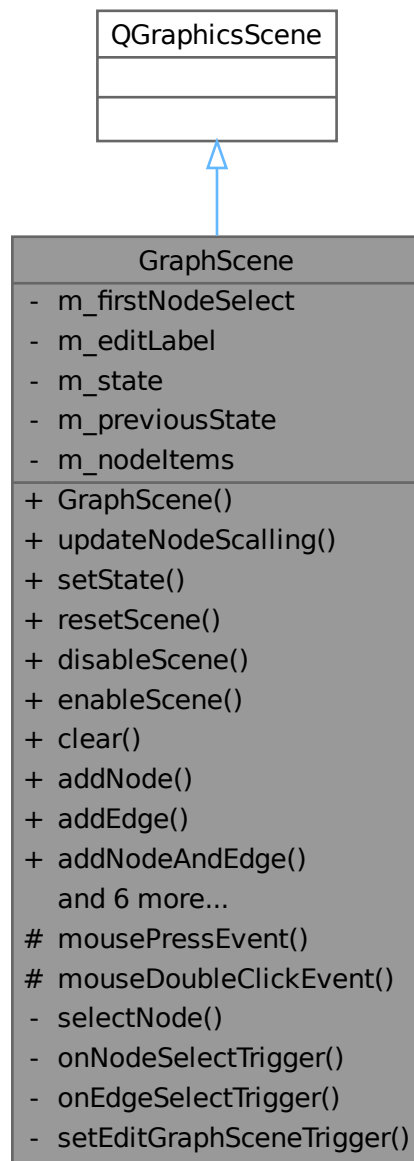
The documentation for this class was generated from the following files:

- [Headers/UI/GraphEditTab.h](#)
- [Source/UI/GraphEditTab.cpp](#)

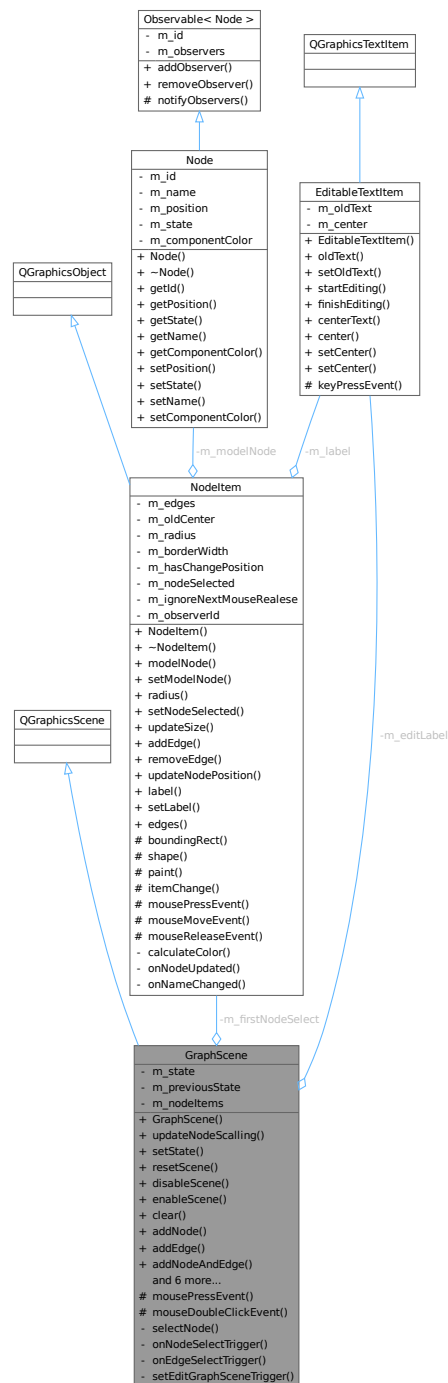
6.30 GraphScene Class Reference

```
#include <GraphScene.h>
```

Inheritance diagram for GraphScene:



Collaboration diagram for GraphScene:



Public Types

- enum class [State](#) { [ADD](#) , [REMOVE](#) , [EDIT](#) , [IDLE](#) }

Signals

- void [addNodeRequest](#) (const QPointF &)

- void [addNodeAndEdgeRequest](#) (const QPointF &, [NodeItem](#) *)
- void [addEdgeRequest](#) ([NodeItem](#) *, [NodeItem](#) *)
- void [removeNodeRequest](#) ([NodeItem](#) *)
- void [removeEdgeRequest](#) ([EdgeItem](#) *)
- void [moveNodeRequest](#) (const [NodeItem](#) *, const QPointF &, const QPointF &)
- void [editNodeNameRequest](#) (const [NodeItem](#) *, const QString &) const
- void [editEdgeWeightRequest](#) (const [EdgeItem](#) *, const QString &) const

Public Member Functions

- [GraphScene](#) (QObject *==nullptr)
- void [updateNodeScalling](#) ()
- void [setState](#) ([GraphScene::State](#))
- void [resetScene](#) ()
- void [disableScene](#) ()
- void [enableScene](#) ()
- void [clear](#) ()
- void [addNode](#) ([Node](#) *)
- void [addEdge](#) ([Edge](#) *, bool, bool)
- void [addNodeAndEdge](#) ([Node](#) *, [Edge](#) *, bool, bool)
- void [removeNode](#) ([NodeItem](#) *)
- void [removeEdge](#) ([EdgeItem](#) *)
- void [applyTheme](#) ([ThemeManager::Theme](#) theme)
- [NodeItem](#) * [findNodeItemById](#) (const unsigned) const
- [EdgeItem](#) * [findEdgeItemByNodes](#) (unsigned, unsigned) const
- [EdgeItem](#) * [findEdgeItemById](#) (unsigned) const

Protected Member Functions

- void [mousePressEvent](#) (QGraphicsSceneMouseEvent *) override
- void [mouseDoubleClickEvent](#) (QGraphicsSceneMouseEvent *) override

Private Slots

- void [onNodeSelectTrigger](#) ([NodeItem](#) *)
- void [onEdgeSelectTrigger](#) ([EdgeItem](#) *)
- void [setEditGraphSceneTrigger](#) (bool, [EditableTextItem](#) *==nullptr)

Private Member Functions

- void [selectNode](#) ([NodeItem](#) *)

Private Attributes

- [NodeItem](#) * [m_firstNodeSelect](#) {nullptr}
- [EditableTextItem](#) * [m_editLabel](#) {nullptr}
- [State](#) [m_state](#) {[GraphScene::State::ADD](#)}
- [State](#) [m_previousState](#)
- std::map< unsigned, [NodeItem](#) * > [m_nodeItems](#)

6.30.1 Member Enumeration Documentation

6.30.1.1 State

```
enum class GraphScene::State [strong]
```

Enumerator

ADD	
REMOVE	
EDIT	
IDLE	

6.30.2 Constructor & Destructor Documentation

6.30.2.1 GraphScene()

```
GraphScene::GraphScene (
    QObject * parent = nullptr ) [explicit]
```

6.30.3 Member Function Documentation

6.30.3.1 addEdge()

```
void GraphScene::addEdge (
    Edge * edgeModel,
    bool isDirected,
    bool isWeighted )
```

6.30.3.2 addEdgeRequest

```
void GraphScene::addEdgeRequest (
    NodeItem * ,
    NodeItem * ) [signal]
```

6.30.3.3 addNode()

```
void GraphScene::addNode (
    Node * nodeModel )
```

6.30.3.4 addNodeAndEdge()

```
void GraphScene::addNodeAndEdge (
    Node * nodeModel,
    Edge * edgeModel,
    bool isDirected,
    bool isWeighted )
```

6.30.3.5 addNodeAndEdgeRequest

```
void GraphScene::addNodeAndEdgeRequest (
    const QPointF & ,
    NodeItem * ) [signal]
```

6.30.3.6 addNodeRequest

```
void GraphScene::addNodeRequest (
    const QPointF & ) [signal]
```

6.30.3.7 applyTheme()

```
void GraphScene::applyTheme (
    ThemeManager::Theme theme )
```

6.30.3.8 clear()

```
void GraphScene::clear ( )
```

6.30.3.9 disableScene()

```
void GraphScene::disableScene ( )
```

6.30.3.10 editEdgeWeightRequest

```
void GraphScene::editEdgeWeightRequest (
    const EdgeItem * ,
    const QString & ) const [signal]
```

6.30.3.11 editNodeNameRequest

```
void GraphScene::editNodeNameRequest (
    const NodeItem * ,
    const QString & ) const [signal]
```

6.30.3.12 enableScene()

```
void GraphScene::enableScene ( )
```

6.30.3.13 findEdgeItemById()

```
EdgeItem * GraphScene::findEdgeItemById (
    unsigned edgeId ) const
```

6.30.3.14 findEdgeItemByNodes()

```
EdgeItem * GraphScene::findEdgeItemByNodes (
    unsigned from,
    unsigned to ) const
```

6.30.3.15 findNodeItemById()

```
NodeItem * GraphScene::findNodeItemById (
    const unsigned id ) const
```

6.30.3.16 mouseDoubleClickEvent()

```
void GraphScene::mouseDoubleClickEvent (
    QGraphicsSceneMouseEvent * event ) [override], [protected]
```

6.30.3.17 mousePressEvent()

```
void GraphScene::mousePressEvent (
    QGraphicsSceneMouseEvent * event ) [override], [protected]
```

6.30.3.18 moveNodeRequest

```
void GraphScene::moveNodeRequest (
    const NodeItem * ,
    const QPointF & ,
    const QPointF & ) [signal]
```

6.30.3.19 onEdgeSelectTrigger

```
void GraphScene::onEdgeSelectTrigger (
    EdgeItem * edge ) [private], [slot]
```

6.30.3.20 onNodeSelectTrigger

```
void GraphScene::onNodeSelectTrigger (
    NodeItem * node ) [private], [slot]
```

6.30.3.21 removeEdge()

```
void GraphScene::removeEdge (
    EdgeItem * edge )
```

6.30.3.22 removeEdgeRequest

```
void GraphScene::removeEdgeRequest (
    EdgeItem * ) [signal]
```


6.30.3.23 removeNode()

```
void GraphScene::removeNode (
    NodeItem * node )
```

6.30.3.24 removeNodeRequest

```
void GraphScene::removeNodeRequest (
    NodeItem * ) [signal]
```

6.30.3.25 resetScene()

```
void GraphScene::resetScene ( )
```

6.30.3.26 selectNode()

```
void GraphScene::selectNode (
    NodeItem * node ) [private]
```

6.30.3.27 setEditGraphSceneTrigger

```
void GraphScene::setEditGraphSceneTrigger (
    bool edit,
    EditableTextItem * label = nullptr ) [private], [slot]
```

6.30.3.28 setState()

```
void GraphScene::setState (
    GraphScene::State state )
```

6.30.3.29 updateNodeScalling()

```
void GraphScene::updateNodeScalling ( )
```

6.30.4 Member Data Documentation

6.30.4.1 m_editLabel

```
EditableTextItem* GraphScene::m_editLabel {nullptr} [private]
```

6.30.4.2 m_firstNodeSelect

```
NodeItem* GraphScene::m_firstNodeSelect {nullptr} [private]
```

6.30.4.3 m_nodeItems

```
std::map<unsigned, NodeItem\*> GraphScene::m_nodeItems [private]
```

6.30.4.4 m_previousState

```
State GraphScene::m_previousState [private]
```

6.30.4.5 m_state

```
State GraphScene::m_state {GraphScene::State::ADD} [private]
```

The documentation for this class was generated from the following files:

- Headers/View/[GraphScene.h](#)
- Source/View/[GraphScene.cpp](#)

6.31 GraphView Class Reference

```
#include <GraphView.h>
```

Inheritance diagram for GraphView:



Collaboration diagram for GraphView:



Public Types

- enum class [State](#) { [IDLE](#) , [PAN_IDLE](#) , [PAN_ACTIVE](#) }

Public Member Functions

- [GraphView](#) (QWidget *=`nullptr`)
- [GraphView](#) (QGraphicsScene *, QWidget *=`nullptr`)
- [GraphView::State](#) state () const
- void [setState](#) ([GraphView::State](#))
- void [resetState](#) ()
- void [zoomIn](#) ()
- void [zoomOut](#) ()

Protected Member Functions

- void [mousePressEvent](#) (QMouseEvent *) override
- void [mouseMoveEvent](#) (QMouseEvent *) override
- void [mouseReleaseEvent](#) (QMouseEvent *) override
- void [mouseDoubleClickEvent](#) (QMouseEvent *event) override
- void [wheelEvent](#) (QWheelEvent *event) override

Private Member Functions

- void [init](#) ()

Private Attributes

- [GraphView::State](#) [m_state](#) {[GraphView::State::IDLE](#)}
- QPoint [m_lastMousePos](#)
- qreal [m_zoomFactor](#) {1.15}
- [GraphView::State](#) [m_oldState](#)

6.31.1 Member Enumeration Documentation**6.31.1.1 State**

```
enum class GraphView::State [strong]
```

Enumerator

IDLE	
PAN_IDLE	
PAN_ACTIVE	

6.31.2 Constructor & Destructor Documentation**6.31.2.1 GraphView() [1/2]**

```
GraphView::GraphView (  
    QWidget * parent = nullptr ) [explicit]
```

6.31.2.2 GraphView() [2/2]

```
GraphView::GraphView (  
    QGraphicsScene * scene,  
    QWidget * parent = nullptr ) [explicit]
```

6.31.3 Member Function Documentation

6.31.3.1 init()

```
void GraphView::init ( ) [private]
```

6.31.3.2 mouseDoubleClickEvent()

```
void GraphView::mouseDoubleClickEvent (
    QMouseEvent * event ) [override], [protected]
```

6.31.3.3 mouseMoveEvent()

```
void GraphView::mouseMoveEvent (
    QMouseEvent * event ) [override], [protected]
```

6.31.3.4 mousePressEvent()

```
void GraphView::mousePressEvent (
    QMouseEvent * event ) [override], [protected]
```

6.31.3.5 mouseReleaseEvent()

```
void GraphView::mouseReleaseEvent (
    QMouseEvent * event ) [override], [protected]
```

6.31.3.6 resetState()

```
void GraphView::resetState ( )
```

6.31.3.7 setState()

```
void GraphView::setState (
    GraphView::State newState )
```

6.31.3.8 state()

```
GraphView::State GraphView::state ( ) const
```

6.31.3.9 wheelEvent()

```
void GraphView::wheelEvent (
    QWheelEvent * event ) [override], [protected]
```

6.31.3.10 zoomIn()

```
void GraphView::zoomIn ( )
```

6.31.3.11 zoomOut()

```
void GraphView::zoomOut ( )
```

6.31.4 Member Data Documentation

6.31.4.1 m_lastMousePos

```
QPoint GraphView::m_lastMousePos [private]
```

6.31.4.2 m_oldState

```
GraphView::State GraphView::m_oldState [private]
```

6.31.4.3 m_state

```
GraphView::State GraphView::m_state {GraphView::State::IDLE} [private]
```

6.31.4.4 m_zoomFactor

```
qreal GraphView::m_zoomFactor {1.15} [private]
```

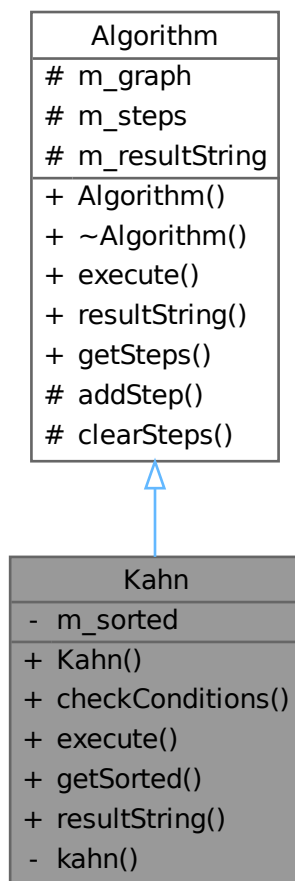
The documentation for this class was generated from the following files:

- Headers/View/[GraphView.h](#)
- Source/View/[GraphView.cpp](#)

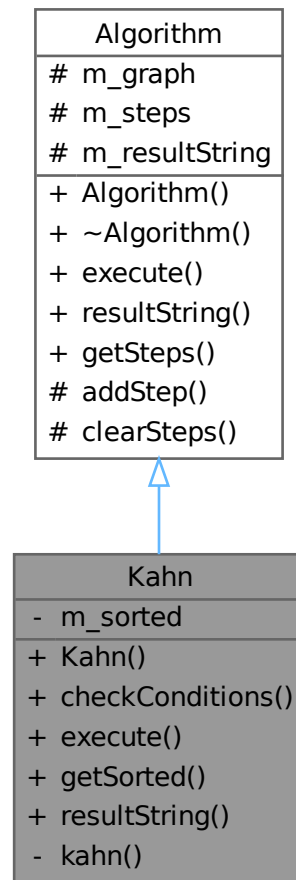
6.32 Kahn Class Reference

```
#include <Kahn.h>
```

Inheritance diagram for Kahn:



Collaboration diagram for Kahn:



Public Member Functions

- [Kahn](#) (const std::shared_ptr< [Graph](#) >)
- std::optional< [AlgorithmError](#) > [checkConditions](#) () const
- std::optional< [AlgorithmError](#) > [execute](#) (unsigned=0, unsigned=0) override
- const std::vector< unsigned > & [getSorted](#) () const
- QString [resultString](#) () const override

Public Member Functions inherited from [Algorithm](#)

- [Algorithm](#) (const std::shared_ptr< [Graph](#) >)
- virtual [~Algorithm](#) ()=default
- const std::vector< [AlgorithmStep](#) > & [getSteps](#) () const

Private Member Functions

- std::optional< [AlgorithmError](#) > [kahn](#) ()

Private Attributes

- `std::vector< unsigned > m_sorted`

Additional Inherited Members

Protected Member Functions inherited from [Algorithm](#)

- `void addStep (const AlgorithmStep &)`
- `void clearSteps ()`

Protected Attributes inherited from [Algorithm](#)

- `std::shared_ptr< Graph > m_graph`
- `std::vector< AlgorithmStep > m_steps`
- `QString m_resultString`

6.32.1 Constructor & Destructor Documentation

6.32.1.1 Kahn()

```
Kahn::Kahn (
    const std::shared_ptr< Graph > g ) [explicit]
```

6.32.2 Member Function Documentation

6.32.2.1 checkConditions()

```
std::optional< AlgorithmError > Kahn::checkConditions ( ) const
```

6.32.2.2 execute()

```
std::optional< AlgorithmError > Kahn::execute (
    unsigned = 0,
    unsigned = 0 ) [override], [virtual]
```

Implements [Algorithm](#).

6.32.2.3 getSorted()

```
const std::vector< unsigned > & Kahn::getSorted ( ) const
```

6.32.2.4 kahn()

```
std::optional< AlgorithmError > Kahn::kahn ( ) [private]
```

6.32.2.5 resultString()

```
QString Kahn::resultString ( ) const [override], [virtual]
```

Implements [Algorithm](#).

6.32.3 Member Data Documentation

6.32.3.1 m_sorted

```
std::vector<unsigned> Kahn::m_sorted [private]
```

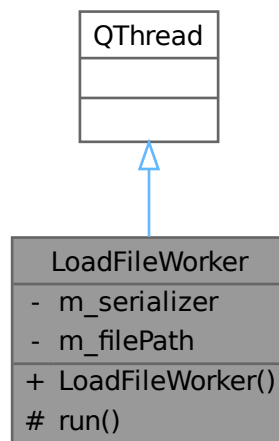
The documentation for this class was generated from the following files:

- Headers/Algorithm/[Kahn.h](#)
- Source/Algorithm/[Kahn.cpp](#)

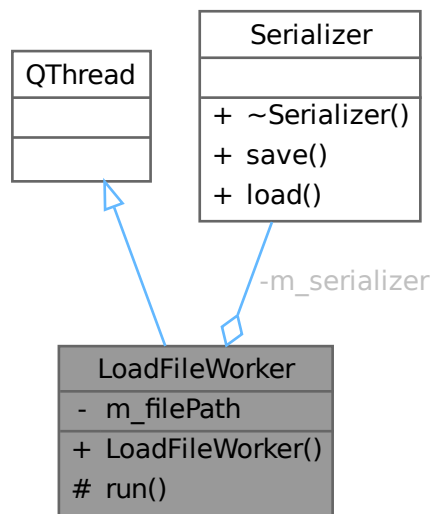
6.33 LoadFileWorker Class Reference

```
#include <LoadFileWorker.h>
```

Inheritance diagram for LoadFileWorker:



Collaboration diagram for LoadFileWorker:



Signals

- void [loaded](#) (const QVariant &, bool, bool, const QString &)
- void [failed](#) (const QString &)

Public Member Functions

- [LoadFileWorker](#) ([Serializer](#) *, const QString &, QObject *==nullptr)

Protected Member Functions

- void [run](#) () override

Private Attributes

- [Serializer](#) * [m_serializer](#) = nullptr
- QString [m_filePath](#)

6.33.1 Constructor & Destructor Documentation

6.33.1.1 LoadFileWorker()

```

LoadFileWorker::LoadFileWorker (
    Serializer * serializer,
    const QString & filePath,
    QObject * parent = nullptr )
  
```

6.33.2 Member Function Documentation

6.33.2.1 failed

```
void LoadFileWorker::failed (
    const QString & ) [signal]
```

6.33.2.2 loaded

```
void LoadFileWorker::loaded (
    const QVariant & ,
    bool ,
    bool ,
    const QString & ) [signal]
```

6.33.2.3 run()

```
void LoadFileWorker::run ( ) [override], [protected]
```

6.33.3 Member Data Documentation

6.33.3.1 m_filePath

```
QString LoadFileWorker::m_filePath [private]
```

6.33.3.2 m_serializer

```
Serializer* LoadFileWorker::m_serializer = nullptr [private]
```

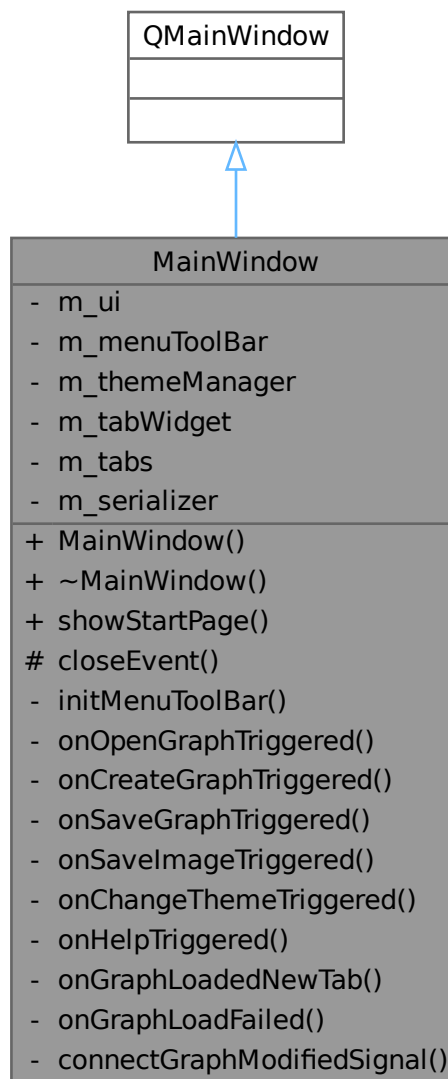
The documentation for this class was generated from the following files:

- Headers/Threads/[LoadFileWorker.h](#)
- Source/Threads/[LoadFileWorker.cpp](#)

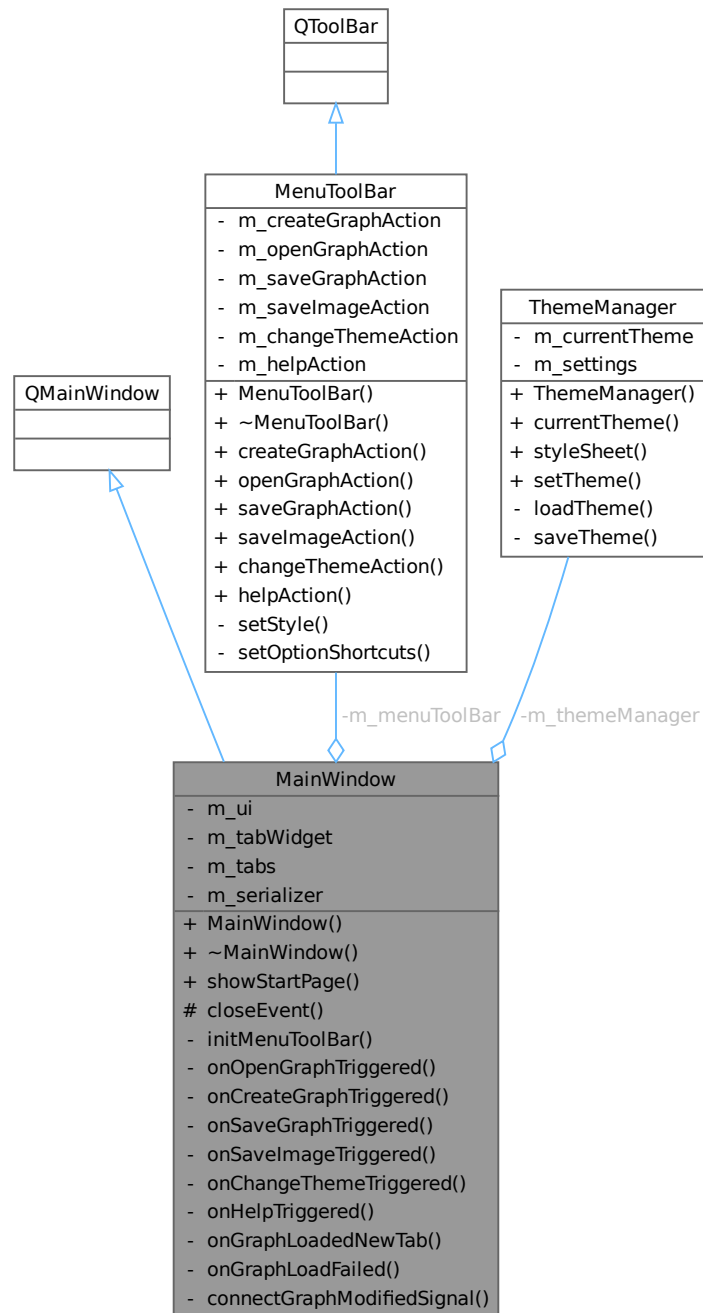
6.34 MainWindow Class Reference

```
#include <MainWindow.h>
```

Inheritance diagram for MainWindow:



Collaboration diagram for MainWindow:



Classes

- struct [TabInfo](#)

Public Member Functions

- [MainWindow](#) (QWidget *=`nullptr`)
- [~MainWindow](#) () override
- void [showStartPage](#) ()

Protected Member Functions

- void [closeEvent](#) (QCloseEvent *) override

Private Slots

- void [onOpenGraphTriggered](#) ()
- void [onCreateGraphTriggered](#) ()
- void [onSaveGraphTriggered](#) ()
- void [onSaveImageTriggered](#) ()
- void [onChangeThemeTriggered](#) ()
- void [onHelpTriggered](#) ()
- void [onGraphLoadedNewTab](#) (const QVariant &, bool, bool, const QString &)
- void [onGraphLoadFailed](#) (const QString &)
- void [connectGraphModifiedSignal](#) (GraphEditor *)

Private Member Functions

- void [initMenuToolBar](#) ()

Private Attributes

- Ui::MainWindow * [m_ui](#)
- [MenuToolBar](#) * [m_menuToolBar](#) = nullptr
- [ThemeManager](#) * [m_themeManager](#)
- QTabWidget * [m_tabWidget](#) = nullptr
- QHash< [GraphEditor](#) *, [TabInfo](#) > [m_tabs](#)
- std::unique_ptr< [Serializer](#) > [m_serializer](#)

6.34.1 Constructor & Destructor Documentation

6.34.1.1 MainWindow()

```
MainWindow::MainWindow (
    QWidget * parent = nullptr ) [explicit]
```

6.34.1.2 ~MainWindow()

```
MainWindow::~MainWindow ( ) [override]
```

6.34.2 Member Function Documentation

6.34.2.1 closeEvent()

```
void MainWindow::closeEvent (
    QCloseEvent * event ) [override], [protected]
```


6.34.2.2 connectGraphModifiedSignal

```
void MainWindow::connectGraphModifiedSignal (
    GraphEditor * editor ) [private], [slot]
```

6.34.2.3 initMenuToolBar()

```
void MainWindow::initMenuToolBar ( ) [private]
```

6.34.2.4 onChangeThemeTriggered

```
void MainWindow::onChangeThemeTriggered ( ) [private], [slot]
```

6.34.2.5 onCreateGraphTriggered

```
void MainWindow::onCreateGraphTriggered ( ) [private], [slot]
```

6.34.2.6 onGraphLoadedNewTab

```
void MainWindow::onGraphLoadedNewTab (
    const QVariant & data,
    bool weighted,
    bool directed,
    const QString & filePath ) [private], [slot]
```

6.34.2.7 onGraphLoadFailed

```
void MainWindow::onGraphLoadFailed (
    const QString & error ) [private], [slot]
```

6.34.2.8 onHelpTriggered

```
void MainWindow::onHelpTriggered ( ) [private], [slot]
```

6.34.2.9 onOpenGraphTriggered

```
void MainWindow::onOpenGraphTriggered ( ) [private], [slot]
```

6.34.2.10 onSaveGraphTriggered

```
void MainWindow::onSaveGraphTriggered ( ) [private], [slot]
```

6.34.2.11 onSaveImageTriggered

```
void MainWindow::onSaveImageTriggered ( ) [private], [slot]
```

6.34.2.12 showStartPage()

```
void MainWindow::showStartPage ( )
```

6.34.3 Member Data Documentation

6.34.3.1 m_menuToolBar

```
MenuToolBar* MainWindow::m_menuToolBar = nullptr [private]
```

6.34.3.2 m_serializer

```
std::unique_ptr<Serializer> MainWindow::m_serializer [private]
```

6.34.3.3 m_tabs

```
QHash<GraphEditor*, TabInfo> MainWindow::m_tabs [private]
```

6.34.3.4 m_tabWidget

```
QTabWidget* MainWindow::m_tabWidget = nullptr [private]
```

6.34.3.5 m_themeManager

```
ThemeManager* MainWindow::m_themeManager [private]
```

6.34.3.6 m_ui

```
Ui::MainWindow* MainWindow::m_ui [private]
```

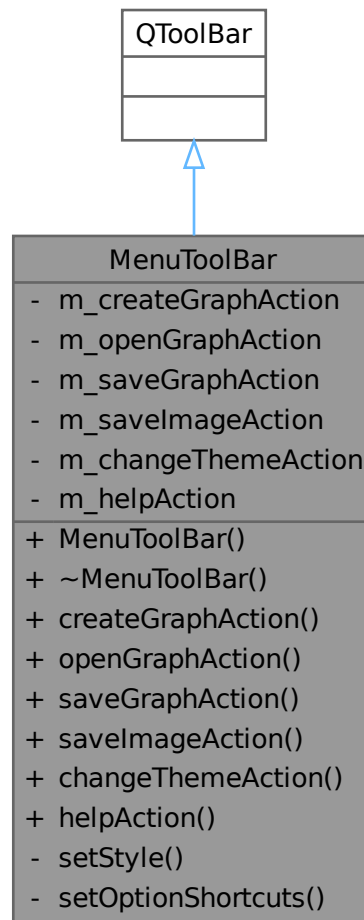
The documentation for this class was generated from the following files:

- Headers/[MainWindow.h](#)
- Source/[MainWindow.cpp](#)

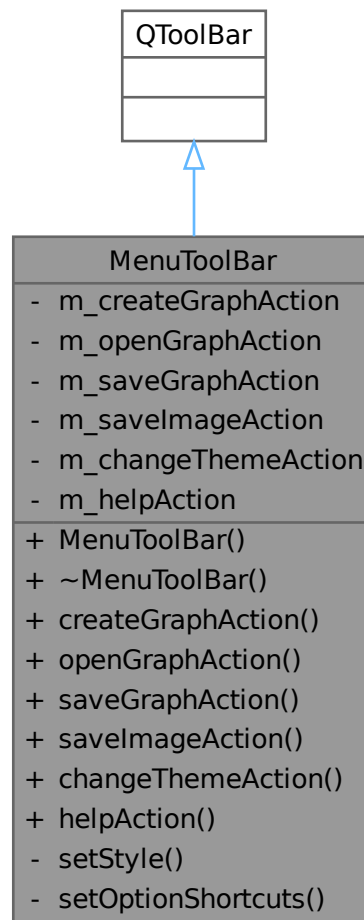
6.35 MenuToolBar Class Reference

```
#include <MenuToolBar.h>
```

Inheritance diagram for MenuToolBar:



Collaboration diagram for MenuToolBar:



Public Member Functions

- [MenuToolBar](#) (QWidget *=`nullptr`)
- [~MenuToolBar](#) () override=default
- QAction * [createGraphAction](#) () const
- QAction * [openGraphAction](#) () const
- QAction * [saveGraphAction](#) () const
- QAction * [saveImageAction](#) () const
- QAction * [changeThemeAction](#) () const
- QAction * [helpAction](#) () const

Private Member Functions

- void [setStyle](#) ()
- void [setOptionShortcuts](#) ()

Private Attributes

- QAction * [m_createGraphAction](#)
- QAction * [m_openGraphAction](#)
- QAction * [m_saveGraphAction](#)
- QAction * [m_saveImageAction](#)
- QAction * [m_changeThemeAction](#)
- QAction * [m_helpAction](#)

6.35.1 Constructor & Destructor Documentation

6.35.1.1 MenuToolBar()

```
MenuToolBar::MenuToolBar (
    QWidget * parent = nullptr ) [explicit]
```

6.35.1.2 ~MenuToolBar()

```
MenuToolBar::~MenuToolBar ( ) [override], [default]
```

6.35.2 Member Function Documentation

6.35.2.1 changeThemeAction()

```
QAction * MenuToolBar::changeThemeAction ( ) const
```

6.35.2.2 createGraphAction()

```
QAction * MenuToolBar::createGraphAction ( ) const
```

6.35.2.3 helpAction()

```
QAction * MenuToolBar::helpAction ( ) const
```

6.35.2.4 openGraphAction()

```
QAction * MenuToolBar::openGraphAction ( ) const
```

6.35.2.5 saveGraphAction()

```
QAction * MenuToolBar::saveGraphAction ( ) const
```

6.35.2.6 saveImageAction()

```
QAction * MenuToolBar::saveImageAction ( ) const
```

6.35.2.7 setOptionShortcuts()

```
void MenuToolBar::setOptionShortcuts ( ) [private]
```

6.35.2.8 setStyle()

```
void MenuToolBar::setStyle ( ) [private]
```

6.35.3 Member Data Documentation

6.35.3.1 m_changeThemeAction

```
QAction* MenuToolBar::m_changeThemeAction [private]
```

6.35.3.2 m_createGraphAction

```
QAction* MenuToolBar::m_createGraphAction [private]
```

6.35.3.3 m_helpAction

```
QAction* MenuToolBar::m_helpAction [private]
```

6.35.3.4 m_openGraphAction

```
QAction* MenuToolBar::m_openGraphAction [private]
```

6.35.3.5 m_saveGraphAction

```
QAction* MenuToolBar::m_saveGraphAction [private]
```

6.35.3.6 m_saveImageAction

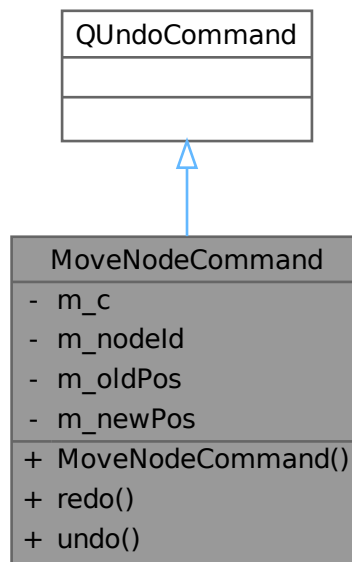
```
QAction* MenuToolBar::m_saveImageAction [private]
```

The documentation for this class was generated from the following files:

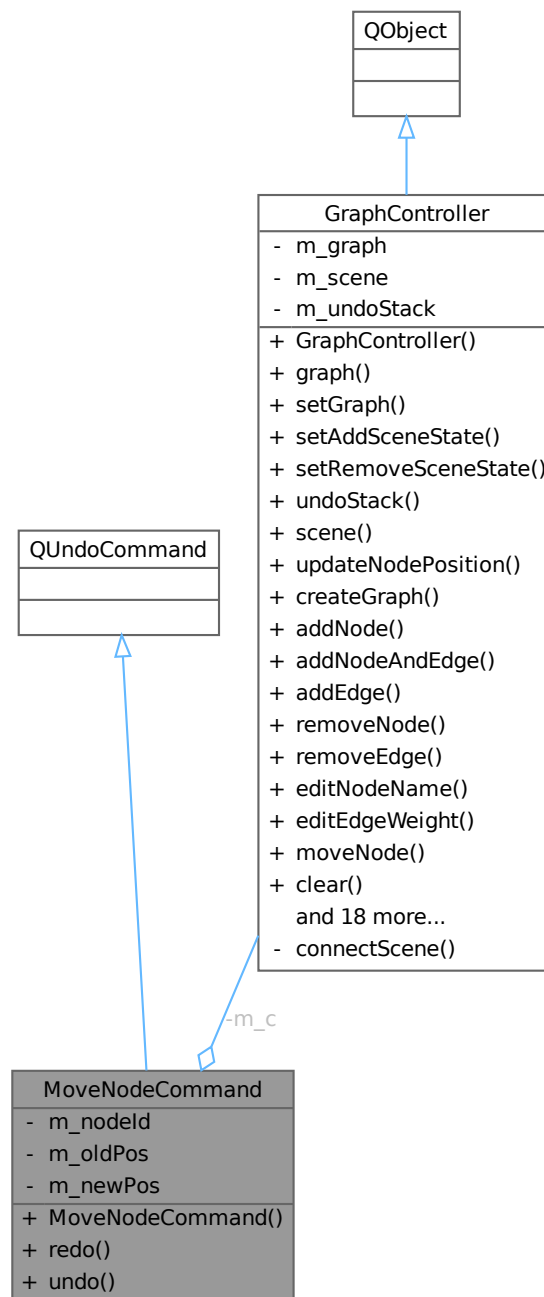
- [Headers/UI/MenuToolBar.h](#)
- [Source/UI/MenuToolBar.cpp](#)

6.36 MoveNodeCommand Class Reference

Inheritance diagram for MoveNodeCommand:



Collaboration diagram for MoveNodeCommand:



Public Member Functions

- [MoveNodeCommand](#) ([GraphController](#) *c, unsigned nodeId, const QPointF &oldPos, const QPointF &newPos)
- void [redo](#) () override
- void [undo](#) () override

Private Attributes

- [GraphController](#) * `m_c` = nullptr
- unsigned `m_nodeId` = 0
- [QPointF](#) `m_oldPos`
- [QPointF](#) `m_newPos`

6.36.1 Constructor & Destructor Documentation

6.36.1.1 MoveNodeCommand()

```
MoveNodeCommand::MoveNodeCommand (
    GraphController * c,
    unsigned nodeId,
    const QPointF & oldPos,
    const QPointF & newPos ) [inline]
```

6.36.2 Member Function Documentation

6.36.2.1 redo()

```
void MoveNodeCommand::redo ( ) [inline], [override]
```

6.36.2.2 undo()

```
void MoveNodeCommand::undo ( ) [inline], [override]
```

6.36.3 Member Data Documentation

6.36.3.1 m_c

```
GraphController* MoveNodeCommand::m_c = nullptr [private]
```

6.36.3.2 m_newPos

```
QPointF MoveNodeCommand::m_newPos [private]
```

6.36.3.3 m_nodeId

```
unsigned MoveNodeCommand::m_nodeId = 0 [private]
```

6.36.3.4 m_oldPos

```
QPointF MoveNodeCommand::m_oldPos [private]
```

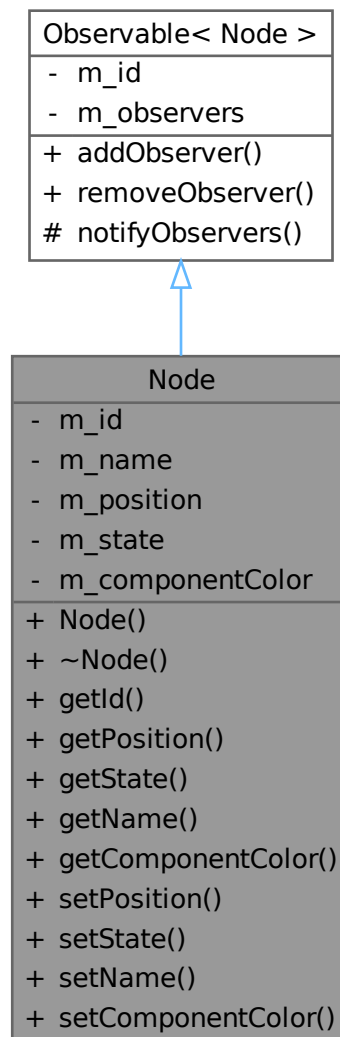
The documentation for this class was generated from the following file:

- Source/Controllers/[GraphController.cpp](#)

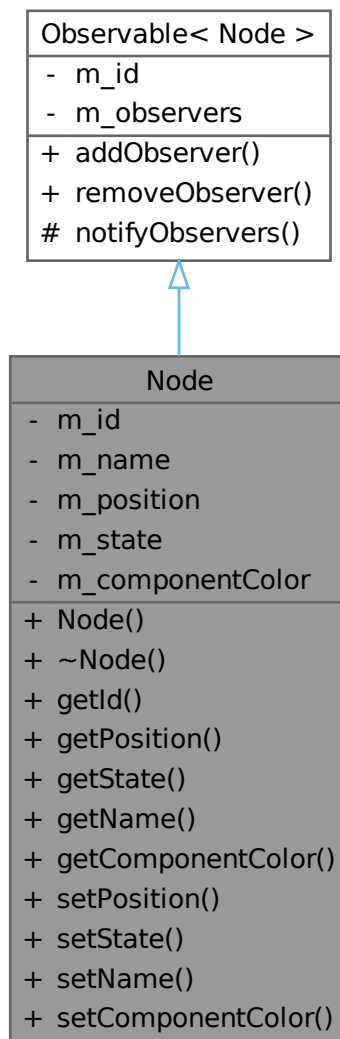
6.37 Node Class Reference

```
#include <Node.h>
```

Inheritance diagram for Node:



Collaboration diagram for Node:



Public Member Functions

- `Node` (unsigned, double=0.0, double=0.0)
- `~Node` ()=default
- unsigned `getId` () const
- const std::pair< double, double > & `getPosition` () const
- `NodeState` `getState` () const
- QString `getName` () const
- int `getComponentColor` () const
- void `setPosition` (double, double)
- void `setState` (`NodeState`)
- void `setName` (const QString &)
- void `setComponentColor` (int)

Public Member Functions inherited from [Observable< Node >](#)

- unsigned [addObserver](#) ([Observer](#) observer)
Adds a new observer.
- void [removeObserver](#) (unsigned id)
Removes an observer by its unique ID.

Private Attributes

- unsigned [m_id](#)
- [QString](#) [m_name](#)
- [std::pair< double, double >](#) [m_position](#)
- [NodeState](#) [m_state](#) = [NodeState::Default](#)
- int [m_componentColor](#) = -1

Additional Inherited Members

Public Types inherited from [Observable< Node >](#)

- using [Observer](#) = [std::function< void\(Node &\)>](#)
Type of the observer function: takes a reference to the subject.

Protected Member Functions inherited from [Observable< Node >](#)

- void [notifyObservers](#) ([Node](#) &subject)
Notifies all registered observers of a state change.

6.37.1 Constructor & Destructor Documentation

6.37.1.1 [Node\(\)](#)

```
Node::Node (
    unsigned id,
    double x = 0.0,
    double y = 0.0 ) [explicit]
```

6.37.1.2 [~Node\(\)](#)

```
Node::~~Node ( ) [default]
```

6.37.2 Member Function Documentation

6.37.2.1 [getComponentColor\(\)](#)

```
int Node::getComponentColor ( ) const
```

6.37.2.2 getId()

```
unsigned Node::getId ( ) const
```

6.37.2.3 getName()

```
QString Node::getName ( ) const
```

6.37.2.4 getPosition()

```
const std::pair< double, double > & Node::getPosition ( ) const
```

6.37.2.5 getState()

```
NodeState Node::getState ( ) const
```

6.37.2.6 setComponentColor()

```
void Node::setComponentColor (
    int colorId )
```

6.37.2.7 setName()

```
void Node::setName (
    const QString & newName )
```

6.37.2.8 setPosition()

```
void Node::setPosition (
    double x,
    double y )
```

6.37.2.9 setState()

```
void Node::setState (
    NodeState state )
```

6.37.3 Member Data Documentation

6.37.3.1 m_componentColor

```
int Node::m_componentColor = -1 [private]
```

6.37.3.2 m_id

```
unsigned Node::m_id [private]
```

6.37.3.3 m_name

```
QString Node::m_name [private]
```

6.37.3.4 m_position

```
std::pair<double, double> Node::m_position [private]
```

6.37.3.5 m_state

```
NodeState Node::m_state = NodeState::Default [private]
```

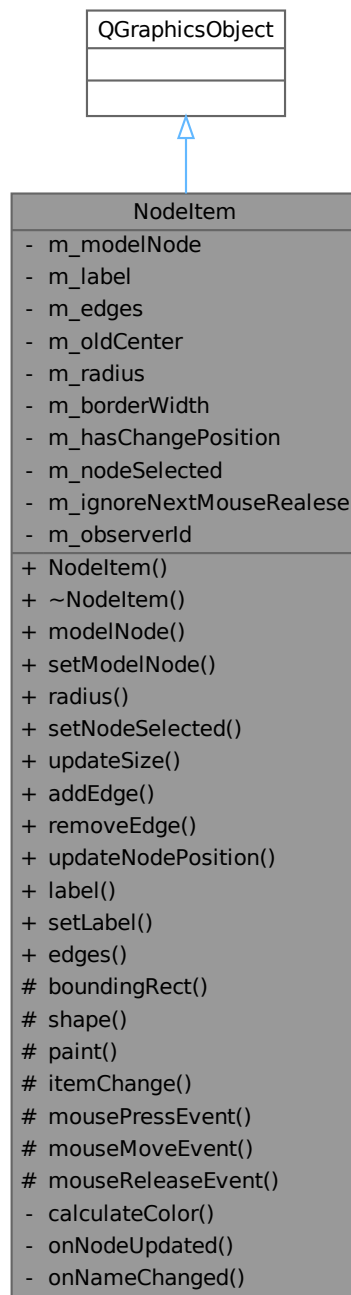
The documentation for this class was generated from the following files:

- Headers/Graph/[Node.h](#)
- Source/Graph/[Node.cpp](#)

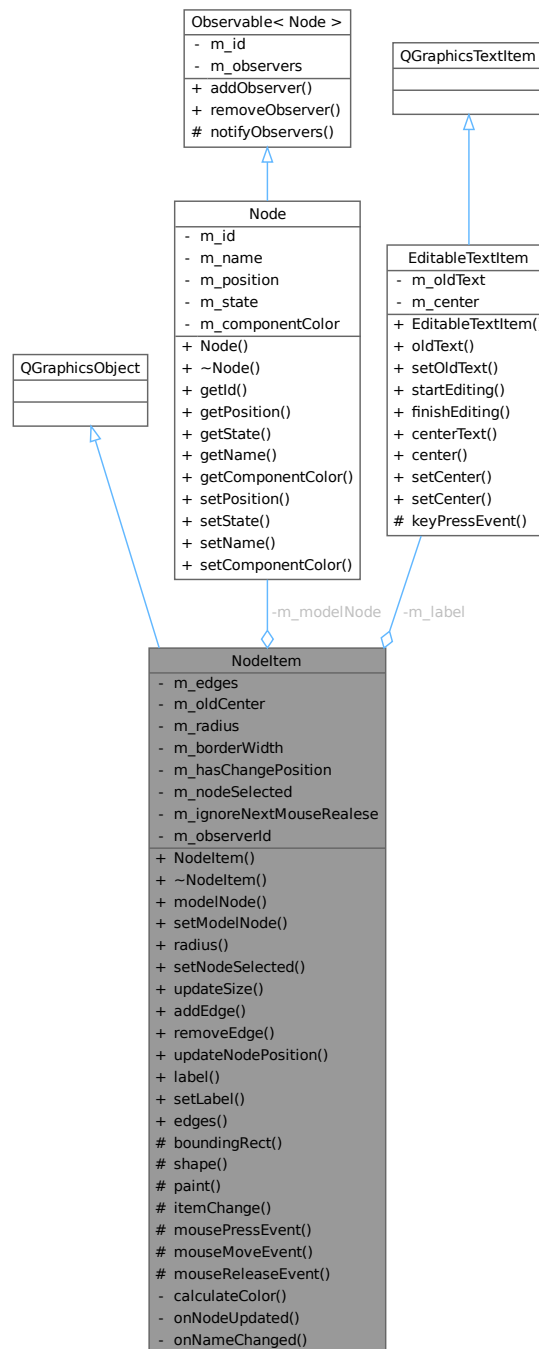
6.38 NodeItem Class Reference

```
#include <NodeItem.h>
```

Inheritance diagram for NodeItem:



Collaboration diagram for NodelItem:



Signals

- void `nodeSelected` (`NodelItem *`)
- void `editNodeNameRequest` (`const NodelItem *`, `const QString &`) `const`
- void `moveNodeRequest` (`const NodelItem *`, `const QPointF &`, `const QPointF &`)

Public Member Functions

- [NodeItem](#) ([Node](#) *)
- [~NodeItem](#) () override
- [Node](#) * [modelNode](#) () const
- void [setModelNode](#) ([Node](#) *)
- qreal [radius](#) () const
- void [setNodeSelected](#) (bool)
- void [updateSize](#) ()
- void [addEdge](#) ([EdgeItem](#) *)
- void [removeEdge](#) ([EdgeItem](#) *)
- void [updateNodePosition](#) ()
- [EditableTextItem](#) * [label](#) () const
- void [setLabel](#) ([EditableTextItem](#) *)
- const QSet< [EdgeItem](#) * > & [edges](#) () const

Protected Member Functions

- QRectF [boundingRect](#) () const override
- QPainterPath [shape](#) () const override
- void [paint](#) (QPainter *, const QStyleOptionGraphicsItem *, QWidget *) override
- QVariant [itemChange](#) (GraphicsItemChange, const QVariant &) override
- void [mousePressEvent](#) (QGraphicsSceneMouseEvent *) override
- void [mouseMoveEvent](#) (QGraphicsSceneMouseEvent *) override
- void [mouseReleaseEvent](#) (QGraphicsSceneMouseEvent *) override

Private Slots

- void [onNameChanged](#) (const QString &) const

Private Member Functions

- const QColor [calculateColor](#) () const
- void [onNodeUpdated](#) ()

Private Attributes

- [Node](#) * [m_modelNode](#)
- [EditableTextItem](#) * [m_label](#)
- QSet< [EdgeItem](#) * > [m_edges](#)
- QPointF [m_oldCenter](#)
- qreal [m_radius](#) {AppConstants::defaultRadius}
- qreal [m_borderWidth](#) {AppConstants::defaultBorderWidth}
- bool [m_hasChangePosition](#) {false}
- bool [m_nodeSelected](#) {false}
- bool [m_ignoreNextMouseRelease](#) {false}
- unsigned [m_observerId](#) {0}

6.38.1 Constructor & Destructor Documentation

6.38.1.1 NodeItem()

```
NodeItem::NodeItem (
    Node * modelNode ) [explicit]
```

6.38.1.2 ~NodeItem()

```
NodeItem::~NodeItem ( ) [override]
```

6.38.2 Member Function Documentation

6.38.2.1 addEdge()

```
void NodeItem::addEdge (
    EdgeItem * edgeItem )
```

6.38.2.2 boundingRect()

```
QRectF NodeItem::boundingRect ( ) const [override], [protected]
```

6.38.2.3 calculateColor()

```
const QColor NodeItem::calculateColor ( ) const [private]
```

6.38.2.4 edges()

```
const QSet< EdgeItem * > & NodeItem::edges ( ) const
```

6.38.2.5 editNodeNameRequest

```
void NodeItem::editNodeNameRequest (
    const NodeItem * ,
    const QString & ) const [signal]
```

6.38.2.6 itemChange()

```
QVariant NodeItem::itemChange (
    GraphicsItemChange change,
    const QVariant & value ) [override], [protected]
```

6.38.2.7 label()

```
EditableTextItem * NodeItem::label ( ) const
```

6.38.2.8 modelNode()

```
Node * NodeItem::modelNode ( ) const
```

6.38.2.9 mouseMoveEvent()

```
void NodeItem::mouseMoveEvent (
    QGraphicsSceneMouseEvent * event ) [override], [protected]
```

6.38.2.10 mousePressEvent()

```
void NodeItem::mousePressEvent (
    QGraphicsSceneMouseEvent * event ) [override], [protected]
```

6.38.2.11 mouseReleaseEvent()

```
void NodeItem::mouseReleaseEvent (
    QGraphicsSceneMouseEvent * event ) [override], [protected]
```

6.38.2.12 moveNodeRequest

```
void NodeItem::moveNodeRequest (
    const NodeItem * ,
    const QPointF & ,
    const QPointF & ) [signal]
```

6.38.2.13 nodeSelected

```
void NodeItem::nodeSelected (
    NodeItem * ) [signal]
```

6.38.2.14 onNameChanged

```
void NodeItem::onNameChanged (
    const QString & name ) const [private], [slot]
```

6.38.2.15 onNodeUpdated()

```
void NodeItem::onNodeUpdated ( ) [private]
```

6.38.2.16 paint()

```
void NodeItem::paint (
    QPainter * painter,
    const QStyleOptionGraphicsItem * option,
    QWidget * widget ) [override], [protected]
```

6.38.2.17 radius()

```
qreal NodeItem::radius ( ) const
```

6.38.2.18 removeEdge()

```
void NodeItem::removeEdge (
    EdgeItem * edgeItem )
```

6.38.2.19 setLabel()

```
void NodeItem::setLabel (
    EditableTextItem * newLabel )
```

6.38.2.20 setModelNode()

```
void NodeItem::setModelNode (
    Node * newModelNode )
```

6.38.2.21 setNodeSelected()

```
void NodeItem::setNodeSelected (
    bool newNodeSelected )
```

6.38.2.22 shape()

```
QPainterPath NodeItem::shape ( ) const [override], [protected]
```

6.38.2.23 updateNodePosition()

```
void NodeItem::updateNodePosition ( )
```

6.38.2.24 updateSize()

```
void NodeItem::updateSize ( )
```

6.38.3 Member Data Documentation

6.38.3.1 m_borderWidth

```
qreal NodeItem::m_borderWidth {AppConstants::defaultBorderWidth} [private]
```

6.38.3.2 m_edges

```
QSet<EdgeItem*> NodeItem::m_edges [private]
```

6.38.3.3 m_hasChangePosition

```
bool NodeItem::m_hasChangePosition {false} [private]
```

6.38.3.4 m_ignoreNextMouseRealese

```
bool NodeItem::m_ignoreNextMouseRealese {false} [private]
```

6.38.3.5 m_label

```
EditableTextItem* NodeItem::m_label [private]
```

6.38.3.6 m_modelNode

```
Node* NodeItem::m_modelNode [private]
```

6.38.3.7 m_nodeSelected

```
bool NodeItem::m_nodeSelected {false} [private]
```

6.38.3.8 m_observerId

```
unsigned NodeItem::m_observerId {0} [private]
```

6.38.3.9 m_oldCenter

```
QPointF NodeItem::m_oldCenter [private]
```

6.38.3.10 m_radius

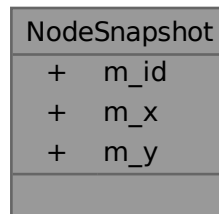
```
qreal NodeItem::m_radius {AppConstants::defaultRadius} [private]
```

The documentation for this class was generated from the following files:

- Headers/View/[NodeItem.h](#)
- Source/View/[NodeItem.cpp](#)

6.39 NodeSnapshot Struct Reference

Collaboration diagram for NodeSnapshot:



Public Attributes

- unsigned [m_id](#)
- double [m_x](#)
- double [m_y](#)

6.39.1 Member Data Documentation

6.39.1.1 m_id

```
unsigned NodeSnapshot::m_id
```

6.39.1.2 m_x

```
double NodeSnapshot::m_x
```

6.39.1.3 m_y

```
double NodeSnapshot::m_y
```

The documentation for this struct was generated from the following file:

- Source/Controllers/[GraphController.cpp](#)

6.40 Observable< T > Class Template Reference

Base class implementing the Observer pattern for any subject type.

```
#include <Observable.h>
```

Collaboration diagram for Observable< T >:

Observable< T >
- m_id
- m_observers
+ addObserver()
+ removeObserver()
notifyObservers()

Public Types

- using [Observer](#) = std::function< void(T &)>
Type of the observer function: takes a reference to the subject.

Public Member Functions

- unsigned [addObserver](#) ([Observer](#) observer)
Adds a new observer.
- void [removeObserver](#) (unsigned id)
Removes an observer by its unique ID.

Protected Member Functions

- void [notifyObservers](#) (T &subject)
Notifies all registered observers of a state change.

Private Attributes

- unsigned `m_id` = 0
- `std::map< unsigned, Observer > m_observers`

6.40.1 Detailed Description

```
template<typename T>
class Observable< T >
```

Base class implementing the Observer pattern for any subject type.

This template class allows objects to maintain a list of observers that are notified when the object's state changes. For example, [Node](#) and [Edge](#) inherit from [Observable](#) so that their corresponding [NodeItem](#) and [EdgeItem](#) can update their visual representation automatically.

Template Parameters

<code>T</code>	Type of the observed subject. Observers receive a reference to this type.
----------------	---

6.40.2 Member Typedef Documentation

6.40.2.1 Observer

```
template<typename T >
using Observable< T >::Observer = std::function<void(T&)>
```

Type of the observer function: takes a reference to the subject.

6.40.3 Member Function Documentation

6.40.3.1 addObserver()

```
template<typename T >
unsigned Observable< T >::addObserver (
    Observer observer ) [inline]
```

Adds a new observer.

Parameters

<code>observer</code>	The function to call when the subject changes.
-----------------------	--

Returns

A unique ID for the observer, which can be used for removal.

6.40.3.2 notifyObservers()

```
template<typename T >
void Observable< T >::notifyObservers (
    T & subject ) [inline], [protected]
```

Notifies all registered observers of a state change.

Parameters

<i>subject</i>	The subject whose state has changed.
----------------	--------------------------------------

This method copies the observer map before iterating to allow safe removal of observers during notification.

6.40.3.3 removeObserver()

```
template<typename T >
void Observable< T >::removeObserver (
    unsigned id ) [inline]
```

Removes an observer by its unique ID.

Parameters

<i>id</i>	The ID of the observer to remove.
-----------	-----------------------------------

6.40.4 Member Data Documentation

6.40.4.1 m_id

```
template<typename T >
unsigned Observable< T >::m_id = 0 [private]
```

Counter for assigning unique observer IDs

6.40.4.2 m_observers

```
template<typename T >
std::map<unsigned, Observer> Observable< T >::m_observers [private]
```

Map of observer ID to observer function

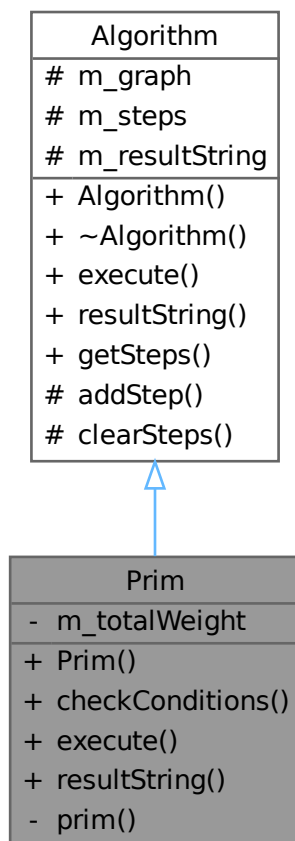
The documentation for this class was generated from the following file:

- Headers/Graph/[Observable.h](#)

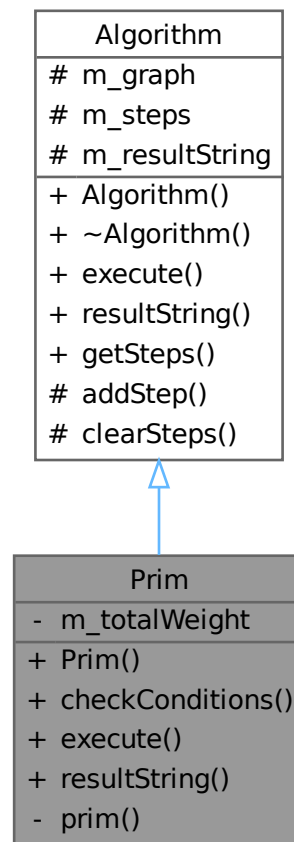
6.41 Prim Class Reference

```
#include <Prim.h>
```

Inheritance diagram for Prim:



Collaboration diagram for Prim:



Public Member Functions

- [Prim](#) (const std::shared_ptr< [Graph](#) >)
- std::optional< [AlgorithmError](#) > [checkConditions](#) () const
- std::optional< [AlgorithmError](#) > [execute](#) (unsigned=0, unsigned=0) override
- QString [resultString](#) () const override

Public Member Functions inherited from [Algorithm](#)

- [Algorithm](#) (const std::shared_ptr< [Graph](#) >)
- virtual [~Algorithm](#) ()=default
- const std::vector< [AlgorithmStep](#) > & [getSteps](#) () const

Private Member Functions

- void [prim](#) ()

Private Attributes

- int `m_totalWeight` = 0

Additional Inherited Members

Protected Member Functions inherited from [Algorithm](#)

- void `addStep` (const [AlgorithmStep](#) &)
- void `clearSteps` ()

Protected Attributes inherited from [Algorithm](#)

- std::shared_ptr< [Graph](#) > `m_graph`
- std::vector< [AlgorithmStep](#) > `m_steps`
- QString `m_resultString`

6.41.1 Constructor & Destructor Documentation

6.41.1.1 Prim()

```
Prim::Prim (  
    const std::shared_ptr< Graph > g ) [explicit]
```

6.41.2 Member Function Documentation

6.41.2.1 checkConditions()

```
std::optional< AlgorithmError > Prim::checkConditions ( ) const
```

6.41.2.2 execute()

```
std::optional< AlgorithmError > Prim::execute (  
    unsigned = 0,  
    unsigned = 0 ) [override], [virtual]
```

Implements [Algorithm](#).

6.41.2.3 prim()

```
void Prim::prim ( ) [private]
```

6.41.2.4 resultString()

```
QString Prim::resultString ( ) const [override], [virtual]
```

Implements [Algorithm](#).

6.41.3 Member Data Documentation

6.41.3.1 m_totalWeight

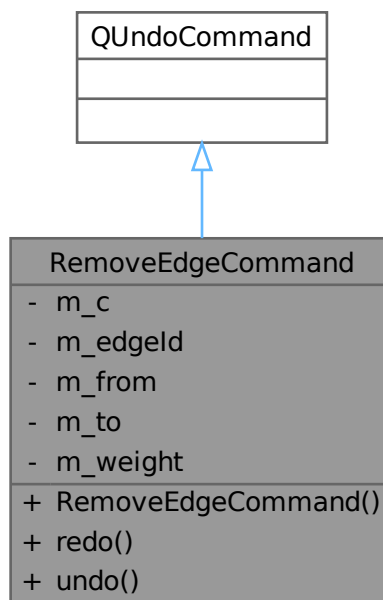
```
int Prim::m_totalWeight = 0 [private]
```

The documentation for this class was generated from the following files:

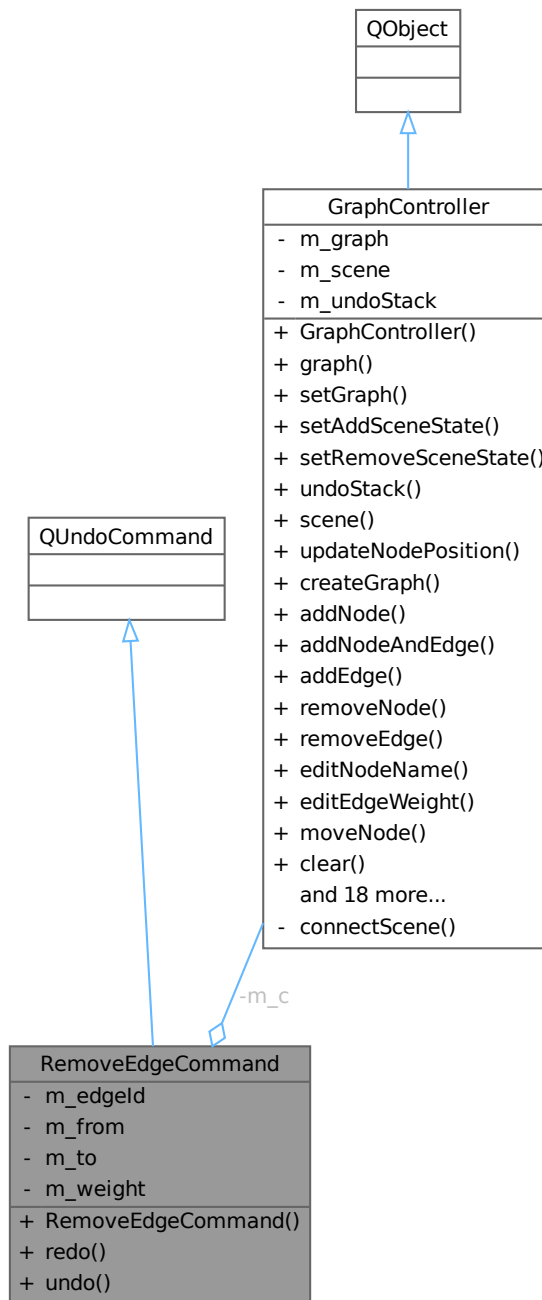
- Headers/Algorithm/[Prim.h](#)
- Source/Algorithm/[Prim.cpp](#)

6.42 RemoveEdgeCommand Class Reference

Inheritance diagram for RemoveEdgeCommand:



Collaboration diagram for RemoveEdgeCommand:



Public Member Functions

- [RemoveEdgeCommand](#) ([GraphController](#) *c, [EdgelItem](#) *edgelitem)
- void [redo](#) () override
- void [undo](#) () override

Private Attributes

- `GraphController * m_c = nullptr`
- `unsigned m_edgId = 0`
- `unsigned m_from = 0`
- `unsigned m_to = 0`
- `int m_weight = 1`

6.42.1 Constructor & Destructor Documentation

6.42.1.1 RemoveEdgeCommand()

```
RemoveEdgeCommand::RemoveEdgeCommand (
    GraphController * c,
    EdgeItem * edgeItem ) [inline]
```

6.42.2 Member Function Documentation

6.42.2.1 redo()

```
void RemoveEdgeCommand::redo ( ) [inline], [override]
```

6.42.2.2 undo()

```
void RemoveEdgeCommand::undo ( ) [inline], [override]
```

6.42.3 Member Data Documentation

6.42.3.1 m_c

```
GraphController* RemoveEdgeCommand::m_c = nullptr [private]
```

6.42.3.2 m_edgId

```
unsigned RemoveEdgeCommand::m_edgeId = 0 [private]
```

6.42.3.3 m_from

```
unsigned RemoveEdgeCommand::m_from = 0 [private]
```

6.42.3.4 m_to

```
unsigned RemoveEdgeCommand::m_to = 0 [private]
```

6.42.3.5 m_weight

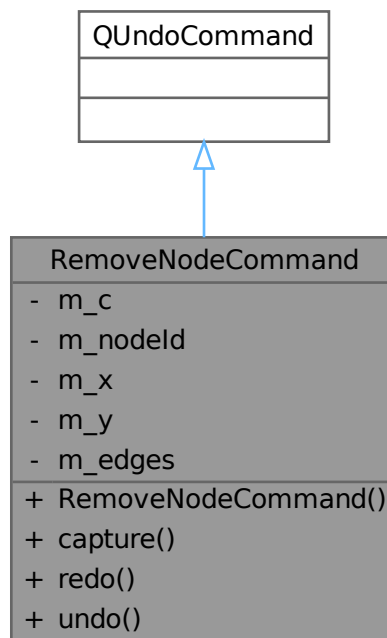
```
int RemoveEdgeCommand::m_weight = 1 [private]
```

The documentation for this class was generated from the following file:

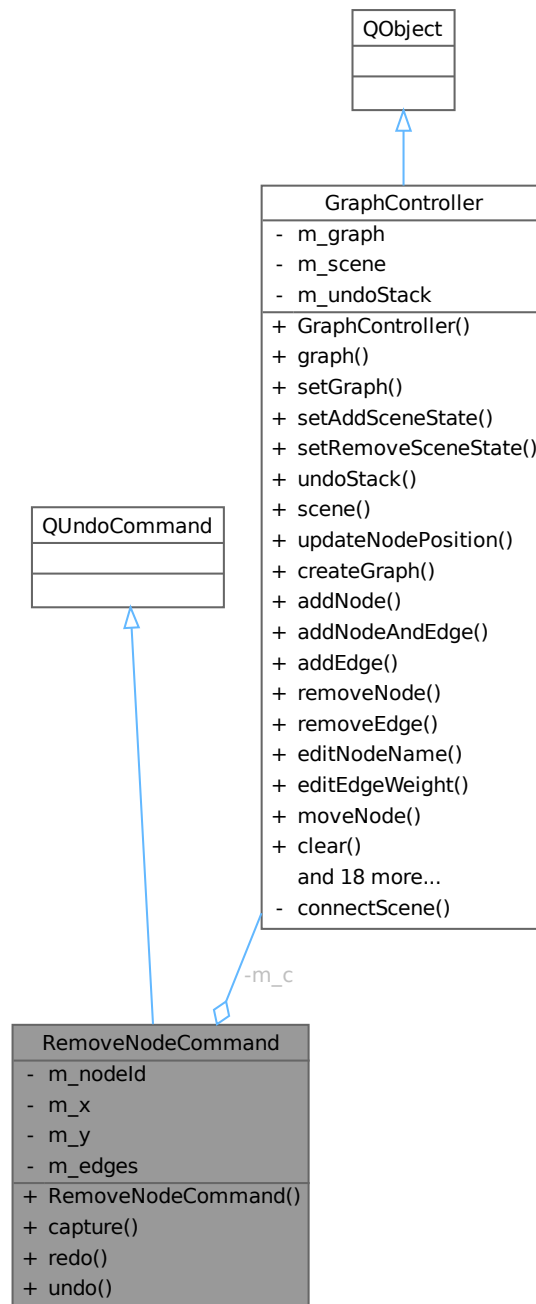
- Source/Controllers/[GraphController.cpp](#)

6.43 RemoveNodeCommand Class Reference

Inheritance diagram for RemoveNodeCommand:



Collaboration diagram for RemoveNodeCommand:



Public Member Functions

- [RemoveNodeCommand](#) ([GraphController](#) *c, unsigned nodeId)
- void [capture](#) ()
- void [redo](#) () override
- void [undo](#) () override

Private Attributes

- [GraphController](#) * [m_c](#)
- unsigned [m_nodeId](#)
- double [m_x](#) = 0
- double [m_y](#) = 0
- std::vector< [EdgeSnapshot](#) > [m_edges](#)

6.43.1 Constructor & Destructor Documentation

6.43.1.1 RemoveNodeCommand()

```
RemoveNodeCommand::RemoveNodeCommand (
    GraphController * c,
    unsigned nodeId ) [inline]
```

6.43.2 Member Function Documentation

6.43.2.1 capture()

```
void RemoveNodeCommand::capture ( ) [inline]
```

6.43.2.2 redo()

```
void RemoveNodeCommand::redo ( ) [inline], [override]
```

6.43.2.3 undo()

```
void RemoveNodeCommand::undo ( ) [inline], [override]
```

6.43.3 Member Data Documentation

6.43.3.1 m_c

```
GraphController* RemoveNodeCommand::m_c [private]
```

6.43.3.2 m_edges

```
std::vector<EdgeSnapshot> RemoveNodeCommand::m_edges [private]
```

6.43.3.3 m_nodeId

```
unsigned RemoveNodeCommand::m_nodeId [private]
```

6.43.3.4 m_x

```
double RemoveNodeCommand::m_x = 0 [private]
```

6.43.3.5 m_y

```
double RemoveNodeCommand::m_y = 0 [private]
```

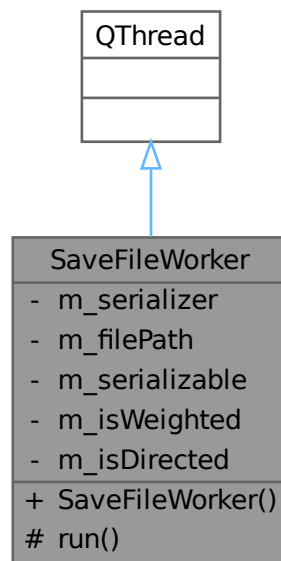
The documentation for this class was generated from the following file:

- Source/Controllers/[GraphController.cpp](#)

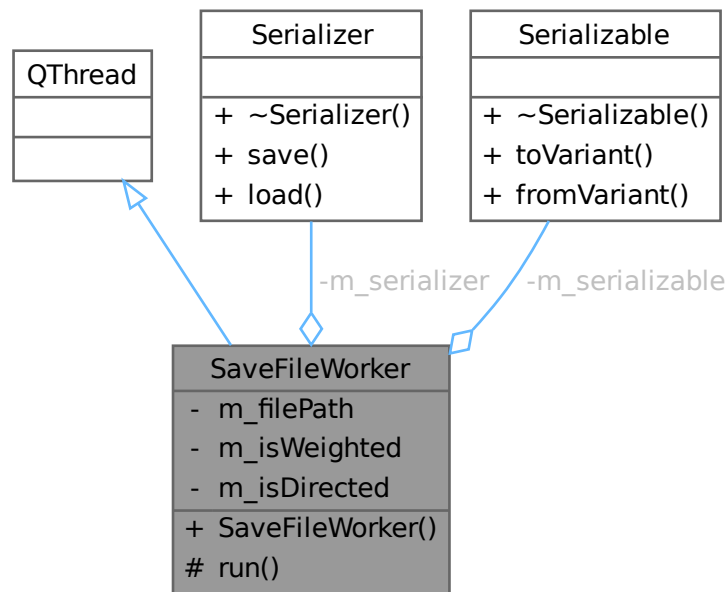
6.44 SaveFileWorker Class Reference

```
#include <SaveFileWorker.h>
```

Inheritance diagram for SaveFileWorker:



Collaboration diagram for SaveFileWorker:



Signals

- void `finished` ()
- void `failed` (const QString &)

Public Member Functions

- `SaveFileWorker` (`Serializer` *, `Serializable` *, const QString &, bool, bool, QObject *!=nullptr)

Protected Member Functions

- void `run` () override

Private Attributes

- `Serializer` * `m_serializer` = nullptr
- QString `m_filePath`
- `Serializable` * `m_serializable` = nullptr
- bool `m_isWeighted` = false
- bool `m_isDirected` = false

6.44.1 Constructor & Destructor Documentation

6.44.1.1 SaveFileWorker()

```
SaveFileWorker::SaveFileWorker (
    Serializer * serializer,
    Serializable * serializable,
    const QString & filePath,
    bool isWeighted,
    bool isDirected,
    QObject * parent = nullptr )
```

6.44.2 Member Function Documentation

6.44.2.1 failed

```
void SaveFileWorker::failed (
    const QString & ) [signal]
```

6.44.2.2 finished

```
void SaveFileWorker::finished ( ) [signal]
```

6.44.2.3 run()

```
void SaveFileWorker::run ( ) [override], [protected]
```

6.44.3 Member Data Documentation

6.44.3.1 m_filePath

```
QString SaveFileWorker::m_filePath [private]
```

6.44.3.2 m_isDirected

```
bool SaveFileWorker::m_isDirected = false [private]
```

6.44.3.3 m_isWeighted

```
bool SaveFileWorker::m_isWeighted = false [private]
```

6.44.3.4 m_serializable

```
Serializable* SaveFileWorker::m_serializable = nullptr [private]
```

6.44.3.5 m_serializer

```
Serializer* SaveFileWorker::m_serializer = nullptr [private]
```

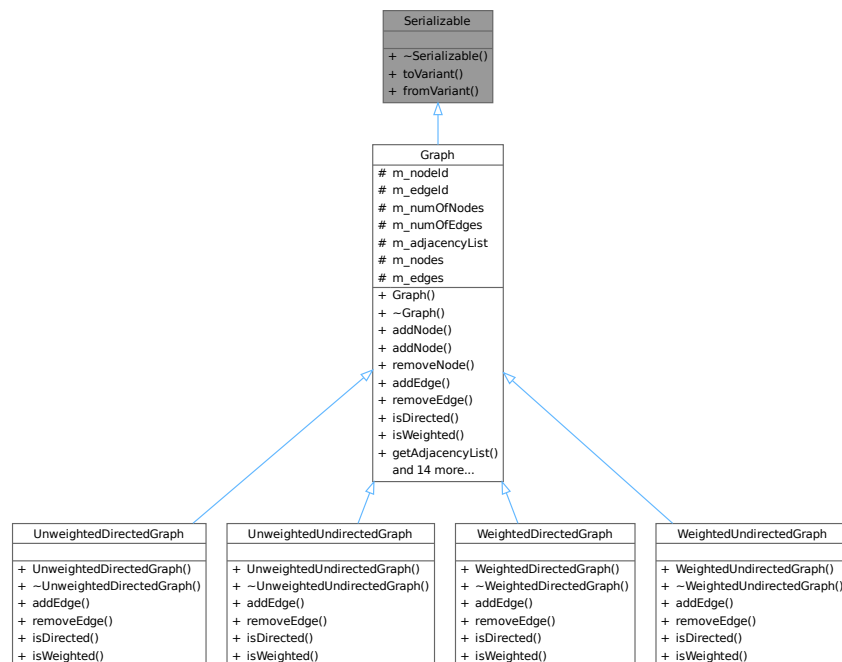
The documentation for this class was generated from the following files:

- Headers/Threads/[SaveFileWorker.h](#)
- Source/Threads/[SaveFileWorker.cpp](#)

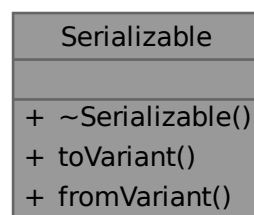
6.45 Serializable Class Reference

```
#include <Serializable.h>
```

Inheritance diagram for Serializable:



Collaboration diagram for Serializable:



Public Member Functions

- virtual [~Serializable](#) ()=default
- virtual QVariant [toVariant](#) () const =0
- virtual void [fromVariant](#) (const QVariant &)=0

6.45.1 Constructor & Destructor Documentation**6.45.1.1 ~Serializable()**

```
virtual Serializable::~Serializable ( ) [virtual], [default]
```

6.45.2 Member Function Documentation**6.45.2.1 fromVariant()**

```
virtual void Serializable::fromVariant (
    const QVariant & ) [pure virtual]
```

Implemented in [Graph](#).

6.45.2.2 toVariant()

```
virtual QVariant Serializable::toVariant ( ) const [pure virtual]
```

Implemented in [Graph](#).

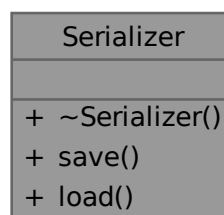
The documentation for this class was generated from the following file:

- Headers/Serialization/[Serializable.h](#)

6.46 Serializer Class Reference

```
#include <Serializer.h>
```

Collaboration diagram for Serializer:



Public Member Functions

- virtual [~Serializer](#) ()=default
- virtual void [save](#) (const [Serializable](#) &, const QString &, bool, bool)
- virtual void [load](#) ([Serializable](#) &, const QVariant &)

6.46.1 Constructor & Destructor Documentation

6.46.1.1 ~Serializer()

```
virtual Serializer::~~Serializer ( ) [virtual], [default]
```

6.46.2 Member Function Documentation

6.46.2.1 load()

```
void Serializer::load (
    Serializable & serializable,
    const QVariant & data ) [virtual]
```

6.46.2.2 save()

```
void Serializer::save (
    const Serializable & s,
    const QString & filePath,
    bool isWeighted,
    bool isDirected ) [virtual]
```

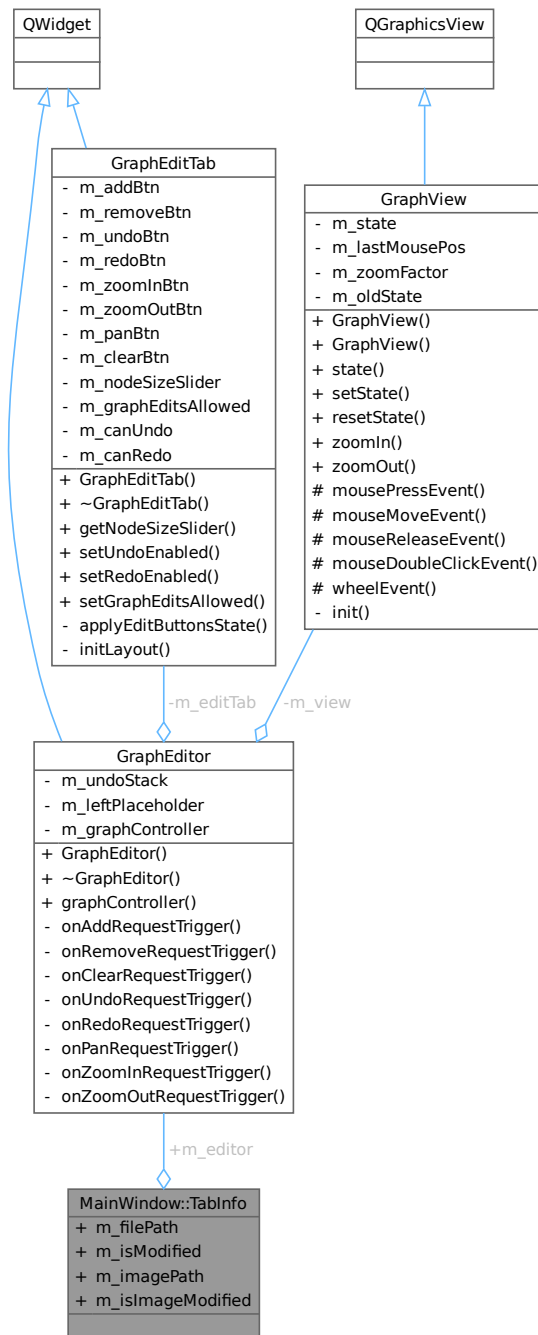
The documentation for this class was generated from the following files:

- Headers/Serialization/[Serializer.h](#)
- Source/Serialization/[Serializer.cpp](#)

6.47 MainWindow::TabInfo Struct Reference

```
#include <MainWindow.h>
```


Collaboration diagram for MainWindow::TabInfo:



Public Attributes

- [GraphEditor](#) * `m_editor`
- `QString` `m_filePath`
- `bool` `m_isModified`
- `QString` `m_imagePath`
- `bool` `m_isImageModified`

6.47.1 Member Data Documentation

6.47.1.1 `m_editor`

`GraphEditor*` `MainWindow::TabInfo::m_editor`

6.47.1.2 `m_filePath`

`QString` `MainWindow::TabInfo::m_filePath`

6.47.1.3 `m_imagePath`

`QString` `MainWindow::TabInfo::m_imagePath`

6.47.1.4 `m_isImageModified`

`bool` `MainWindow::TabInfo::m_isImageModified`

6.47.1.5 `m_isModified`

`bool` `MainWindow::TabInfo::m_isModified`

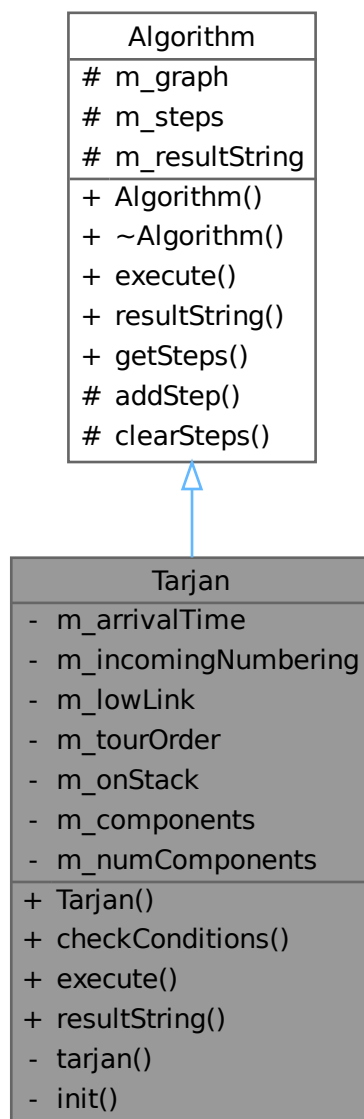
The documentation for this struct was generated from the following file:

- Headers/[MainWindow.h](#)

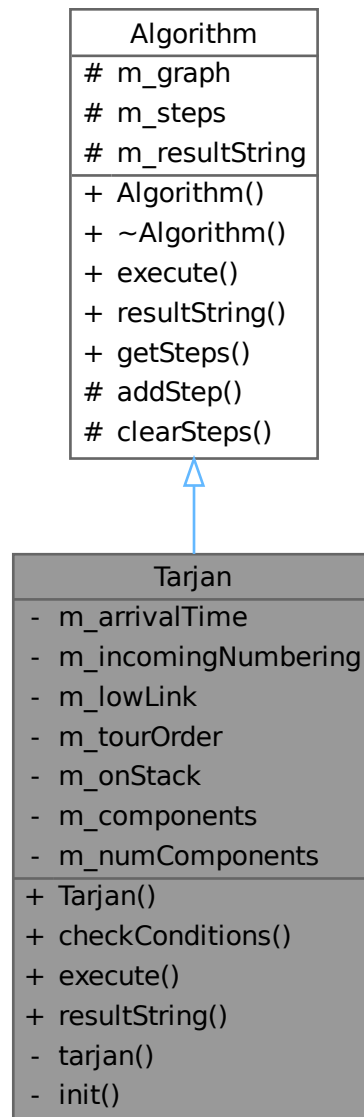
6.48 Tarjan Class Reference

```
#include <Tarjan.h>
```

Inheritance diagram for Tarjan:



Collaboration diagram for Tarjan:



Public Member Functions

- [Tarjan](#) (const std::shared_ptr< [Graph](#) >)
- std::optional< [AlgorithmError](#) > [checkConditions](#) () const
- std::optional< [AlgorithmError](#) > [execute](#) (unsigned=0, unsigned=0) override
- QString [resultString](#) () const override

Public Member Functions inherited from [Algorithm](#)

- [Algorithm](#) (const std::shared_ptr< [Graph](#) >)
- virtual [~Algorithm](#) ()=default
- const std::vector< [AlgorithmStep](#) > & [getSteps](#) () const

Private Member Functions

- void [tarjan](#) (unsigned)
- void [init](#) ()

Private Attributes

- int [m_arrivalTime](#) = 0
- std::map< unsigned, int > [m_incomingNumbering](#)
- std::map< unsigned, int > [m_lowLink](#)
- std::stack< unsigned > [m_tourOrder](#)
- std::map< unsigned, bool > [m_onStack](#)
- std::map< unsigned, int > [m_components](#)
- int [m_numComponents](#) = 0

Additional Inherited Members**Protected Member Functions inherited from [Algorithm](#)**

- void [addStep](#) (const [AlgorithmStep](#) &)
- void [clearSteps](#) ()

Protected Attributes inherited from [Algorithm](#)

- std::shared_ptr< [Graph](#) > [m_graph](#)
- std::vector< [AlgorithmStep](#) > [m_steps](#)
- QString [m_resultString](#)

6.48.1 Constructor & Destructor Documentation**6.48.1.1 [Tarjan\(\)](#)**

```
Tarjan::Tarjan (
    const std::shared_ptr< Graph > g ) [explicit]
```

6.48.2 Member Function Documentation**6.48.2.1 [checkConditions\(\)](#)**

```
std::optional< AlgorithmError > Tarjan::checkConditions ( ) const
```

6.48.2.2 [execute\(\)](#)

```
std::optional< AlgorithmError > Tarjan::execute (
    unsigned = 0,
    unsigned = 0 ) [override], [virtual]
```

Implements [Algorithm](#).

6.48.2.3 init()

```
void Tarjan::init ( ) [private]
```

6.48.2.4 resultString()

```
QString Tarjan::resultString ( ) const [override], [virtual]
```

Implements [Algorithm](#).

6.48.2.5 tarjan()

```
void Tarjan::tarjan (
    unsigned nodeId ) [private]
```

6.48.3 Member Data Documentation

6.48.3.1 m_arrivalTime

```
int Tarjan::m_arrivalTime = 0 [private]
```

6.48.3.2 m_components

```
std::map<unsigned, int> Tarjan::m_components [private]
```

6.48.3.3 m_incomingNumbering

```
std::map<unsigned, int> Tarjan::m_incomingNumbering [private]
```

6.48.3.4 m_lowLink

```
std::map<unsigned, int> Tarjan::m_lowLink [private]
```

6.48.3.5 m_numComponents

```
int Tarjan::m_numComponents = 0 [private]
```

6.48.3.6 m_onStack

```
std::map<unsigned, bool> Tarjan::m_onStack [private]
```

6.48.3.7 m_tourOrder

```
std::stack<unsigned> Tarjan::m_tourOrder [private]
```

The documentation for this class was generated from the following files:

- Headers/Algorithm/[Tarjan.h](#)
- Source/Algorithm/[Tarjan.cpp](#)

6.49 ThemeManager Class Reference

```
#include <ThemeManager.h>
```

Collaboration diagram for ThemeManager:

ThemeManager
- m_currentTheme
- m_settings
+ ThemeManager()
+ currentTheme()
+ styleSheet()
+ setTheme()
- loadTheme()
- saveTheme()

Public Types

- enum class [Theme](#) { [LIGHT](#) , [DARK](#) , [PURPLE](#) }

Public Member Functions

- [ThemeManager](#) ()
- [Theme](#) [currentTheme](#) () const
- QString [styleSheet](#) () const
- void [setTheme](#) ([Theme](#))

Private Member Functions

- void [loadTheme](#) ()
- void [saveTheme](#) ()

Private Attributes

- Theme `m_currentTheme` {Theme::DARK}
- QSettings `m_settings`

6.49.1 Member Enumeration Documentation

6.49.1.1 Theme

```
enum class ThemeManager::Theme [strong]
```

Enumerator

LIGHT	
DARK	
PURPLE	

6.49.2 Constructor & Destructor Documentation

6.49.2.1 ThemeManager()

```
ThemeManager::ThemeManager ( ) [explicit]
```

6.49.3 Member Function Documentation

6.49.3.1 currentTheme()

```
ThemeManager::Theme ThemeManager::currentTheme ( ) const
```

6.49.3.2 loadTheme()

```
void ThemeManager::loadTheme ( ) [private]
```

6.49.3.3 saveTheme()

```
void ThemeManager::saveTheme ( ) [private]
```

6.49.3.4 setTheme()

```
void ThemeManager::setTheme (
    Theme theme )
```


6.49.3.5 `styleSheet()`

```
QString ThemeManager::styleSheet ( ) const
```

6.49.4 Member Data Documentation

6.49.4.1 `m_currentTheme`

```
Theme ThemeManager::m_currentTheme {Theme::DARK} [private]
```

6.49.4.2 `m_settings`

```
QSettings ThemeManager::m_settings [private]
```

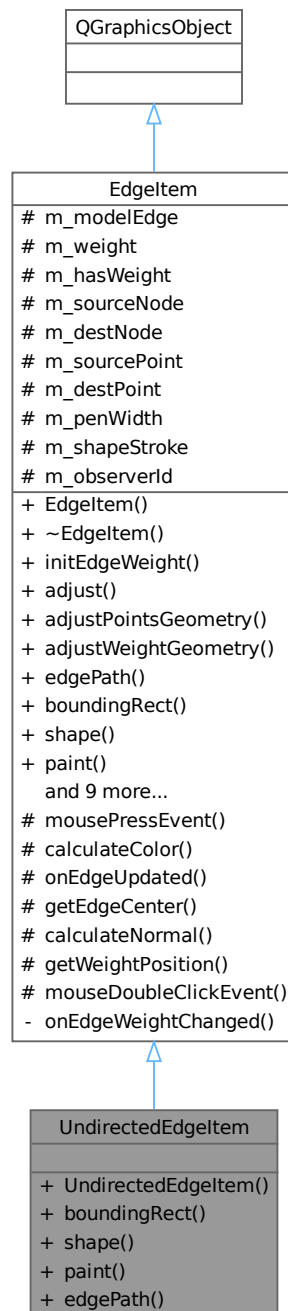
The documentation for this class was generated from the following files:

- Headers/UI/[ThemeManager.h](#)
- Source/UI/[ThemeManager.cpp](#)

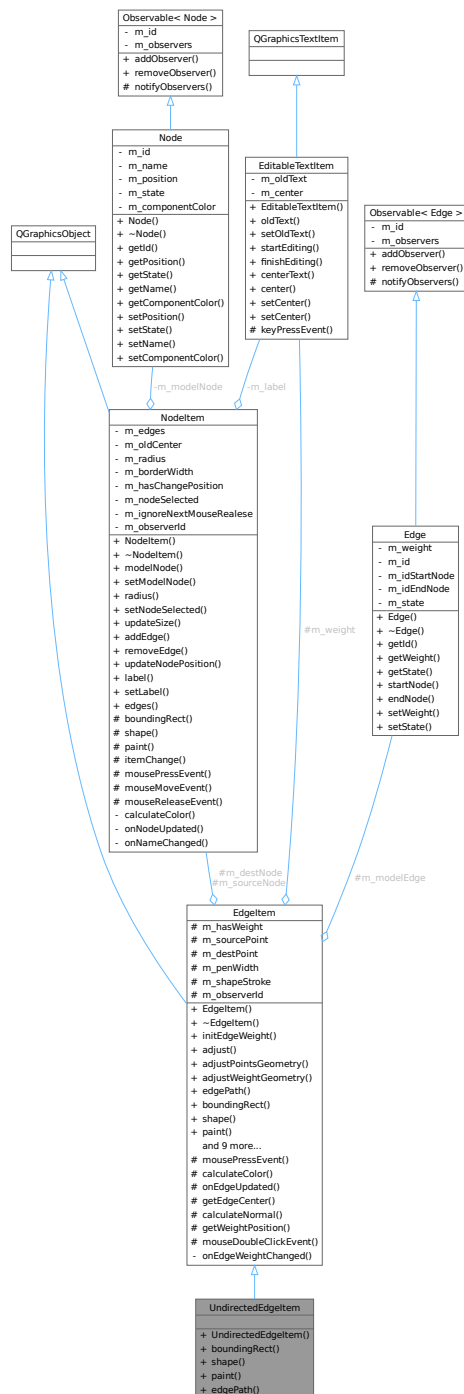
6.50 UndirectedEdgeItem Class Reference

```
#include <UndirectedEdgeItem.h>
```

Inheritance diagram for UndirectedEdgeItem:



Collaboration diagram for UndirectedEdgeItem:



Public Member Functions

- [UndirectedEdgeItem](#) ([Edge](#) *, [NodeItem](#) *, [NodeItem](#) *, bool=false)
- [QRectF](#) [boundingRect](#) () const override
- [QPainterPath](#) [shape](#) () const override
- void [paint](#) ([QPainter](#) *, const [QStyleOptionGraphicsItem](#) *, [QWidget](#) *) override
- [QPainterPath](#) [edgePath](#) () const override

Public Member Functions inherited from [EdgeItem](#)

- [EdgeItem](#) ([Edge](#) *, [NodeItem](#) *, [NodeItem](#) *, bool=false)
- [~EdgeItem](#) () override
- void [initEdgeWeight](#) ()
- void [adjust](#) ()
- void [adjustPointsGeometry](#) ()
- void [adjustWeightGeometry](#) () const
- void [connectNodes](#) ()
- void [disconnectNodes](#) ()
- [Edge](#) * [modelEdge](#) () const
- void [setModelEdge](#) ([Edge](#) *)
- bool [hasWeight](#) () const
- void [setHasWeight](#) (bool)
- [EditableTextItem](#) * [weight](#) () const
- void [setWeight](#) ([EditableTextItem](#) *)
- void [updateSize](#) ()

Additional Inherited Members

Signals inherited from [EdgeItem](#)

- void [edgeSelected](#) ([EdgeItem](#) *)
- void [editEdgeWeightRequest](#) (const [EdgeItem](#) *, const QString &) const

Protected Member Functions inherited from [EdgeItem](#)

- void [mousePressEvent](#) (QGraphicsSceneMouseEvent *) override
- const QColor [calculateColor](#) () const
- void [onEdgeUpdated](#) ()
- QPointF [getEdgeCenter](#) () const
- QPointF [calculateNormal](#) () const
- QPointF [getWeightPosition](#) () const
- void [mouseDoubleClickEvent](#) (QGraphicsSceneMouseEvent *) override

Protected Attributes inherited from [EdgeItem](#)

- [Edge](#) * [m_modelEdge](#)
- [EditableTextItem](#) * [m_weight](#) {nullptr}
- bool [m_hasWeight](#)
- [NodeItem](#) * [m_sourceNode](#)
- [NodeItem](#) * [m_destNode](#)
- QPointF [m_sourcePoint](#)
- QPointF [m_destPoint](#)
- qreal [m_penWidth](#) {2}
- qreal [m_shapeStroke](#) {20.0}
- unsigned [m_observerId](#) {0}

6.50.1 Constructor & Destructor Documentation

6.50.1.1 UndirectedEdgeItem()

```
UndirectedEdgeItem::UndirectedEdgeItem (
    Edge * modelEdge,
    NodeItem * sourceNode,
    NodeItem * destNode,
    bool hasWeight = false )
```

6.50.2 Member Function Documentation

6.50.2.1 boundingRect()

```
QRectF UndirectedEdgeItem::boundingRect ( ) const [override], [virtual]
```

Implements [EdgeItem](#).

6.50.2.2 edgePath()

```
QPainterPath UndirectedEdgeItem::edgePath ( ) const [override], [virtual]
```

Implements [EdgeItem](#).

6.50.2.3 paint()

```
void UndirectedEdgeItem::paint (
    QPainter * painter,
    const QStyleOptionGraphicsItem * option,
    QWidget * widget ) [override], [virtual]
```

Implements [EdgeItem](#).

6.50.2.4 shape()

```
QPainterPath UndirectedEdgeItem::shape ( ) const [override], [virtual]
```

Implements [EdgeItem](#).

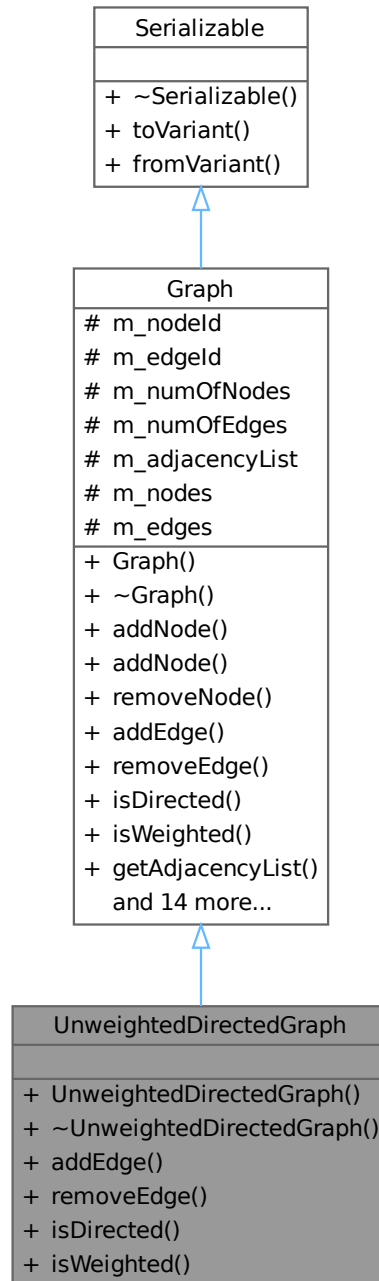
The documentation for this class was generated from the following files:

- Headers/View/[UndirectedEdgeItem.h](#)
- Source/View/[UndirectedEdgeItem.cpp](#)

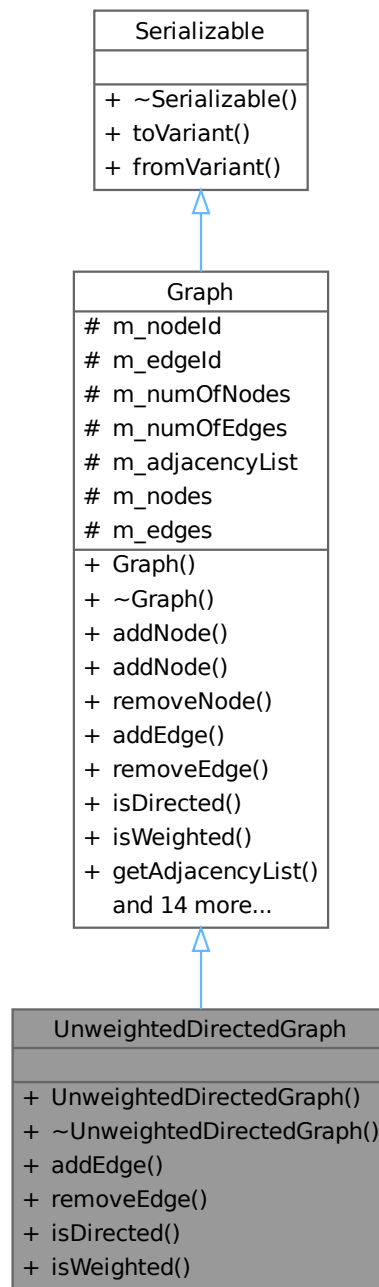
6.51 UnweightedDirectedGraph Class Reference

```
#include <UnweightedDirectedGraph.h>
```

Inheritance diagram for UnweightedDirectedGraph:



Collaboration diagram for UnweightedDirectedGraph:



Public Member Functions

- [UnweightedDirectedGraph](#) ()=default
- [~UnweightedDirectedGraph](#) () override=default
- void [addEdge](#) (unsigned, unsigned, int=1) override
- void [removeEdge](#) (unsigned) override
- bool [isDirected](#) () const override
- bool [isWeighted](#) () const override

Public Member Functions inherited from [Graph](#)

- [Graph](#) ()=default
- [~Graph](#) () override=default
- [Node](#) * [addNode](#) (double, double)
- void [addNode](#) (unsigned, double=0.0, double=0.0)
- void [removeNode](#) (unsigned)
- std::map< unsigned, std::map< unsigned, unsigned > > [getAdjacencyList](#) () const
- std::map< unsigned, [Node](#) > [getNodes](#) () const
- std::map< unsigned, [Edge](#) > [getEdges](#) () const
- std::vector< [Node](#) * > [getNodesMutable](#) ()
- std::vector< [Edge](#) * > [getEdgesMutable](#) ()
- [Node](#) * [getNode](#) (unsigned)
- const [Node](#) * [getNode](#) (unsigned) const
- [Edge](#) * [getEdge](#) (unsigned)
- const [Edge](#) * [getEdge](#) (unsigned) const
- [Edge](#) * [getEdge](#) (unsigned, unsigned)
- const [Edge](#) * [getEdge](#) (unsigned, unsigned) const
- void [addEdgeSerialized](#) (unsigned, unsigned, unsigned, int)
- QVariant [toVariant](#) () const override
- void [fromVariant](#) (const QVariant &) override
- void [clear](#) ()

Public Member Functions inherited from [Serializable](#)

- virtual [~Serializable](#) ()=default

Additional Inherited Members

Protected Attributes inherited from [Graph](#)

- unsigned [m_nodeId](#) = 0
- unsigned [m_edgeId](#) = 0
- unsigned [m_numOfNodes](#) = 0
- unsigned [m_numOfEdges](#) = 0
- std::map< unsigned, std::map< unsigned, unsigned > > [m_adjacencyList](#)
- std::map< unsigned, [Node](#) > [m_nodes](#)
- std::map< unsigned, [Edge](#) > [m_edges](#)

6.51.1 Constructor & Destructor Documentation

6.51.1.1 UnweightedDirectedGraph()

```
UnweightedDirectedGraph::UnweightedDirectedGraph ( ) [default]
```

6.51.1.2 ~UnweightedDirectedGraph()

```
UnweightedDirectedGraph::~~UnweightedDirectedGraph ( ) [override], [default]
```


6.51.2 Member Function Documentation

6.51.2.1 addEdge()

```
void UnweightedDirectedGraph::addEdge (
    unsigned from,
    unsigned to,
    int w = 1 ) [override], [virtual]
```

Implements [Graph](#).

6.51.2.2 isDirected()

```
bool UnweightedDirectedGraph::isDirected ( ) const [override], [virtual]
```

Implements [Graph](#).

6.51.2.3 isWeighted()

```
bool UnweightedDirectedGraph::isWeighted ( ) const [override], [virtual]
```

Implements [Graph](#).

6.51.2.4 removeEdge()

```
void UnweightedDirectedGraph::removeEdge (
    unsigned edgeId ) [override], [virtual]
```

Implements [Graph](#).

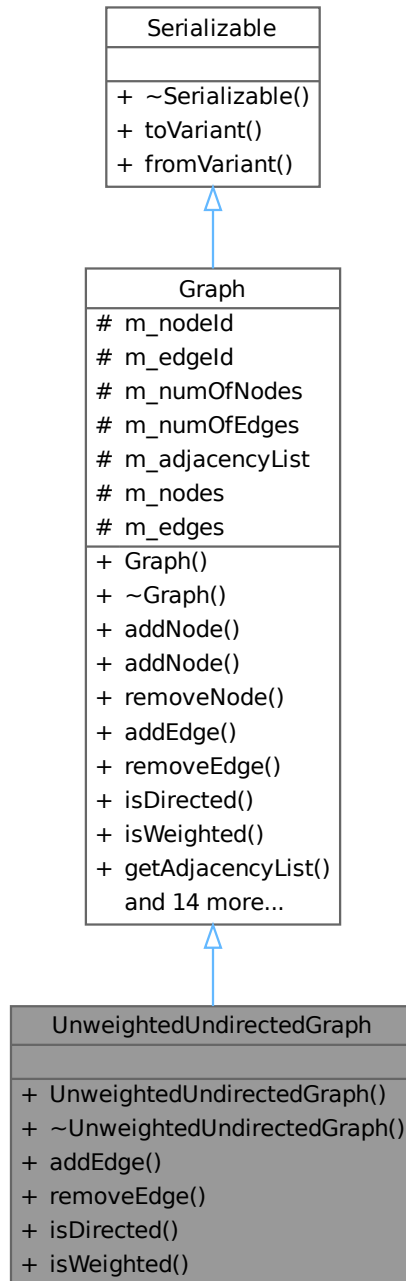
The documentation for this class was generated from the following files:

- Headers/Graph/[UnweightedDirectedGraph.h](#)
- Source/Graph/[UnweightedDirectedGraph.cpp](#)

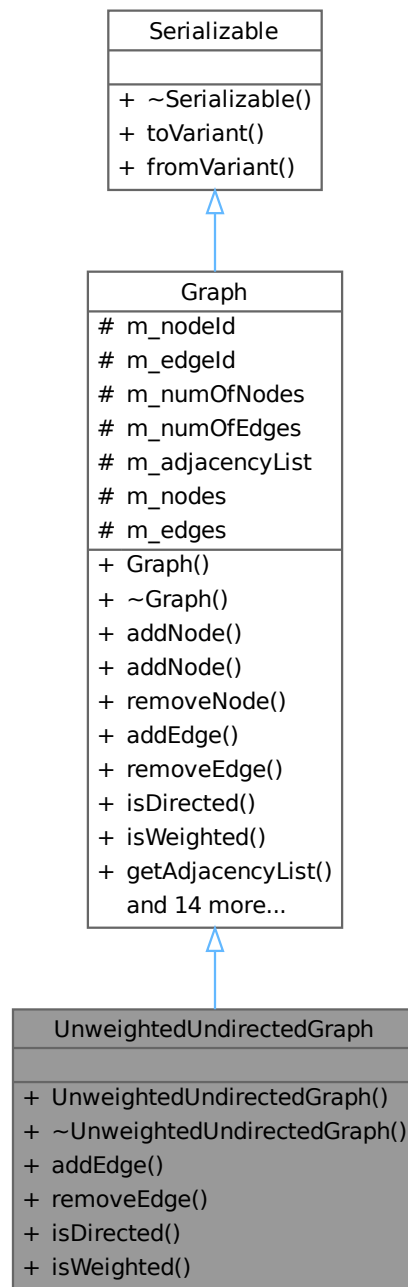
6.52 UnweightedUndirectedGraph Class Reference

```
#include <UnweightedUndirectedGraph.h>
```

Inheritance diagram for UnweightedUndirectedGraph:



Collaboration diagram for UnweightedUndirectedGraph:



Public Member Functions

- [UnweightedUndirectedGraph](#) ()=default
- [~UnweightedUndirectedGraph](#) () override=default
- void [addEdge](#) (unsigned, unsigned, int=1) override
- void [removeEdge](#) (unsigned) override
- bool [isDirected](#) () const override
- bool [isWeighted](#) () const override

Public Member Functions inherited from [Graph](#)

- [Graph](#) ()=default
- [~Graph](#) () override=default
- [Node](#) * [addNode](#) (double, double)
- void [addNode](#) (unsigned, double=0.0, double=0.0)
- void [removeNode](#) (unsigned)
- std::map< unsigned, std::map< unsigned, unsigned > > [getAdjacencyList](#) () const
- std::map< unsigned, [Node](#) > [getNodes](#) () const
- std::map< unsigned, [Edge](#) > [getEdges](#) () const
- std::vector< [Node](#) * > [getNodesMutable](#) ()
- std::vector< [Edge](#) * > [getEdgesMutable](#) ()
- [Node](#) * [getNode](#) (unsigned)
- const [Node](#) * [getNode](#) (unsigned) const
- [Edge](#) * [getEdge](#) (unsigned)
- const [Edge](#) * [getEdge](#) (unsigned) const
- [Edge](#) * [getEdge](#) (unsigned, unsigned)
- const [Edge](#) * [getEdge](#) (unsigned, unsigned) const
- void [addEdgeSerialized](#) (unsigned, unsigned, unsigned, int)
- QVariant [toVariant](#) () const override
- void [fromVariant](#) (const QVariant &) override
- void [clear](#) ()

Public Member Functions inherited from [Serializable](#)

- virtual [~Serializable](#) ()=default

Additional Inherited Members

Protected Attributes inherited from [Graph](#)

- unsigned [m_nodeId](#) = 0
- unsigned [m_edgeId](#) = 0
- unsigned [m_numOfNodes](#) = 0
- unsigned [m_numOfEdges](#) = 0
- std::map< unsigned, std::map< unsigned, unsigned > > [m_adjacencyList](#)
- std::map< unsigned, [Node](#) > [m_nodes](#)
- std::map< unsigned, [Edge](#) > [m_edges](#)

6.52.1 Constructor & Destructor Documentation

6.52.1.1 UnweightedUndirectedGraph()

```
UnweightedUndirectedGraph::UnweightedUndirectedGraph ( ) [default]
```

6.52.1.2 ~UnweightedUndirectedGraph()

```
UnweightedUndirectedGraph::~UnweightedUndirectedGraph ( ) [override], [default]
```

6.52.2 Member Function Documentation

6.52.2.1 addEdge()

```
void UnweightedUndirectedGraph::addEdge (
    unsigned from,
    unsigned to,
    int w = 1 ) [override], [virtual]
```

Implements [Graph](#).

6.52.2.2 isDirected()

```
bool UnweightedUndirectedGraph::isDirected ( ) const [override], [virtual]
```

Implements [Graph](#).

6.52.2.3 isWeighted()

```
bool UnweightedUndirectedGraph::isWeighted ( ) const [override], [virtual]
```

Implements [Graph](#).

6.52.2.4 removeEdge()

```
void UnweightedUndirectedGraph::removeEdge (
    unsigned edgeId ) [override], [virtual]
```

Implements [Graph](#).

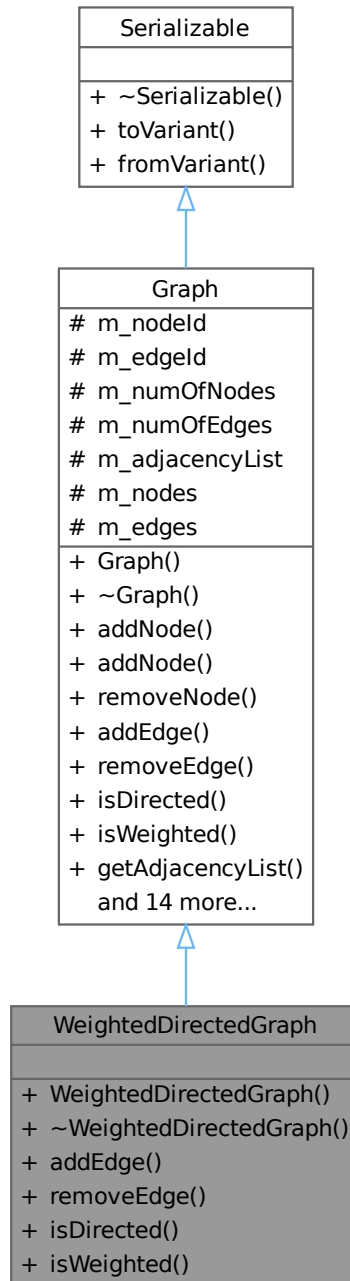
The documentation for this class was generated from the following files:

- Headers/Graph/[UnweightedUndirectedGraph.h](#)
- Source/Graph/[UnweightedUndirectedGraph.cpp](#)

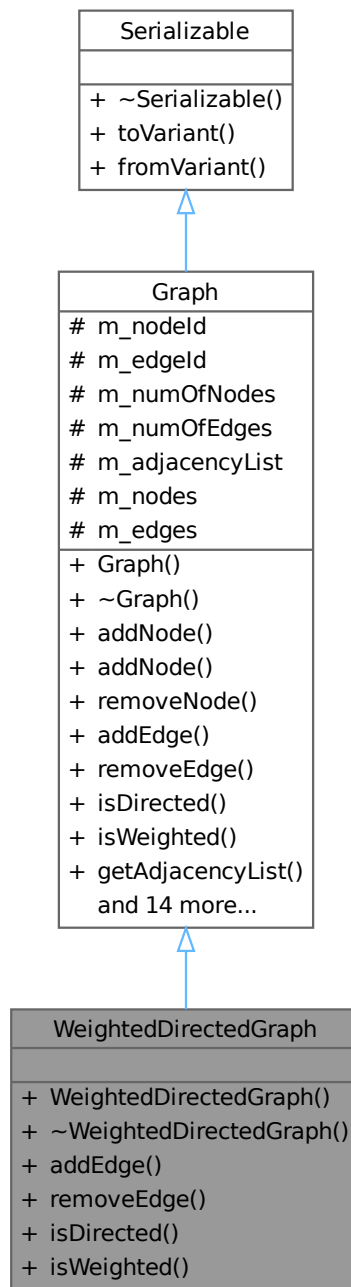
6.53 WeightedDirectedGraph Class Reference

```
#include <WeightedDirectedGraph.h>
```

Inheritance diagram for WeightedDirectedGraph:



Collaboration diagram for WeightedDirectedGraph:



Public Member Functions

- [WeightedDirectedGraph](#) ()=default
- [~WeightedDirectedGraph](#) () override=default
- void [addEdge](#) (unsigned, unsigned, int) override
- void [removeEdge](#) (unsigned) override
- bool [isDirected](#) () const override
- bool [isWeighted](#) () const override

Public Member Functions inherited from [Graph](#)

- [Graph](#) ()=default
- [~Graph](#) () override=default
- [Node](#) * [addNode](#) (double, double)
- void [addNode](#) (unsigned, double=0.0, double=0.0)
- void [removeNode](#) (unsigned)
- std::map< unsigned, std::map< unsigned, unsigned > > [getAdjacencyList](#) () const
- std::map< unsigned, [Node](#) > [getNodes](#) () const
- std::map< unsigned, [Edge](#) > [getEdges](#) () const
- std::vector< [Node](#) * > [getNodesMutable](#) ()
- std::vector< [Edge](#) * > [getEdgesMutable](#) ()
- [Node](#) * [getNode](#) (unsigned)
- const [Node](#) * [getNode](#) (unsigned) const
- [Edge](#) * [getEdge](#) (unsigned)
- const [Edge](#) * [getEdge](#) (unsigned) const
- [Edge](#) * [getEdge](#) (unsigned, unsigned)
- const [Edge](#) * [getEdge](#) (unsigned, unsigned) const
- void [addEdgeSerialized](#) (unsigned, unsigned, unsigned, int)
- QVariant [toVariant](#) () const override
- void [fromVariant](#) (const QVariant &) override
- void [clear](#) ()

Public Member Functions inherited from [Serializable](#)

- virtual [~Serializable](#) ()=default

Additional Inherited Members

Protected Attributes inherited from [Graph](#)

- unsigned [m_nodeId](#) = 0
- unsigned [m_edgeId](#) = 0
- unsigned [m_numOfNodes](#) = 0
- unsigned [m_numOfEdges](#) = 0
- std::map< unsigned, std::map< unsigned, unsigned > > [m_adjacencyList](#)
- std::map< unsigned, [Node](#) > [m_nodes](#)
- std::map< unsigned, [Edge](#) > [m_edges](#)

6.53.1 Constructor & Destructor Documentation

6.53.1.1 [WeightedDirectedGraph](#)()

`WeightedDirectedGraph::WeightedDirectedGraph ( ) [default]`

6.53.1.2 [~WeightedDirectedGraph](#)()

`WeightedDirectedGraph::~~WeightedDirectedGraph ( ) [override], [default]`

6.53.2 Member Function Documentation

6.53.2.1 addEdge()

```
void WeightedDirectedGraph::addEdge (
    unsigned from,
    unsigned to,
    int w ) [override], [virtual]
```

Implements [Graph](#).

6.53.2.2 isDirected()

```
bool WeightedDirectedGraph::isDirected ( ) const [override], [virtual]
```

Implements [Graph](#).

6.53.2.3 isWeighted()

```
bool WeightedDirectedGraph::isWeighted ( ) const [override], [virtual]
```

Implements [Graph](#).

6.53.2.4 removeEdge()

```
void WeightedDirectedGraph::removeEdge (
    unsigned edgeId ) [override], [virtual]
```

Implements [Graph](#).

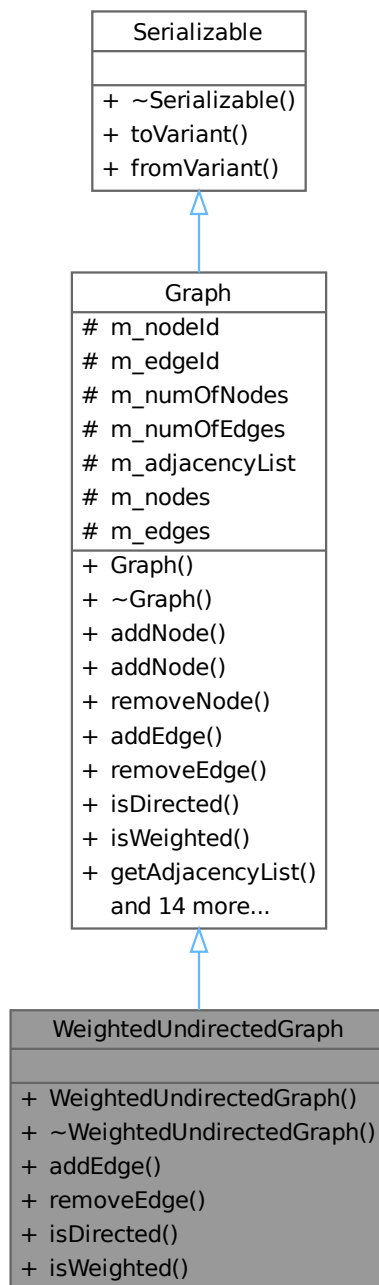
The documentation for this class was generated from the following files:

- Headers/Graph/[WeightedDirectedGraph.h](#)
- Source/Graph/[WeightedDirectedGraph.cpp](#)

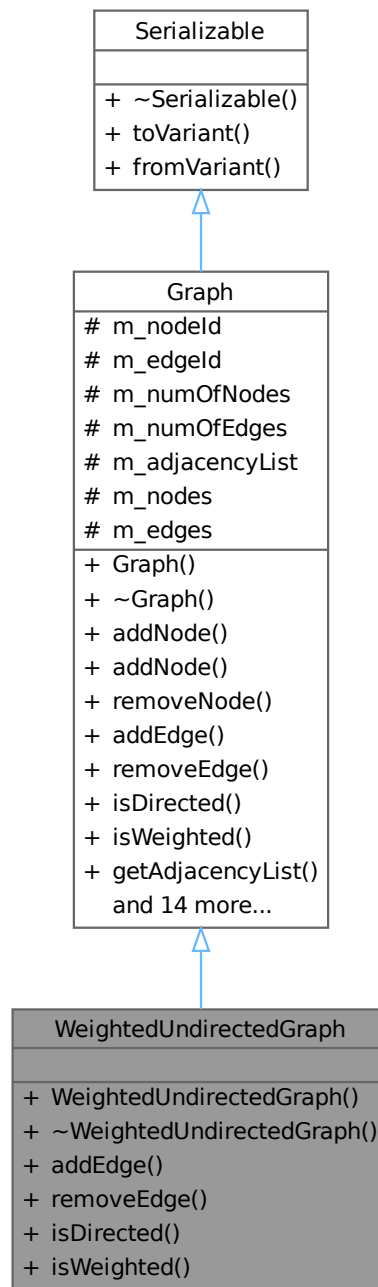
6.54 WeightedUndirectedGraph Class Reference

```
#include <WeightedUndirectedGraph.h>
```

Inheritance diagram for WeightedUndirectedGraph:



Collaboration diagram for WeightedUndirectedGraph:



Public Member Functions

- [WeightedUndirectedGraph](#) ()=default
- [~WeightedUndirectedGraph](#) () override=default
- void [addEdge](#) (unsigned, unsigned, int) override
- void [removeEdge](#) (unsigned) override
- bool [isDirected](#) () const override
- bool [isWeighted](#) () const override

Public Member Functions inherited from [Graph](#)

- [Graph](#) ()=default
- [~Graph](#) () override=default
- [Node](#) * [addNode](#) (double, double)
- void [addNode](#) (unsigned, double=0.0, double=0.0)
- void [removeNode](#) (unsigned)
- std::map< unsigned, std::map< unsigned, unsigned > > [getAdjacencyList](#) () const
- std::map< unsigned, [Node](#) > [getNodes](#) () const
- std::map< unsigned, [Edge](#) > [getEdges](#) () const
- std::vector< [Node](#) * > [getNodesMutable](#) ()
- std::vector< [Edge](#) * > [getEdgesMutable](#) ()
- [Node](#) * [getNode](#) (unsigned)
- const [Node](#) * [getNode](#) (unsigned) const
- [Edge](#) * [getEdge](#) (unsigned)
- const [Edge](#) * [getEdge](#) (unsigned) const
- [Edge](#) * [getEdge](#) (unsigned, unsigned)
- const [Edge](#) * [getEdge](#) (unsigned, unsigned) const
- void [addEdgeSerialized](#) (unsigned, unsigned, unsigned, int)
- QVariant [toVariant](#) () const override
- void [fromVariant](#) (const QVariant &) override
- void [clear](#) ()

Public Member Functions inherited from [Serializable](#)

- virtual [~Serializable](#) ()=default

Additional Inherited Members

Protected Attributes inherited from [Graph](#)

- unsigned [m_nodeId](#) = 0
- unsigned [m_edgeId](#) = 0
- unsigned [m_numOfNodes](#) = 0
- unsigned [m_numOfEdges](#) = 0
- std::map< unsigned, std::map< unsigned, unsigned > > [m_adjacencyList](#)
- std::map< unsigned, [Node](#) > [m_nodes](#)
- std::map< unsigned, [Edge](#) > [m_edges](#)

6.54.1 Constructor & Destructor Documentation

6.54.1.1 [WeightedUndirectedGraph](#)()

`WeightedUndirectedGraph::WeightedUndirectedGraph ( ) [default]`

6.54.1.2 [~WeightedUndirectedGraph](#)()

`WeightedUndirectedGraph::~~WeightedUndirectedGraph ( ) [override], [default]`

6.54.2 Member Function Documentation

6.54.2.1 addEdge()

```
void WeightedUndirectedGraph::addEdge (
    unsigned from,
    unsigned to,
    int w ) [override], [virtual]
```

Implements [Graph](#).

6.54.2.2 isDirected()

```
bool WeightedUndirectedGraph::isDirected ( ) const [override], [virtual]
```

Implements [Graph](#).

6.54.2.3 isWeighted()

```
bool WeightedUndirectedGraph::isWeighted ( ) const [override], [virtual]
```

Implements [Graph](#).

6.54.2.4 removeEdge()

```
void WeightedUndirectedGraph::removeEdge (
    unsigned edgeId ) [override], [virtual]
```

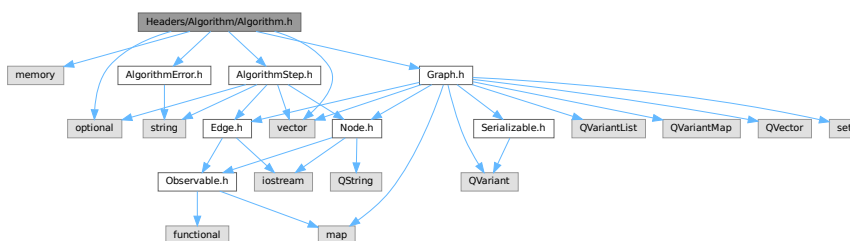
Implements [Graph](#).

The documentation for this class was generated from the following files:

- Headers/Graph/[WeightedUndirectedGraph.h](#)
- Source/Graph/[WeightedUndirectedGraph.cpp](#)

File Documentation

```
#include <memory>
#include <optional>
#include <vector>
#include "AlgorithmError.h"
#include "AlgorithmStep.h"
#include "Graph.h"
Include dependency graph for Algorithm.h:
```

[illegible]

- class Algorithm

7.2 Algorithm.h

[Go to the documentation of this file.](#)

```

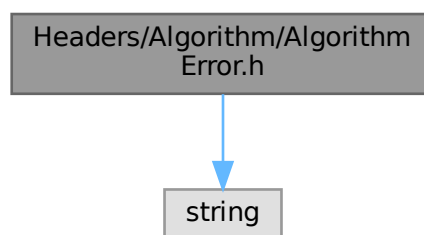
00001 #ifndef ALGORITHM_H
00002 #define ALGORITHM_H
00003
00004 #include <memory>
00005 #include <optional>
00006 #include <vector>
00007
00008 #include "AlgorithmError.h"
00009 #include "AlgorithmStep.h"
00010 #include "Graph.h"
00011
00012 class Algorithm {
00013 public:
00014     explicit Algorithm(const std::shared_ptr<Graph>);
00015     virtual ~Algorithm() = default;
00016
00017     virtual std::optional<AlgorithmError> execute(unsigned = 0, unsigned = 0) = 0;
00018
00019     virtual QString resultString() const = 0;
00020
00021     const std::vector<AlgorithmStep>& getSteps() const;
00022
00023 protected:
00024     std::shared_ptr<Graph> m_graph;
00025     std::vector<AlgorithmStep> m_steps;
00026     QString m_resultString;
00027
00028     void addStep(const AlgorithmStep&);
00029     void clearSteps();
00030 };
00031
00032 #endif // ALGORITHM_H

```

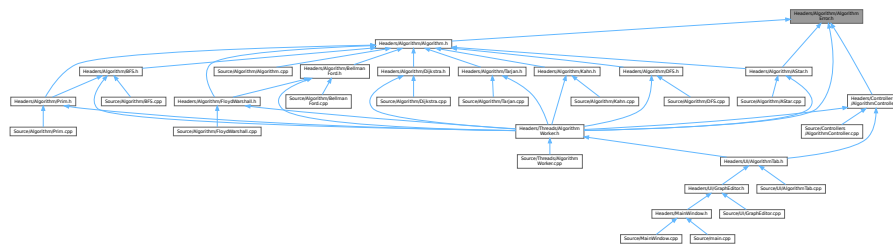
7.3 Headers/Algorithm/AlgorithmError.h File Reference

```
#include <string>
```

Include dependency graph for AlgorithmError.h:



This graph shows which files directly or indirectly include this file:



Classes

- struct [AlgorithmError](#)
Represents an error that occurred during algorithm execution.

Enumerations

- enum class [AlgorithmErrorType](#) {
[GraphNotInitialized](#) , [GraphTypeInvalid](#) , [StartNodeMissing](#) , [GoalNodeMissing](#) ,
[NegativeEdgeWeights](#) , [NoPathFound](#) , [GraphHasNegativeCycle](#) , [GraphHasCycle](#) ,
[GraphNotConnected](#) }
Represents different types of errors that can occur during algorithm execution.

7.3.1 Enumeration Type Documentation

7.3.1.1 AlgorithmErrorType

```
enum class AlgorithmErrorType [strong]
```

Represents different types of errors that can occur during algorithm execution.

These errors are used to indicate problems like missing nodes, invalid graph type, or algorithm-specific failures such as negative cycles or unreachable paths.

Enumerator

GraphNotInitialized	
GraphTypeInvalid	
StartNodeMissing	
GoalNodeMissing	
NegativeEdgeWeights	
NoPathFound	
GraphHasNegativeCycle	
GraphHasCycle	
GraphNotConnected	

7.4 AlgorithmError.h

[Go to the documentation of this file.](#)

```

00001 #ifndef ALGORITHMERROR_H
00002 #define ALGORITHMERROR_H
00003
00004 #include <string>
00005
00014 enum class AlgorithmErrorType {
00015     GraphNotInitialized,
00016     GraphTypeInvalid,
00017     StartNodeMissing,
00018     GoalNodeMissing,
00019     NegativeEdgeWeights,
00020     NoPathFound,
00021     GraphHasNegativeCycle,
00022     GraphHasCycle,
00023     GraphNotConnected
00024 };
00025
00033 struct AlgorithmError {
00034     AlgorithmErrorType m_type;
00035     std::string m_message;
00036 };
00037
00038 #endif // ALGORITHMERROR_H

```

7.5 Headers/Algorithm/AlgorithmStep.h File Reference

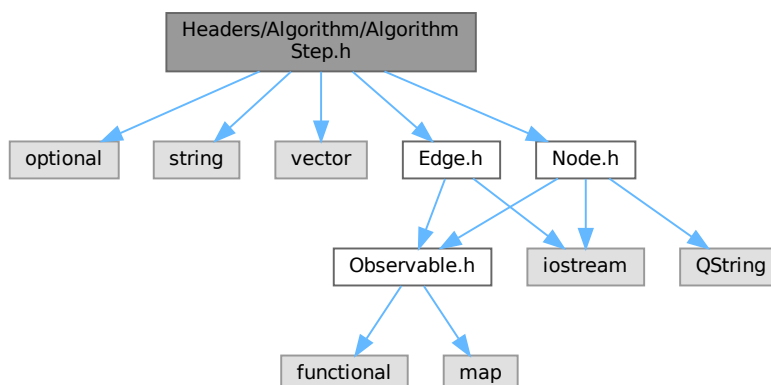
Defines structures and enums for representing and visualizing steps of graph algorithms.

```

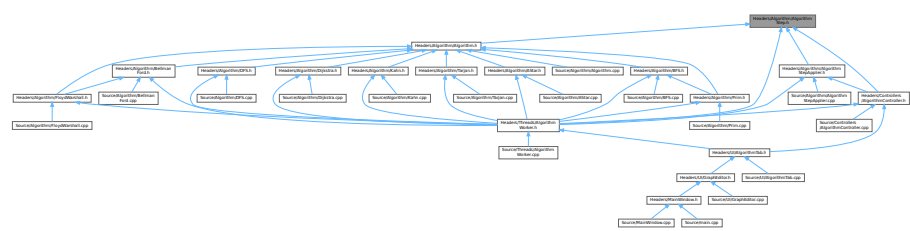
#include <optional>
#include <string>
#include <vector>
#include "Edge.h"
#include "Node.h"

```

Include dependency graph for AlgorithmStep.h:



This graph shows which files directly or indirectly include this file:



Classes

- struct [AlgorithmStep](#)
Represents a single step of an algorithm for visualization.

Enumerations

- enum class [StepType](#) {
 DefaultType , VisitNode , ProcessNode , ExamineEdge ,
 RelaxEdge , SelectEdge , UpdateDistance , AddToPath ,
 AddToTopologicalOrder , AssignComponent }
Enumerates the types of steps in an algorithm for visualization.

7.5.1 Detailed Description

Defines structures and enums for representing and visualizing steps of graph algorithms.

This file provides the [AlgorithmStep](#) struct and StepType enum, which are used to record changes to nodes and edges during algorithm execution for visualization purposes.

7.5.2 Enumeration Type Documentation

7.5.2.1 StepType

```
enum class StepType [strong]
```

Enumerates the types of steps in an algorithm for visualization.

Each step represents a meaningful action on a node or edge during algorithm execution.

Enumerator

DefaultType	
VisitNode	
ProcessNode	
ExamineEdge	
RelaxEdge	
SelectEdge	
UpdateDistance	
AddToPath	
AddToTopologicalOrder	
AssignComponent	

7.6 AlgorithmStep.h

[Go to the documentation of this file.](#)

```

00001 #ifndef ALGORITHMSTEP_H
00002 #define ALGORITHMSTEP_H
00003
00012 #include <optional>
00013 #include <string>
00014 #include <vector>
00015
00016 #include "Edge.h"
00017 #include "Node.h"
00018
00027 enum class StepType {
00028     DefaultType,
00029
00030     VisitNode,
00031     ProcessNode,
00032
00033     ExamineEdge,
00034     RelaxEdge,
00035     SelectEdge,
00036
00037     UpdateDistance,
00038     AddToPath,
00039     AddToTopologicalOrder,
00040     AssignComponent
00041 };
00042
00051 struct AlgorithmStep {
00052     StepType m_type{StepType::DefaultType};
00053
00054     std::optional<unsigned> m_node;
00055     std::optional<unsigned> m_from;
00056     std::optional<unsigned> m_to;
00057
00058     std::optional<NodeState> m_prevNodeState;
00059     std::optional<EdgeState> m_prevEdgeState;
00060 };
00061
00062 #endif // ALGORITHMSTEP_H

```

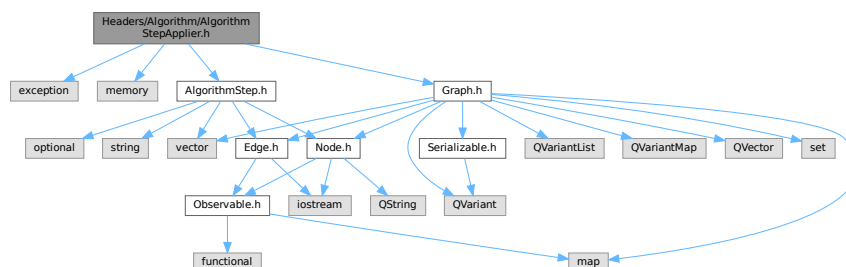
7.7 Headers/Algorithm/AlgorithmStepApplier.h File Reference

```

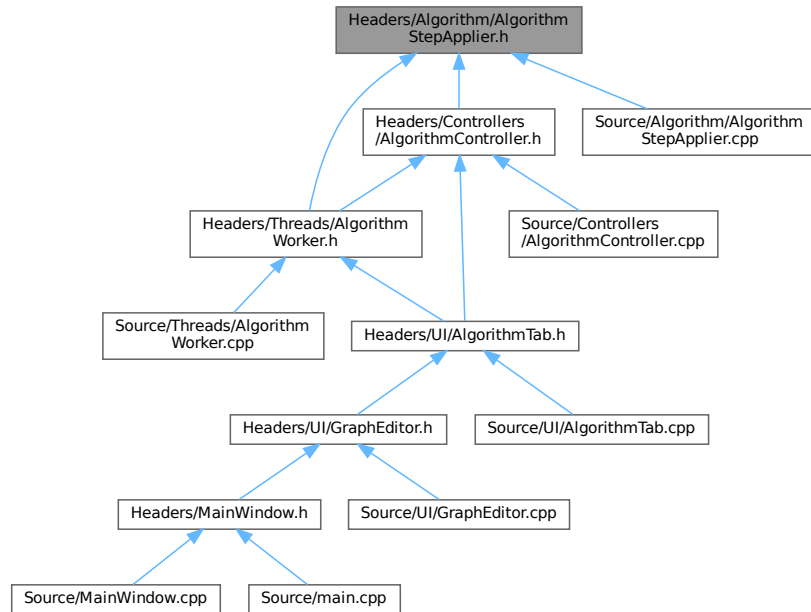
#include <exception>
#include <memory>
#include "AlgorithmStep.h"
#include "Graph.h"

```

Include dependency graph for AlgorithmStepApplier.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [AlgorithmStepApplier](#)
Applies algorithm steps to the graph.

7.8 AlgorithmStepApplier.h

[Go to the documentation of this file.](#)

```

00001 #ifndef ASA_H
00002 #define ASA_H
00003
00004 #include <exception>
00005 #include <memory>
00006
00007 #include "AlgorithmStep.h"
00008 #include "Graph.h"
00009
00010 class AlgorithmStepApplier {
00011 public:
00012     explicit AlgorithmStepApplier(const std::shared_ptr<Graph>);
00013
00014     void apply(AlgorithmStep&);
00015     void undo(const AlgorithmStep&);
00016
00017     NodeState stepToNodeState(const StepType) const;
00018     EdgeState stepToEdgeState(const StepType) const;
00019
00020     void setGraph(std::shared_ptr<Graph>);
00021     void resetGraphState();
00022
00023 private:
00024     std::shared_ptr<Graph> m_graph;
00025     void check(const AlgorithmStep&) const;
00026 };
00027 #endif // ASA_H
  
```



```

00002 #define ASTAR_H
00003
00004 #include <cmath>
00005 #include <iostream>
00006 #include <limits>
00007 #include <map>
00008 #include <memory>
00009 #include <optional>
00010 #include <queue>
00011 #include <vector>
00012
00013 #include "Algorithm.h"
00014 #include "AlgorithmError.h"
00015
00016 class AStar : public Algorithm {
00017 public:
00018     explicit AStar(const std::shared_ptr<Graph>);
00019
00020     std::optional<AlgorithmError> checkConditions(unsigned, unsigned) const;
00021     std::optional<AlgorithmError> execute(unsigned, unsigned) override;
00022
00023     const std::vector<unsigned>& getPath() const;
00024     int getTotalCost() const;
00025     QString resultString() const override;
00026
00027 private:
00028     std::vector<unsigned> m_path;
00029     int m_totalCost = 0;
00030
00031     std::optional<AlgorithmError> aStar(unsigned, unsigned);
00032     int heuristic(unsigned, unsigned) const;
00033 };
00034
00035 #endif // ASTAR_H

```

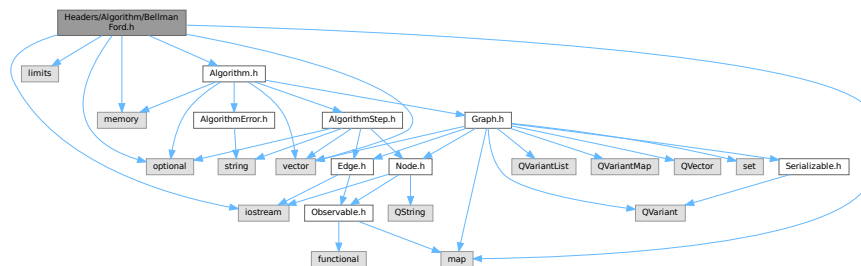
7.11 Headers/Algorithm/BellmanFord.h File Reference

```

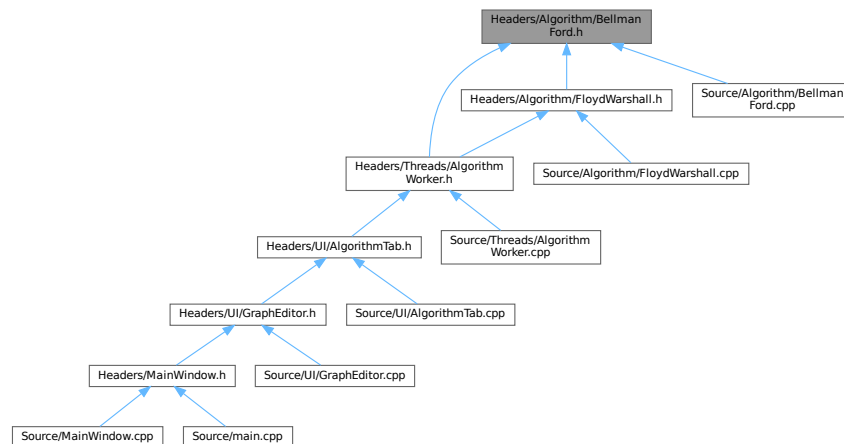
#include <iostream>
#include <limits>
#include <map>
#include <memory>
#include <optional>
#include <vector>
#include "Algorithm.h"

```

Include dependency graph for BellmanFord.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [BellmanFord](#)

7.12 BellmanFord.h

[Go to the documentation of this file.](#)

```

00001 #ifndef BF_H
00002 #define BF_H
00003
00004 #include <iostream>
00005 #include <limits>
00006 #include <map>
00007 #include <memory>
00008 #include <optional>
00009 #include <vector>
00010
00011 #include "Algorithm.h"
00012
00013 class BellmanFord : public Algorithm {
00014 public:
00015     explicit BellmanFord(const std::shared_ptr<Graph>);
00016
00017     std::optional<AlgorithmError> checkConditions(unsigned) const;
00018     std::optional<AlgorithmError> execute(unsigned, unsigned = 0) override;
00019
00020     bool hasNegativeCycle() const;
00021     QString resultString() const override;
00022
00023 private:
00024     bool m_hasNegativeCycle = false;
00025     std::map<unsigned, int> m_minDistance;
00026     QString m_resultString;
00027
00028     void bellmanFord(unsigned);
00029 };
00030
00031 #endif // BF_H
  
```

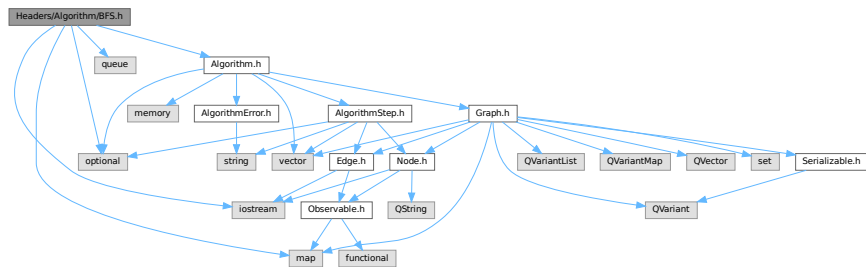
7.13 Headers/Algorithm/BFS.h File Reference

```

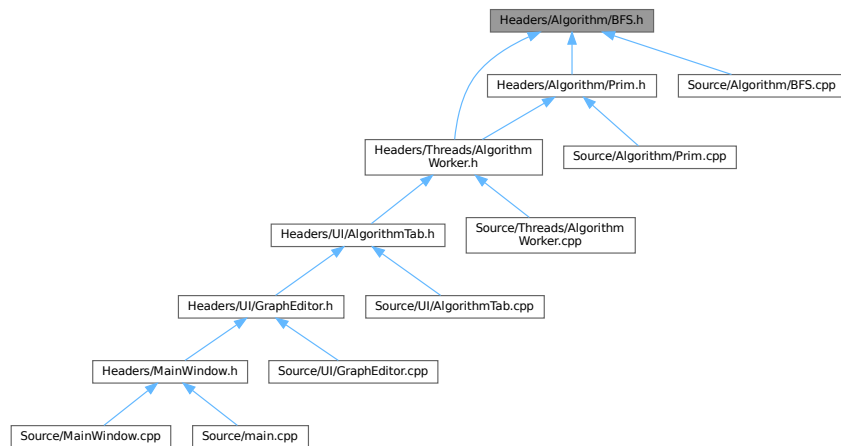
#include <iostream>
#include <map>
  
```



```
#include <optional>
#include <queue>
#include "Algorithm.h"
Include dependency graph for BFS.h:
```



This graph shows which files directly or indirectly include this file:



Classes

- class [BFS](#)

7.14 BFS.h

[Go to the documentation of this file.](#)

```
00001 #ifndef BFS_H
00002 #define BFS_H
00003
00004 #include <iostream>
00005 #include <map>
00006 #include <optional>
00007 #include <queue>
00008
00009 #include "Algorithm.h"
00010
00011 class BFS : public Algorithm {
00012 public:
00013     explicit BFS(const std::shared_ptr<Graph>);
```

```

00014
00015     std::optional<AlgorithmError> checkConditions(unsigned) const;
00016     std::optional<AlgorithmError> execute(unsigned, unsigned = 0) override;
00017
00018     const std::map<unsigned, bool>& getVisited() const;
00019     QString resultString() const override;
00020
00021 private:
00022     std::map<unsigned, bool> m_visited;
00023     std::vector<unsigned> m_order;
00024
00025     void bfs(unsigned);
00026 };
00027
00028 #endif // BFS_H

```

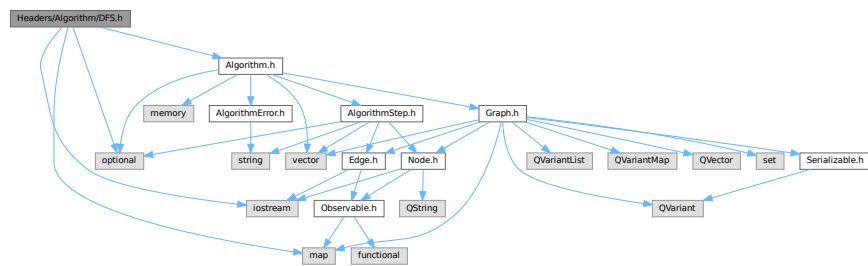
7.15 Headers/Algorithm/DFS.h File Reference

```

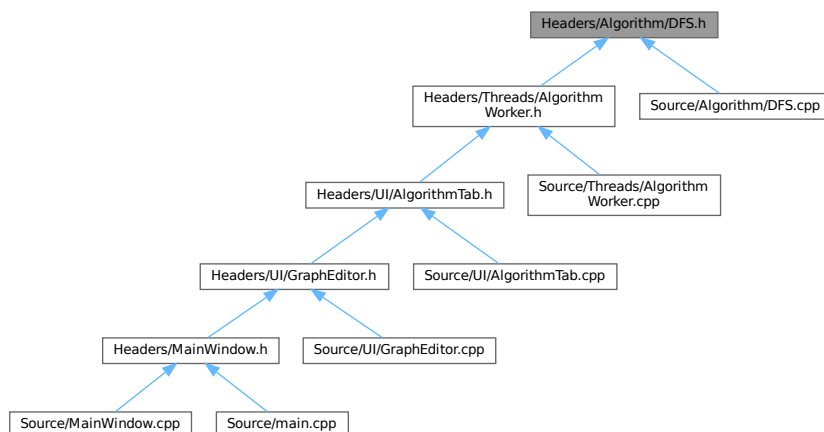
#include <iostream>
#include <map>
#include <optional>
#include "Algorithm.h"

```

Include dependency graph for DFS.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [DFS](#)

7.16 DFS.h

[Go to the documentation of this file.](#)

```

00001 #ifndef DFS_H
00002 #define DFS_H
00003
00004 #include <iostream>
00005 #include <map>
00006 #include <optional>
00007
00008 #include "Algorithm.h"
00009
00010 class DFS : public Algorithm {
00011 public:
00012     explicit DFS(const std::shared_ptr<Graph>);
00013
00014     std::optional<AlgorithmError> checkConditions(unsigned) const;
00015     std::optional<AlgorithmError> execute(unsigned, unsigned = 0) override;
00016
00017     const std::map<unsigned, bool>& getVisited() const;
00018     QString resultString() const override;
00019 private:
00020     std::map<unsigned, bool> m_visited;
00021     std::vector<unsigned> m_order;
00022
00023     void dfs(unsigned);
00024 };
00025
00026
00027 #endif // DFS_H

```

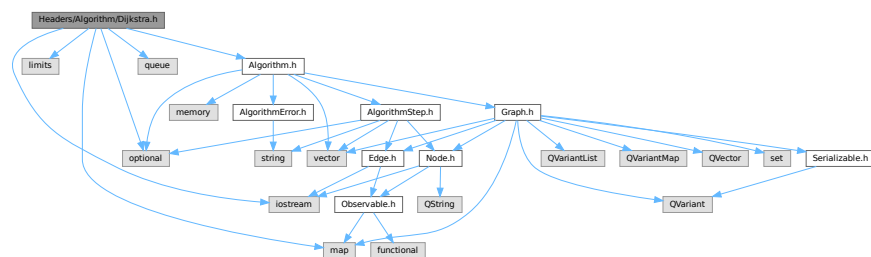
7.17 Headers/Algorithm/Dijkstra.h File Reference

```

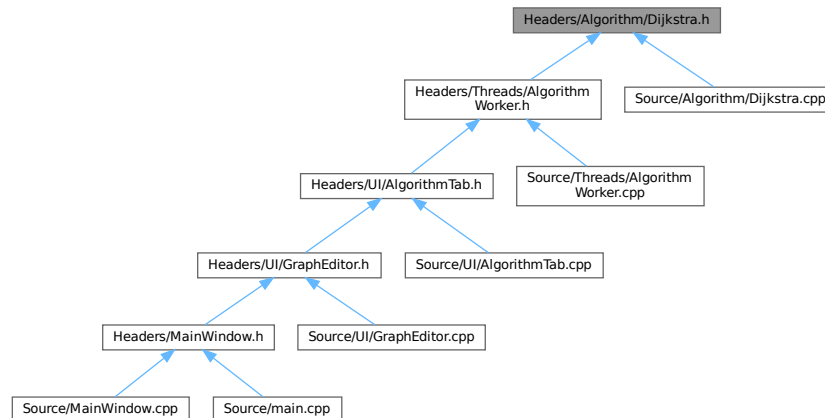
#include <iostream>
#include <limits>
#include <map>
#include <optional>
#include <queue>
#include "Algorithm.h"

```

Include dependency graph for Dijkstra.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [Dijkstra](#)

7.18 Dijkstra.h

[Go to the documentation of this file.](#)

```

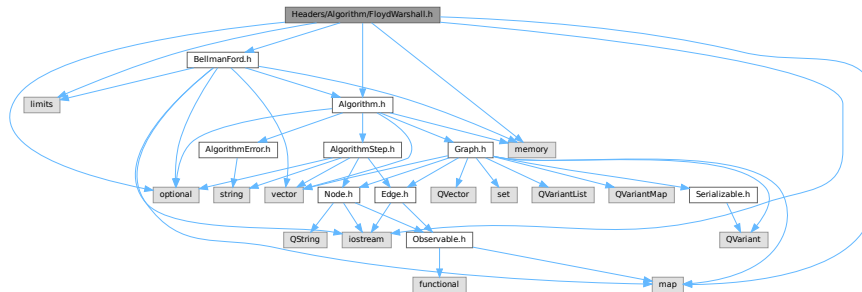
00001 #ifndef DIJKSTRA_H
00002 #define DIJKSTRA_H
00003
00004 #include <iostream>
00005 #include <limits>
00006 #include <map>
00007 #include <optional>
00008 #include <queue>
00009
00010 #include "Algorithm.h"
00011
00012 class Dijkstra : public Algorithm {
00013 public:
00014     explicit Dijkstra(const std::shared_ptr<Graph>);
00015
00016     std::optional<AlgorithmError> checkConditions(unsigned) const;
00017     std::optional<AlgorithmError> execute(unsigned, unsigned = 0) override;
00018
00019     QString resultString() const override;
00020
00021 private:
00022     QString m_resultString;
00023
00024     void dijkstra(unsigned);
00025 };
00026
00027 #endif // DIJKSTRA_H
  
```

7.19 Headers/Algorithm/FloydWarshall.h File Reference

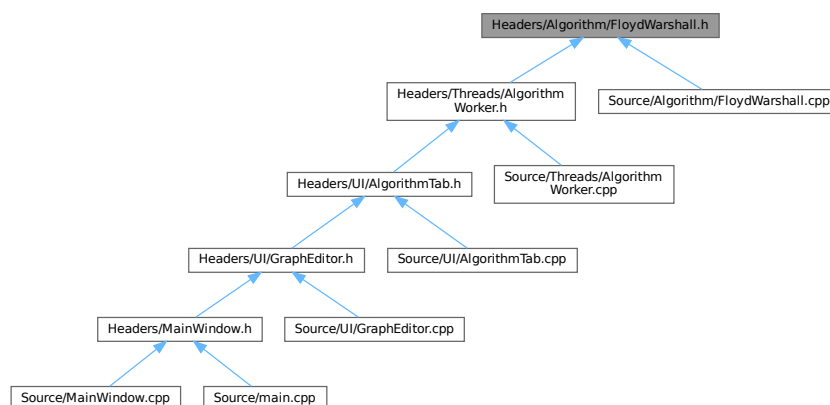
```

#include <iostream>
#include <limits>
#include <map>
  
```

```
#include <memory>
#include <optional>
#include "Algorithm.h"
#include "BellmanFord.h"
Include dependency graph for FloydWarshall.h:
```



This graph shows which files directly or indirectly include this file:



Classes

- class `FloydWarshall`

7.20 FloydWarshall.h

[Go to the documentation of this file.](#)

```
00001 #ifndef FW_H
00002 #define FW_H
00003
00004 #include <iostream>
00005 #include <limits>
00006 #include <map>
00007 #include <memory>
00008 #include <optional>
00009
00010 #include "Algorithm.h"
00011 #include "BellmanFord.h"
00012
```

```

00013 class FloydWarshall : public Algorithm {
00014 public:
00015     explicit FloydWarshall(const std::shared_ptr<Graph>);
00016
00017     std::optional<AlgorithmError> checkConditions() const;
00018     std::optional<AlgorithmError> execute(unsigned = 0, unsigned = 0) override;
00019
00020     QString resultString() const override;
00021
00022 private:
00023     std::map<unsigned, std::map<unsigned, int> m_distances;
00024
00025     std::optional<AlgorithmError> floydWarshall();
00026 };
00027
00028 #endif // FW_H

```

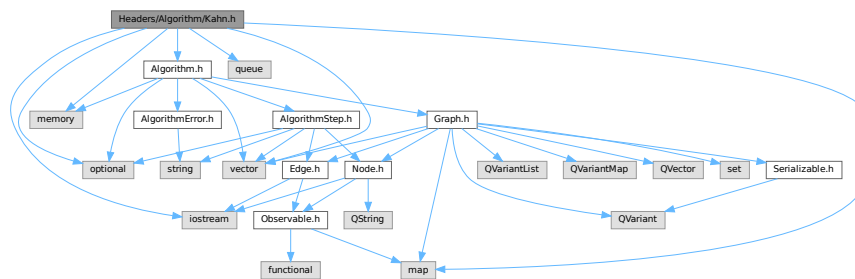
7.21 Headers/Algorithm/Kahn.h File Reference

```

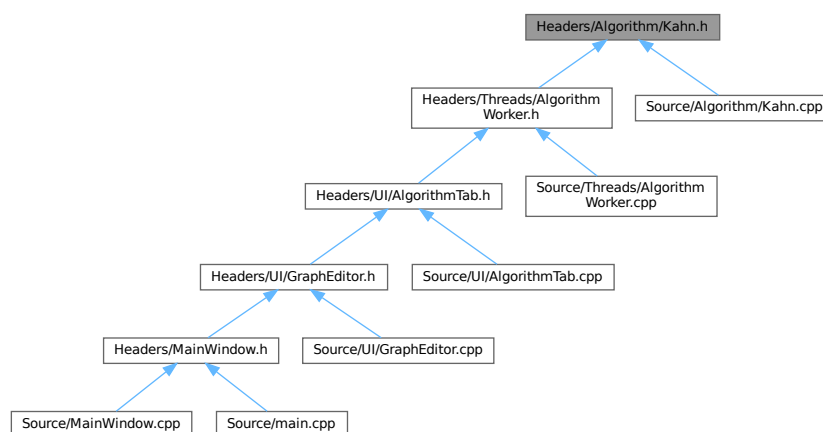
#include <iostream>
#include <map>
#include <memory>
#include <optional>
#include <queue>
#include <vector>
#include "Algorithm.h"

```

Include dependency graph for Kahn.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [Kahn](#)

7.22 Kahn.h

[Go to the documentation of this file.](#)

```

00001 #ifndef KAHN_H
00002 #define KAHN_H
00003
00004 #include <iostream>
00005 #include <map>
00006 #include <memory>
00007 #include <optional>
00008 #include <queue>
00009 #include <vector>
00010
00011 #include "Algorithm.h"
00012
00013 class Kahn : public Algorithm {
00014 public:
00015     explicit Kahn(const std::shared_ptr<Graph>);
00016
00017     std::optional<AlgorithmError> checkConditions() const;
00018     std::optional<AlgorithmError> execute(unsigned = 0, unsigned = 0) override;
00019
00020     const std::vector<unsigned>& getSorted() const;
00021     QString resultString() const override;
00022 private:
00023     std::vector<unsigned> m_sorted;
00024
00025     std::optional<AlgorithmError> kahn();
00026 };
00027
00028 #endif // KAHN_H

```

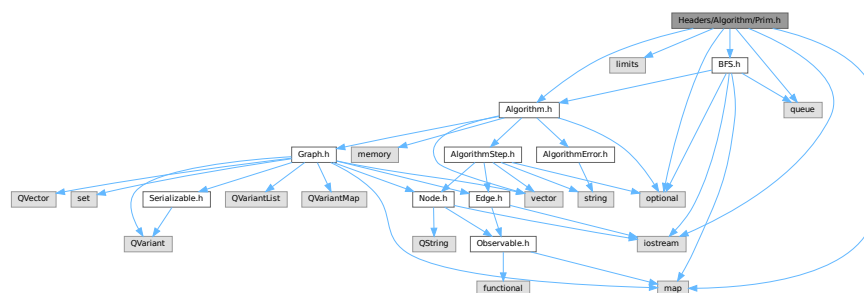
7.23 Headers/Algorithm/Prim.h File Reference

```

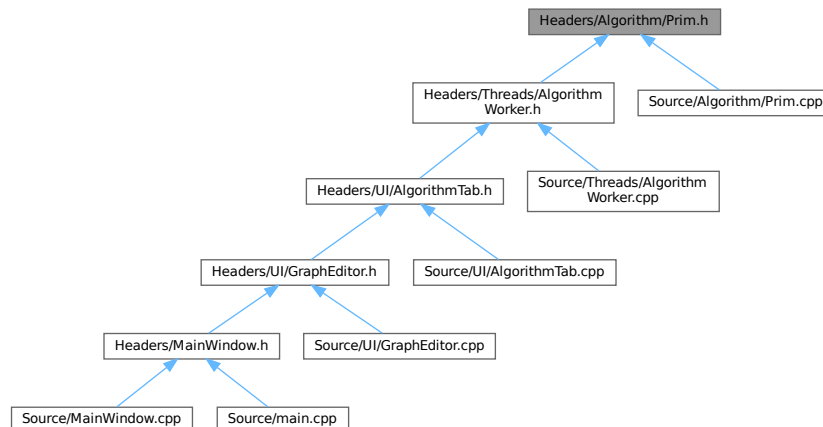
#include <iostream>
#include <limits>
#include <map>
#include <optional>
#include <queue>
#include "Algorithm.h"
#include "BFS.h"

```

Include dependency graph for Prim.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [Prim](#)

7.24 Prim.h

[Go to the documentation of this file.](#)

```

00001 #ifndef PRIM_H
00002 #define PRIM_H
00003
00004 #include <iostream>
00005 #include <limits>
00006 #include <map>
00007 #include <optional>
00008 #include <queue>
00009
00010 #include "Algorithm.h"
00011 #include "BFS.h"
00012
00013 class Prim : public Algorithm {
00014 public:
00015     explicit Prim(const std::shared_ptr<Graph>);
00016
00017     std::optional<AlgorithmError> checkConditions() const;
00018     std::optional<AlgorithmError> execute(unsigned = 0, unsigned = 0) override;
00019
00020     QString resultString() const override;
00021
00022 private:
00023     int m_totalWeight = 0;
00024
00025     void prim();
00026 };
00027
00028 #endif // PRIM_H

```

7.25 Headers/Algorithm/Tarjan.h File Reference

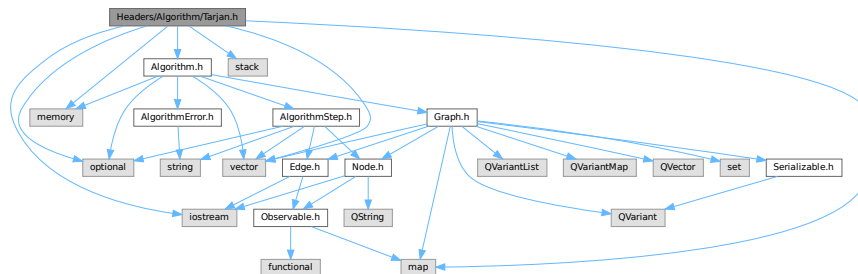
```

#include <iostream>
#include <map>

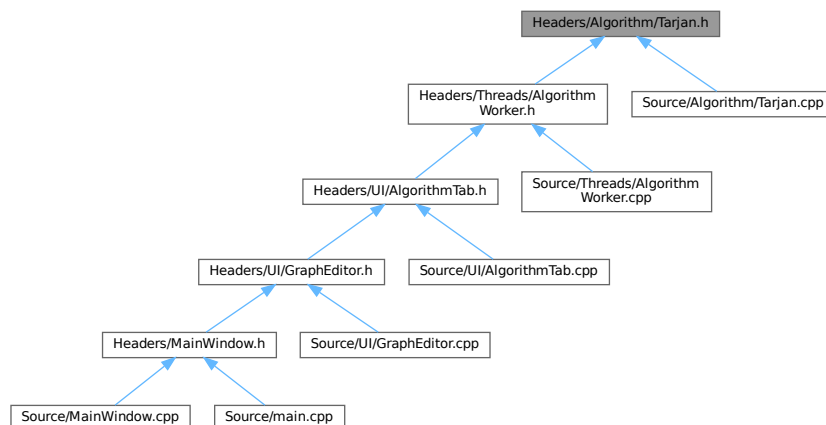
```



```
#include <memory>
#include <optional>
#include <stack>
#include <vector>
#include "Algorithm.h"
Include dependency graph for Tarjan.h:
```



This graph shows which files directly or indirectly include this file:



Classes

- class [Tarjan](#)

7.26 Tarjan.h

[Go to the documentation of this file.](#)

```
00001 #ifndef TARJAN_H
00002 #define TARJAN_H
00003
00004 #include <iostream>
00005 #include <map>
00006 #include <memory>
00007 #include <optional>
00008 #include <stack>
00009 #include <vector>
00010
```

```

00011 #include "Algorithm.h"
00012
00013 class Tarjan : public Algorithm {
00014 public:
00015     explicit Tarjan(const std::shared_ptr<Graph>);
00016
00017     std::optional<AlgorithmError> checkConditions() const;
00018     std::optional<AlgorithmError> execute(unsigned = 0, unsigned = 0) override;
00019
00020     QString resultString() const override;
00021
00022 private:
00023     int m_arrivalTime = 0;
00024     std::map<unsigned, int> m_incomingNumbering;
00025     std::map<unsigned, int> m_lowLink;
00026     std::stack<unsigned> m_tourOrder; // traversal order
00027     std::map<unsigned, bool> m_onStack;
00028     std::map<unsigned, int> m_components; // node id -> component
00029     int m_numComponents = 0; // num of SCC
00030
00031     void tarjan(unsigned);
00032     void init();
00033 };
00034
00035 #endif // TARJAN_H

```

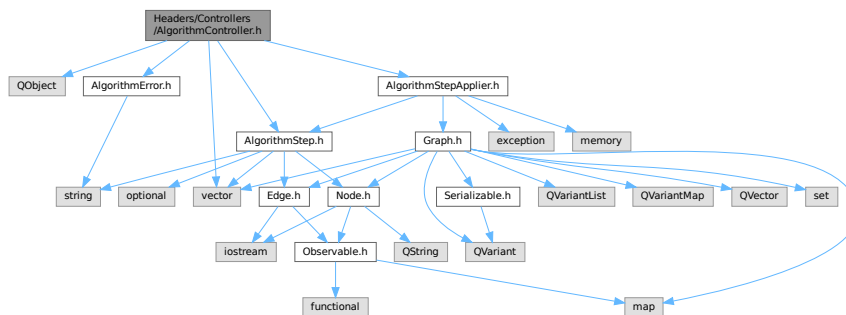
7.27 Headers/Controllers/AlgorithmController.h File Reference

```

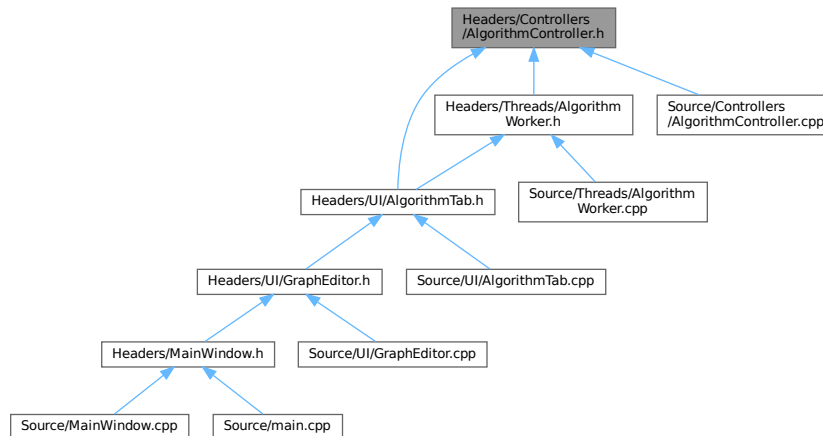
#include <QObject>
#include <vector>
#include "AlgorithmError.h"
#include "AlgorithmStep.h"
#include "AlgorithmStepApplier.h"

```

Include dependency graph for AlgorithmController.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [AlgorithmController](#)

Controller that connects algorithm execution with the UI.

7.28 AlgorithmController.h

[Go to the documentation of this file.](#)

```

00001 #ifndef ALGORITHM_CONTROLLER_H
00002 #define ALGORITHM_CONTROLLER_H
00003
00004 #include <QObject>
00005
00006 #include <vector>
00007
00008 #include "AlgorithmError.h"
00009 #include "AlgorithmStep.h"
00010 #include "AlgorithmStepApplier.h"
00011
00021 class AlgorithmController : public QObject {
00022     Q_OBJECT
00023 public:
00024     explicit AlgorithmController(AlgorithmStepApplier&, QObject* = nullptr);
00025     ~AlgorithmController() override = default;
00026
00027     void clear();
00028     QString resultString() const;
00029
00030 public slots:
00031     void loadSteps(const std::vector<AlgorithmStep>&);
00032
00033     void nextStep();
00034     void prevStep();
00035     void reset();
00036     bool isFinished() const;
00037     void setResultString(const QString&);
00038
00039     void onAlgorithmError(const AlgorithmError&);
00040
00041 signals:
00042     void requestErrorDialog(const AlgorithmError&);
00043
00044 private:
00045     AlgorithmStepApplier& m_applier;
00046     std::vector<AlgorithmStep> m_steps;
00047     int m_currentIndex = -1;

```

```

00048
00049     std::vector<AlgorithmStep> m_undoStack;
00050     std::vector<AlgorithmStep> m_redoStack;
00051
00052     QString m_resultString;
00053
00054     void undo();
00055     void redo();
00056 };
00057
00058 #endif // ALGORITHM_CONTROLLER_H

```

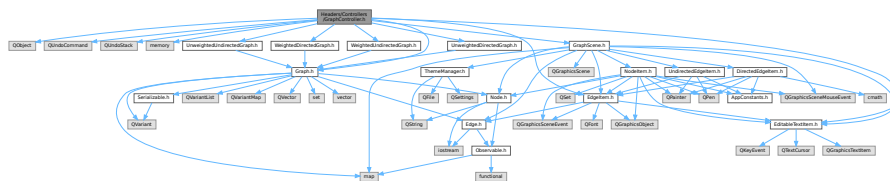
7.29 Headers/Controllers/GraphController.h File Reference

```

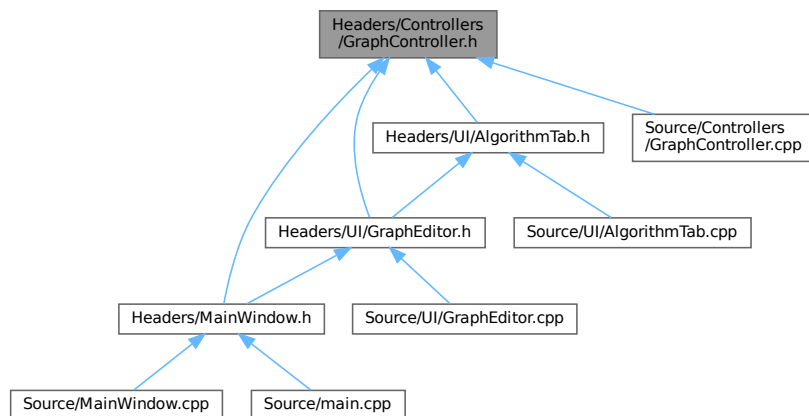
#include <QObject>
#include <QUndoCommand>
#include <QUndoStack>
#include <memory>
#include "EdgeItem.h"
#include "EditableTextItem.h"
#include "Graph.h"
#include "GraphScene.h"
#include "UnweightedDirectedGraph.h"
#include "UnweightedUndirectedGraph.h"
#include "WeightedDirectedGraph.h"
#include "WeightedUndirectedGraph.h"

```

Include dependency graph for GraphController.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [GraphController](#)

7.30 GraphController.h

[Go to the documentation of this file.](#)

```

00001 #ifndef GRAPH_CONTROLLER_H
00002 #define GRAPH_CONTROLLER_H
00003
00004 #include <QObject>
00005 #include <QUndoCommand>
00006 #include <QUndoStack>
00007
00008 #include <memory>
00009
00010 #include "EdgeItem.h"
00011 #include "EditableTextItem.h"
00012 #include "Graph.h"
00013 #include "GraphScene.h"
00014 #include "UnweightedDirectedGraph.h"
00015 #include "UnweightedUndirectedGraph.h"
00016 #include "WeightedDirectedGraph.h"
00017 #include "WeightedUndirectedGraph.h"
00018
00019 class GraphController : public QObject {
00020     Q_OBJECT
00021
00022 public:
00023     explicit GraphController(QObject* = nullptr);
00024
00025     std::shared_ptr<Graph> graph() const;
00026     void setGraph(const std::shared_ptr<Graph>&);
00027
00028     void setAddSceneState() const;
00029     void setRemoveSceneState() const;
00030
00031     QUndoStack* undoStack() const;
00032
00033     GraphScene* scene() const;
00034
00035     void updateNodePosition(NodeItem*, const QPointF&);
00036
00037 signals:
00038     void sceneModified();
00039
00040 public slots:
00041     void createGraph(bool, bool);
00042     void addNode(const QPointF&);
00043     void addNodeAndEdge(const QPointF&, NodeItem*);
00044     void addEdge(NodeItem*, NodeItem*);
00045     void removeNode(NodeItem*);
00046     void removeEdge(EdgeItem*);
00047     void editNodeName(const NodeItem*, const QString&);
00048     void editEdgeWeight(const EdgeItem*, const QString&);
00049     void moveNode(const NodeItem*, const QPointF&, const QPointF&);
00050
00051     // clears nodes and edges in both model and view
00052     void clear();
00053     // clears nodes and edges only in view
00054     void clearScene() const;
00055     // builds scene from already loaded graph
00056     void buildScene() const;
00057
00058     // handle methods for GraphScene methods
00059     void disableScene() const;
00060     void enableScene() const;
00061     void resetScene() const;
00062
00063     void addNodeNoHistory(const QPointF&, unsigned&);
00064     void addNodeWithIdNoHistory(unsigned, const QPointF&);
00065     void removeNodeNoHistory(unsigned);
00066     void addEdgeNoHistory(unsigned, unsigned, int);
00067     void removeEdgeNoHistory(unsigned, unsigned);
00068     void setEdgeWeightNoHistory(unsigned, unsigned, int, const QString&);
00069     void setNodeNameNoHistory(unsigned, const QString&);
00070     void removeEdgeNoHistoryById(unsigned);
00071     void addEdgeNoHistoryById(unsigned);
00072     void restoreEdgeNoHistory(unsigned, unsigned, unsigned, int);
00073     void setEdgeWeightNoHistoryById(unsigned, int, const QString&);

```

```

00074     void moveNodeNoHistory(unsigned, const QPointF&);
00075     void clearNoHistory();
00076
00077 private:
00078     std::shared_ptr<Graph> m_graph;
00079     std::unique_ptr<GraphScene> m_scene = std::make_unique<GraphScene>();
00080     QUndoStack* m_undoStack = nullptr;
00081
00082     void connectScene() const;
00083 };
00084
00085 #endif // GRAPH_CONTROLLER_H

```

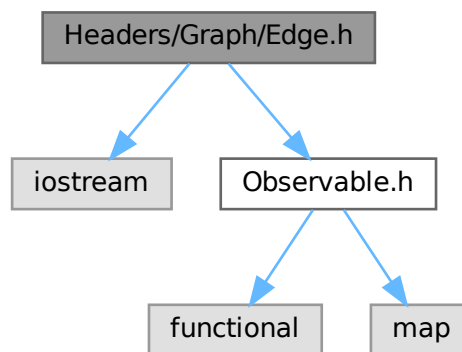
7.31 Headers/Graph/Edge.h File Reference

```

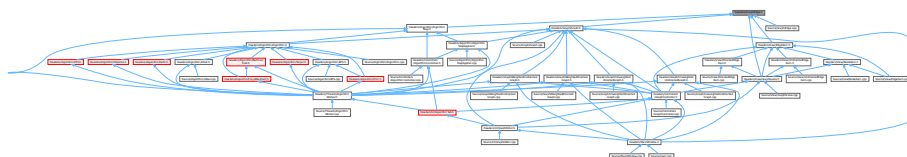
#include <iostream>
#include "Observable.h"

```

Include dependency graph for Edge.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [Edge](#)

Enumerations

- enum class [EdgeState](#) { [Default](#) , [Examined](#) , [Relaxed](#) , [Selected](#) }

7.31.1 Enumeration Type Documentation

7.31.1.1 EdgeState

```
enum class EdgeState [strong]
```

Enumerator

Default	
Examined	
Relaxed	
Selected	

7.32 Edge.h

[Go to the documentation of this file.](#)

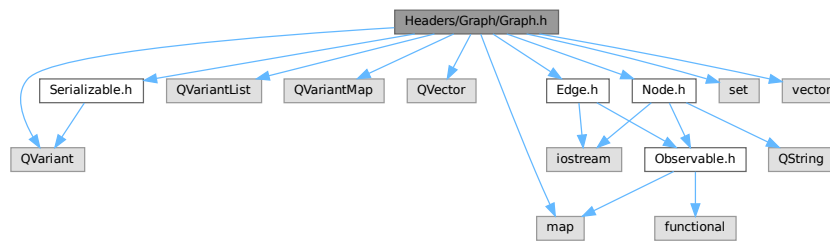
```
00001 #ifndef EDGE_H
00002 #define EDGE_H
00003
00004 #include <iostream>
00005
00006 #include "Observable.h"
00007
00008 enum class EdgeState { Default, Examined, Relaxed, Selected };
00009
00010 class Edge : public Observable<Edge> {
00011 public:
00012     Edge(unsigned, unsigned, unsigned, int = 1);
00013     ~Edge() = default;
00014
00015     unsigned getId() const;
00016     int getWeight() const;
00017     EdgeState getState() const;
00018     unsigned startNode() const;
00019     unsigned endNode() const;
00020
00021     void setWeight(int);
00022     void setState(EdgeState);
00023
00024 private:
00025     int m_weight;
00026     unsigned m_id;
00027     unsigned m_idStartNode;
00028     unsigned m_idEndNode;
00029     EdgeState m_state = EdgeState::Default;
00030 };
00031
00032 #endif // EDGE_H
```

7.33 Headers/Graph/Graph.h File Reference

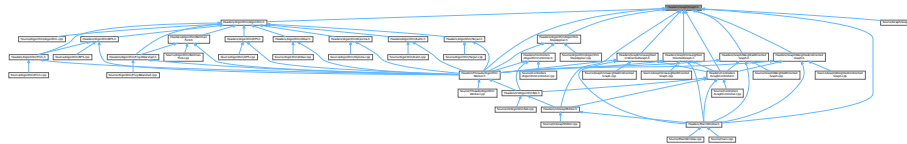
```
#include <QVariant>
#include <QVariantList>
#include <QVariantMap>
#include <QVector>
#include <map>
#include <set>
#include <vector>
#include "Edge.h"
#include "Node.h"
```

```
#include "Serializable.h"
```

Include dependency graph for Graph.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [Graph](#)

7.34 Graph.h

[Go to the documentation of this file.](#)

```

00001 #ifndef GRAPH_H
00002 #define GRAPH_H
00003
00004 #include <QVariant>
00005 #include <QVariantList>
00006 #include <QVariantMap>
00007 #include <QVector>
00008
00009 #include <map>
00010 #include <set>
00011 #include <vector>
00012
00013 #include "Edge.h"
00014 #include "Node.h"
00015 #include "Serializable.h"
00016
00017 class Graph : public Serializable {
00018 public:
00019     Graph() = default;
00020     ~Graph() override = default;
00021
00022     Node* addNode(double, double);
00023     void addNode(unsigned, double = 0.0, double = 0.0);
00024     void removeNode(unsigned);
00025
00026     virtual void addEdge(unsigned, unsigned, int = 1) = 0;
00027     virtual void removeEdge(unsigned) = 0;
00028
00029     virtual bool isDirected() const = 0;
00030     virtual bool isWeighted() const = 0;
00031
00032     std::map<unsigned, std::map<unsigned, unsigned> > getAdjacencyList() const;
00033     std::map<unsigned, Node> getNodes() const;

```



```

00034     std::map<unsigned, Edge>                               getEdges() const;
00035
00036     std::vector<Node*>   getNodesMutable();
00037     std::vector<Edge*>   getEdgesMutable();
00038
00039     Node*               getNode(unsigned);
00040     const Node*         getNode(unsigned) const;
00041
00042     Edge*               getEdge(unsigned);
00043     const Edge*         getEdge(unsigned) const;
00044     Edge*               getEdge(unsigned, unsigned);
00045     const Edge*         getEdge(unsigned, unsigned) const;
00046
00047     void addEdgeSerialized(unsigned, unsigned, unsigned, int);
00048
00049     QVariant toVariant() const override;
00050     void     fromVariant(const QVariant&) override;
00051
00052     void clear();
00053
00054 protected:
00055     unsigned          m_nodeId      = 0;
00056     unsigned          m_edgeId      = 0;
00057     unsigned          m_numOfNodes   = 0;
00058     unsigned          m_numOfEdges  = 0;
00059     std::map<unsigned, std::map<unsigned, unsigned> > m_adjacencyList;
00060     std::map<unsigned, Node> m_nodes;
00061     std::map<unsigned, Edge> m_edges;
00062 };
00063
00064 #endif // GRAPH_H

```

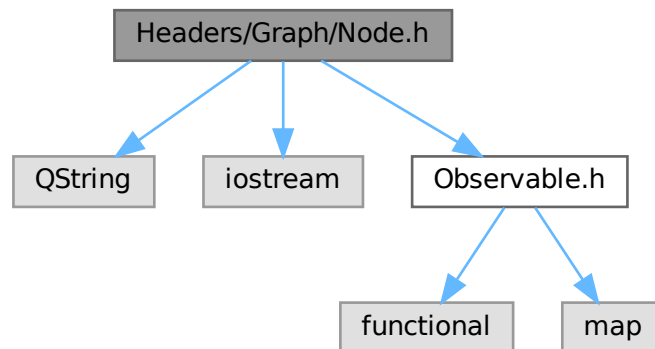
7.35 Headers/Graph/Node.h File Reference

```

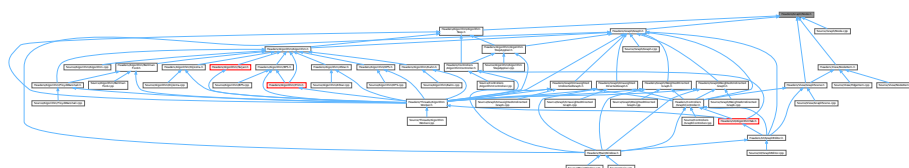
#include <QString>
#include <iostream>
#include "Observable.h"

```

Include dependency graph for Node.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [Node](#)

Enumerations

- enum class [NodeState](#) {
[Default](#) , [Active](#) , [Visited](#) , [UpdatedDistance](#) ,
[InPath](#) , [TopologicalOrder](#) , [AssignedComponent](#) }

7.35.1 Enumeration Type Documentation

7.35.1.1 NodeState

```
enum class NodeState [strong]
```

Enumerator

Default	
Active	
Visited	
UpdatedDistance	
InPath	
TopologicalOrder	
AssignedComponent	

7.36 Node.h

[Go to the documentation of this file.](#)

```
00001 #ifndef NODE_H
00002 #define NODE_H
00003
00004 #include <QString>
00005 #include <iostream>
00006
00007 #include "Observable.h"
00008
00009 enum class NodeState {
00010     Default,
00011     Active,
00012     Visited,
00013     UpdatedDistance,
00014     InPath,
00015     TopologicalOrder,
00016     AssignedComponent
00017 };
00018
00019 class Node : public Observable<Node> {
00020 public:
00021     explicit Node(unsigned, double = 0.0, double = 0.0);
00022     ~Node() = default;
00023
00024     unsigned getId() const;
00025     const std::pair<double, double>& getPosition() const;
00026     NodeState getState() const;
00027     QString getName() const;
00028     int getComponentColor() const;
00029
00030     void setPosition(double, double);
```

```

00031     void setState(NodeState);
00032     void setName(const QString&);
00033     void setComponentColor(int);
00034
00035 private:
00036     unsigned                m_id;
00037     QString                 m_name;
00038     std::pair<double, double> m_position; // node coordinates
00039     NodeState               m_state      = NodeState::Default;
00040     int                     m_componentColor = -1;
00041 };
00042
00043 #endif // NODE_H

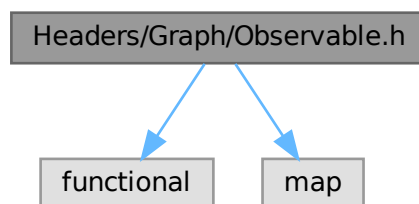
```

7.37 Headers/Graph/Observable.h File Reference

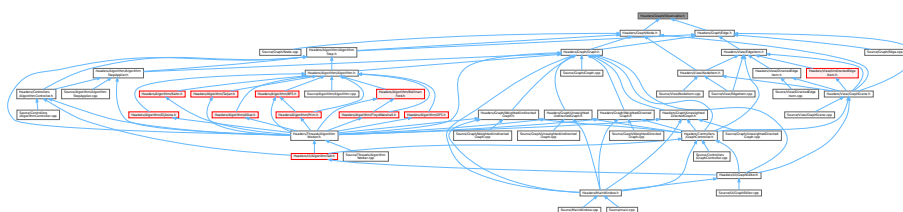
```
#include <functional>
```

```
#include <map>
```

Include dependency graph for Observable.h:



This graph shows which files directly or indirectly include this file:



Classes

- class `Observable< T >`

Base class implementing the Observer pattern for any subject type.

7.38 Observable.h

[Go to the documentation of this file.](#)

```

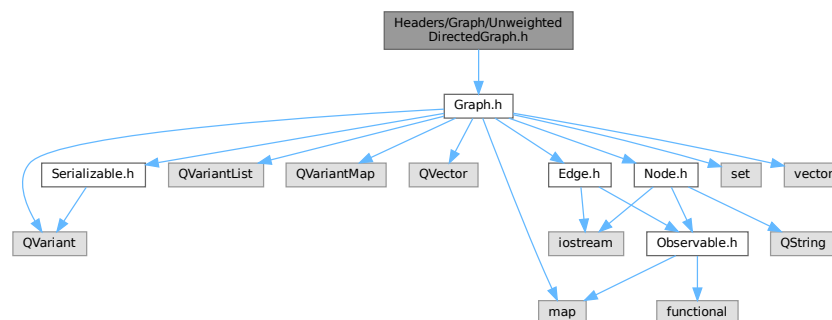
00001 #ifndef OBSERVABLE_H
00002 #define OBSERVABLE_H
00003
00004 #include <functional>
00005 #include <map>
00006
00019 template <typename T> class Observable {
00020 public:
00022     using Observer = std::function<void(T&);>;
00023
00029     unsigned addObserver(Observer observer) {
00030         unsigned id = ++m_id;
00031         m_observers[id] = std::move(observer);
00032         return id;
00033     }
00034
00039     void removeObserver(unsigned id) {
00040         m_observers.erase(id);
00041     }
00042
00043 protected:
00051     void notifyObservers(T& subject) {
00052         auto observersCopy = m_observers;
00053         for(auto& [_ , observer]: observersCopy) {
00054             if(observer) {
00055                 observer(subject);
00056             }
00057         }
00058     }
00059
00060 private:
00061     unsigned m_id = 0;
00062     std::map<unsigned, Observer> m_observers;
00063 };
00064
00065 #endif // OBSERVABLE_H

```

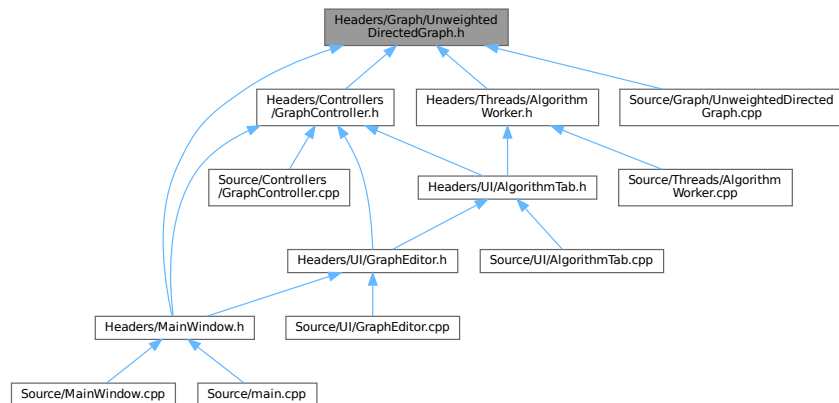
7.39 Headers/Graph/UnweightedDirectedGraph.h File Reference

```
#include "Graph.h"
```

Include dependency graph for UnweightedDirectedGraph.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [UnweightedDirectedGraph](#)

7.40 UnweightedDirectedGraph.h

[Go to the documentation of this file.](#)

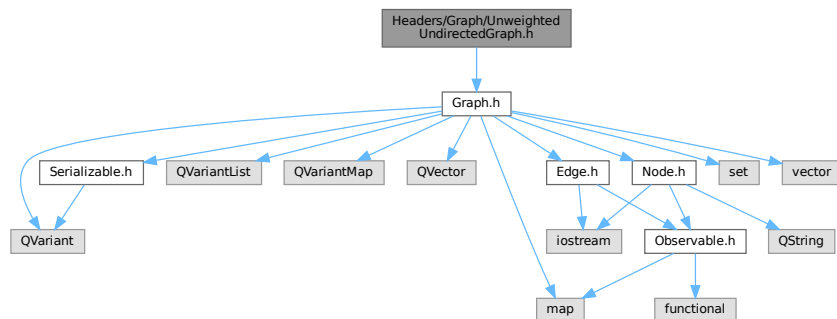
```

00001 #ifndef UDG_H
00002 #define UDG_H
00003
00004 #include "Graph.h"
00005
00006 class UnweightedDirectedGraph : public Graph {
00007 public:
00008     UnweightedDirectedGraph() = default;
00009     ~UnweightedDirectedGraph() override = default;
00010
00011     void addEdge(unsigned, unsigned, int = 1) override;
00012     void removeEdge(unsigned) override;
00013
00014     bool isDirected() const override;
00015     bool isWeighted() const override;
00016 };
00017
00018 #endif // UDG_H
  
```

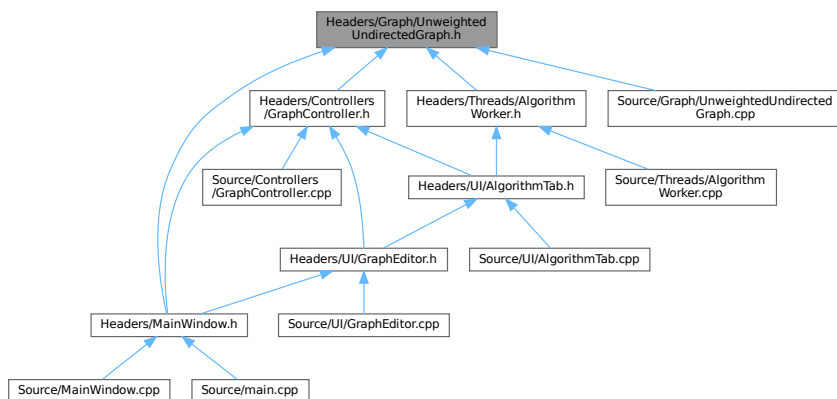
7.41 Headers/Graph/UnweightedUndirectedGraph.h File Reference

```
#include "Graph.h"
```

Include dependency graph for UnweightedUndirectedGraph.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [UnweightedUndirectedGraph](#)

7.42 UnweightedUndirectedGraph.h

[Go to the documentation of this file.](#)

```

00001 #ifndef UUG_H
00002 #define UUG_H
00003
00004 #include "Graph.h"
00005
00006 class UnweightedUndirectedGraph : public Graph {
00007 public:
00008     UnweightedUndirectedGraph() = default;
00009     ~UnweightedUndirectedGraph() override = default;
00010

```

```

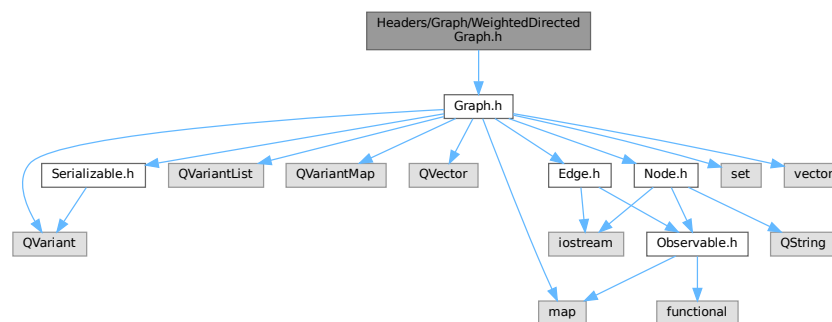
00011     void addEdge(unsigned, unsigned, int = 1) override;
00012     void removeEdge(unsigned) override;
00013
00014     bool isDirected() const override;
00015     bool isWeighted() const override;
00016 };
00017
00018 #endif // UUG_H

```

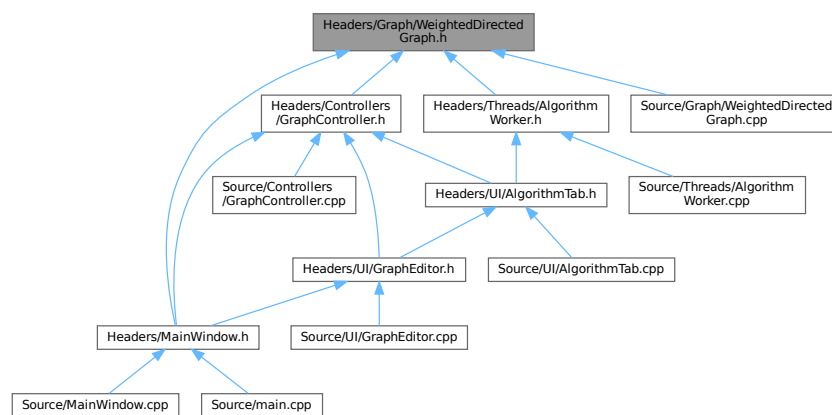
7.43 Headers/Graph/WeightedDirectedGraph.h File Reference

```
#include "Graph.h"
```

Include dependency graph for WeightedDirectedGraph.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [WeightedDirectedGraph](#)

7.44 WeightedDirectedGraph.h

[Go to the documentation of this file.](#)

```

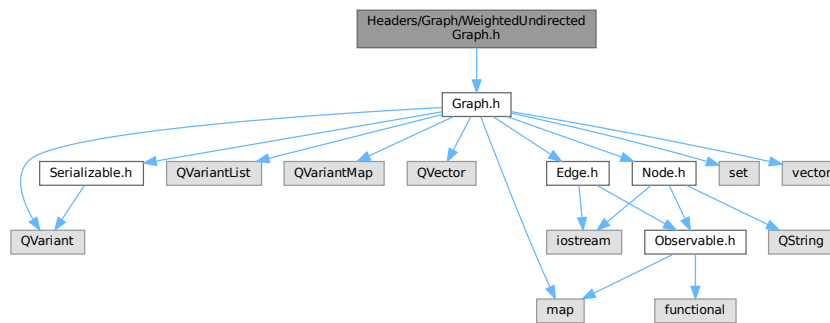
00001 #ifndef WDG_H
00002 #define WDG_H
00003
00004 #include "Graph.h"
00005
00006 class WeightedDirectedGraph : public Graph {
00007 public:
00008     WeightedDirectedGraph() = default;
00009     ~WeightedDirectedGraph() override = default;
00010
00011     void addEdge(unsigned, unsigned, int) override;
00012     void removeEdge(unsigned) override;
00013
00014     bool isDirected() const override;
00015     bool isWeighted() const override;
00016 };
00017
00018 #endif // WDG_H

```

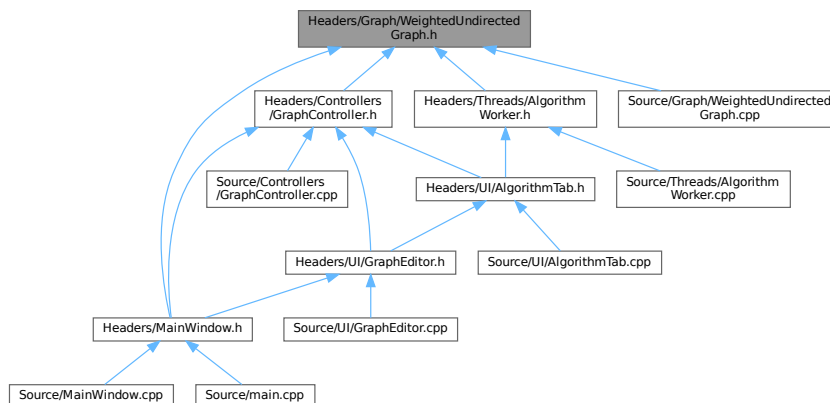
7.45 Headers/Graph/WeightedUndirectedGraph.h File Reference

```
#include "Graph.h"
```

Include dependency graph for WeightedUndirectedGraph.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [WeightedUndirectedGraph](#)

7.46 WeightedUndirectedGraph.h

[Go to the documentation of this file.](#)

```

00001 #ifndef WUG_H
00002 #define WUG_H
00003
00004 #include "Graph.h"
00005
00006 class WeightedUndirectedGraph : public Graph {
00007 public:
00008     WeightedUndirectedGraph() = default;
00009     ~WeightedUndirectedGraph() override = default;
00010
00011     void addEdge(unsigned, unsigned, int) override;
00012     void removeEdge(unsigned) override;
00013
00014     bool isDirected() const override;
00015     bool isWeighted() const override;
00016 };
00017
00018 #endif // WUG_H

```

7.47 Headers/MainWindow.h File Reference

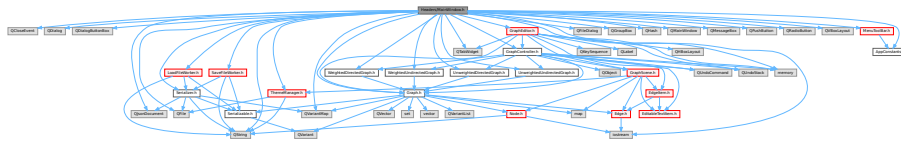
```

#include <QCloseEvent>
#include <QDialog>
#include <QDialogButtonBox>
#include <QFile>
#include <QFileDialog>
#include <QGroupBox>
#include <QHash>
#include <QJsonDocument>
#include <QMainWindow>
#include <QMessageBox>
#include <QPushButton>
#include <QRadioButton>
#include <QString>
#include <QTabWidget>
#include <QVBoxLayout>
#include <QVariantMap>
#include <iostream>
#include <memory>
#include "AppConstants.h"
#include "Graph.h"
#include "GraphController.h"
#include "GraphEditor.h"
#include "LoadFileWorker.h"
#include "MenuToolBar.h"
#include "SaveFileWorker.h"
#include "Serializer.h"
#include "ThemeManager.h"
#include "UnweightedDirectedGraph.h"
#include "UnweightedUndirectedGraph.h"
#include "WeightedDirectedGraph.h"

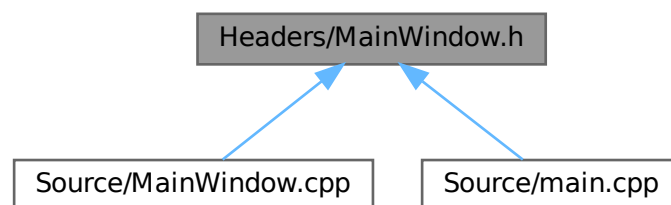
```

```
#include "WeightedUndirectedGraph.h"
```

Include dependency graph for MainWindow.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [MainWindow](#)
- struct [MainWindow::TabInfo](#)

Namespaces

- namespace [Ui](#)

7.48 MainWindow.h

[Go to the documentation of this file.](#)

```
00001 #ifndef MAINWINDOW_H
00002 #define MAINWINDOW_H
00003
00004 #include <QCloseEvent>
00005 #include <QDialog>
00006 #include <QDialogButtonBox>
00007 #include <QFile>
00008 #include <QFileDialog>
00009 #include <QGroupBox>
00010 #include <QHash>
00011 #include <QJsonDocument>
00012 #include <QMainWindow>
00013 #include <QMessageBox>
00014 #include <QPushButton>
00015 #include <QRadioButton>
00016 #include <QString>
00017 #include <QTabWidget>
00018 #include <QVBoxLayout>
00019 #include <QVariantMap>
00020
```

```

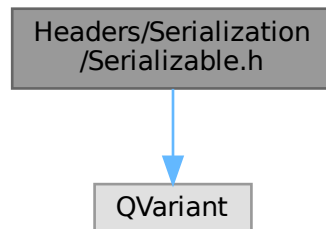
00021 #include <iostream>
00022 #include <memory>
00023
00024 #include "AppConstants.h"
00025 #include "Graph.h"
00026 #include "GraphController.h"
00027 #include "GraphEditor.h"
00028 #include "LoadFileWorker.h"
00029 #include "MenuToolBar.h"
00030 #include "SaveFileWorker.h"
00031 #include "Serializer.h"
00032 #include "ThemeManager.h"
00033 #include "UnweightedDirectedGraph.h"
00034 #include "UnweightedUndirectedGraph.h"
00035 #include "WeightedDirectedGraph.h"
00036 #include "WeightedUndirectedGraph.h"
00037
00038 QT_BEGIN_NAMESPACE
00039 namespace Ui {
00040     class MainWindow;
00041 }
00042 QT_END_NAMESPACE
00043
00044 class MainWindow : public QMainWindow {
00045     Q_OBJECT
00046
00047 public:
00048     struct TabInfo {
00049         GraphEditor* m_editor;
00050         QString      m_filePath;
00051         bool         m_isModified;
00052         QString      m_imagePath;
00053         bool         m_isImageModified;
00054     };
00055
00056     explicit MainWindow(QWidget* = nullptr);
00057     ~MainWindow() override;
00058
00059     void showStartPage();
00060
00061 private slots:
00062     void onOpenGraphTriggered();
00063     void onCreateGraphTriggered();
00064     void onSaveGraphTriggered();
00065     void onSaveImageTriggered();
00066     void onChangeThemeTriggered();
00067     void onHelpTriggered();
00068
00069     void onGraphLoadedNewTab(const QVariant&, bool, bool, const QString&);
00070     void onGraphLoadFailed(const QString&);
00071     void connectGraphModifiedSignal(GraphEditor*);
00072
00073 protected:
00074     void closeEvent(QCloseEvent*) override;
00075
00076 private:
00077     Ui::MainWindow*          m_ui;
00078     MenuToolBar*            m_menuToolBar = nullptr;
00079     ThemeManager*           m_themeManager;
00080     QTabWidget*             m_tabWidget = nullptr;
00081     QHash<GraphEditor*, TabInfo> m_tabs;
00082
00083     std::unique_ptr<Serializer> m_serializer;
00084
00085     void initMenuToolBar();
00086 };
00087
00088 #endif // MAINWINDOW_H

```

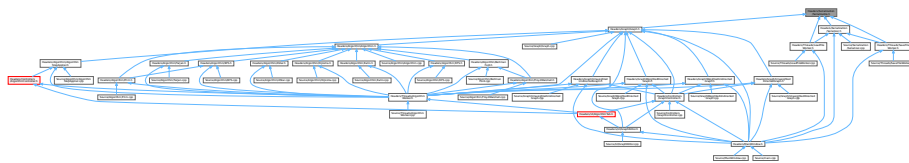
7.49 Headers/Serialization/Serializable.h File Reference

```
#include <QVariant>
```

Include dependency graph for Serializable.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [Serializable](#)

7.50 Serializable.h

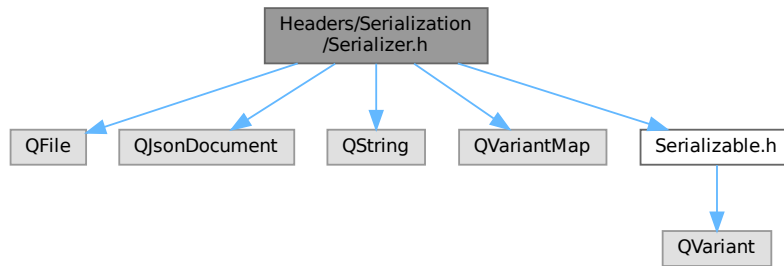
[Go to the documentation of this file.](#)

```

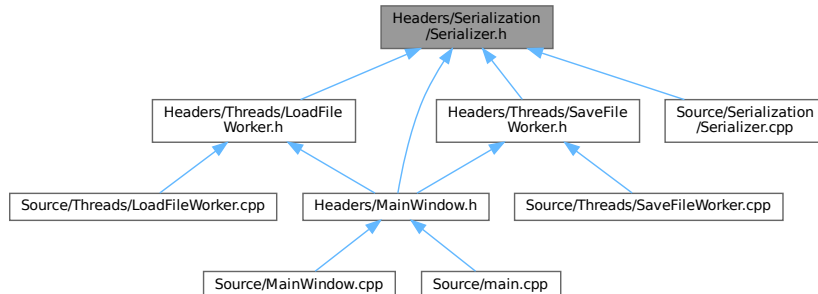
00001 #ifndef SERIALIZABLE_H
00002 #define SERIALIZABLE_H
00003
00004 #include <QVariant>
00005
00006 class Serializable {
00007 public:
00008     virtual ~Serializable() = default;
00009
00010     virtual QVariant toVariant() const = 0;
00011     virtual void fromVariant(const QVariant&) = 0;
00012 };
00013
00014 #endif // SERIALIZABLE_H
  
```

7.51 Headers/Serialization/Serializer.h File Reference

```
#include <QFile>
#include <QJsonDocument>
#include <QString>
#include <QVariantMap>
#include "Serializable.h"
Include dependency graph for Serializer.h:
```



This graph shows which files directly or indirectly include this file:



Classes

- class [Serializer](#)

7.52 Serializer.h

[Go to the documentation of this file.](#)

```
00001 #ifndef SERIALIZER_H
00002 #define SERIALIZER_H
00003
00004 #include <QFile>
00005 #include <QJsonDocument>
00006 #include <QString>
00007 #include <QVariantMap>
00008
```

```

00009 #include "Serializable.h"
00010
00011 class Serializer {
00012 public:
00013     virtual ~Serializer() = default;
00014
00015     virtual void save(const Serializable&, const QString&, bool, bool);
00016     virtual void load(Serializable&, const QVariant&);
00017 };
00018
00019 #endif // SERIALIZER_H

```

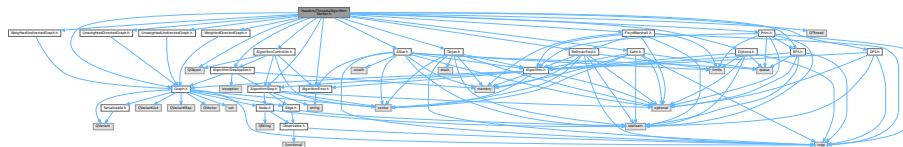
7.53 Headers/Threads/AlgorithmWorker.h File Reference

```

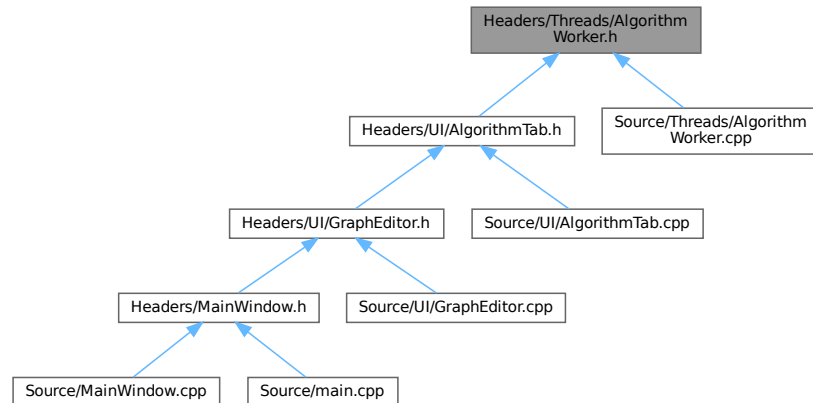
#include <QObject>
#include <QThread>
#include "AStar.h"
#include "AlgorithmController.h"
#include "AlgorithmError.h"
#include "AlgorithmStep.h"
#include "AlgorithmStepApplier.h"
#include "BFS.h"
#include "BellmanFord.h"
#include "DFS.h"
#include "Dijkstra.h"
#include "FloydWarshall.h"
#include "Graph.h"
#include "Kahn.h"
#include "Prim.h"
#include "Tarjan.h"
#include "UnweightedDirectedGraph.h"
#include "UnweightedUndirectedGraph.h"
#include "WeightedDirectedGraph.h"
#include "WeightedUndirectedGraph.h"

```

Include dependency graph for AlgorithmWorker.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [AlgorithmWorker](#)

Executes graph algorithms in a separate thread and generates visualization steps.

7.54 AlgorithmWorker.h

[Go to the documentation of this file.](#)

```

00001 #ifndef WORKER_H
00002 #define WORKER_H
00003
00004 #include <QObject>
00005 #include <QThread>
00006
00007 #include "AStar.h"
00008 #include "AlgorithmController.h"
00009 #include "AlgorithmError.h"
00010 #include "AlgorithmStep.h"
00011 #include "AlgorithmStepApplier.h"
00012 #include "BFS.h"
00013 #include "BellmanFord.h"
00014 #include "DFS.h"
00015 #include "Dijkstra.h"
00016 #include "FloydWarshall.h"
00017 #include "Graph.h"
00018 #include "Kahn.h"
00019 #include "Prim.h"
00020 #include "Tarjan.h"
00021 #include "UnweightedDirectedGraph.h"
00022 #include "UnweightedUndirectedGraph.h"
00023 #include "WeightedDirectedGraph.h"
00024 #include "WeightedUndirectedGraph.h"
00025
00038 class AlgorithmWorker : public QThread {
00039     Q_OBJECT
00040 public:
00041     AlgorithmWorker(const QString&, const std::shared_ptr<Graph>, unsigned = 0, unsigned = 0);
00042
00043 signals:
00044     void stepsReady(const std::vector<AlgorithmStep>&);
00045     void resultReady(const QString&);
00046     void algorithmErrorOccurred(const AlgorithmError&);
00047
00048 protected:
00049     void run() override;
00050
00051 private:

```

```

00052     QString                m_algorithm;
00053     std::shared_ptr<Graph> m_graph;
00054     unsigned                m_start;
00055     unsigned                m_end;
00056 };
00057
00058 #endif // WORKER_H

```

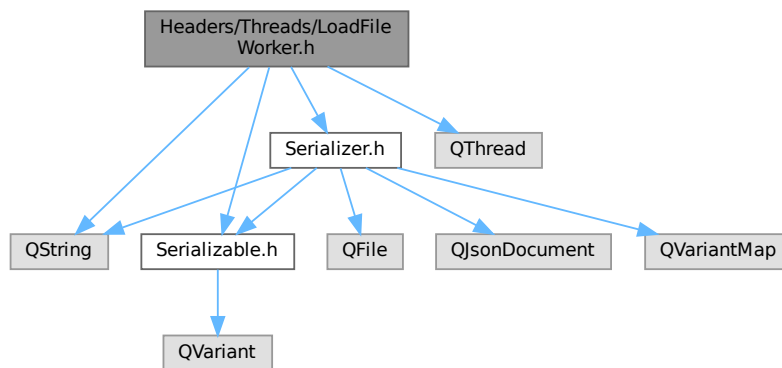
7.55 Headers/Threads/LoadFileWorker.h File Reference

```

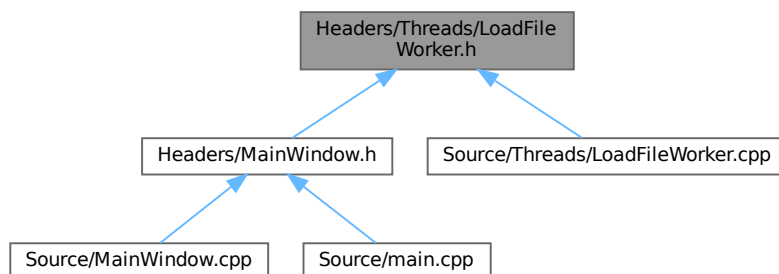
#include <QString>
#include <QThread>
#include "Serializable.h"
#include "Serializer.h"

```

Include dependency graph for LoadFileWorker.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [LoadFileWorker](#)

7.56 LoadFileWorker.h

[Go to the documentation of this file.](#)

```

00001 #ifndef LOADFILEWORKER_H
00002 #define LOADFILEWORKER_H
00003
00004 #include <QString>
00005 #include <QThread>
00006
00007 #include "Serializable.h"
00008 #include "Serializer.h"
00009
00010 class LoadFileWorker : public QThread {
00011     Q_OBJECT
00012 public:
00013     LoadFileWorker(Serializer*, const QString&, QObject* = nullptr);
00014
00015 signals:
00016     void loaded(const QVariant&, bool, bool, const QString&);
00017     void failed(const QString&);
00018
00019 protected:
00020     void run() override;
00021
00022 private:
00023     Serializer* m_serializer = nullptr;
00024     QString     m_filePath;
00025 };
00026
00027 #endif // LOADFILEWORKER_H

```

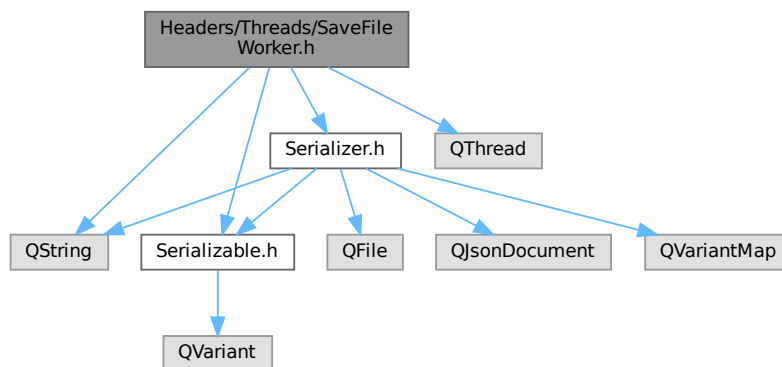
7.57 Headers/Threads/SaveFileWorker.h File Reference

```

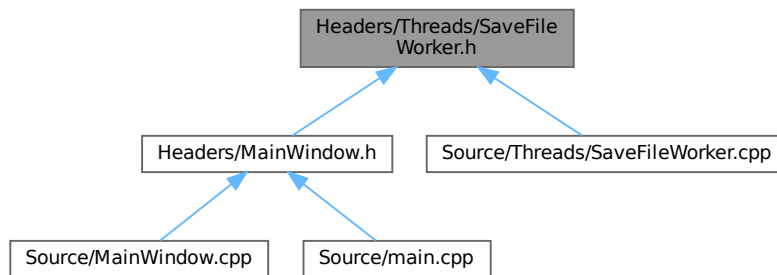
#include <QString>
#include <QThread>
#include "Serializable.h"
#include "Serializer.h"

```

Include dependency graph for SaveFileWorker.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [SaveFileWorker](#)

7.58 SaveFileWorker.h

[Go to the documentation of this file.](#)

```

00001 #ifndef SAVEFILEWORKER_H
00002 #define SAVEFILEWORKER_H
00003
00004 #include <QString>
00005 #include <QThread>
00006
00007 #include "Serializable.h"
00008 #include "Serializer.h"
00009
00010 class SaveFileWorker : public QThread {
00011     Q_OBJECT
00012 public:
00013     SaveFileWorker(Serializer*, Serializable*, const QString&, bool, bool, QObject* = nullptr);
00014
00015 signals:
00016     void finished();
00017     void failed(const QString&);
00018
00019 protected:
00020     void run() override;
00021
00022 private:
00023     Serializer* m_serializer = nullptr;
00024     QString m_filePath;
00025
00026     Serializable* m_serializable = nullptr;
00027
00028     bool m_isWeighted = false;
00029     bool m_isDirected = false;
00030 };
00031
00032 #endif // SAVEFILEWORKER_H

```

7.59 Headers/UI/AlgorithmTab.h File Reference

```

#include <QAbstractItemView>
#include <QColor>
#include <QComboBox>

```

```

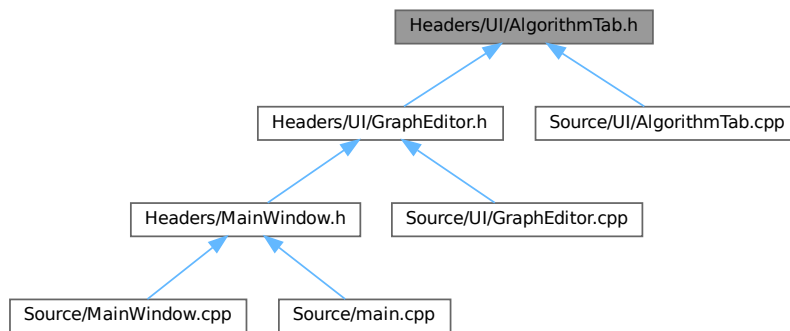
#include <QFrame>
#include <QGroupBox>
#include <QHBoxLayout>
#include <QLabel>
#include <QLineEdit>
#include <QMessageBox>
#include <QPushButton>
#include <QScrollArea>
#include <QStandardItem>
#include <QStandardItemModel>
#include <QString>
#include <QStyle>
#include <QTimer>
#include <QToolButton>
#include <QVBoxLayout>
#include <QWidget>
#include <optional>
#include "AlgorithmController.h"
#include "AlgorithmWorker.h"
#include "GraphController.h"

```

Include dependency graph for AlgorithmTab.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [AlgorithmTab](#)
- struct [AlgorithmTab::AlgorithmConfig](#)

7.60 AlgorithmTab.h

[Go to the documentation of this file.](#)

```

00001 #ifndef ALGORITHMHTAB_H
00002 #define ALGORITHMHTAB_H
00003
00004 #include <QAbstractItemView>
00005 #include <QColor>
00006 #include <QComboBox>
00007 #include <QFrame>
00008 #include <QGroupBox>
00009 #include <QHBoxLayout>
00010 #include <QLabel>
00011 #include <QLineEdit>
00012 #include <QMessageBox>
00013 #include <QPushButton>
00014 #include <QScrollArea>
00015 #include <QStandardItem>
00016 #include <QStandardItemModel>
00017 #include <QString>
00018 #include <QStyle>
00019 #include <QTimer>
00020 #include <QToolButton>
00021 #include <QVBoxLayout>
00022 #include <QWidget>
00023
00024 #include <optional>
00025
00026 #include "AlgorithmController.h"
00027 #include "AlgorithmWorker.h"
00028 #include "GraphController.h"
00029
00030 class AlgorithmTab : public QWidget {
00031     Q_OBJECT
00032
00033 public:
00034     explicit AlgorithmTab(std::shared_ptr<GraphController>, QWidget* = nullptr);
00035     ~AlgorithmTab() override = default;
00036
00037 public slots:
00038     void showAlgorithmErrorDialog(const AlgorithmError&);
00039
00040 signals:
00041     void graphEditAllowedChanged(bool);
00042
00043 private:
00044     struct AlgorithmConfig {
00045         QString m_algorithmName;
00046         int m_startNode;
00047         int m_endNode;
00048
00049         bool operator==(const AlgorithmConfig&) const;
00050         bool operator!=(const AlgorithmConfig&) const;
00051     };
00052
00053     AlgorithmConfig selectedConfig() const;
00054
00055     enum class RunState { Idle, Playing, Paused, Finished };
00056
00057     QComboBox* m_algorithmCombo;
00058
00059     QWidget* m_startRow;
00060     QLabel* m_startLabel;
00061     QLineEdit* m_startNodeEdit;
00062
00063     QWidget* m_endRow;
00064     QLabel* m_endLabel;
00065     QLineEdit* m_endNodeEdit;
00066
00067     QLabel* m_noInputLabel;
00068
00069     QScrollArea* m_legendScroll;
00070
00071     QVBoxLayout* m_legendLayout;
00072     QWidget* m_legendContainer;
00073
00074     QPushButton* m_helpBtn;
00075
00076     QToolButton* m_prevBtn;
00077     QToolButton* m_playBtn;
00078     QToolButton* m_pauseBtn;
00079     QToolButton* m_nextBtn;
00080     QToolButton* m_restartBtn;
00081
00082     std::shared_ptr<GraphController> m_graphController;
00083     AlgorithmStepApplier m_applier;
00084     AlgorithmController m_algorithmController;
00085     AlgorithmWorker* m_worker = nullptr;
00086     QTimer* m_timer = nullptr;
00087     RunState m_state = RunState::Idle;

```

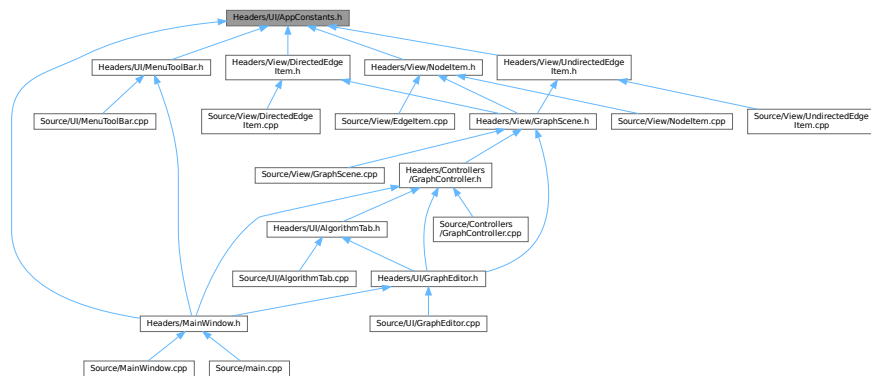
```

00088     std::optional<AlgorithmConfig>    m_currentConfig;
00089     bool                               m_graphEditAllowed = true;
00090
00091     void    initLayout ();
00092     void    initIcons ();
00093     void    updateUiForAlgorithm(const QString&);
00094     void    updateLegendForAlgorithm(const QString&);
00095     QWidget* makeLegendItem(const QColor&, const QString&);
00096
00097     void startTimerForPlay ();
00098     void updateControls ();
00099 };
00100
00101 #endif // ALGORITHMHTAB_H

```

7.61 Headers/UI/AppConstants.h File Reference

This graph shows which files directly or indirectly include this file:



Namespaces

- namespace [AppConstants](#)

Variables

- constexpr int [AppConstants::windowMinWidth](#) = 800
- constexpr int [AppConstants::windowMinHeight](#) = 600
- constexpr int [AppConstants::windowWidth](#) = 1200
- constexpr int [AppConstants::windowHeight](#) = 900
- constexpr int [AppConstants::btnSize](#) = 120
- constexpr int [AppConstants::btnSpacing](#) = 24
- constexpr auto [AppConstants::startPageTitle](#) = "AlgoVision"
- constexpr int [AppConstants::optionBtnSize](#) = 90
- constexpr int [AppConstants::optionIconSize](#) = 64
- constexpr auto [AppConstants::graphPageDefaultTitle](#) = "Untitled"
- constexpr double [AppConstants::defaultRadius](#) = 30.0
- constexpr double [AppConstants::defaultBorderWidth](#) = 3.0
- double [AppConstants::NodeScale](#) = 1.0
- constexpr double [AppConstants::MinNodeScale](#) = 0.4
- constexpr double [AppConstants::MaxNodeScale](#) = 3.0
- constexpr double [AppConstants::EdgePadding](#) = 6.0

- constexpr double [AppConstants::BaseEdgeWidth](#) = 2.0
- constexpr int [AppConstants::createGraphDialogMinWidth](#) = 400
- constexpr int [AppConstants::createGraphDialogMinHeight](#) = 250
- constexpr int [AppConstants::graphTypeDialogErrorMinWidth](#) = 500
- constexpr int [AppConstants::graphTypeDialogErrorMinHeight](#) = 250
- constexpr int [AppConstants::legendMinHeight](#) = 400

7.62 AppConstants.h

[Go to the documentation of this file.](#)

```

00001 #ifndef APPCONSTANT_H
00002 #define APPCONSTANT_H
00003
00004 namespace AppConstants {
00005     // window size
00006     constexpr int windowMinWidth = 800;
00007     constexpr int windowMinHeight = 600;
00008     constexpr int windowWidth = 1200;
00009     constexpr int windowHeight = 900;
00010
00011     // start page element's attributes
00012     constexpr int btnSize = 120;
00013     constexpr int btnSpacing = 24;
00014     constexpr auto startPageTitle = "AlgoVision";
00015
00016     // menubar element's attributes
00017     constexpr int optionBtnSize = 90;
00018     constexpr int optionIconSize = 64;
00019
00020     // graph page element's attributes
00021     constexpr auto graphPageDefaultTitle = "Untitled";
00022
00023     // node view
00024     constexpr double defaultRadius = 30.0;
00025     constexpr double defaultBorderWidth = 3.0;
00026
00027     inline double NodeScale = 1.0;
00028
00029     constexpr double MinNodeScale = 0.4;
00030     constexpr double MaxNodeScale = 3.0;
00031
00032     constexpr double EdgePadding = 6.0;
00033     constexpr double BaseEdgeWidth = 2.0;
00034
00035     static constexpr double BaseFontSize = 10.0;
00036
00037     // dialogs
00038     constexpr int createGraphDialogMinWidth = 400;
00039     constexpr int createGraphDialogMinHeight = 250;
00040
00041     constexpr int graphTypeDialogErrorMinWidth = 500;
00042     constexpr int graphTypeDialogErrorMinHeight = 250;
00043
00044     // legend
00045     constexpr int legendMinHeight = 400;
00046 } // namespace AppConstants
00047
00048 #endif // APPCONSTANT

```

7.63 Headers/UI/GraphEditor.h File Reference

```

#include <QHBoxLayout>
#include <QKeySequence>
#include <QLabel>
#include <QShortcut>
#include <QSlider>
#include <QSplitter>
#include <QTabWidget>
#include <QUndoCommand>

```

```

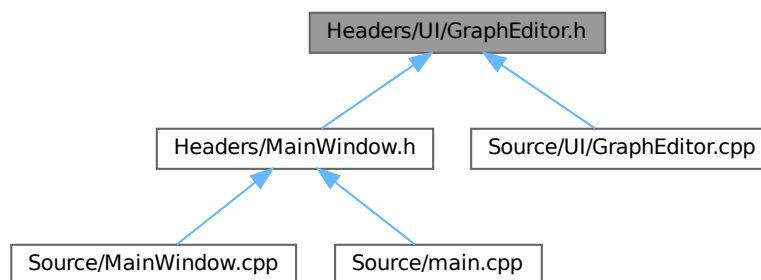
#include <QUndoStack>
#include <QWidget>
#include <functional>
#include <memory>
#include "AlgorithmTab.h"
#include "Graph.h"
#include "GraphController.h"
#include "GraphEditTab.h"
#include "GraphScene.h"
#include "GraphView.h"

```

Include dependency graph for GraphEditor.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [GraphEditor](#)

7.64 GraphEditor.h

[Go to the documentation of this file.](#)

```

00001 #ifndef GRAPHEDITOR_H
00002 #define GRAPHEDITOR_H
00003
00004 #include <QHBoxLayout>
00005 #include <QKeySequence>
00006 #include <QLabel>
00007 #include <QShortcut>
00008 #include <QSlider>
00009 #include <QSplitter>
00010 #include <QTabWidget>
00011 #include <QUndoCommand>
00012 #include <QUndoStack>
00013 #include <QWidget>
00014
00015 #include <functional>
00016 #include <memory>

```

```

00017
00018 #include "AlgorithmTab.h"
00019 #include "Graph.h"
00020 #include "GraphController.h"
00021 #include "GraphEditTab.h"
00022 #include "GraphScene.h"
00023 #include "GraphView.h"
00024
00025 class GraphEditor : public QWidget {
00026     Q_OBJECT
00027
00028 public:
00029     explicit GraphEditor(const std::shared_ptr<GraphController>&, QWidget* = nullptr);
00030     ~GraphEditor() override = default;
00031
00032     std::shared_ptr<GraphController> graphController() const;
00033
00034 signals:
00035     void undoRequested();
00036     void redoRequested();
00037
00038 private slots:
00039     void onAddRequestTrigger();
00040     void onRemoveRequestTrigger();
00041     void onClearRequestTrigger();
00042     void onUndoRequestTrigger();
00043     void onRedoRequestTrigger();
00044     void onPanRequestTrigger();
00045     void onZoomInRequestTrigger();
00046     void onZoomOutRequestTrigger();
00047
00048 private:
00049     QUndoStack*          m_undoStack      = nullptr;
00050     GraphEditTab*        m_editTab       = nullptr;
00051     QLabel*             m_leftPlaceholder = nullptr;
00052     GraphView*          m_view           = nullptr;
00053     std::shared_ptr<GraphController> m_graphController;
00054 };
00055
00056 #endif // GRAPHEDITOR_H

```

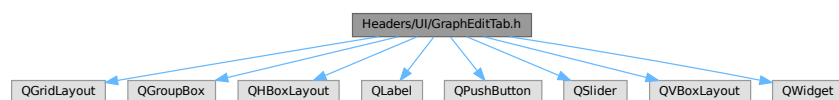
7.65 Headers/UI/GraphEditTab.h File Reference

```

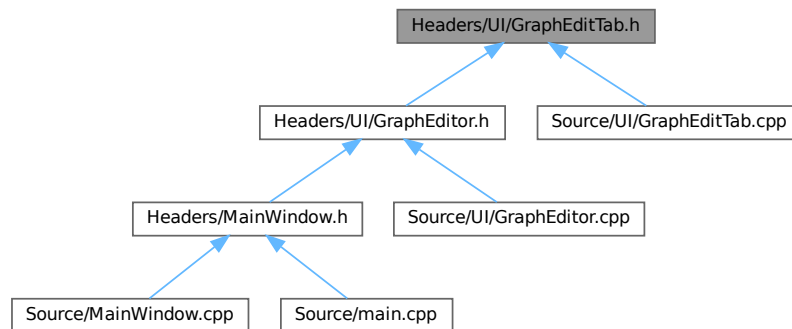
#include <QGridLayout>
#include <QGroupBox>
#include <QHBoxLayout>
#include <QLabel>
#include <QPushButton>
#include <QSlider>
#include <QVBoxLayout>
#include <QWidget>

```

Include dependency graph for GraphEditTab.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [GraphEditTab](#)

7.66 GraphEditTab.h

[Go to the documentation of this file.](#)

```

00001 #ifndef GRAPHEEDITTAB_H
00002 #define GRAPHEEDITTAB_H
00003
00004 #include <QGridLayout>
00005 #include <QGroupBox>
00006 #include <QHBoxLayout>
00007 #include <QLabel>
00008 #include <QPushButton>
00009 #include <QSlider>
00010 #include <QVBoxLayout>
00011 #include <QWidget>
00012
00013 class GraphEditTab : public QWidget {
00014     Q_OBJECT
00015
00016 public:
00017     explicit GraphEditTab(QWidget* = nullptr);
00018     ~GraphEditTab() override = default;
00019     QSlider* getNodeSizeSlider() const;
00020
00021 signals:
00022     void undoRequested();
00023     void redoRequested();
00024     void addRequested();
00025     void removeRequested();
00026     void clearRequested();
00027
00028     void panRequested();
00029     void zoomInRequested();
00030     void zoomOutRequested();
00031
00032 public slots:
00033     void setUndoEnabled(bool);
00034     void setRedoEnabled(bool);
00035     void setGraphEditsAllowed(bool);
00036
00037 private:
00038     QPushButton* m_addBtn;
00039     QPushButton* m_removeBtn;
00040     QPushButton* m_undoBtn;
00041     QPushButton* m_redoBtn;
00042     QPushButton* m_zoomInBtn;
00043     QPushButton* m_zoomOutBtn;
00044     QPushButton* m_panBtn;
  
```

```

00045     QPushButton* m_clearBtn;
00046     QSlider*      m_nodeSizeSlider;
00047
00048     bool m_graphEditsAllowed = true;
00049     bool m_canUndo           = false;
00050     bool m_canRedo           = false;
00051
00052     void applyEditButtonsState();
00053     void initLayout();
00054 };
00055
00056 #endif // GRAPHEDITTAB_H

```

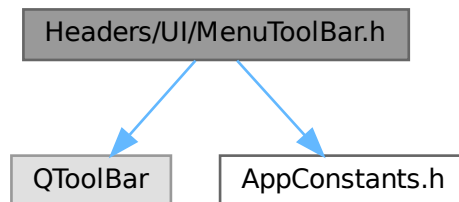
7.67 Headers/UI/MenuToolBar.h File Reference

```

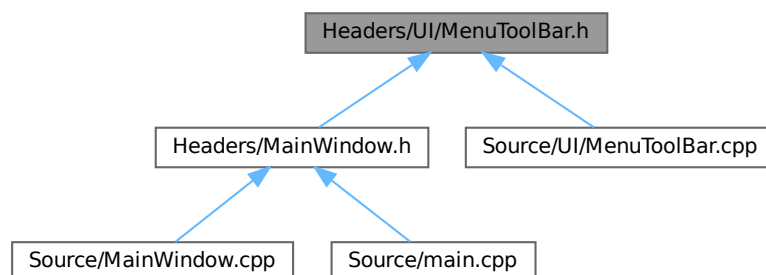
#include <QToolBar>
#include "AppConstants.h"

```

Include dependency graph for MenuToolBar.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [MenuToolBar](#)

7.68 MenuToolBar.h

[Go to the documentation of this file.](#)

```

00001 #ifndef MENU TOOLBAR_H
00002 #define MENU TOOLBAR_H
00003
00004 #include <QToolBar>
00005
00006 #include "AppConstants.h"
00007
00008 class QAction;
00009
00010 class MenuToolBar : public QToolBar {
00011     Q_OBJECT
00012
00013 public:
00014     explicit MenuToolBar(QWidget* = nullptr);
00015     ~MenuToolBar() override = default;
00016
00017     QAction* createGraphAction() const;
00018     QAction* openGraphAction() const;
00019     QAction* saveGraphAction() const;
00020     QAction* saveImageAction() const;
00021     QAction* changeThemeAction() const;
00022     QAction* helpAction() const;
00023
00024 private:
00025     QAction* m_createGraphAction;
00026     QAction* m_openGraphAction;
00027     QAction* m_saveGraphAction;
00028     QAction* m_saveImageAction;
00029     QAction* m_changeThemeAction;
00030     QAction* m_helpAction;
00031
00032     void setStyle();
00033     void setOptionShortcuts();
00034 };
00035
00036 #endif // MENU TOOLBAR_H

```

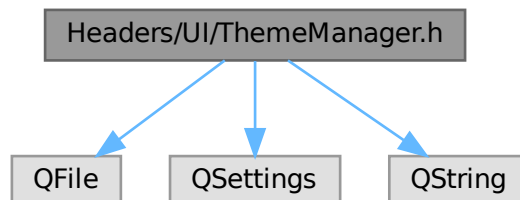
7.69 Headers/UI/ThemeManager.h File Reference

```

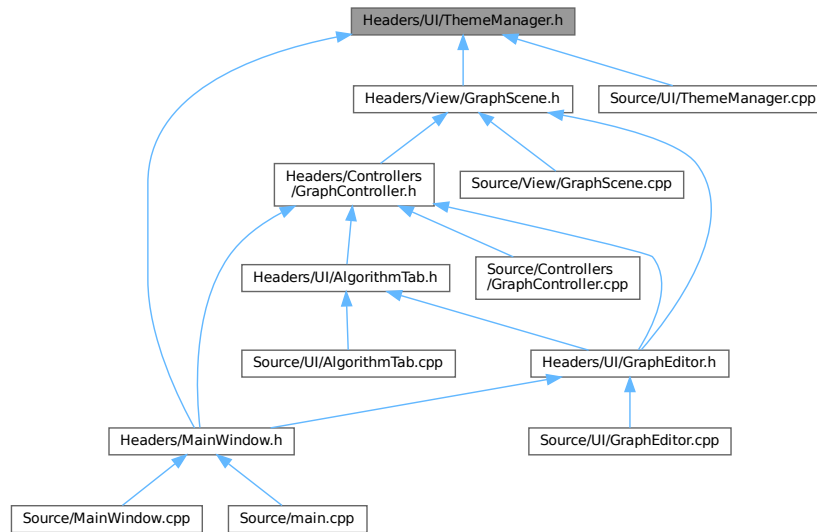
#include <QFile>
#include <QSettings>
#include <QString>

```

Include dependency graph for ThemeManager.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [ThemeManager](#)

7.70 ThemeManager.h

[Go to the documentation of this file.](#)

```

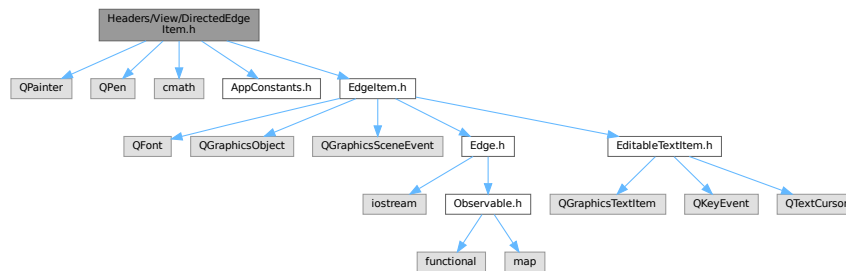
00001 #ifndef THEME_H
00002 #define THEME_H
00003
00004 #include <QFile>
00005 #include <QSettings>
00006 #include <QString>
00007
00008 class ThemeManager {
00009 public:
00010     enum class Theme { LIGHT, DARK, PURPLE };
00011
00012     explicit ThemeManager();
00013
00014     Theme    currentTheme() const;
00015     QString  styleSheet() const;
00016
00017     void    setTheme(Theme);
00018
00019 private:
00020     Theme    m_currentTheme{Theme::DARK};
00021     QSettings m_settings;
00022
00023     void    loadTheme();
00024     void    saveTheme();
00025 };
00026
00027 #endif // THEME_H

```

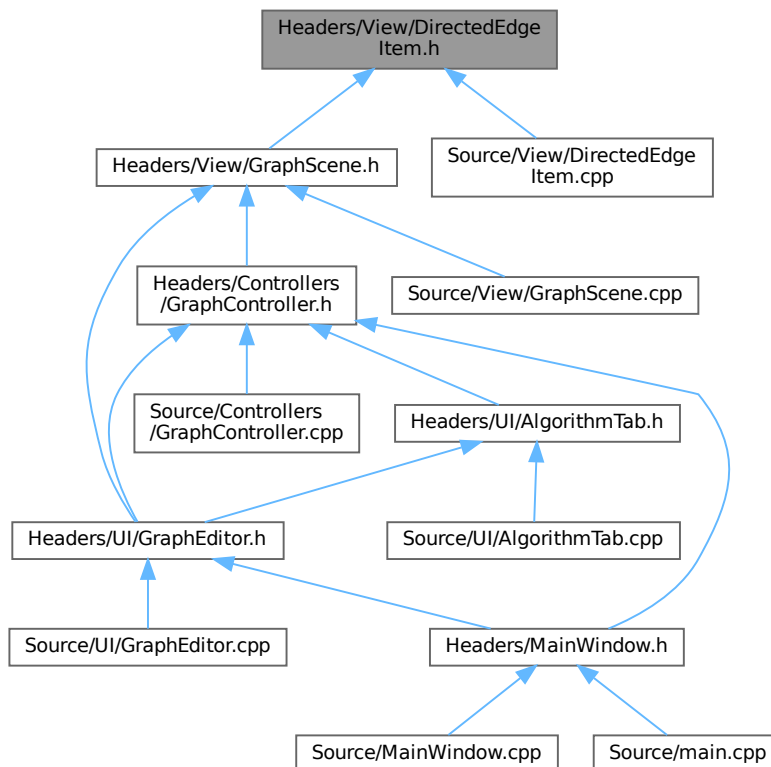
7.71 Headers/View/DirectedEdgeItem.h File Reference

```
#include <QPainter>
#include <QPen>
#include <cmath>
#include "AppConstants.h"
#include "EdgeItem.h"
```

Include dependency graph for DirectedEdgeItem.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [DirectedEdgeItem](#)

7.72 DirectedEdgeItem.h

[Go to the documentation of this file.](#)

```

00001 #ifndef DIRECTED_EDGEITEM_H
00002 #define DIRECTED_EDGEITEM_H
00003
00004 #include <QPainter>
00005 #include <QPen>
00006 #include <cmath>
00007
00008 #include "AppConstants.h"
00009 #include "EdgeItem.h"
00010
00011 class DirectedEdgeItem : public EdgeItem {
00012 public:
00013     DirectedEdgeItem(Edge*, NodeItem*, NodeItem*, bool = false);
00014
00015     QRectF boundingRect() const override;
00016     QPainterPath shape() const override;
00017     void paint(QPainter*, const QStyleOptionGraphicsItem*, QWidget*) override;
00018
00019     QPainterPath edgePath() const override;
00020 private:
00021     qreal m_skewness{70.0};
00022     qreal m_arrowSize{20.0};
00023     qreal m_arrowAngle{M_PI / 6.0};
00024
00025     QPainterPath arrowPath(const QPainterPath&) const;
00026 };
00027
00028 #endif // DIRECTED_EDGEITEM_H

```

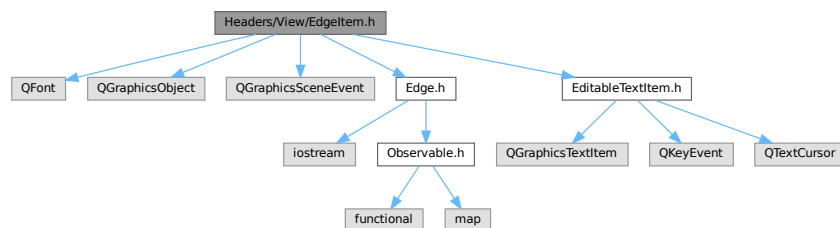
7.73 Headers/View/EdgeItem.h File Reference

```

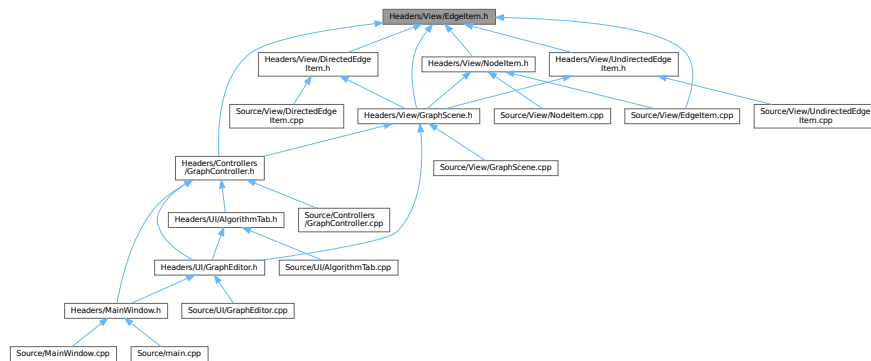
#include <QFont>
#include <QGraphicsObject>
#include <QGraphicsSceneEvent>
#include "Edge.h"
#include "EditableTextItem.h"

```

Include dependency graph for EdgeItem.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [EdgItem](#)

7.74 EdgItem.h

[Go to the documentation of this file.](#)

```

00001 #ifndef EDGITEM_H
00002 #define EDGITEM_H
00003
00004 #include <QFont>
00005 #include <QGraphicsObject>
00006 #include <QGraphicsSceneEvent>
00007
00008 #include "Edge.h"
00009 #include "EditableTextItem.h"
00010
00011 class NodeItem;
00012
00013 class EdgItem : public QGraphicsObject {
00014     Q_OBJECT
00015
00016 public:
00017     EdgItem(Edge*, NodeItem*, NodeItem*, bool = false);
00018     ~EdgItem() override;
00019
00020     void initEdgeWeight();
00021     void adjust();
00022     void adjustPointsGeometry();
00023     void adjustWeightGeometry() const;
00024     virtual QPainterPath edgePath() const = 0;
00025     QRectF boundingRect() const override = 0;
00026     QPainterPath shape() const override = 0;
00027     void paint(QPainter*, const QStyleOptionGraphicsItem*, QWidget*) override = 0;
00028
00029     void connectNodes();
00030     void disconnectNodes();
00031
00032     Edge* modelEdge() const;
00033     void setModelEdge(Edge*);
00034
00035     bool hasWeight() const;
00036     void setHasWeight(bool);
00037
00038     EditableTextItem* weight() const;
00039     void setWeight(EditableTextItem*);
00040
00041     void updateSize();
00042
00043 signals:
00044     void edgeSelected(EdgItem*);
00045     void editEdgeWeightRequest(const EdgItem*, const QString&) const;
00046

```

```

00047 private slots:
00048     void onEdgeWeightChanged(const QString&) const;
00049
00050 protected:
00051     Edge* m_modelEdge;
00052     EditableTextItem* m_weight{nullptr};
00053     bool m_hasWeight;
00054
00055     NodeItem* m_sourceNode;
00056     NodeItem* m_destNode;
00057     QPointF m_sourcePoint;
00058     QPointF m_destPoint;
00059
00060     qreal m_penWidth{2};
00061     qreal m_shapeStroke{20.0};
00062
00063     unsigned m_observerId{0};
00064
00065     void mousePressEvent(QGraphicsSceneMouseEvent*) override;
00066     const QColor calculateColor() const;
00067     void onEdgeUpdated();
00068
00069     QPointF getEdgeCenter() const;
00070     QPointF calculateNormal() const;
00071     QPointF getWeightPosition() const;
00072
00073     void mouseDoubleClickEvent(QGraphicsSceneMouseEvent*) override;
00074 };
00075
00076 #endif // EDGEITEM_H

```

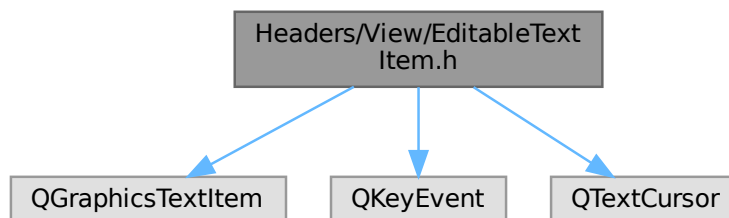
7.75 Headers/View/EditableTextItem.h File Reference

```

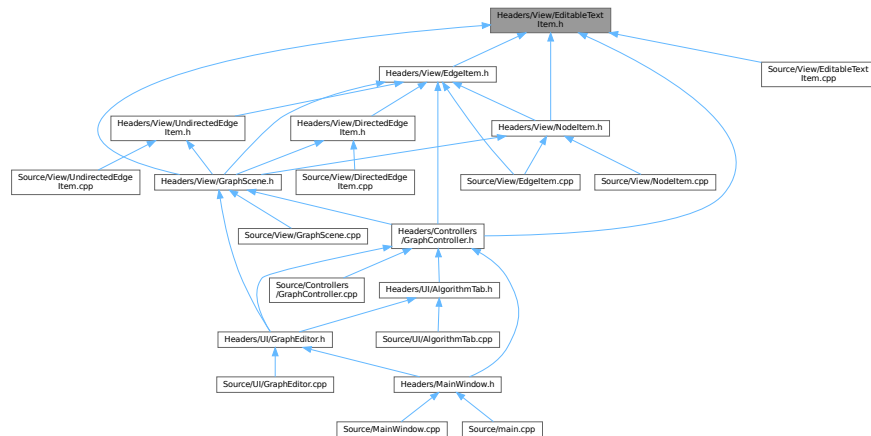
#include <QGraphicsTextItem>
#include <QKeyEvent>
#include <QTextCursor>

```

Include dependency graph for EditableTextItem.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [EditableTextItem](#)

7.76 EditableTextItem.h

[Go to the documentation of this file.](#)

```

00001 #ifndef EDITABLE_TEXTITEM_H
00002 #define EDITABLE_TEXTITEM_H
00003
00004 #include <QGraphicsTextItem>
00005 #include <QKeyEvent>
00006 #include <QTextCursor>
00007
00008 class EditableTextItem : public QGraphicsTextItem {
00009     Q_OBJECT
00010
00011 public:
00012     explicit EditableTextItem(QGraphicsItem* = nullptr);
00013
00014     QString oldText() const;
00015     void setOldText(const QString&);
00016
00017     void startEditing();
00018     void finishEditing(bool);
00019     void centerText();
00020
00021     QPointF center() const;
00022     void setCenter(const QPointF&);
00023     void setCenter(qreal, qreal);
00024
00025 protected:
00026     void keyPressEvent(QKeyEvent*) override;
00027
00028 signals:
00029     void textCommitted(const QString& text);
00030
00031     // notify graphscene if it's EDIT state or not (then it's ADD state)
00032     void setEditGraphSceneState(bool, EditableTextItem* = nullptr);
00033
00034 private:
00035     QString m_oldText;
00036     QPointF m_center;
00037 };
00038
00039 #endif

```


7.78 GraphScene.h

[Go to the documentation of this file.](#)

```

00001 #ifndef GRAPHSCENE_H
00002 #define GRAPHSCENE_H
00003
00004 #include <QGraphicsScene>
00005 #include <QGraphicsSceneMouseEvent>
00006
00007 #include <map>
00008
00009 #include "DirectedEdgeItem.h"
00010 #include "Edge.h"
00011 #include "EdgeItem.h"
00012 #include "EditableTextItem.h"
00013 #include "Node.h"
00014 #include "NodeItem.h"
00015 #include "ThemeManager.h"
00016 #include "UndirectedEdgeItem.h"
00017
00018 class GraphScene : public QGraphicsScene {
00019     Q_OBJECT
00020
00021 public:
00022     explicit GraphScene(QObject* = nullptr);
00023
00024     /*
00025         ADD -> adding nodes and edges
00026         REMOVE -> removing nodes and edges
00027         EDIT -> changing edge weights, and changing node name, which is supported and can be
00028         activated IDLE -> all scene interactions are disabled
00029     */
00030     enum class State { ADD, REMOVE, EDIT, IDLE };
00031
00032     void updateNodeScaling();
00033
00034     void setState(GraphScene::State);
00035     // returns the scene in initial state (clears all selections and ends all edits)
00036     void resetScene();
00037     // disables scene interactions (adding, removing, updating and moving elements)
00038     void disableScene();
00039     // reenables scene, returns it in state before disable
00040     void enableScene();
00041
00042     void clear();
00043
00044     void addNode(Node*);
00045     void addEdge(Edge*, bool, bool);
00046     // additional command that encapsulates adding node and edge into one command that is revertable
00047     // via undo/redo
00048     void addNodeAndEdge(Node*, Edge*, bool, bool);
00049     void removeNode(NodeItem*);
00050     void removeEdge(EdgeItem*);
00051     void applyTheme(ThemeManager::Theme theme);
00052
00053     NodeItem* findNodeItemById(const unsigned) const;
00054     EdgeItem* findEdgeItemByNodes(unsigned, unsigned) const;
00055     EdgeItem* findEdgeItemById(unsigned) const;
00056
00057 signals:
00058     void addNodeRequest(const QPointF&);
00059     void addNodeAndEdgeRequest(const QPointF&, NodeItem*);
00060     void addEdgeRequest(NodeItem*, NodeItem*);
00061     void removeNodeRequest(NodeItem*);
00062     void removeEdgeRequest(EdgeItem*);
00063     void moveNodeRequest(const NodeItem*, const QPointF&, const QPointF&);
00064     void editNodeNameRequest(const NodeItem*, const QString&) const;
00065     void editEdgeWeightRequest(const EdgeItem*, const QString&) const;
00066
00067 protected:
00068     void mousePressEvent(QGraphicsSceneMouseEvent*) override;
00069     void mouseDoubleClickEvent(QGraphicsSceneMouseEvent*) override;
00070
00071 private slots:
00072     void onNodeSelectTrigger(NodeItem*);
00073     void onEdgeSelectTrigger(EdgeItem*);
00074
00075     // reacts to editabletextitem scene change
00076     void setEditGraphSceneTrigger(bool, EditableTextItem* = nullptr);
00077
00078 private:
00079     NodeItem* m_firstNodeSelect{nullptr};
00080     EditableTextItem* m_editLabel{nullptr};
00081     State m_state{GraphScene::State::ADD};
00082     State m_previousState;

```

```

00083
00084     // structure only used as helper for operations, not owning elements
00085     std::map<unsigned, NodeItem*> m_nodeItems;
00086
00087     void selectNode(NodeItem*);
00088 };
00089
00090 #endif // GRAPHSCENE_H

```

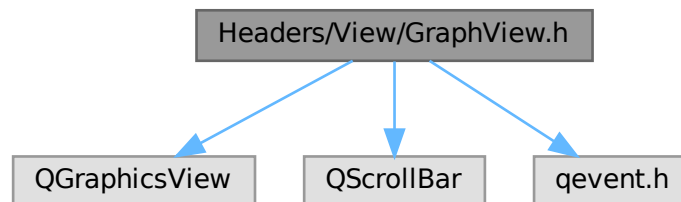
7.79 Headers/View/GraphView.h File Reference

```

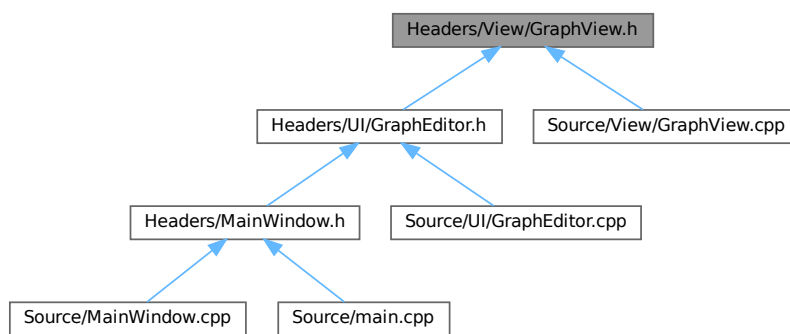
#include <QGraphicsView>
#include <QScrollBar>
#include <qevent.h>

```

Include dependency graph for GraphView.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [GraphView](#)

7.80 GraphView.h

[Go to the documentation of this file.](#)

```

00001 #ifndef GRAPH_VIEW_H
00002 #define GRAPH_VIEW_H
00003
00004 #include <QGraphicsView>
00005 #include <QScrollBar>
00006 #include <qevent.h>
00007
00008 class GraphView : public QGraphicsView {
00009 public:
00010     enum class State { IDLE, PAN_IDLE, PAN_ACTIVE };
00011
00012     explicit GraphView(QWidget* = nullptr);
00013     explicit GraphView(QGraphicsScene*, QWidget* = nullptr);
00014
00015     GraphView::State state() const;
00016     void setState(GraphView::State);
00017     void resetState();
00018
00019     void zoomIn();
00020     void zoomOut();
00021
00022 protected:
00023     void mousePressEvent(QMouseEvent*) override;
00024     void mouseMoveEvent(QMouseEvent*) override;
00025     void mouseReleaseEvent(QMouseEvent*) override;
00026     void mouseDoubleClickEvent(QMouseEvent* event) override;
00027     void wheelEvent(QWheelEvent* event) override;
00028
00029 private:
00030     GraphView::State m_state{GraphView::State::IDLE};
00031
00032     // pan state (in scene coordinates)
00033     QPoint m_lastMousePos;
00034
00035     qreal m_zoomFactor{1.15};
00036
00037     // used for returning to previous state after fast pan (right click drag)
00038     GraphView::State m_oldState;
00039
00040     void init();
00041 };
00042
00043 #endif // GRAPH_VIEW

```

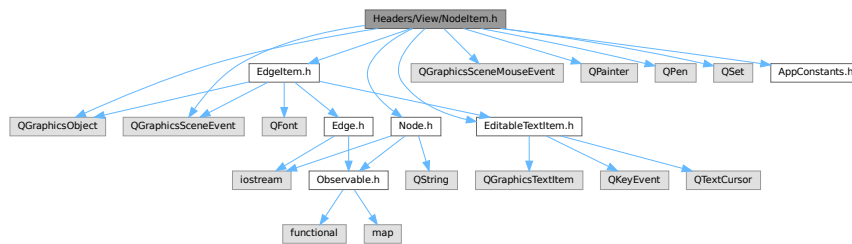
7.81 Headers/View/NodeItem.h File Reference

```

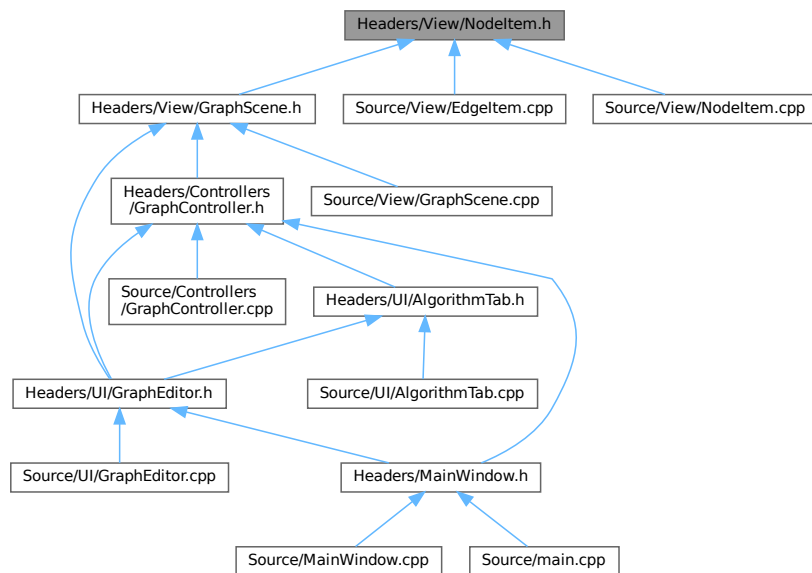
#include <QGraphicsObject>
#include <QGraphicsSceneEvent>
#include <QGraphicsSceneMouseEvent>
#include <QPainter>
#include <QPen>
#include <QSet>
#include "AppConstants.h"
#include "EdgeItem.h"
#include "EditableTextItem.h"
#include "Node.h"

```

Include dependency graph for Nodeltem.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [Nodeltem](#)

7.82 Nodeltem.h

[Go to the documentation of this file.](#)

```

00001 #ifndef NODEITEM_H
00002 #define NODEITEM_H
00003
00004 #include <QGraphicsObject>
00005 #include <QGraphicsSceneEvent>
00006 #include <QGraphicsSceneMouseEvent>
00007 #include <QPainter>
00008 #include <QPen>
00009 #include <QSet>
00010
00011 #include "AppConstants.h"
  
```

```

00012 #include "EdgeItem.h"
00013 #include "EditableTextItem.h"
00014 #include "Node.h"
00015
00016 class EdgeItem;
00017
00018 class NodeItem : public QGraphicsObject {
00019     Q_OBJECT
00020
00021 public:
00022     explicit NodeItem(Node*);
00023     ~NodeItem() override;
00024
00025     Node* modelNode() const;
00026     void setModelNode(Node*);
00027     qreal radius() const;
00028
00029     void setNodeSelected(bool);
00030     void updateSize();
00031
00032     void addEdge(EdgeItem*);
00033     void removeEdge(EdgeItem*);
00034
00035     void updateNodePosition();
00036
00037     EditableTextItem* label() const;
00038     void setLabel(EditableTextItem*);
00039
00040     const QSet<EdgeItem*> & edges() const;
00041
00042 signals:
00043     void nodeSelected(NodeItem*);
00044     void editNodeNameRequest(const NodeItem*, const QString&) const;
00045     void moveNodeRequest(const NodeItem*, const QPointF&, const QPointF&);
00046
00047 protected:
00048     QRectF boundingRect() const override;
00049     QPainterPath shape() const override;
00050     void paint(QPainter*, const QStyleOptionGraphicsItem*, QWidget*) override;
00051     QVariant itemChange(GraphicsItemChange, const QVariant&) override;
00052     void mousePressEvent(QGraphicsSceneMouseEvent*) override;
00053     void mouseMoveEvent(QGraphicsSceneMouseEvent*) override;
00054     void mouseReleaseEvent(QGraphicsSceneMouseEvent*) override;
00055
00056     // removed feature
00057     // it will remain in code all connected, so the feature can easily be activated again, or used
00058     // for other purposes
00059     // void mouseDoubleClickEvent(QGraphicsSceneMouseEvent*) override;
00060
00061 private slots:
00062     void onNameChanged(const QString&) const;
00063
00064 private:
00065     Node* m_modelNode;
00066     EditableTextItem* m_label;
00067     QSet<EdgeItem*> m_edges;
00068     // in scene coordinates, used to save old node position(center) before move command
00069     QPointF m_oldCenter;
00070
00071     qreal m_radius{AppConstants::defaultRadius};
00072     qreal m_borderWidth{AppConstants::defaultBorderWidth};
00073     bool m_hasChangePosition{false};
00074     bool m_nodeSelected{false};
00075     bool m_ignoreNextMouseRelease{false};
00076
00077     unsigned m_observerId{0};
00078
00079     const QColor calculateColor() const;
00080     void onNodeUpdated();
00081 };
00082
00083 #endif // NODEITEM_H

```

7.83 Headers/View/UndirectedEdgeItem.h File Reference

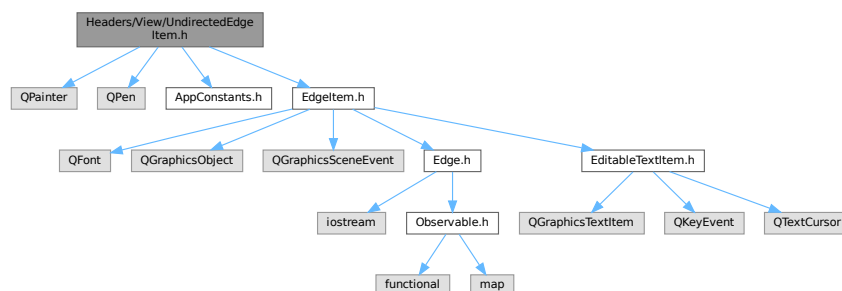
```

#include <QPainter>
#include <QPen>
#include "AppConstants.h"

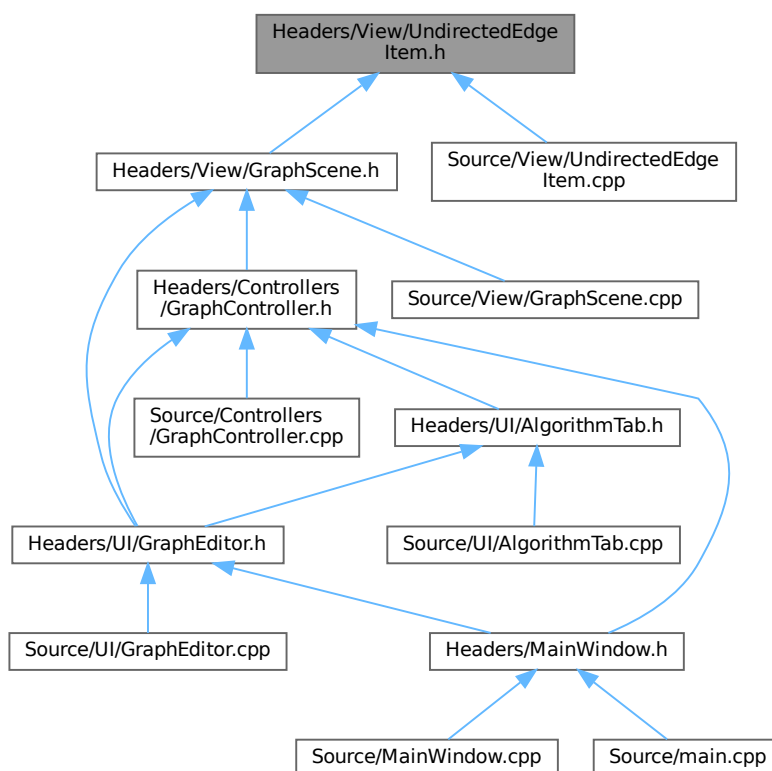
```

```
#include "EdgeItem.h"
```

Include dependency graph for UndirectedEdgeItem.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [UndirectedEdgeItem](#)

7.84 UndirectedEdgeItem.h

[Go to the documentation of this file.](#)

```

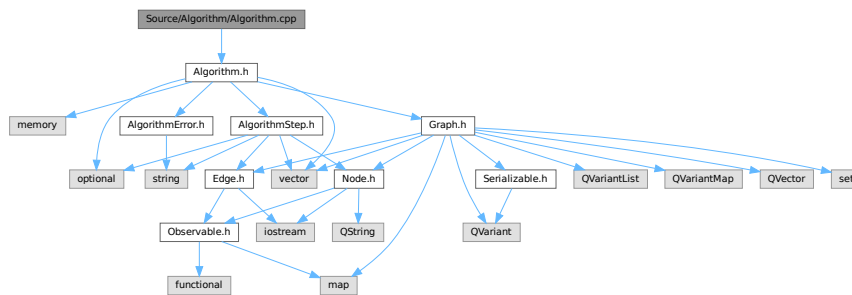
00001 #ifndef UNDIRECTED_EDGEITEM_H
00002 #define UNDIRECTED_EDGEITEM_H
00003
00004 #include <QPainter>
00005 #include <QPen>
00006
00007 #include "AppConstants.h"
00008 #include "EdgeItem.h"
00009
00010 class UndirectedEdgeItem : public EdgeItem {
00011 public:
00012     UndirectedEdgeItem(Edge*, NodeItem*, NodeItem*, bool = false);
00013
00014     QRectF      boundingRect() const override;
00015     QPainterPath shape() const override;
00016     void        paint(QPainter*, const QStyleOptionGraphicsItem*, QWidget*) override;
00017     QPainterPath edgePath() const override;
00018 };
00019
00020 #endif // UNDIRECTED_EDGEITEM_H

```

7.85 Source/Algorithm/Algorithm.cpp File Reference

```
#include "Algorithm.h"
```

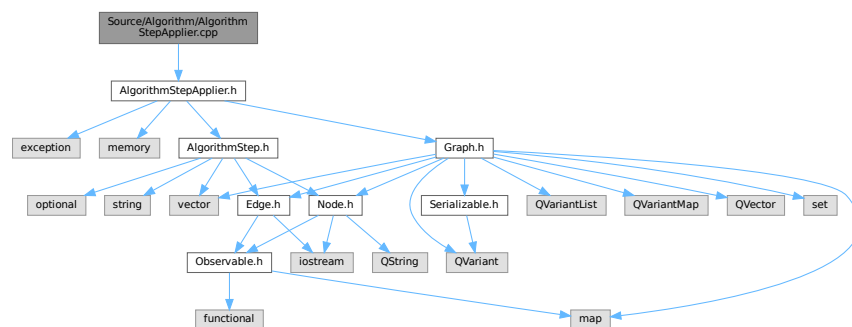
Include dependency graph for Algorithm.cpp:



7.86 Source/Algorithm/AlgorithmStepApplier.cpp File Reference

```
#include "AlgorithmStepApplier.h"
```

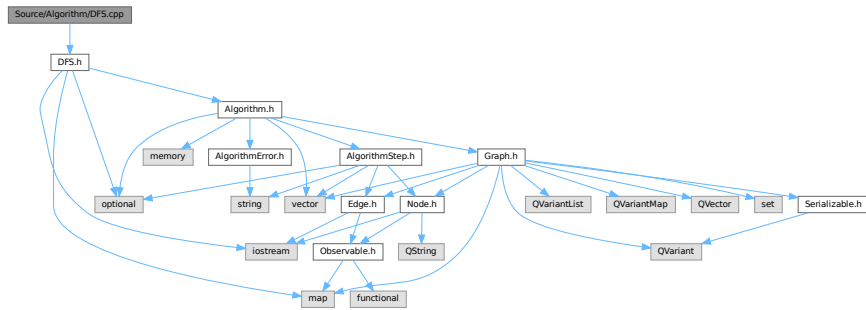
Include dependency graph for AlgorithmStepApplier.cpp:



7.90 Source/Algorithm/DFS.cpp File Reference

```
#include "DFS.h"
```

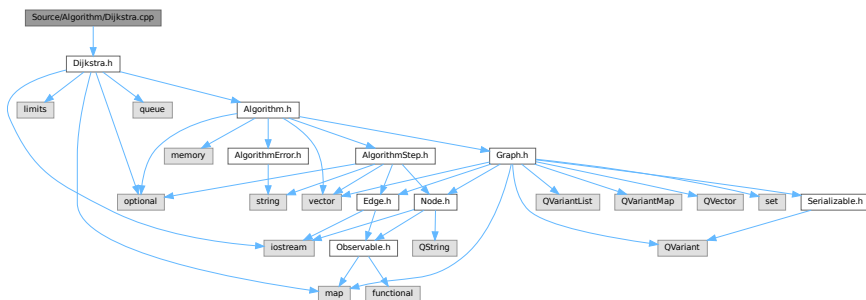
Include dependency graph for DFS.cpp:



7.91 Source/Algorithm/Dijkstra.cpp File Reference

```
#include "Dijkstra.h"
```

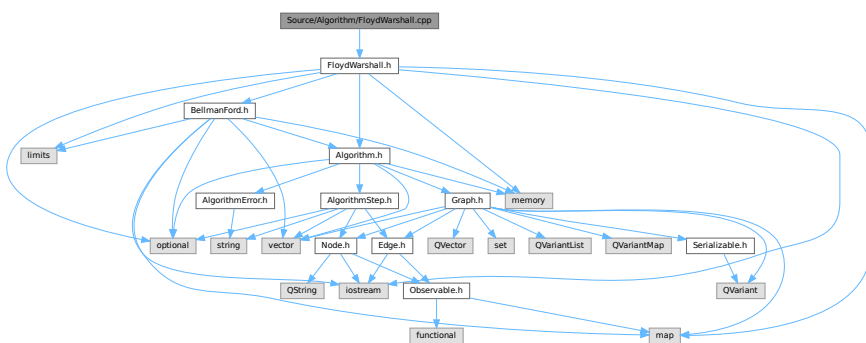
Include dependency graph for Dijkstra.cpp:



7.92 Source/Algorithm/FloydWarshall.cpp File Reference

```
#include "FloydWarshall.h"
```

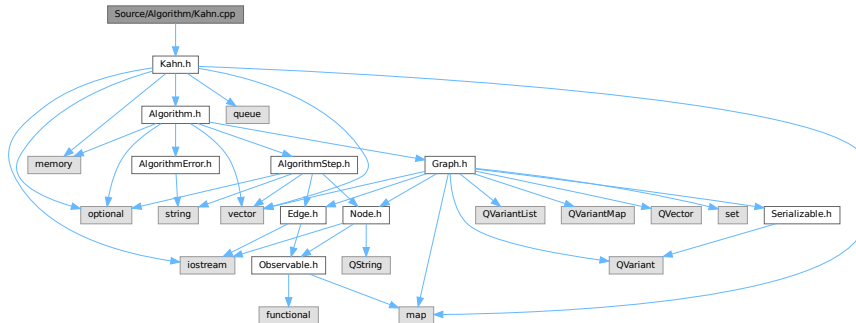
Include dependency graph for FloydWarshall.cpp:



7.93 Source/Algorithm/Kahn.cpp File Reference

```
#include "Kahn.h"
```

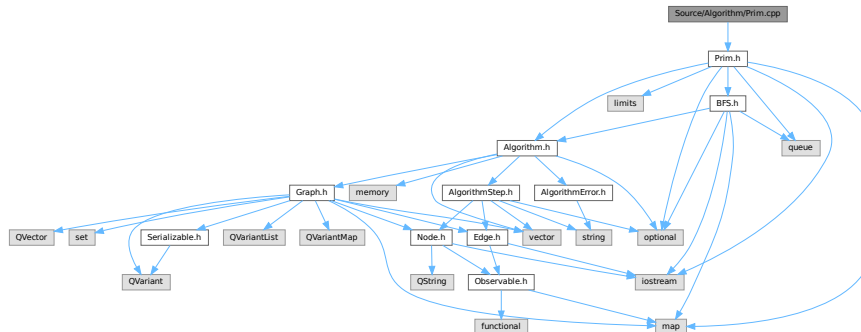
Include dependency graph for Kahn.cpp:



7.94 Source/Algorithm/Prim.cpp File Reference

```
#include "Prim.h"
```

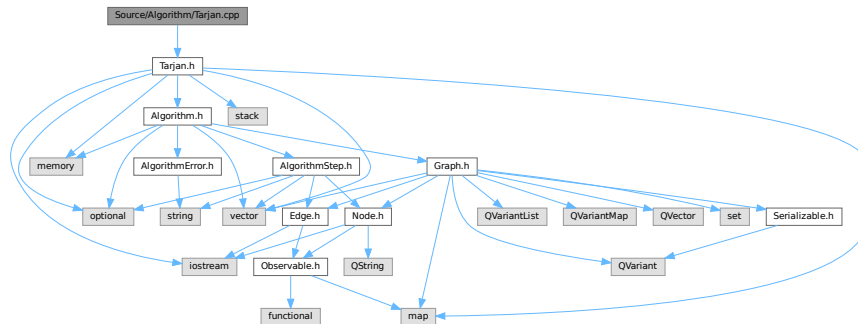
Include dependency graph for Prim.cpp:



7.95 Source/Algorithm/Tarjan.cpp File Reference

```
#include "Tarjan.h"
```

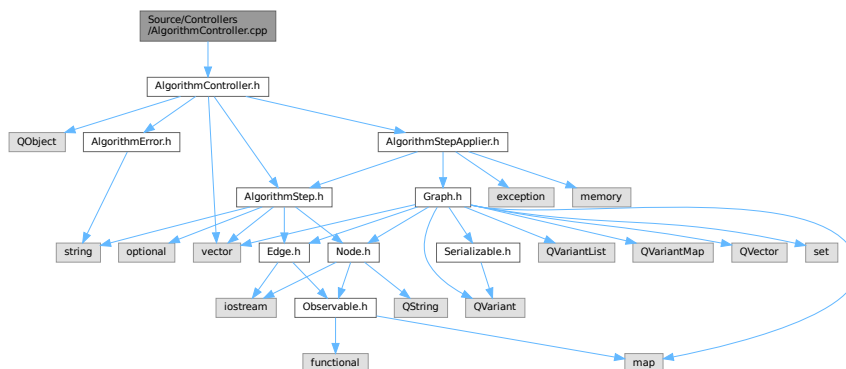
Include dependency graph for Tarjan.cpp:



7.96 Source/Controllers/AlgorithmController.cpp File Reference

```
#include "AlgorithmController.h"
```

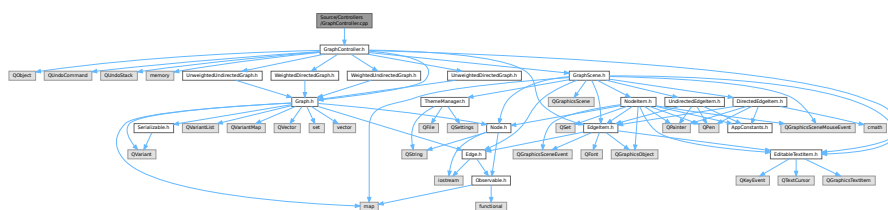
Include dependency graph for AlgorithmController.cpp:



7.97 Source/Controllers/GraphController.cpp File Reference

```
#include "GraphController.h"
```

Include dependency graph for GraphController.cpp:



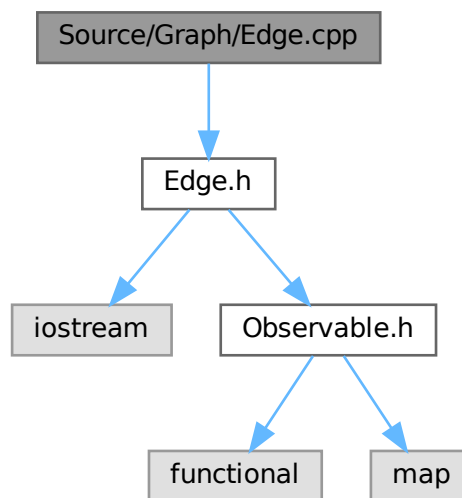
Classes

- struct [EdgeSnapshot](#)
- struct [NodeSnapshot](#)
- class [AddNodeCommand](#)
- class [RemoveNodeCommand](#)
- class [AddEdgeCommand](#)
- class [RemoveEdgeCommand](#)
- class [EditEdgeWeightCommand](#)
- class [EditNodeNameCommand](#)
- class [AddNodeAndEdgeCommand](#)
- class [MoveNodeCommand](#)
- class [ClearGraphCommand](#)

7.98 Source/Graph/Edge.cpp File Reference

```
#include "Edge.h"
```

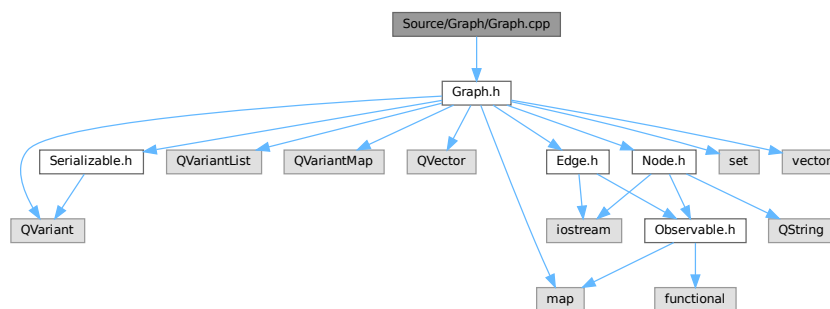
Include dependency graph for Edge.cpp:



7.99 Source/Graph/Graph.cpp File Reference

```
#include "Graph.h"
```

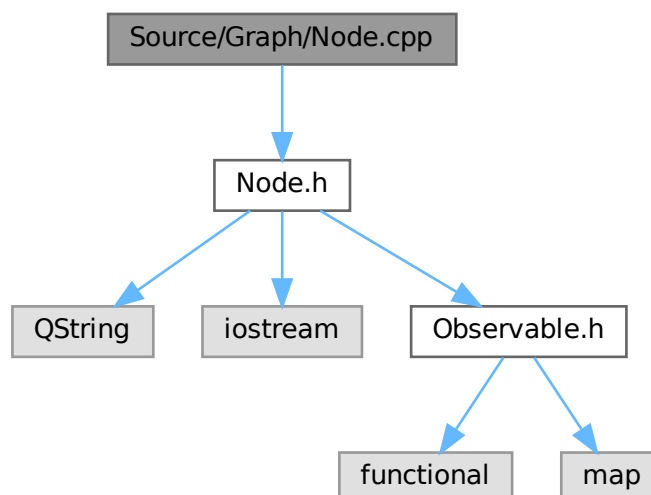
Include dependency graph for Graph.cpp:



7.100 Source/Graph/Node.cpp File Reference

```
#include "Node.h"
```

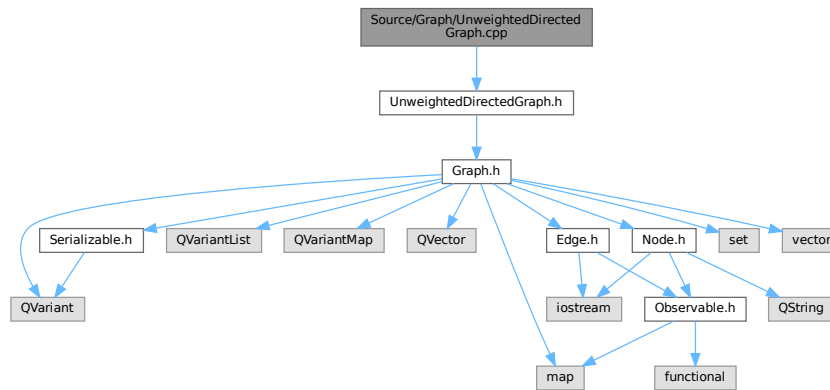
Include dependency graph for Node.cpp:



7.101 Source/Graph/UnweightedDirectedGraph.cpp File Reference

```
#include "UnweightedDirectedGraph.h"
```

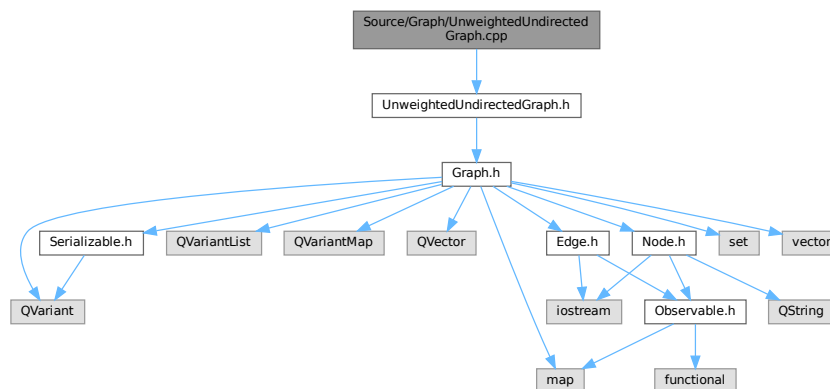
Include dependency graph for UnweightedDirectedGraph.cpp:



7.102 Source/Graph/UnweightedUndirectedGraph.cpp File Reference

```
#include "UnweightedUndirectedGraph.h"
```

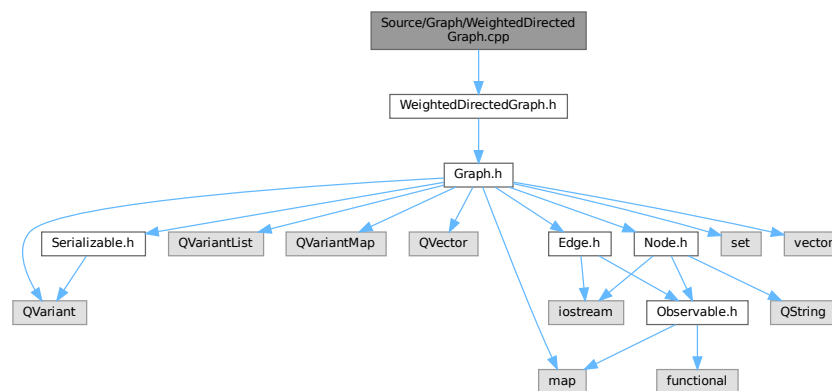
Include dependency graph for UnweightedUndirectedGraph.cpp:



7.103 Source/Graph/WeightedDirectedGraph.cpp File Reference

```
#include "WeightedDirectedGraph.h"
```

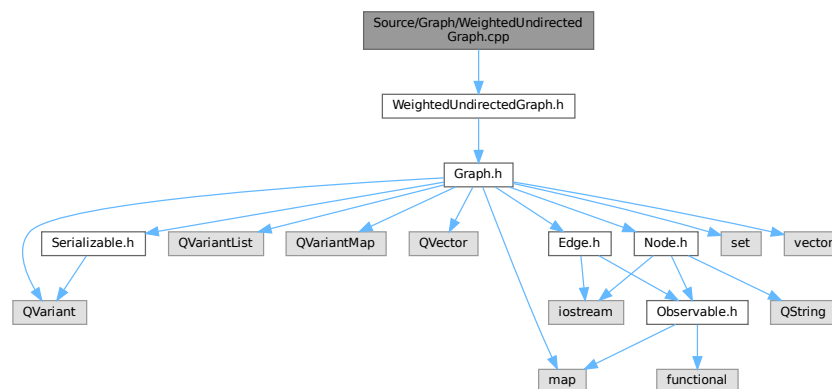

Include dependency graph for WeightedDirectedGraph.cpp:



7.104 Source/Graph/WeightedUndirectedGraph.cpp File Reference

```
#include "WeightedUndirectedGraph.h"
```

Include dependency graph for WeightedUndirectedGraph.cpp:



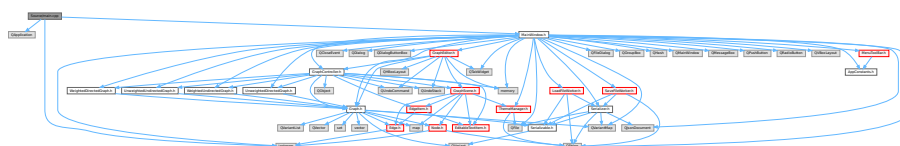
7.105 Source/main.cpp File Reference

```
#include <QApplication>
```

```
#include <iostream>
```

```
#include "MainWindow.h"
```

Include dependency graph for main.cpp:



Functions

- `int main (int argc, char *argv[ ])`

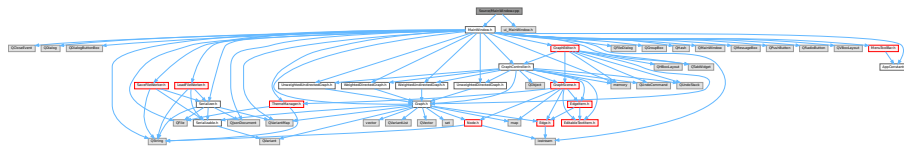
7.105.1 Function Documentation

7.105.1.1 main()

```
int main (
    int argc,
    char * argv[ ] )
```

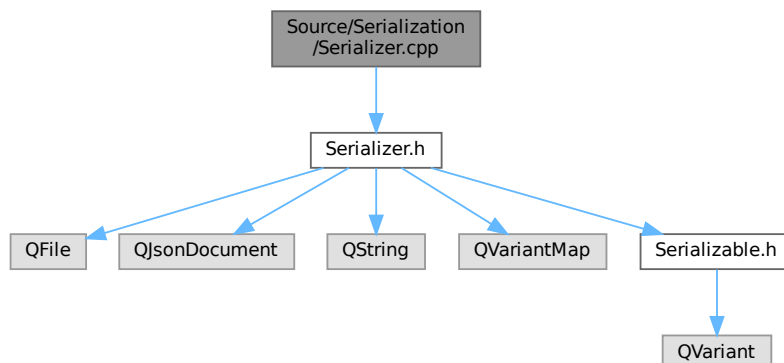
7.106 Source/MainWindow.cpp File Reference

```
#include "MainWindow.h"
#include "ui_MainWindow.h"
Include dependency graph for MainWindow.cpp:
```



7.107 Source/Serialization/Serializer.cpp File Reference

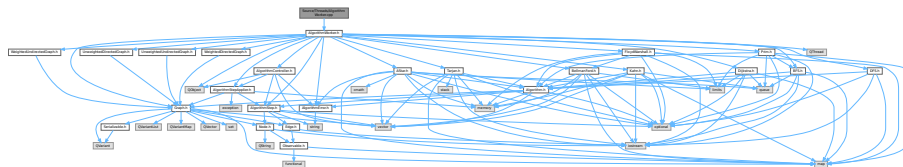
```
#include "Serializer.h"
Include dependency graph for Serializer.cpp:
```



7.108 Source/Threads/AlgorithmWorker.cpp File Reference

```
#include "AlgorithmWorker.h"
```

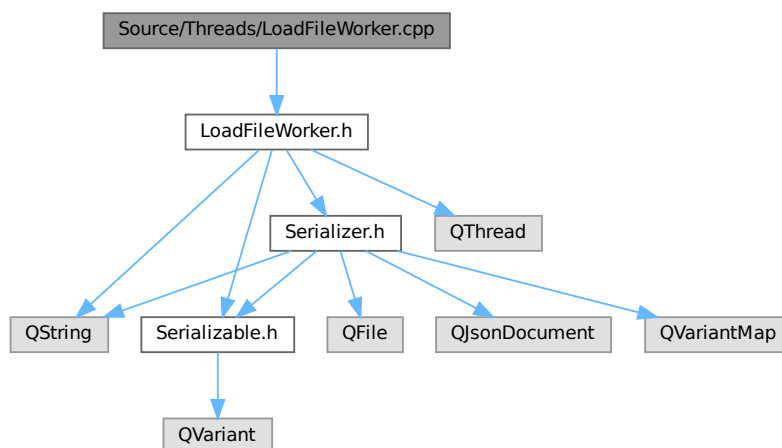
Include dependency graph for AlgorithmWorker.cpp:



7.109 Source/Threads/LoadFileWorker.cpp File Reference

```
#include "LoadFileWorker.h"
```

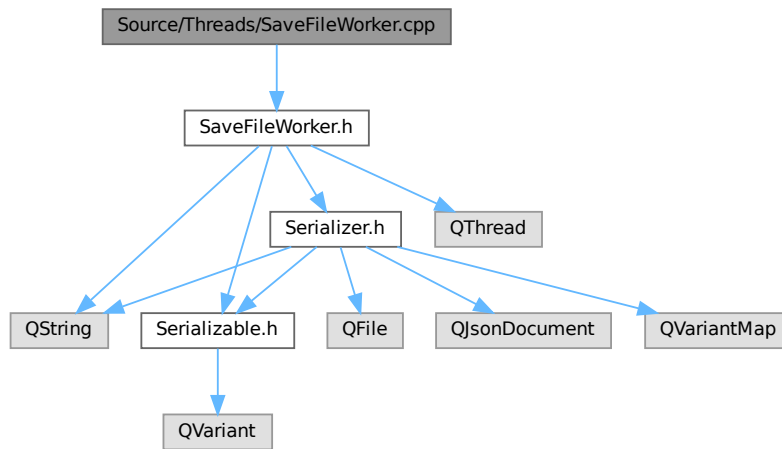
Include dependency graph for LoadFileWorker.cpp:



7.110 Source/Threads/SaveFileWorker.cpp File Reference

```
#include "SaveFileWorker.h"
```

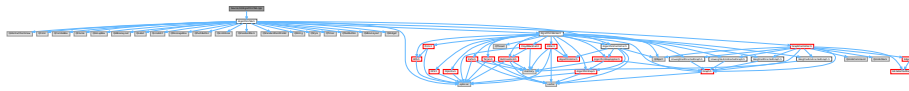
Include dependency graph for SaveFileWorker.cpp:



7.111 Source/UI/AlgorithmTab.cpp File Reference

```
#include "AlgorithmTab.h"
```

Include dependency graph for AlgorithmTab.cpp:



7.112 Source/UI/GraphEditor.cpp File Reference

```
#include "GraphEditor.h"
```

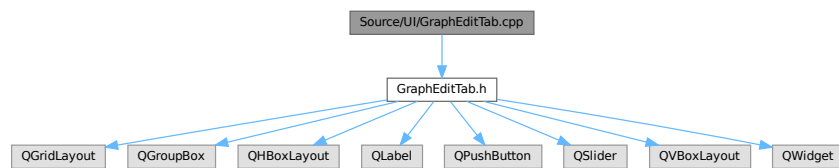
Include dependency graph for GraphEditor.cpp:



7.113 Source/UI/GraphEditTab.cpp File Reference

```
#include "GraphEditTab.h"
```

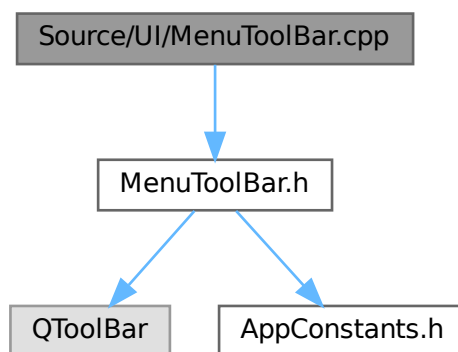
Include dependency graph for GraphEditTab.cpp:



7.114 Source/UI/MenuToolBar.cpp File Reference

```
#include "MenuToolBar.h"
```

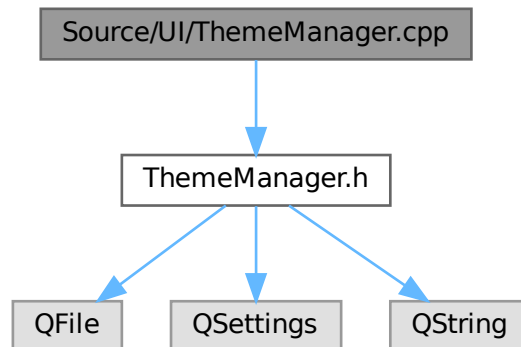
Include dependency graph for MenuToolBar.cpp:



7.115 Source/UI/ThemeManager.cpp File Reference

```
#include "ThemeManager.h"
```

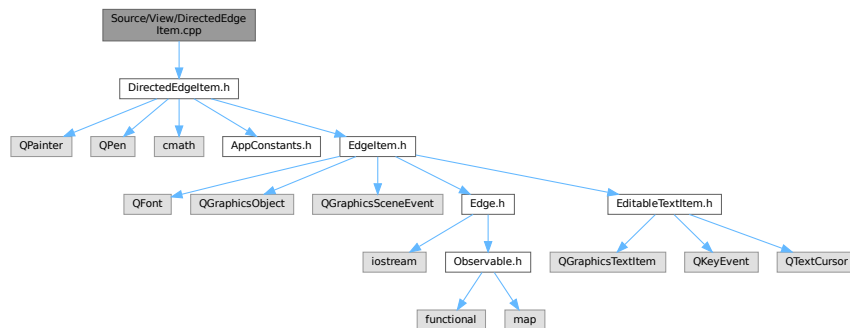
Include dependency graph for ThemeManager.cpp:



7.116 Source/View/DirectedEdgeItem.cpp File Reference

```
#include "DirectedEdgeItem.h"
```

Include dependency graph for DirectedEdgeItem.cpp:

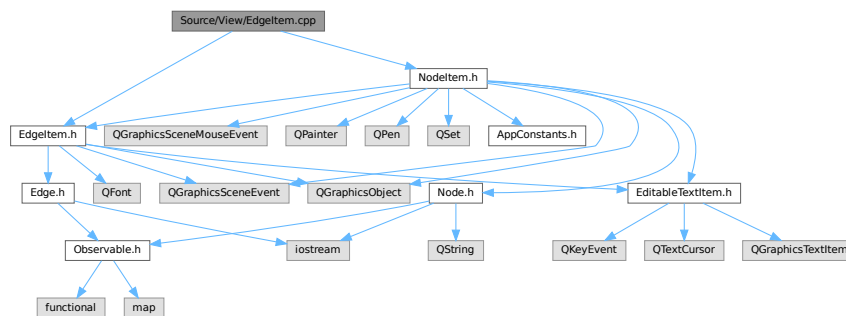


7.117 Source/View/EdgeItem.cpp File Reference

```
#include "EdgeItem.h"
```

```
#include "NodeItem.h"
```

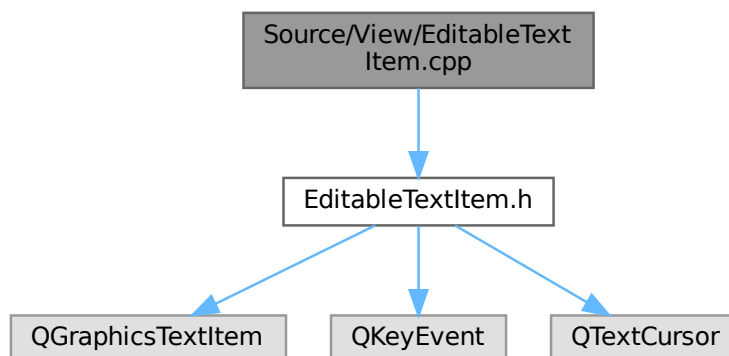
Include dependency graph for Edgeltem.cpp:



7.118 Source/View/EditableTextItem.cpp File Reference

```
#include "EditableTextItem.h"
```

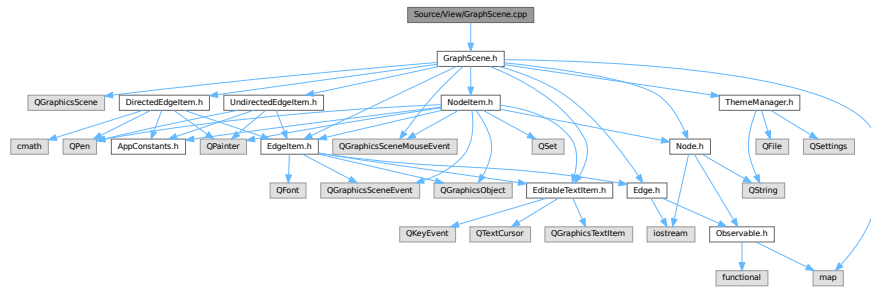
Include dependency graph for EditableTextItem.cpp:



7.119 Source/View/GraphScene.cpp File Reference

```
#include "GraphScene.h"
```

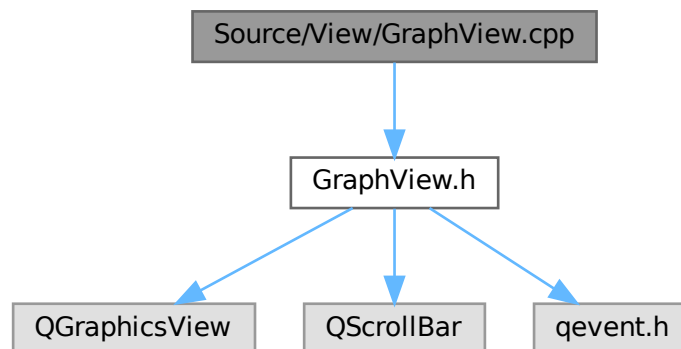
Include dependency graph for GraphScene.cpp:



7.120 Source/View/GraphView.cpp File Reference

```
#include "GraphView.h"
```

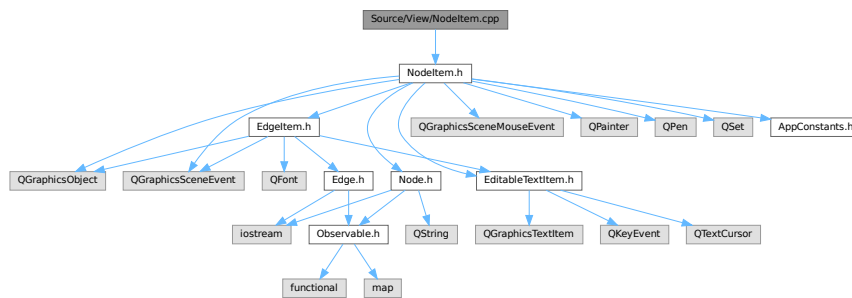
Include dependency graph for GraphView.cpp:



7.121 Source/View/NodeItem.cpp File Reference

```
#include "NodeItem.h"
```

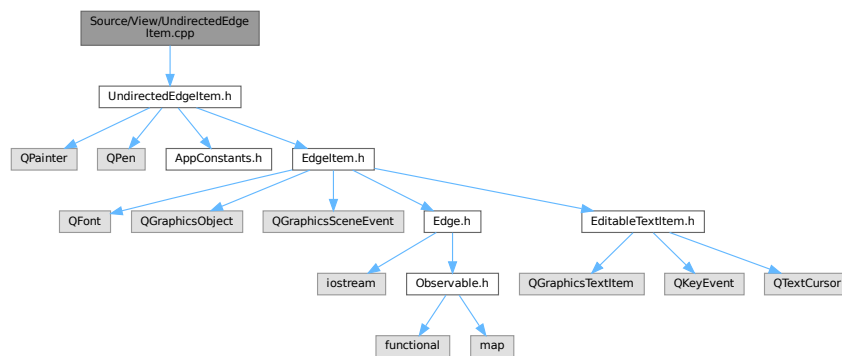

Include dependency graph for NodeItem.cpp:



7.122 Source/View/UndirectedEdgeItem.cpp File Reference

```
#include "UndirectedEdgeItem.h"
```

Include dependency graph for UndirectedEdgeItem.cpp:



Index

- ~Algorithm
 - Algorithm, [23](#)
- ~AlgorithmController
 - AlgorithmController, [28](#)
- ~AlgorithmTab
 - AlgorithmTab, [39](#)
- ~Edge
 - Edge, [76](#)
- ~EdgeItem
 - EdgeItem, [82](#)
- ~Graph
 - Graph, [103](#)
- ~GraphEditTab
 - GraphEditTab, [123](#)
- ~GraphEditor
 - GraphEditor, [118](#)
- ~MainWindow
 - MainWindow, [148](#)
- ~MenuToolBar
 - MenuToolBar, [153](#)
- ~Node
 - Node, [160](#)
- ~NodeItem
 - NodeItem, [166](#)
- ~Serializable
 - Serializable, [187](#)
- ~Serializer
 - Serializer, [188](#)
- ~UnweightedDirectedGraph
 - UnweightedDirectedGraph, [204](#)
- ~UnweightedUndirectedGraph
 - UnweightedUndirectedGraph, [208](#)
- ~WeightedDirectedGraph
 - WeightedDirectedGraph, [212](#)
- ~WeightedUndirectedGraph
 - WeightedUndirectedGraph, [216](#)
- Active
 - Node.h, [246](#)
- ADD
 - GraphScene, [130](#)
- addEdge
 - Graph, [103](#)
 - GraphController, [111](#)
 - GraphScene, [130](#)
 - NodeItem, [166](#)
 - UnweightedDirectedGraph, [205](#)
 - UnweightedUndirectedGraph, [209](#)
 - WeightedDirectedGraph, [213](#)
 - WeightedUndirectedGraph, [217](#)
- AddEdgeCommand, [13](#)
 - AddEdgeCommand, [15](#)
 - m_c, [15](#)
 - m_edgId, [15](#)
 - m_from, [15](#)
 - m_hasId, [15](#)
 - m_to, [16](#)
 - m_weight, [16](#)
 - redo, [15](#)
 - undo, [15](#)
- addEdgeNoHistory
 - GraphController, [111](#)
- addEdgeNoHistoryById
 - GraphController, [111](#)
- addEdgeRequest
 - GraphScene, [130](#)
- addEdgeSerialized
 - Graph, [103](#)
- addNode
 - Graph, [104](#)
 - GraphController, [111](#)
 - GraphScene, [130](#)
- addNodeAndEdge
 - GraphController, [111](#)
 - GraphScene, [130](#)
- AddNodeAndEdgeCommand, [16](#)
 - AddNodeAndEdgeCommand, [18](#)
 - m_c, [18](#)
 - m_fromId, [18](#)
 - m_hasId, [18](#)
 - m_newId, [18](#)
 - m_pos, [18](#)
 - redo, [18](#)
 - undo, [18](#)
- addNodeAndEdgeRequest
 - GraphScene, [130](#)
- AddNodeCommand, [19](#)
 - AddNodeCommand, [21](#)
 - m_c, [21](#)
 - m_hasId, [21](#)
 - m_id, [21](#)
 - m_pos, [21](#)
 - redo, [21](#)
 - undo, [21](#)
- addNodeNoHistory
 - GraphController, [111](#)
- addNodeRequest
 - GraphScene, [130](#)
- addNodeWithIdNoHistory

- GraphController, 111
- addObserver
 - Observable< T >, 172
- addRequested
 - GraphEditTab, 123
- addStep
 - Algorithm, 23
- AddToPath
 - AlgorithmStep.h, 223
- AddToTopologicalOrder
 - AlgorithmStep.h, 223
- adjust
 - EdgeItem, 82
- adjustPointsGeometry
 - EdgeItem, 82
- adjustWeightGeometry
 - EdgeItem, 82
- Algorithm, 22
 - ~Algorithm, 23
 - addStep, 23
 - Algorithm, 23
 - clearSteps, 23
 - execute, 23
 - getSteps, 23
 - m_graph, 24
 - m_resultString, 24
 - m_steps, 24
 - resultString, 23
- AlgorithmController, 25
 - ~AlgorithmController, 28
 - AlgorithmController, 28
 - clear, 29
 - isFinished, 29
 - loadSteps, 29
 - m_applier, 30
 - m_currentIndex, 30
 - m_redoStack, 30
 - m_resultString, 30
 - m_steps, 30
 - m_undoStack, 30
 - nextStep, 29
 - onAlgorithmError, 29
 - prevStep, 29
 - redo, 29
 - requestErrorDialog, 29
 - reset, 29
 - resultString, 29
 - setResultString, 30
 - undo, 30
- AlgorithmError, 31
 - m_message, 31
 - m_type, 31
- AlgorithmError.h
 - AlgorithmErrorType, 221
 - GoalNodeMissing, 221
 - GraphHasCycle, 221
 - GraphHasNegativeCycle, 221
 - GraphNotConnected, 221
 - GraphNotInitialized, 221
 - GraphTypeInvalid, 221
 - NegativeEdgeWeights, 221
 - NoPathFound, 221
 - StartNodeMissing, 221
- algorithmErrorOccurred
 - AlgorithmWorker, 45
- AlgorithmErrorType
 - AlgorithmError.h, 221
- AlgorithmStep, 32
 - m_from, 32
 - m_node, 32
 - m_prevEdgeState, 33
 - m_prevNodeState, 33
 - m_to, 33
 - m_type, 33
- AlgorithmStep.h
 - AddToPath, 223
 - AddToTopologicalOrder, 223
 - AssignComponent, 223
 - DefaultType, 223
 - ExamineEdge, 223
 - ProcessNode, 223
 - RelaxEdge, 223
 - SelectEdge, 223
 - StepType, 223
 - UpdateDistance, 223
 - VisitNode, 223
- AlgorithmStepApplier, 33
 - AlgorithmStepApplier, 34
 - apply, 34
 - check, 34
 - m_graph, 35
 - resetGraphState, 34
 - setGraph, 34
 - stepToEdgeState, 35
 - stepToNodeState, 35
 - undo, 35
- AlgorithmTab, 35
 - ~AlgorithmTab, 39
 - AlgorithmTab, 39
 - Finished, 39
 - graphEditAllowedChanged, 39
 - Idle, 39
 - initIcons, 39
 - initLayout, 39
 - m_algorithmCombo, 40
 - m_algorithmController, 40
 - m_applier, 40
 - m_currentConfig, 40
 - m_endLabel, 40
 - m_endNodeEdit, 41
 - m_endRow, 41
 - m_graphController, 41
 - m_graphEditAllowed, 41
 - m_helpBtn, 41
 - m_legendContainer, 41
 - m_legendLayout, 41

- m_legendScroll, 41
- m_nextBtn, 41
- m_noInputLabel, 41
- m_pauseBtn, 42
- m_playBtn, 42
- m_prevBtn, 42
- m_restartBtn, 42
- m_startLabel, 42
- m_startNodeEdit, 42
- m_startRow, 42
- m_state, 42
- m_timer, 42
- m_worker, 42
- makeLegendItem, 39
- Paused, 39
- Playing, 39
- RunState, 38
- selectedConfig, 39
- showAlgorithmErrorDialog, 39
- startTimerForPlay, 40
- updateControls, 40
- updateLegendForAlgorithm, 40
- updateUiForAlgorithm, 40
- AlgorithmTab::AlgorithmConfig, 24
 - m_algorithmName, 25
 - m_endNode, 25
 - m_startNode, 25
 - operator!=, 25
 - operator==, 25
- AlgorithmWorker, 43
 - algorithmErrorOccurred, 45
 - AlgorithmWorker, 45
 - m_algorithm, 46
 - m_end, 46
 - m_graph, 46
 - m_start, 46
 - resultReady, 45
 - run, 45
 - stepsReady, 45
- AppConstants, 9
 - BaseEdgeWidth, 9
 - btnSize, 9
 - btnSpacing, 10
 - createGraphDialogMinHeight, 10
 - createGraphDialogMinWidth, 10
 - defaultBorderWidth, 10
 - defaultRadius, 10
 - EdgePadding, 10
 - graphPageDefaultTitle, 10
 - graphTypeDialogErrorMinHeight, 10
 - graphTypeDialogErrorMinWidth, 10
 - legendMinHeight, 10
 - MaxNodeScale, 11
 - MinNodeScale, 11
 - NodeScale, 11
 - optionBtnSize, 11
 - optionIconSize, 11
 - startPageTitle, 11
 - windowHeight, 11
 - windowMinHeight, 11
 - windowMinWidth, 11
 - windowWidth, 11
- apply
 - AlgorithmStepApplier, 34
- applyEditButtonsState
 - GraphEditTab, 123
- applyTheme
 - GraphScene, 131
- arrowPath
 - DirectedEdgeItem, 72
- AssignComponent
 - AlgorithmStep.h, 223
- AssignedComponent
 - Node.h, 246
- AStar, 46
 - AStar, 49
 - aStar, 49
 - checkConditions, 49
 - execute, 49
 - getPath, 50
 - getTotalCost, 50
 - heuristic, 50
 - m_path, 50
 - m_totalCost, 50
 - resultString, 50
- aStar
 - AStar, 49
- BaseEdgeWidth
 - AppConstants, 9
- BellmanFord, 51
 - BellmanFord, 53
 - bellmanFord, 53
 - checkConditions, 53
 - execute, 53
 - hasNegativeCycle, 54
 - m_hasNegativeCycle, 54
 - m_minDistance, 54
 - m_resultString, 54
 - resultString, 54
- bellmanFord
 - BellmanFord, 53
- BFS, 55
 - BFS, 57
 - bfs, 57
 - checkConditions, 57
 - execute, 57
 - getVisited, 57
 - m_order, 58
 - m_visited, 58
 - resultString, 58
- bfs
 - BFS, 57
- boundingRect
 - DirectedEdgeItem, 72
 - EdgeItem, 82
 - NodeItem, 166

- UndirectedEdgeItem, 201
- btnSize
 - AppConstants, 9
- btnSpacing
 - AppConstants, 10
- buildScene
 - GraphController, 112
- calculateColor
 - EdgeItem, 82
 - NodeItem, 166
- calculateNormal
 - EdgeItem, 82
- capture
 - ClearGraphCommand, 60
 - RemoveNodeCommand, 182
- center
 - EditableTextItem, 90
- centerText
 - EditableTextItem, 90
- changeThemeAction
 - MenuToolBar, 153
- check
 - AlgorithmStepApplier, 34
- checkConditions
 - AStar, 49
 - BellmanFord, 53
 - BFS, 57
 - DFS, 63
 - Dijkstra, 67
 - FloydWarshall, 100
 - Kahn, 142
 - Prim, 176
 - Tarjan, 193
- clear
 - AlgorithmController, 29
 - Graph, 104
 - GraphController, 112
 - GraphScene, 131
- ClearGraphCommand, 58
 - capture, 60
 - ClearGraphCommand, 60
 - m_c, 60
 - m_edges, 60
 - m_nodes, 60
 - redo, 60
 - undo, 60
- clearNoHistory
 - GraphController, 112
- clearRequested
 - GraphEditTab, 124
- clearScene
 - GraphController, 112
- clearSteps
 - Algorithm, 23
- closeEvent
 - MainWindow, 148
- connectGraphModifiedSignal
 - MainWindow, 148
- connectNodes
 - EdgeItem, 82
- connectScene
 - GraphController, 112
- createGraph
 - GraphController, 112
- createGraphAction
 - MenuToolBar, 153
- createGraphDialogMinHeight
 - AppConstants, 10
- createGraphDialogMinWidth
 - AppConstants, 10
- currentTheme
 - ThemeManager, 196
- DARK
 - ThemeManager, 196
- Default
 - Edge.h, 243
 - Node.h, 246
- defaultBorderWidth
 - AppConstants, 10
- defaultRadius
 - AppConstants, 10
- DefaultType
 - AlgorithmStep.h, 223
- DFS, 61
 - checkConditions, 63
 - DFS, 63
 - dfs, 63
 - execute, 63
 - getVisited, 63
 - m_order, 64
 - m_visited, 64
 - resultString, 64
- dfs
 - DFS, 63
- Dijkstra, 64
 - checkConditions, 67
 - Dijkstra, 67
 - dijkstra, 67
 - execute, 67
 - m_resultString, 68
 - resultString, 67
- dijkstra
 - Dijkstra, 67
- DirectedEdgeItem, 68
 - arrowPath, 72
 - boundingRect, 72
 - DirectedEdgeItem, 72
 - edgePath, 72
 - m_arrowAngle, 73
 - m_arrowSize, 73
 - m_skewness, 73
 - paint, 72
 - shape, 72
- disableScene
 - GraphController, 112
 - GraphScene, 131

- disconnectNodes
 - EdgeItem, 83
- Edge, 73
 - ~Edge, 76
 - Edge, 76
 - endNode, 76
 - getId, 76
 - getState, 77
 - getWeight, 77
 - m_id, 77
 - m_idEndNode, 77
 - m_idStartNode, 77
 - m_state, 77
 - m_weight, 78
 - setState, 77
 - setWeight, 77
 - startNode, 77
- Edge.h
 - Default, 243
 - EdgeState, 243
 - Examined, 243
 - Relaxed, 243
 - Selected, 243
- EdgeItem, 78
 - ~EdgeItem, 82
 - adjust, 82
 - adjustPointsGeometry, 82
 - adjustWeightGeometry, 82
 - boundingRect, 82
 - calculateColor, 82
 - calculateNormal, 82
 - connectNodes, 82
 - disconnectNodes, 83
 - EdgeItem, 82
 - edgePath, 83
 - edgeSelected, 83
 - editEdgeWeightRequest, 83
 - getEdgeCenter, 83
 - getWeightPosition, 83
 - hasWeight, 83
 - initEdgeWeight, 83
 - m_destNode, 85
 - m_destPoint, 85
 - m_hasWeight, 85
 - m_modelEdge, 85
 - m_observerId, 85
 - m_penWidth, 85
 - m_shapeStroke, 86
 - m_sourceNode, 86
 - m_sourcePoint, 86
 - m_weight, 86
 - modelEdge, 83
 - mouseDoubleClickEvent, 84
 - mousePressEvent, 84
 - onEdgeUpdated, 84
 - onEdgeWeightChanged, 84
 - paint, 84
 - setHasWeight, 84
 - setModelEdge, 84
 - setWeight, 84
 - shape, 85
 - updateSize, 85
 - weight, 85
- EdgePadding
 - AppConstants, 10
- edgePath
 - DirectedEdgeItem, 72
 - EdgeItem, 83
 - UndirectedEdgeItem, 201
- edges
 - NodeItem, 166
- edgeSelected
 - EdgeItem, 83
- EdgeSnapshot, 86
 - m_from, 87
 - m_id, 87
 - m_to, 87
 - m_weight, 87
- EdgeState
 - Edge.h, 243
- EDIT
 - GraphScene, 130
- EditableTextItem, 87
 - center, 90
 - centerText, 90
 - EditableTextItem, 90
 - finishEditing, 90
 - keyPressEvent, 90
 - m_center, 91
 - m_oldText, 91
 - oldText, 90
 - setCenter, 90
 - setEditGraphSceneState, 91
 - setOldText, 91
 - startEditing, 91
 - textCommitted, 91
- editEdgeWeight
 - GraphController, 112
- EditEdgeWeightCommand, 92
 - EditEdgeWeightCommand, 94
 - m_afterText, 94
 - m_afterW, 94
 - m_beforeText, 94
 - m_beforeW, 94
 - m_c, 95
 - m_edgId, 95
 - redo, 94
 - undo, 94
- editEdgeWeightRequest
 - EdgeItem, 83
 - GraphScene, 131
- editNodeName
 - GraphController, 112
- EditNodeNameCommand, 95
 - EditNodeNameCommand, 97
 - m_afterName, 97

- m_beforeName, 97
 - m_c, 97
 - m_nodId, 97
 - redo, 97
 - undo, 97
- editNodeNameRequest
 - GraphScene, 131
 - NodeItem, 166
- enableScene
 - GraphController, 113
 - GraphScene, 131
- endNode
 - Edge, 76
- Examined
 - Edge.h, 243
- ExamineEdge
 - AlgorithmStep.h, 223
- execute
 - Algorithm, 23
 - AStar, 49
 - BellmanFord, 53
 - BFS, 57
 - DFS, 63
 - Dijkstra, 67
 - FloydWarshall, 100
 - Kahn, 142
 - Prim, 176
 - Tarjan, 193
- failed
 - LoadFileWorker, 145
 - SaveFileWorker, 185
- findEdgeItemById
 - GraphScene, 131
- findEdgeItemByNodes
 - GraphScene, 131
- findNodeItemById
 - GraphScene, 131
- Finished
 - AlgorithmTab, 39
- finished
 - SaveFileWorker, 185
- finishEditing
 - EditableTextItem, 90
- FloydWarshall, 98
 - checkConditions, 100
 - execute, 100
 - FloydWarshall, 100
 - floydWarshall, 100
 - m_distances, 101
 - resultString, 100
- floydWarshall
 - FloydWarshall, 100
- fromVariant
 - Graph, 104
 - Serializable, 187
- getAdjacencyList
 - Graph, 104
- getComponentColor
 - Node, 160
- getEdge
 - Graph, 104, 105
- getEdgeCenter
 - EdgeItem, 83
- getEdges
 - Graph, 105
- getEdgesMutable
 - Graph, 105
- getId
 - Edge, 76
 - Node, 160
- getName
 - Node, 161
- getNode
 - Graph, 105
- getNodes
 - Graph, 105
- getNodeSizeSlider
 - GraphEditTab, 124
- getNodesMutable
 - Graph, 105
- getPath
 - AStar, 50
- getPosition
 - Node, 161
- getSorted
 - Kahn, 142
- getState
 - Edge, 77
 - Node, 161
- getSteps
 - Algorithm, 23
- getTotalCost
 - AStar, 50
- getVisited
 - BFS, 57
 - DFS, 63
- getWeight
 - Edge, 77
- getWeightPosition
 - EdgeItem, 83
- GoalNodeMissing
 - AlgorithmError.h, 221
- Graph, 101
 - ~Graph, 103
 - addEdge, 103
 - addEdgeSerialized, 103
 - addNode, 104
 - clear, 104
 - fromVariant, 104
 - getAdjacencyList, 104
 - getEdge, 104, 105
 - getEdges, 105
 - getEdgesMutable, 105
 - getNode, 105
 - getNodes, 105

- getNodesMutable, 105
- Graph, 103
- isDirected, 105
- isWeighted, 105
- m_adjacencyList, 106
- m_edgId, 106
- m_edges, 106
- m_nodeId, 106
- m_nodes, 106
- m_numOfEdges, 107
- m_numOfNodes, 107
- removeEdge, 106
- removeNode, 106
- toVariant, 106
- graph
 - GraphController, 113
- GraphController, 107
 - addEdge, 111
 - addEdgeNoHistory, 111
 - addEdgeNoHistoryById, 111
 - addNode, 111
 - addNodeAndEdge, 111
 - addNodeNoHistory, 111
 - addNodeWithIdNoHistory, 111
 - buildScene, 112
 - clear, 112
 - clearNoHistory, 112
 - clearScene, 112
 - connectScene, 112
 - createGraph, 112
 - disableScene, 112
 - editEdgeWeight, 112
 - editNodeName, 112
 - enableScene, 113
 - graph, 113
 - GraphController, 111
 - m_graph, 115
 - m_scene, 115
 - m_undoStack, 115
 - moveNode, 113
 - moveNodeNoHistory, 113
 - removeEdge, 113
 - removeEdgeNoHistory, 113
 - removeEdgeNoHistoryById, 113
 - removeNode, 113
 - removeNodeNoHistory, 114
 - resetScene, 114
 - restoreEdgeNoHistory, 114
 - scene, 114
 - sceneModified, 114
 - setAddSceneState, 114
 - setEdgeWeightNoHistory, 114
 - setEdgeWeightNoHistoryById, 114
 - setGraph, 115
 - setNodeNameNoHistory, 115
 - setRemoveSceneState, 115
 - undoStack, 115
 - updateNodePosition, 115
- graphController
 - GraphEditor, 118
- graphEditAllowedChanged
 - AlgorithmTab, 39
- GraphEditor, 116
 - ~GraphEditor, 118
 - graphController, 118
 - GraphEditor, 118
 - m_editTab, 119
 - m_graphController, 119
 - m_leftPlaceholder, 120
 - m_undoStack, 120
 - m_view, 120
 - onAddRequestTrigger, 118
 - onClearRequestTrigger, 118
 - onPanRequestTrigger, 119
 - onRedoRequestTrigger, 119
 - onRemoveRequestTrigger, 119
 - onUndoRequestTrigger, 119
 - onZoomInRequestTrigger, 119
 - onZoomOutRequestTrigger, 119
 - redoRequested, 119
 - undoRequested, 119
- GraphEditTab, 120
 - ~GraphEditTab, 123
 - addRequested, 123
 - applyEditButtonsState, 123
 - clearRequested, 124
 - getNodeSizeSlider, 124
 - GraphEditTab, 123
 - initLayout, 124
 - m_addBtn, 125
 - m_canRedo, 125
 - m_canUndo, 125
 - m_clearBtn, 125
 - m_graphEditsAllowed, 125
 - m_nodeSizeSlider, 125
 - m_panBtn, 125
 - m_redoBtn, 125
 - m_removeBtn, 126
 - m_undoBtn, 126
 - m_zoomInBtn, 126
 - m_zoomOutBtn, 126
 - panRequested, 124
 - redoRequested, 124
 - removeRequested, 124
 - setGraphEditsAllowed, 124
 - setRedoEnabled, 124
 - setUndoEnabled, 124
 - undoRequested, 124
 - zoomInRequested, 125
 - zoomOutRequested, 125
- GraphHasCycle
 - AlgorithmError.h, 221
- GraphHasNegativeCycle
 - AlgorithmError.h, 221
- GraphNotConnected
 - AlgorithmError.h, 221

- GraphNotInitialized
 - AlgorithmError.h, 221
- graphPageDefaultTitle
 - AppConstants, 10
- GraphScene, 126
 - ADD, 130
 - addEdge, 130
 - addEdgeRequest, 130
 - addNode, 130
 - addNodeAndEdge, 130
 - addNodeAndEdgeRequest, 130
 - addNodeRequest, 130
 - applyTheme, 131
 - clear, 131
 - disableScene, 131
 - EDIT, 130
 - editEdgeWeightRequest, 131
 - editNodeNameRequest, 131
 - enableScene, 131
 - findEdgeItemById, 131
 - findEdgeItemByNodes, 131
 - findNodeItemById, 131
 - GraphScene, 130
 - IDLE, 130
 - m_editLabel, 133
 - m_firstNodeSelect, 133
 - m_nodeItems, 133
 - m_previousState, 134
 - m_state, 134
 - mouseDoubleClickEvent, 132
 - mousePressEvent, 132
 - moveNodeRequest, 132
 - onEdgeSelectTrigger, 132
 - onNodeSelectTrigger, 132
 - REMOVE, 130
 - removeEdge, 132
 - removeEdgeRequest, 132
 - removeNode, 132
 - removeNodeRequest, 133
 - resetScene, 133
 - selectNode, 133
 - setEditGraphSceneTrigger, 133
 - setState, 133
 - State, 129
 - updateNodeScaling, 133
- graphTypeDialogErrorMinHeight
 - AppConstants, 10
- graphTypeDialogErrorMinWidth
 - AppConstants, 10
- GraphTypeInvalid
 - AlgorithmError.h, 221
- GraphView, 134
 - GraphView, 137
 - IDLE, 137
 - init, 138
 - m_lastMousePos, 139
 - m_oldState, 139
 - m_state, 139
 - m_zoomFactor, 139
 - mouseDoubleClickEvent, 138
 - mouseMoveEvent, 138
 - mousePressEvent, 138
 - mouseReleaseEvent, 138
 - PAN_ACTIVE, 137
 - PAN_IDLE, 137
 - resetState, 138
 - setState, 138
 - State, 137
 - state, 138
 - wheelEvent, 138
 - zoomIn, 138
 - zoomOut, 139
- hasNegativeCycle
 - BellmanFord, 54
- hasWeight
 - EdgeItem, 83
- Headers/Algorithm/Algorithm.h, 219, 220
- Headers/Algorithm/AlgorithmError.h, 220, 222
- Headers/Algorithm/AlgorithmStep.h, 222, 224
- Headers/Algorithm/AlgorithmStepApplier.h, 224, 225
- Headers/Algorithm/AStar.h, 226
- Headers/Algorithm/BellmanFord.h, 227, 228
- Headers/Algorithm/BFS.h, 228, 229
- Headers/Algorithm/DFS.h, 230, 231
- Headers/Algorithm/Dijkstra.h, 231, 232
- Headers/Algorithm/FloydWarshall.h, 232, 233
- Headers/Algorithm/Kahn.h, 234, 235
- Headers/Algorithm/Prim.h, 235, 236
- Headers/Algorithm/Tarjan.h, 236, 237
- Headers/Controllers/AlgorithmController.h, 238, 239
- Headers/Controllers/GraphController.h, 240, 241
- Headers/Graph/Edge.h, 242, 243
- Headers/Graph/Graph.h, 243, 244
- Headers/Graph/Node.h, 245, 246
- Headers/Graph/Observable.h, 247, 248
- Headers/Graph/UnweightedDirectedGraph.h, 248, 249
- Headers/Graph/UnweightedUndirectedGraph.h, 250
- Headers/Graph/WeightedDirectedGraph.h, 251, 252
- Headers/Graph/WeightedUndirectedGraph.h, 252, 253
- Headers/MainWindow.h, 253, 254
- Headers/Serialization/Serializable.h, 256
- Headers/Serialization/Serializer.h, 257
- Headers/Threads/AlgorithmWorker.h, 258, 259
- Headers/Threads/LoadFileWorker.h, 260, 261
- Headers/Threads/SaveFileWorker.h, 261, 262
- Headers/UI/AlgorithmTab.h, 262, 263
- Headers/UI/AppConstants.h, 265, 266
- Headers/UI/GraphEditor.h, 266, 267
- Headers/UI/GraphEditTab.h, 268, 269
- Headers/UI/MenuToolBar.h, 270, 271
- Headers/UI/ThemeManager.h, 271, 272
- Headers/View/DirectedEdgeItem.h, 273, 274
- Headers/View/EdgeItem.h, 274, 275
- Headers/View/EditableTextItem.h, 276, 277
- Headers/View/GraphScene.h, 278, 279
- Headers/View/GraphView.h, 280, 281

- Headers/View/NodeItem.h, [281](#), [282](#)
- Headers/View/UndirectedEdgeItem.h, [283](#), [285](#)
- helpAction
 - MenuToolBar, [153](#)
- heuristic
 - AStar, [50](#)
- IDLE
 - GraphScene, [130](#)
 - GraphView, [137](#)
- Idle
 - AlgorithmTab, [39](#)
- init
 - GraphView, [138](#)
 - Tarjan, [193](#)
- initEdgeWeight
 - EdgeItem, [83](#)
- initIcons
 - AlgorithmTab, [39](#)
- initLayout
 - AlgorithmTab, [39](#)
 - GraphEditTab, [124](#)
- initMenuToolBar
 - MainWindow, [149](#)
- InPath
 - Node.h, [246](#)
- isDirected
 - Graph, [105](#)
 - UnweightedDirectedGraph, [205](#)
 - UnweightedUndirectedGraph, [209](#)
 - WeightedDirectedGraph, [213](#)
 - WeightedUndirectedGraph, [217](#)
- isFinished
 - AlgorithmController, [29](#)
- isWeighted
 - Graph, [105](#)
 - UnweightedDirectedGraph, [205](#)
 - UnweightedUndirectedGraph, [209](#)
 - WeightedDirectedGraph, [213](#)
 - WeightedUndirectedGraph, [217](#)
- itemChange
 - NodeItem, [166](#)
- Kahn, [140](#)
 - checkConditions, [142](#)
 - execute, [142](#)
 - getSorted, [142](#)
 - Kahn, [142](#)
 - kahn, [142](#)
 - m_sorted, [143](#)
 - resultString, [142](#)
- kahn
 - Kahn, [142](#)
- keyPressEvent
 - EditableTextItem, [90](#)
- label
 - NodeItem, [166](#)
- legendMinHeight
 - AppConstants, [10](#)
- LIGHT
 - ThemeManager, [196](#)
- load
 - Serializer, [188](#)
- loaded
 - LoadFileWorker, [145](#)
- LoadFileWorker, [143](#)
 - failed, [145](#)
 - loaded, [145](#)
 - LoadFileWorker, [144](#)
 - m_filePath, [145](#)
 - m_serializer, [145](#)
 - run, [145](#)
- loadSteps
 - AlgorithmController, [29](#)
- loadTheme
 - ThemeManager, [196](#)
- m_addBtn
 - GraphEditTab, [125](#)
- m_adjacencyList
 - Graph, [106](#)
- m_afterName
 - EditNodeNameCommand, [97](#)
- m_afterText
 - EditEdgeWeightCommand, [94](#)
- m_afterW
 - EditEdgeWeightCommand, [94](#)
- m_algorithm
 - AlgorithmWorker, [46](#)
- m_algorithmCombo
 - AlgorithmTab, [40](#)
- m_algorithmController
 - AlgorithmTab, [40](#)
- m_algorithmName
 - AlgorithmTab::AlgorithmConfig, [25](#)
- m_applier
 - AlgorithmController, [30](#)
 - AlgorithmTab, [40](#)
- m_arrivalTime
 - Tarjan, [194](#)
- m_arrowAngle
 - DirectedEdgeItem, [73](#)
- m_arrowSize
 - DirectedEdgeItem, [73](#)
- m_beforeName
 - EditNodeNameCommand, [97](#)
- m_beforeText
 - EditEdgeWeightCommand, [94](#)
- m_beforeW
 - EditEdgeWeightCommand, [94](#)
- m_borderWidth
 - NodeItem, [169](#)
- m_c
 - AddEdgeCommand, [15](#)
 - AddNodeAndEdgeCommand, [18](#)
 - AddNodeCommand, [21](#)
 - ClearGraphCommand, [60](#)

- EditEdgeWeightCommand, 95
 - EditNodeNameCommand, 97
 - MoveNodeCommand, 157
 - RemoveEdgeCommand, 179
 - RemoveNodeCommand, 182
- m_canRedo
 - GraphEditTab, 125
- m_canUndo
 - GraphEditTab, 125
- m_center
 - EditableTextItem, 91
- m_changeThemeAction
 - MenuToolBar, 154
- m_clearBtn
 - GraphEditTab, 125
- m_componentColor
 - Node, 161
- m_components
 - Tarjan, 194
- m_createGraphAction
 - MenuToolBar, 154
- m_currentConfig
 - AlgorithmTab, 40
- m_currentIndex
 - AlgorithmController, 30
- m_currentTheme
 - ThemeManager, 197
- m_destNode
 - EdgeItem, 85
- m_destPoint
 - EdgeItem, 85
- m_distances
 - FloydWarshall, 101
- m_edgedId
 - AddEdgeCommand, 15
 - EditEdgeWeightCommand, 95
 - Graph, 106
 - RemoveEdgeCommand, 179
- m_edges
 - ClearGraphCommand, 60
 - Graph, 106
 - NodeItem, 169
 - RemoveNodeCommand, 182
- m_editLabel
 - GraphScene, 133
- m_editor
 - MainWindow::TabInfo, 190
- m_editTab
 - GraphEditor, 119
- m_end
 - AlgorithmWorker, 46
- m_endLabel
 - AlgorithmTab, 40
- m_endNode
 - AlgorithmTab::AlgorithmConfig, 25
- m_endNodeEdit
 - AlgorithmTab, 41
- m_endRow
 - AlgorithmTab, 41
- m_filePath
 - LoadFileWorker, 145
 - MainWindow::TabInfo, 190
 - SaveFileWorker, 185
- m_firstNodeSelect
 - GraphScene, 133
- m_from
 - AddEdgeCommand, 15
 - AlgorithmStep, 32
 - EdgeSnapshot, 87
 - RemoveEdgeCommand, 179
- m_fromId
 - AddNodeAndEdgeCommand, 18
- m_graph
 - Algorithm, 24
 - AlgorithmStepApplier, 35
 - AlgorithmWorker, 46
 - GraphController, 115
- m_graphController
 - AlgorithmTab, 41
 - GraphEditor, 119
- m_graphEditAllowed
 - AlgorithmTab, 41
- m_graphEditsAllowed
 - GraphEditTab, 125
- m_hasChangePosition
 - NodeItem, 169
- m_hasId
 - AddEdgeCommand, 15
 - AddNodeAndEdgeCommand, 18
 - AddNodeCommand, 21
- m_hasNegativeCycle
 - BellmanFord, 54
- m_hasWeight
 - EdgeItem, 85
- m_helpAction
 - MenuToolBar, 154
- m_helpBtn
 - AlgorithmTab, 41
- m_id
 - AddNodeCommand, 21
 - Edge, 77
 - EdgeSnapshot, 87
 - Node, 161
 - NodeSnapshot, 170
 - Observable< T >, 173
- m_idEndNode
 - Edge, 77
- m_idStartNode
 - Edge, 77
- m_ignoreNextMouseRelease
 - NodeItem, 169
- m_imagePath
 - MainWindow::TabInfo, 190
- m_incomingNumbering
 - Tarjan, 194
- m_isDirected

- SaveFileWorker, 185
- m_isImageModified
 - MainWindow::TabInfo, 190
- m_isModified
 - MainWindow::TabInfo, 190
- m_isWeighted
 - SaveFileWorker, 185
- m_label
 - NodelItem, 169
- m_lastMousePos
 - GraphView, 139
- m_leftPlaceholder
 - GraphEditor, 120
- m_legendContainer
 - AlgorithmTab, 41
- m_legendLayout
 - AlgorithmTab, 41
- m_legendScroll
 - AlgorithmTab, 41
- m_lowLink
 - Tarjan, 194
- m_menuToolBar
 - MainWindow, 150
- m_message
 - AlgorithmError, 31
- m_minDistance
 - BellmanFord, 54
- m_modelEdge
 - EdgelItem, 85
- m_modelNode
 - NodelItem, 169
- m_name
 - Node, 162
- m_newId
 - AddNodeAndEdgeCommand, 18
- m_newPos
 - MoveNodeCommand, 157
- m_nextBtn
 - AlgorithmTab, 41
- m_node
 - AlgorithmStep, 32
- m_nodeId
 - EditNodeNameCommand, 97
 - Graph, 106
 - MoveNodeCommand, 157
 - RemoveNodeCommand, 182
- m_nodelItems
 - GraphScene, 133
- m_nodes
 - ClearGraphCommand, 60
 - Graph, 106
- m_nodeSelected
 - NodelItem, 169
- m_nodeSizeSlider
 - GraphEditTab, 125
- m_noInputLabel
 - AlgorithmTab, 41
- m_numComponents
 - Tarjan, 194
- m_numOfEdges
 - Graph, 107
- m_numOfNodes
 - Graph, 107
- m_observerId
 - EdgelItem, 85
 - NodelItem, 169
- m_observers
 - Observable< T >, 173
- m_oldCenter
 - NodelItem, 169
- m_oldPos
 - MoveNodeCommand, 157
- m_oldState
 - GraphView, 139
- m_oldText
 - EditableTextItem, 91
- m_onStack
 - Tarjan, 194
- m_openGraphAction
 - MenuToolBar, 154
- m_order
 - BFS, 58
 - DFS, 64
- m_panBtn
 - GraphEditTab, 125
- m_path
 - AStar, 50
- m_pauseBtn
 - AlgorithmTab, 42
- m_penWidth
 - EdgelItem, 85
- m_playBtn
 - AlgorithmTab, 42
- m_pos
 - AddNodeAndEdgeCommand, 18
 - AddNodeCommand, 21
- m_position
 - Node, 162
- m_prevBtn
 - AlgorithmTab, 42
- m_prevEdgeState
 - AlgorithmStep, 33
- m_previousState
 - GraphScene, 134
- m_prevNodeState
 - AlgorithmStep, 33
- m_radius
 - NodelItem, 169
- m_redoBtn
 - GraphEditTab, 125
- m_redoStack
 - AlgorithmController, 30
- m_removeBtn
 - GraphEditTab, 126
- m_restartBtn
 - AlgorithmTab, 42

- m_resultString
 - Algorithm, 24
 - AlgorithmController, 30
 - BellmanFord, 54
 - Dijkstra, 68
- m_saveGraphAction
 - MenuToolBar, 154
- m_saveImageAction
 - MenuToolBar, 154
- m_scene
 - GraphController, 115
- m_serializable
 - SaveFileWorker, 185
- m_serializer
 - LoadFileWorker, 145
 - MainWindow, 150
 - SaveFileWorker, 185
- m_settings
 - ThemeManager, 197
- m_shapeStroke
 - EdgeItem, 86
- m_skewness
 - DirectedEdgeItem, 73
- m_sorted
 - Kahn, 143
- m_sourceNode
 - EdgeItem, 86
- m_sourcePoint
 - EdgeItem, 86
- m_start
 - AlgorithmWorker, 46
- m_startLabel
 - AlgorithmTab, 42
- m_startNode
 - AlgorithmTab::AlgorithmConfig, 25
- m_startNodeEdit
 - AlgorithmTab, 42
- m_startRow
 - AlgorithmTab, 42
- m_state
 - AlgorithmTab, 42
 - Edge, 77
 - GraphScene, 134
 - GraphView, 139
 - Node, 162
- m_steps
 - Algorithm, 24
 - AlgorithmController, 30
- m_tabs
 - MainWindow, 150
- m_tabWidget
 - MainWindow, 150
- m_themeManager
 - MainWindow, 150
- m_timer
 - AlgorithmTab, 42
- m_to
 - AddEdgeCommand, 16
 - AlgorithmStep, 33
 - EdgeSnapshot, 87
 - RemoveEdgeCommand, 179
- m_totalCost
 - AStar, 50
- m_totalWeight
 - Prim, 177
- m_tourOrder
 - Tarjan, 194
- m_type
 - AlgorithmError, 31
 - AlgorithmStep, 33
- m_ui
 - MainWindow, 150
- m_undoBtn
 - GraphEditTab, 126
- m_undoStack
 - AlgorithmController, 30
 - GraphController, 115
 - GraphEditor, 120
- m_view
 - GraphEditor, 120
- m_visited
 - BFS, 58
 - DFS, 64
- m_weight
 - AddEdgeCommand, 16
 - Edge, 78
 - EdgeItem, 86
 - EdgeSnapshot, 87
 - RemoveEdgeCommand, 179
- m_worker
 - AlgorithmTab, 42
- m_x
 - NodeSnapshot, 170
 - RemoveNodeCommand, 182
- m_y
 - NodeSnapshot, 170
 - RemoveNodeCommand, 183
- m_zoomFactor
 - GraphView, 139
- m_zoomInBtn
 - GraphEditTab, 126
- m_zoomOutBtn
 - GraphEditTab, 126
- main
 - main.cpp, 294
- main.cpp
 - main, 294
- MainWindow, 146
 - ~MainWindow, 148
 - closeEvent, 148
 - connectGraphModifiedSignal, 148
 - initMenuToolBar, 149
 - m_menuToolBar, 150
 - m_serializer, 150
 - m_tabs, 150
 - m_tabWidget, 150

- [m_themeManager](#), 150
 - [m_ui](#), 150
 - [MainWindow](#), 148
 - [onChangeThemeTriggered](#), 149
 - [onCreateGraphTriggered](#), 149
 - [onGraphLoadedNewTab](#), 149
 - [onGraphLoadFailed](#), 149
 - [onHelpTriggered](#), 149
 - [onOpenGraphTriggered](#), 149
 - [onSaveGraphTriggered](#), 149
 - [onSaveImageTriggered](#), 149
 - [showStartPage](#), 150
- [MainWindow::TabInfo](#), 188
 - [m_editor](#), 190
 - [m_filePath](#), 190
 - [m_imagePath](#), 190
 - [m_isImageModified](#), 190
 - [m_isModified](#), 190
- [makeLegendItem](#)
 - [AlgorithmTab](#), 39
- [MaxNodeScale](#)
 - [AppConstants](#), 11
- [MenuToolBar](#), 151
 - [~MenuToolBar](#), 153
 - [changeThemeAction](#), 153
 - [createGraphAction](#), 153
 - [helpAction](#), 153
 - [m_changeThemeAction](#), 154
 - [m_createGraphAction](#), 154
 - [m_helpAction](#), 154
 - [m_openGraphAction](#), 154
 - [m_saveGraphAction](#), 154
 - [m_saveImageAction](#), 154
 - [MenuToolBar](#), 153
 - [openGraphAction](#), 153
 - [saveGraphAction](#), 153
 - [saveImageAction](#), 153
 - [setOptionShortcuts](#), 154
 - [setStyle](#), 154
- [MinNodeScale](#)
 - [AppConstants](#), 11
- [modelEdge](#)
 - [EdgeItem](#), 83
- [modelNode](#)
 - [NodeItem](#), 167
- [mouseDoubleClickEvent](#)
 - [EdgeItem](#), 84
 - [GraphScene](#), 132
 - [GraphView](#), 138
- [mouseMoveEvent](#)
 - [GraphView](#), 138
 - [NodeItem](#), 167
- [mousePressEvent](#)
 - [EdgeItem](#), 84
 - [GraphScene](#), 132
 - [GraphView](#), 138
 - [NodeItem](#), 167
- [mouseReleaseEvent](#)
 - [GraphView](#), 138
 - [NodeItem](#), 167
- [moveNode](#)
 - [GraphController](#), 113
- [MoveNodeCommand](#), 155
 - [m_c](#), 157
 - [m_newPos](#), 157
 - [m_nodeId](#), 157
 - [m_oldPos](#), 157
 - [MoveNodeCommand](#), 157
 - [redo](#), 157
 - [undo](#), 157
- [moveNodeNoHistory](#)
 - [GraphController](#), 113
- [moveNodeRequest](#)
 - [GraphScene](#), 132
 - [NodeItem](#), 167
- [NegativeEdgeWeights](#)
 - [AlgorithmError.h](#), 221
- [nextStep](#)
 - [AlgorithmController](#), 29
- [Node](#), 158
 - [~Node](#), 160
 - [getComponentColor](#), 160
 - [getId](#), 160
 - [getName](#), 161
 - [getPosition](#), 161
 - [getState](#), 161
 - [m_componentColor](#), 161
 - [m_id](#), 161
 - [m_name](#), 162
 - [m_position](#), 162
 - [m_state](#), 162
 - [Node](#), 160
 - [setComponentColor](#), 161
 - [setName](#), 161
 - [setPosition](#), 161
 - [setState](#), 161
- [Node.h](#)
 - [Active](#), 246
 - [AssignedComponent](#), 246
 - [Default](#), 246
 - [InPath](#), 246
 - [NodeState](#), 246
 - [TopologicalOrder](#), 246
 - [UpdatedDistance](#), 246
 - [Visited](#), 246
- [NodeItem](#), 162
 - [~NodeItem](#), 166
 - [addEdge](#), 166
 - [boundingRect](#), 166
 - [calculateColor](#), 166
 - [edges](#), 166
 - [editNodeNameRequest](#), 166
 - [itemChange](#), 166
 - [label](#), 166
 - [m_borderWidth](#), 169
 - [m_edges](#), 169

- m_hasChangePosition, 169
- m_ignoreNextMouseRelease, 169
- m_label, 169
- m_modelNode, 169
- m_nodeSelected, 169
- m_observerId, 169
- m_oldCenter, 169
- m_radius, 169
- modelNode, 167
- mouseMoveEvent, 167
- mousePressEvent, 167
- mouseReleaseEvent, 167
- moveNodeRequest, 167
- NodeItem, 166
- nodeSelected, 167
- onNameChanged, 167
- onNodeUpdated, 167
- paint, 167
- radius, 168
- removeEdge, 168
- setLabel, 168
- setModelNode, 168
- setNodeSelected, 168
- shape, 168
- updateNodePosition, 168
- updateSize, 168
- NodeScale
 - AppConstants, 11
- nodeSelected
 - NodeItem, 167
- NodeSnapshot, 170
 - m_id, 170
 - m_x, 170
 - m_y, 170
- NodeState
 - Node.h, 246
- NoPathFound
 - AlgorithmError.h, 221
- notifyObservers
 - Observable< T >, 172
- Observable< T >, 171
 - addObserver, 172
 - m_id, 173
 - m_observers, 173
 - notifyObservers, 172
 - Observer, 172
 - removeObserver, 173
- Observer
 - Observable< T >, 172
- oldText
 - EditableTextItem, 90
- onAddRequestTrigger
 - GraphEditor, 118
- onAlgorithmError
 - AlgorithmController, 29
- onChangeThemeTriggered
 - MainWindow, 149
- onClearRequestTrigger
 - GraphEditor, 118
- onCreateGraphTriggered
 - MainWindow, 149
- onEdgeSelectTrigger
 - GraphScene, 132
- onEdgeUpdated
 - EdgeItem, 84
- onEdgeWeightChanged
 - EdgeItem, 84
- onGraphLoadedNewTab
 - MainWindow, 149
- onGraphLoadFailed
 - MainWindow, 149
- onHelpTriggered
 - MainWindow, 149
- onNameChanged
 - NodeItem, 167
- onNodeSelectTrigger
 - GraphScene, 132
- onNodeUpdated
 - NodeItem, 167
- onOpenGraphTriggered
 - MainWindow, 149
- onPanRequestTrigger
 - GraphEditor, 119
- onRedoRequestTrigger
 - GraphEditor, 119
- onRemoveRequestTrigger
 - GraphEditor, 119
- onSaveGraphTriggered
 - MainWindow, 149
- onSaveImageTriggered
 - MainWindow, 149
- onUndoRequestTrigger
 - GraphEditor, 119
- onZoomInRequestTrigger
 - GraphEditor, 119
- onZoomOutRequestTrigger
 - GraphEditor, 119
- openGraphAction
 - MenuToolBar, 153
- operator!=
 - AlgorithmTab::AlgorithmConfig, 25
- operator==
 - AlgorithmTab::AlgorithmConfig, 25
- optionBtnSize
 - AppConstants, 11
- optionIconSize
 - AppConstants, 11
- paint
 - DirectedEdgeItem, 72
 - EdgeItem, 84
 - NodeItem, 167
 - UndirectedEdgeItem, 201
- PAN_ACTIVE
 - GraphView, 137
- PAN_IDLE
 - GraphView, 137

- panRequested
 - GraphEditTab, 124
- Paused
 - AlgorithmTab, 39
- Playing
 - AlgorithmTab, 39
- prevStep
 - AlgorithmController, 29
- Prim, 174
 - checkConditions, 176
 - execute, 176
 - m_totalWeight, 177
 - Prim, 176
 - prim, 176
 - resultString, 176
- prim
 - Prim, 176
- ProcessNode
 - AlgorithmStep.h, 223
- PURPLE
 - ThemeManager, 196
- radius
 - NodeItem, 168
- redo
 - AddEdgeCommand, 15
 - AddNodeAndEdgeCommand, 18
 - AddNodeCommand, 21
 - AlgorithmController, 29
 - ClearGraphCommand, 60
 - EditEdgeWeightCommand, 94
 - EditNodeNameCommand, 97
 - MoveNodeCommand, 157
 - RemoveEdgeCommand, 179
 - RemoveNodeCommand, 182
- redoRequested
 - GraphEditor, 119
 - GraphEditTab, 124
- Relaxed
 - Edge.h, 243
- RelaxEdge
 - AlgorithmStep.h, 223
- REMOVE
 - GraphScene, 130
- removeEdge
 - Graph, 106
 - GraphController, 113
 - GraphScene, 132
 - NodeItem, 168
 - UnweightedDirectedGraph, 205
 - UnweightedUndirectedGraph, 209
 - WeightedDirectedGraph, 213
 - WeightedUndirectedGraph, 217
- RemoveEdgeCommand, 177
 - m_c, 179
 - m_edgeld, 179
 - m_from, 179
 - m_to, 179
 - m_weight, 179
 - redo, 179
 - RemoveEdgeCommand, 179
 - undo, 179
- removeEdgeNoHistory
 - GraphController, 113
- removeEdgeNoHistoryById
 - GraphController, 113
- removeEdgeRequest
 - GraphScene, 132
- removeNode
 - Graph, 106
 - GraphController, 113
 - GraphScene, 132
- RemoveNodeCommand, 180
 - capture, 182
 - m_c, 182
 - m_edges, 182
 - m_nodeld, 182
 - m_x, 182
 - m_y, 183
 - redo, 182
 - RemoveNodeCommand, 182
 - undo, 182
- removeNodeNoHistory
 - GraphController, 114
- removeNodeRequest
 - GraphScene, 133
- removeObserver
 - Observable< T >, 173
- removeRequested
 - GraphEditTab, 124
- requestErrorDialog
 - AlgorithmController, 29
- reset
 - AlgorithmController, 29
- resetGraphState
 - AlgorithmStepApplier, 34
- resetScene
 - GraphController, 114
 - GraphScene, 133
- resetState
 - GraphView, 138
- restoreEdgeNoHistory
 - GraphController, 114
- resultReady
 - AlgorithmWorker, 45
- resultString
 - Algorithm, 23
 - AlgorithmController, 29
 - AStar, 50
 - BellmanFord, 54
 - BFS, 58
 - DFS, 64
 - Dijkstra, 67
 - FloydWarshall, 100
 - Kahn, 142
 - Prim, 176
 - Tarjan, 194

- run
 - AlgorithmWorker, 45
 - LoadFileWorker, 145
 - SaveFileWorker, 185
- RunState
 - AlgorithmTab, 38
- save
 - Serializer, 188
- SaveFileWorker, 183
 - failed, 185
 - finished, 185
 - m_filePath, 185
 - m_isDirected, 185
 - m_isWeighted, 185
 - m_serializable, 185
 - m_serializer, 185
 - run, 185
 - SaveFileWorker, 185
- saveGraphAction
 - MenuToolBar, 153
- savelmageAction
 - MenuToolBar, 153
- saveTheme
 - ThemeManager, 196
- scene
 - GraphController, 114
- sceneModified
 - GraphController, 114
- Selected
 - Edge.h, 243
- selectedConfig
 - AlgorithmTab, 39
- SelectEdge
 - AlgorithmStep.h, 223
- selectNode
 - GraphScene, 133
- Serializable, 186
 - ~Serializable, 187
 - fromVariant, 187
 - toVariant, 187
- Serializer, 187
 - ~Serializer, 188
 - load, 188
 - save, 188
- setAddSceneState
 - GraphController, 114
- setCenter
 - EditableTextItem, 90
- setComponentColor
 - Node, 161
- setEdgeWeightNoHistory
 - GraphController, 114
- setEdgeWeightNoHistoryByld
 - GraphController, 114
- setEditGraphSceneState
 - EditableTextItem, 91
- setEditGraphSceneTrigger
 - GraphScene, 133
- setGraph
 - AlgorithmStepApplier, 34
 - GraphController, 115
- setGraphEditsAllowed
 - GraphEditTab, 124
- setHasWeight
 - EdgeItem, 84
- setLabel
 - NodeItem, 168
- setModelEdge
 - EdgeItem, 84
- setModelNode
 - NodeItem, 168
- setName
 - Node, 161
- setNodeNameNoHistory
 - GraphController, 115
- setNodeSelected
 - NodeItem, 168
- setOldText
 - EditableTextItem, 91
- setOptionShortcuts
 - MenuToolBar, 154
- setPosition
 - Node, 161
- setRedoEnabled
 - GraphEditTab, 124
- setRemoveSceneState
 - GraphController, 115
- setResultString
 - AlgorithmController, 30
- setState
 - Edge, 77
 - GraphScene, 133
 - GraphView, 138
 - Node, 161
- setStyle
 - MenuToolBar, 154
- setTheme
 - ThemeManager, 196
- setUndoEnabled
 - GraphEditTab, 124
- setWeight
 - Edge, 77
 - EdgeItem, 84
- shape
 - DirectedEdgeItem, 72
 - EdgeItem, 85
 - NodeItem, 168
 - UndirectedEdgeItem, 201
- showAlgorithmErrorDialog
 - AlgorithmTab, 39
- showStartPage
 - MainWindow, 150
- Source/Algorithm/Algorithm.cpp, 285
- Source/Algorithm/AlgorithmStepApplier.cpp, 285
- Source/Algorithm/AStar.cpp, 286
- Source/Algorithm/BellmanFord.cpp, 286

- Source/Algorithm/BFS.cpp, 286
- Source/Algorithm/DFS.cpp, 287
- Source/Algorithm/Dijkstra.cpp, 287
- Source/Algorithm/FloydWarshall.cpp, 287
- Source/Algorithm/Kahn.cpp, 288
- Source/Algorithm/Prim.cpp, 288
- Source/Algorithm/Tarjan.cpp, 289
- Source/Controllers/AlgorithmController.cpp, 289
- Source/Controllers/GraphController.cpp, 289
- Source/Graph/Edge.cpp, 290
- Source/Graph/Graph.cpp, 290
- Source/Graph/Node.cpp, 291
- Source/Graph/UnweightedDirectedGraph.cpp, 291
- Source/Graph/UnweightedUndirectedGraph.cpp, 292
- Source/Graph/WeightedDirectedGraph.cpp, 292
- Source/Graph/WeightedUndirectedGraph.cpp, 293
- Source/main.cpp, 293
- Source/MainWindow.cpp, 294
- Source/Serialization/Serializer.cpp, 294
- Source/Threads/AlgorithmWorker.cpp, 295
- Source/Threads/LoadFileWorker.cpp, 295
- Source/Threads/SaveFileWorker.cpp, 295
- Source/UI/AlgorithmTab.cpp, 296
- Source/UI/GraphEditor.cpp, 296
- Source/UI/GraphEditTab.cpp, 296
- Source/UI/MenuToolBar.cpp, 297
- Source/UI/ThemeManager.cpp, 297
- Source/View/DirectedEdgeItem.cpp, 298
- Source/View/EdgeItem.cpp, 298
- Source/View/EditableTextItem.cpp, 299
- Source/View/GraphScene.cpp, 299
- Source/View/GraphView.cpp, 300
- Source/View/NodeItem.cpp, 300
- Source/View/UndirectedEdgeItem.cpp, 301
- startEditing
 - EditableTextItem, 91
- startNode
 - Edge, 77
- StartNodeMissing
 - AlgorithmError.h, 221
- startPageTitle
 - AppConstants, 11
- startTimerForPlay
 - AlgorithmTab, 40
- State
 - GraphScene, 129
 - GraphView, 137
- state
 - GraphView, 138
- stepsReady
 - AlgorithmWorker, 45
- stepToEdgeState
 - AlgorithmStepApplier, 35
- stepToNodeState
 - AlgorithmStepApplier, 35
- StepType
 - AlgorithmStep.h, 223
- styleSheet
 - ThemeManager, 196
- Tarjan, 190
 - checkConditions, 193
 - execute, 193
 - init, 193
 - m_arrivalTime, 194
 - m_components, 194
 - m_incomingNumbering, 194
 - m_lowLink, 194
 - m_numComponents, 194
 - m_onStack, 194
 - m_tourOrder, 194
 - resultString, 194
 - Tarjan, 193
 - tarjan, 194
- tarjan
 - Tarjan, 194
- textCommitted
 - EditableTextItem, 91
- Theme
 - ThemeManager, 196
- ThemeManager, 195
 - currentTheme, 196
 - DARK, 196
 - LIGHT, 196
 - loadTheme, 196
 - m_currentTheme, 197
 - m_settings, 197
 - PURPLE, 196
 - saveTheme, 196
 - setTheme, 196
 - styleSheet, 196
 - Theme, 196
 - ThemeManager, 196
- TopologicalOrder
 - Node.h, 246
- toVariant
 - Graph, 106
 - Serializable, 187
- Ui, 11
- UndirectedEdgeItem, 197
 - boundingRect, 201
 - edgePath, 201
 - paint, 201
 - shape, 201
 - UndirectedEdgeItem, 201
- undo
 - AddEdgeCommand, 15
 - AddNodeAndEdgeCommand, 18
 - AddNodeCommand, 21
 - AlgorithmController, 30
 - AlgorithmStepApplier, 35
 - ClearGraphCommand, 60
 - EditEdgeWeightCommand, 94
 - EditNodeNameCommand, 97
 - MoveNodeCommand, 157
 - RemoveEdgeCommand, 179

- RemoveNodeCommand, 182
- undoRequested
 - GraphEditor, 119
 - GraphEditTab, 124
- undoStack
 - GraphController, 115
- UnweightedDirectedGraph, 202
 - ~UnweightedDirectedGraph, 204
 - addEdge, 205
 - isDirected, 205
 - isWeighted, 205
 - removeEdge, 205
 - UnweightedDirectedGraph, 204
- UnweightedUndirectedGraph, 206
 - ~UnweightedUndirectedGraph, 208
 - addEdge, 209
 - isDirected, 209
 - isWeighted, 209
 - removeEdge, 209
 - UnweightedUndirectedGraph, 208
- updateControls
 - AlgorithmTab, 40
- UpdatedDistance
 - Node.h, 246
- UpdateDistance
 - AlgorithmStep.h, 223
- updateLegendForAlgorithm
 - AlgorithmTab, 40
- updateNodePosition
 - GraphController, 115
 - NodeItem, 168
- updateNodeScaling
 - GraphScene, 133
- updateSize
 - EdgeItem, 85
 - NodeItem, 168
- updateUiForAlgorithm
 - AlgorithmTab, 40
- Visited
 - Node.h, 246
- VisitNode
 - AlgorithmStep.h, 223
- weight
 - EdgeItem, 85
- WeightedDirectedGraph, 210
 - ~WeightedDirectedGraph, 212
 - addEdge, 213
 - isDirected, 213
 - isWeighted, 213
 - removeEdge, 213
 - WeightedDirectedGraph, 212
- WeightedUndirectedGraph, 214
 - ~WeightedUndirectedGraph, 216
 - addEdge, 217
 - isDirected, 217
 - isWeighted, 217
 - removeEdge, 217
 - WeightedUndirectedGraph, 216
- wheelEvent
 - GraphView, 138
- windowHeight
 - AppConstants, 11
- windowMinHeight
 - AppConstants, 11
- windowMinWidth
 - AppConstants, 11
- windowWidth
 - AppConstants, 11
- zoomIn
 - GraphView, 138
- zoomInRequested
 - GraphEditTab, 125
- zoomOut
 - GraphView, 139
- zoomOutRequested
 - GraphEditTab, 125